

# Contradiction Fields: CRT Rigidity, Prime Matrix Geometry, and Zeta-Zero Escape

Project manuscript

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# Abstract

This monograph draft unifies two contradiction-field research programs. The first studies prime existence in CRT prime matrices through row, column, diagonal, anchor, and rough-composite rigidity. The second studies an off-critical zero of the Riemann zeta function through a global contradiction-field ledger for prime anomalies. The common method is to convert a hypothetical counterexample into a structured covering field, decompose it into local rigidity layers, and exclude all free escape channels by capacity, periodicity, and no-cycle bookkeeping.

This is a synthesis and dependency manuscript. It does not assert that the Riemann Hypothesis or the prime-matrix row/column theorem has already been accepted as a final unconditional theorem. Each claim is marked by status in the accompanying status table: proved in manuscript, reduced to a structured input, external, computationally certified, or requiring independent referee verification.



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# Chapter 1

## Guiding Principle: The Contradiction Field

A contradiction field is a structured ledger attached to a hypothetical counterexample. Instead of estimating a single exceptional set, it records every possible explanation for the counterexample as a named channel. A proof closes when every channel is either absorbed, forced to descend in a finite potential, transferred with source deletion, or contradicts a global capacity lower/upper bound.

The common rigidities used throughout this manuscript are:

1. CRT skeletons and nonzero-residue balance;
2. local short-window non-reuse of large prime factors;
3. small-prime shell migration and diagonal locking;
4. rough-composite and double-anchor decompositions;
5. capacity ledgers after baseline subtraction;
6. no-cycle potentials for internal escape mechanisms.





## Chapter 2

# Critical-Density Feedback Bridge

This chapter records the critical-density principle used to organize the current frontier. It is a bridge from heuristic insight to proof obligations, not a replacement for the remaining row/column, two-point, or RH exit proofs.

**Definition 2.1** (Self-sieving density model). Let a self-sieving set have local density

$$\rho(x) \sim \frac{C}{(\log x)^\alpha}.$$

Its previous elements impose a first-order one-residue sieve intensity

$$S(x) = \sum_{a \leq x, a \in A} \frac{1}{a} \sim \int_2^x \frac{\rho(t)}{t} dt.$$

The associated model survivor density is  $\exp(-S(x))$ .

**Lemma 2.2** (Critical fixed point). *Inside the self-sieving density model, the only logarithmic fixed-point exponent is*

$$\rho(x) \asymp \frac{1}{\log x}.$$

*More precisely, if  $\rho(x) \sim C/(\log x)^\alpha$  and the model survivor density is required to have the same logarithmic order as  $\rho(x)$ , then  $\alpha = 1$ ; at that layer, exponent matching gives  $C = 1$ .*

*Proof.* If  $\alpha > 1$ , then

$$\int_2^x \frac{C dt}{t(\log t)^\alpha}$$

converges, so the model survivor density stays bounded away from zero while  $C/(\log x)^\alpha$  tends to zero. This is not self-consistent.

If  $\alpha < 1$ , then the same integral is

$$\frac{C}{1-\alpha} (\log x)^{1-\alpha} + O(1),$$

so the survivor density is

$$\exp\{-c(\log x)^{1-\alpha} + O(1)\},$$

which is smaller than any fixed negative power of  $\log x$ . This is again not of order  $C/(\log x)^\alpha$ .

Thus  $\alpha = 1$ . Then  $S(x) \sim C \log \log x$ , and the survivor density is  $(\log x)^{-C+o(1)}$ . Matching the logarithmic exponent with  $C/\log x$  forces  $C = 1$  at the model level.  $\square$

**Remark 2.3** (Proof boundary). The lemma is a fixed-point calculation for the self-sieving model. It does not by itself prove a short-interval prime, a prime pair, or RH. To become a theorem, the fixed-point pressure must be upgraded into local lower bounds, new-modulus anti-concentration estimates, or controlled-exit inequalities.

**Reduction-closed Statement 2.4** (Critical feedback bridge for the three programs). *The critical-density principle routes the three current frontiers as follows.*

1. *For the prime-matrix row/column program, a fully covered row or column is a local collapse below the  $1/\log x$  fixed point. Such a collapse must be explained by fixed CRT concentration, by moving high-factor slots that pay Rankin/SAE mass, or by a failure of the local survivor lower bound. The active narrow gate is*

*AffineTwinFillResidueArrivalBoundOrFillCatchUpMassPDECExclusion.*

2. *For the two-point secondary sieve, the corresponding fixed point is the two-residue product*

$$\prod_{q \leq y} \left(1 - \frac{\nu_q(w)}{q}\right) \asymp \frac{\mathfrak{S}(w)}{(\log y)^2}.$$

*The remaining theorem-level input is not a one-point density estimate but the true-residual new-modulus correlation bound in the I3-Core/TRC package.*

3. *For the RH contradiction field, an off-critical zero creates a smooth prime anomaly relative to the same  $1/\log x$  fixed point. The anomaly must pass through the recorded sparse, dense, tail, internal, and global exits; the manuscript can be promoted only after those controlled exits are independently verified under one load convention.*

*Status and reference.* This is a routing statement. The detailed version is archived as

docs/monograph/three-claims-critical-density-feedback-frontier.md.

It identifies the next prime-matrix input as a critical fill-arrival anti-overfilling theorem: persistent AffineTwin threshold crossing must either reuse a residue and become a reset/ColumnCRT/PDEC certificate, or introduce new fill residues whose Rankin/SAE mass exceeds the critical-density carrying capacity. That final carrying-capacity inequality is the new proof obligation; it is not asserted here as already proved.  $\square$

**Reduction-closed Statement 2.5** (Critical fill-arrival pressure packet frontier). *In the AffineTwin branch, write  $g = q - 2$ ,  $f = q$ , and let  $A_g, A_f$  be the generator-side and fill-side residue counts. The safe integer gate is*

$$A_g A_f \leq T_q := \lfloor \sqrt{gf} \rfloor.$$

*For a threshold-crossing route with final counts  $A'_g, A'_f$ , crossing is equivalent to*

$$A'_g A'_f > T_q, \quad \text{hence} \quad \frac{(A'_g)^2}{g} \frac{(A'_f)^2}{f} > 1.$$

*Thus fill catch-up is not merely a raw residue-arrival event. It is a threshold-relevant pressure packet: for fixed  $A'_g$  the minimum fill target is*

$$A'_{f,\min}(A'_g) = \left\lceil \frac{T_q}{A'_g} \right\rceil + 1.$$

*Consequently a persistent row/column counterexample in this branch must enter one of four named exits: repeated-residue reset/ColumnCRT PDEC, fixed pressure packet recurrence PDEC, moving pressure packet SAE/Rankin, or non-sparse AffineTwin pressure demand.*

*Status and reference.* The product gate and the pressure form are algebraic equivalents. The minimum fill target follows by solving  $A'_g A'_f > T_q$  in integers. The repeated-residue exit is the previous fill-catchup dichotomy; the fixed-packet exit is the AffineTwin slot phase lock, where the combined modulus  $q(q-2)$  exceeds the common phase support; and the moving-packet exit is the existing moving-family SAE/ColumnCRT split. The remaining unproved input is the global pressure-packet carrying ceiling, recorded in

`docs/monograph/prime-matrix-critical-fill-arrival-anti-overfilling-router.md.`

□

**Reduction-closed Statement 2.6** (Pressure-packet carrying ceiling). *Let  $M_q = A_g A_f$  be the pressure-packet multiplicity for an AffineTwin modulus  $q$ , so that  $q$  and  $q-2$  are prime. The packet SAE mass is*

$$\mu_q = \frac{M_q}{q(q-2)}.$$

*If the square-root packet gate*

$$M_q^2 \leq q(q-2)$$

*holds, then*

$$\mu_q \leq \frac{1}{\sqrt{q(q-2)}} \leq \frac{1}{q-2}.$$

*Consequently, any standard Brun–Selberg upper-bound sieve of the form*

$$\Pi_2(X) := \#\{q \leq X : q, q-2 \text{ prime}\} \leq C \frac{X}{(\log X)^2}$$

*implies by partial summation that the moving pressure-packet tail is summable. If the square-root packet gate fails, then*

$$M_q^2 > q(q-2),$$

*which is exactly the named SuperSqrt/PressureProduct PDEC exit.*

*Status and reference.* The first inequalities are immediate from the square-root gate. The Brun–Selberg bound gives

$$\sum_{\substack{q \geq Q \\ q, q-2 \text{ prime}}} \frac{1}{q} \leq \frac{\Pi_2(Q)}{Q} + \int_Q^\infty \frac{\Pi_2(t)}{t^2} dt \ll \frac{1}{(\log Q)^2} + \frac{1}{\log Q},$$

and  $1/(q-2) \ll 1/q$  for  $q \geq 5$ . Thus all packets satisfying the square-root gate are absorbed by the reciprocal ceiling. The complementary case is the previous pressure-product inequality with product  $> 1$ , hence the registered PDEC/ColumnCRT exit. The detailed route is recorded in the pressure-packet carrying-ceiling router under `docs/monograph/`. This still leaves two explicit tasks before final row/column promotion. First, internalize or cite the Brun–Selberg reciprocal ceiling under the monograph’s convention. Second, exclude the SuperSqrt/PressureProduct PDEC family. □

**Reduction-closed Statement 2.7** (Critical-error frontier). *After the critical-density fixed point has been fixed, the remaining frontier is best measured by a normalized critical error*

$$\mathcal{E} = \frac{\text{actual load}}{\text{critical capacity}} - 1.$$

*A positive critical error is not allowed to remain an anonymous remainder in the contradiction field. In the prime-matrix AffineTwin branch it is the pressure product*

$$\frac{A_g^2}{g} \frac{A_f^2}{f} - 1;$$

*in the two-point sieve it is the excess of the large-incidence mean above the Buchstab semiprime transition constant  $K(\alpha)$ , together with the terminal gap to the threshold 1; in the RH branch it is the off-critical zero load after zero-frequency CRT subtraction, normalized by the capacity of each controlled exit. Thus every positive critical error must become a fixed CRT/PDEC defect, a moving SAE/Rankin or analytic-average packet, a denominator-floor defect, or a named controlled exit.*

*Status and reference.* This is a frontier reduction and audit rule, not a final theorem. The detailed ledger is recorded in

`docs/monograph/three-claims-critical-error-frontier-audit.md`.

It sharpens the current next tasks to three interfaces: a non-PDEC square-root phase-support bound for the prime-matrix pressure packets, a critical denominator-floor check for the two-point BMD-to-TLI transfer, and a common baseline-subtracted load convention for every RH controlled exit.  $\square$

**Reduction-closed Statement 2.8** (Actual-packet square-root phase support). *In the prime-matrix AffineTwin branch, distinguish the formal product*

$$M_q^{\text{form}} = A_g A_f$$

*from the actual non-PDEC packet count  $N_q$ . The latter counts only generator-fill packets whose two slot conditions are compatible, whose common phase support is nonempty, and which have not already become a reset or ColumnCRT/PDEC event. For a primitive AffineTwin slot pair with  $q \geq 13$ , the recorded depth identities give common support width*

$$W_q = \frac{q+9}{2}.$$

*Since*

$$W_q \leq \sqrt{q(q-2)}$$

*for all  $q \geq 13$ , and since a repeated actual packet projection inside this support is by definition a PDEC/ColumnCRT event, every non-PDEC primitive packet family satisfies*

$$N_q \leq W_q \leq \sqrt{q(q-2)}.$$

*Thus the square-root gate is closed for actual primitive packets. A remaining formal SuperSqrt failure must be either product-accounting looseness ( $M_q^{\text{form}}$  overcounts  $N_q$ ), a projection-collision PDEC, or a primitive support escape.*

*Status and reference.* The inequality follows by squaring:

$$4q(q-2) - (q+9)^2 = 3q^2 - 26q - 81 \geq 0$$

for  $q \geq 13$ . The injection of actual packets into the primitive support is the non-PDEC projection rule: two actual packets with the same primitive projection would be a repeated fixed CRT phase and therefore a reset/ColumnCRT/PDEC event. The detailed audit is archived in

`docs/monograph/prime-matrix-nonpdec-sqrt-phase-support-reduction.md`.

This does not yet prove the row/column theorem. It reduces the remaining Prime-Matrix obligation to product-accounting tightening and to excluding or routing primitive-support escape packets.  $\square$

**Reduction-closed Statement 2.9** (Formal envelope versus actual load). *Across the three programs, a critical contradiction can only be triggered by actual load, not by a formal envelope. If  $F$  is a formal upper envelope,  $A$  is the actual load, and  $C$  is the critical capacity, then*

$$0 \leq A \leq F.$$

*The case  $F \leq C$  is absorbed. The case  $F > C$  but  $A \leq C$  is only an accounting issue and requires tightening the envelope to the actual load. Only the case  $A > C$  can force a structural exit. In the prime-matrix AffineTwin branch this is the distinction between  $(A_g A_f)^2$  and  $N_q^2$ ; in the two-point sieve it is the distinction between model numerator/denominator envelopes and the actual ratio*

$$|U_Y(I)|^{-1} \sum_{x \in U_Y(I)} D_Y^P(x);$$

*in the RH branch it is the distinction between routed atom envelopes and source-deleted final loads in the GEE ledger.*

*Status and reference.* This is a status-discipline and reduction rule. It prevents any of the three frontiers from closing by a coarse upper envelope alone. The detailed cross-program audit is recorded in

`docs/monograph/three-claims-formal-to-actual-critical-load-frontier.md`.

The resulting active interfaces are actual packet enumeration for the prime-matrix branch, actual denominator-floor control for the two-point BMD-to-TLI transfer, and actual final-load normalization for the RH controlled exits.  $\square$

**Reduction-closed Statement 2.10** (Actual-load closure contracts). *The current frontier is organized into three actual-load contracts. In the prime-matrix branch, PM-ALC requires that the actual AffineTwin packet count  $N_q$  replace the formal product  $A_g A_f$  in the critical load; the current machine contract records candidate  $q = 31, 43, 103$ , total formal product upper bound 40, but only one actual packet in the current sweep. In the two-point branch, TP-ALC requires both a BMD/KLS numerator bound and a denominator floor for  $|U_Y(I)|$  in the same Buchstab convention, with*

$$K(\alpha) + \varepsilon(\alpha) < 1 - \delta(\alpha).$$

*In the RH branch, RH-ALC requires every controlled exit to be written with baseline-subtracted source-deleted final load. Failure of any contract is not an anonymous error; it is a named PDEC/SAE/parity/control exit obligation.*

*Status and reference.* The PM current-sweep contract is generated by

`experiments/prime_matrix_affine_twin_actual_packet_contract.py.`

The unified contract ledger is archived in

`docs/monograph/three-claims-actual-load-closure-contracts.md.`

This is a reduction and evidence-organizing step. It does not close the row/column theorem, the two-point theorem, or RH.  $\square$

**Reduction-closed Statement 2.11** (AffineTwin formal-pair pruning). *In the current prime-matrix AffineTwin sweep, the formal product universe has total size 40, while the actual packet count is 1. The difference is now fully explained at the current level: 11 formal residue pairs are removed by an exact CRT-window emptiness check, and 28 formal residue pairs have no matching gap-fill source materialization in the required orientation. Thus the current formal-to-actual gap is not an unaccounted critical load.*

*Status and reference.* The pruning audit is generated by

`experiments/prime_matrix_affine_twin_formal_pair_pruning_audit.py.`

For  $q = 31$ , the audit enumerates the  $3 \cdot 4$  formal residue pairs and finds only the pair  $(19, 8)$  with CRT representative 2687 in the support interval  $[2669, 2688]$ . The remaining current candidates  $q = 43, 103$  have no matching gap-fill source in the required direction. The global obligation remains: future formal pairs must be routed to CRT-window emptiness, source materialization failure as PDEC/SAE, or primitive-support escape. This does not close the row/column theorem.  $\square$

**Reduction-closed Statement 2.12** (AffineTwin source materialization gate). *The current source-materialization failures in the AffineTwin audit are not unclassified missing mass. An actual AffineTwin packet must have a matching gap-fill source with gap  $q$ , generator modulus  $q - 2$ , fill modulus  $q$ , the prescribed orientation, and affine delay*

$$p_{\text{fill}} - p_{\text{gen}} = \frac{11q - 21}{4}.$$

*In the current sweep,  $q = 31$  is the only candidate passing this gate. The candidate  $q = 43$  has a same-gap source, but it uses generator 47, orientation plus-to-minus, and delay 74 instead of generator 41, orientation minus-to-plus, and delay 113. The candidate  $q = 103$  has no gap-fill source. Thus the 28 source-unmaterialized formal pairs from the pruning audit are fully classified at the current level.*

*Status and reference.* The source gate audit is generated by

`experiments/prime_matrix_affine_twin_source_materialization_gate_audit.py.`

It records 16 blocked formal pairs for the wrong-source  $q = 43$  row and 12 blocked formal pairs for the no-source  $q = 103$  row. The remaining global obligation is to prove that every future source-gate failure routes to SameGapWrongSource-PDEC/SAE or NoGapSource-PDEC/SAE, rather than contributing to actual load. This does not close the row/column theorem.  $\square$

**Reduction-closed Statement 2.13** (AffineTwin CRT window gap). *After the source gate, the remaining current AffineTwin formal pairs are controlled by a CRT window test. For  $q = 31$ , the exact source has generator modulus 29, fill modulus 31, combined modulus 899, and common phase support  $[2669, 2688]$  of width 20. Among the 12 formal residue pairs, exactly one pair,  $(19, 8)$ , has a CRT representative in the support, namely 2687. The other 11 pairs have nearest CRT representatives at positive distance from the support; the minimum such distance is 40.*

*Status and reference.* The CRT-window distance audit is generated by

`experiments/prime_matrix_affine_twin_crt_window_gap_audit.py`.

It upgrades current CRTWindowEmpty rows from a Boolean non-hit to positive phase-gap data. The global obligation remains to prove that future source-materialized non-hits have a uniform CRT window gap, or otherwise route the failure to the named PDEC/SAE exits recorded in the audit. This does not close the row/column theorem.  $\square$

**Reduction-closed Statement 2.14** (Window edge collision). *The current CRT-window non-hits can be expressed as residue-grid displacement requirements. For  $q = 31$ , the phase window  $[2669, 2688]$  induces 20 target residue pairs. Each of the 11 non-hit formal pairs has a nearest target pair; the least  $L^1$  displacement is 1, realized by*

$$(19, 9) \longrightarrow (19, 8).$$

*Thus the narrowest current edge-collision atom is a one-step fill-residue collision with the already realized actual pair  $(19, 8)$ .*

*Status and reference.* The edge-collision displacement audit is generated by

`experiments/prime_matrix_affine_twin_window_edge_collision_audit.py`.

It records two non-hit pairs whose nearest target is the existing actual pair, and nine non-hit pairs whose nearest target requires unused target-residue arrival. The global obligation remains to prove that such residue movements cannot be supplied persistently by a counterexample chain, or else to route the failure to residue-collision, target-arrival, or support-motion exits. This does not close the row/column theorem.  $\square$

**Reduction-closed Statement 2.15** (Existing actual CRT jump). *In the current AffineTwin edge-collision audit, the existing-actual branch is not a small boundary slip. The two non-hit formal pairs whose nearest target is the already realized pair  $(19, 8)$  have CRT phase jumps 120 and 435, whereas the common phase support has width 20. In particular the narrowest residue atom*

$$(19, 9) \longrightarrow (19, 8)$$

*has  $L^1$  residue displacement 1, but the fill-residue unit step in the combined CRT modulo  $29 \cdot 31$  is 435.*

*Status and reference.* The CRT jump audit is generated by

`experiments/prime_matrix_affine_twin_existing_actual_collision_jump_audit.py`.

It records generator and fill CRT unit steps 465 and 435, respectively, and verifies the exact signed jumps from the empty-pair nearest representatives to the existing actual point. This closes the current-sweep existing-actual collision jump ledger. The global obligation remains to prove that such jumps cannot be reset persistently by moving support, or to route the failure to repeated-residue, ColumnCRT, PDEC, or SAE exits. This does not close the row/column theorem.  $\square$

**Reduction-closed Statement 2.16** (Unused target arrival). *In the current AffineTwin edge-collision audit, the unused-target branch also has no hidden actual load. The 9 non-hit formal pairs whose nearest targets are unused collapse to 5 target residue pairs:*

$$(10, 30), (16, 5), (17, 6), (18, 7), (20, 9).$$

*None of these pairs belongs to the current formal product. To make them available, one must introduce new generator residues*

$$10, 16, 17, 18, 20$$

*and new fill residues*

$$5, 6, 7, 30.$$

*Moreover every corresponding CRT jump has absolute size at least 59, while the common phase support has width 20.*

*Status and reference.* The unused-target arrival audit is generated by

`experiments/prime_matrix_affine_twin_unused_target_arrival_audit.py.`

It records that all unused targets are outside the current formal product, all need a new side residue, all need a new generator residue, and all CRT jump identities exceed the support width in the current sweep. The global obligation remains to control new generator/fill residue arrival or route the failure to support-motion, PDEC, or SAE exits. This does not close the row/column theorem.  $\square$

**Reduction-closed Statement 2.17** (Support motion depth). *In the current AffineTwin CRT-window audit, support motion is not a free escape. The common support is the intersection of*

$$[2669, 2693] \quad \text{and} \quad [2659, 2688],$$

*namely [2669, 2688]. Therefore an empty CRT representative can enter by support motion only if the corresponding generator endpoint and shifted-fill endpoint are both released. In the current sweep all 11 empty pairs require such a two-endpoint release, and the common side depth required ranges from 58 to 435.*

*Status and reference.* The support-motion depth audit is generated by

`experiments/prime_matrix_affine_twin_support_motion_depth_audit.py.`

For the narrowest row (19, 12), the nearest CRT representative is 2629. To include it, the generator left depth must increase by 40 and the fill left depth by 30, so the total endpoint release is 70. The audit records the analogous depth-inflation vector for every empty row. This closes the current-sweep support-motion depth ledger. The global obligation remains to prove non-persistence of such synchronized depth inflation, or to route it to moving-support, endpoint-coupling, PDEC, or SAE exits. This does not close the row/column theorem.  $\square$

**Reduction-closed Statement 2.18** (Support motion primitive defect). *The current support-motion branch cannot be absorbed by the fixed  $q = 31$  AffineTwin primitive key. The slot-lock identities give*

$$d_g^- = 18, \quad d_g^+ = 6, \quad d_f^- = 28, \quad d_f^+ = 1.$$



*Every support-motion candidate requires a common side depth different from the corresponding generator and fill depths. The smallest defect is the row (19,12): it requires left common depth 58, producing defects*

$$58 - 18 = 40, \quad 58 - 28 = 30.$$

*Thus even the narrowest support-motion atom breaks both fixed primitive depth identities.*

*Status and reference.* The primitive-depth defect audit is generated by

`experiments/prime_matrix_affine_twin_support_motion_primitive_defect_audit.py`.

It records 11 support-motion candidates, all of which break both the generator and fill affine-depth identities for the fixed  $q = 31$  primitive key. This closes the current-sweep fixed-key absorption route. The global obligation remains to exclude moving primitive keys, or to route them to moving-key, depth-inflation, endpoint-coupling, PDEC, or SAE exits. This does not close the row/column theorem.  $\square$

**Reduction-closed Statement 2.19** (Moving-key depth formula). *Even if the AffineTwin key is allowed to move, same-orientation depth absorption is blocked in the current support-motion rows. On the lower side, a common depth  $D$  would have to satisfy*

$$(q + 5)/2 = D, \quad q - 3 = D,$$

*so  $q = 2D - 5$  and  $q = D + 3$  must agree. This occurs only at  $D = 8$ . The narrowest lower support-motion row has  $D = 58$ , giving  $q = 111$  from the generator formula and  $q = 61$  from the fill formula. On the upper side, the fill depth is fixed at 1, while the current required depths are at least 342.*

*Status and reference.* The moving-key depth formula audit is generated by

`experiments/prime_matrix_affine_twin_moving_key_depth_formula_audit.py`.

It records 11 support-motion candidates. None admits a common moving  $q$  with the same orientation and both depth formulas. This closes the current-sweep same-orientation moving-key absorption route. The global obligation remains to exclude orientation changes, source rematerialization, or other primitive-key migration, or to route them to PDEC/SAE/endpoint-coupling exits. This does not close the row/column theorem.  $\square$

**Reduction-closed Statement 2.20** (Moving-key source rematerialization). *The one-sided  $q$  values produced by the moving-key depth formulas also fail to become AffineTwin sources with the same orientation in the current sweep. Across the 11 support-motion rows the formulas produce 19 distinct one-sided candidate keys. Thirteen are composite. The six prime candidates*

$$61, 181, 293, 761, 1499, 1747$$

*all fail the AffineTwin prime/source gate: none satisfies the full same-orientation requirement that  $q$  and  $q - 2$  are prime,  $q \equiv 3 \pmod{4}$ , the delay  $(11q - 21)/4$  is integral, and a matching gap-fill source with generator  $q - 2$ , fill  $q$ , and sides minus-to-plus exists.*

*Status and reference.* The moving-key source-rematerialization audit is generated by

`experiments/prime_matrix_affine_twin_moving_key_source_remateriazation_audit.py`.

It records no exact rematerialized candidate. The narrowest prior depth atom 19:12 gives  $q = 111$  and  $q = 61$ ; 111 is composite, while 61 has  $61 \equiv 1 \pmod{4}$  and hence a nonintegral AffineTwin delay. The nearest available gap source has gap 59, with generator 61, fill 59, and the wrong orientation for a  $q = 61$  AffineTwin source. Thus the current-sweep same-orientation moving-key source-rematerialization route is closed. The global obligation remains to exclude orientation changes or persistent source rematerialization, or to route them to the named PDEC/SAE exits. This does not close the row/column theorem.  $\square$

**Reduction-closed Statement 2.21** (Endpoint-release critical error). *In the current AffineTwin sweep, support motion has no anonymous critical-error channel left. If an empty CRT representative is pulled into the primitive support by endpoint motion, the actual load is the total generator and shifted fill endpoint release. The critical capacity is the current common support width, which is 20. All 11 support-motion candidates are supercritical: the total endpoint-release load is 4929 against total capacity 220, and the narrowest atom 19:12 already has load 70, hence critical error*

$$\frac{70}{20} - 1 = \frac{50}{20} = 2.5.$$

*Status and reference.* The endpoint-release critical-error audit is generated by

`experiments/prime_matrix_affine_twin_endpoint_release_critical_error_audit.py`.

It records that every support-motion row requires both endpoints to release, that every endpoint-release load exceeds the support width, and that the fixed primitive-depth defect total matches the endpoint-release load row by row. Together with the moving-key source-rematerialization audit, this eliminates same-orientation source rematerialization as a hidden carrier for the positive local error. Thus, in the current sweep, the positive endpoint-release critical error is routed to EndpointReleaseCoupling-PDEC. The global obligation remains to exclude persistent endpoint-release coupling, orientation changes, and source rematerialization, or to route them to named exits. This does not close the row/column theorem.  $\square$

**Reduction-closed Statement 2.22** (Endpoint-release feedback loss). *Even after the current support-motion rows are given full geometric credit for the larger endpoint movement, a supercritical feedback loss remains. For each row, write*

$$L = \text{generator release} + \text{fill release}, \quad G = \max(\text{generator release}, \text{fill release}).$$

*The credited support gain is  $G$ , so the unrecovered feedback loss is*

$$L - G = \min(\text{generator release}, \text{fill release}).$$

*In the current audit, the total release load is 4929, the total geometric credit is 2512, and the remaining feedback loss is 2417 against support-width budget 220. The narrowest atom 19:12 has load 70, credit 40, and feedback loss 30, still larger than the support width 20.*

*Status and reference.* The endpoint-release feedback-loss audit is generated by

`experiments/prime_matrix_affine_twin_endpoint_release_feedback_loss_audit.py`.

It records that all 11 feedback losses are positive, equal to the smaller synchronized endpoint release, and still exceed the original support width. Thus the current sweep has no self-feedback absorption channel after geometric support gain is fully credited. The remaining positive residual is routed to EndpointReleaseFeedbackLoss-PDEC. The global obligation is to exclude that persistent feedback-loss family, or route it to the named moving-key, orientation-change, source-rematerialization, or SAE exits. This does not close the row/column theorem.  $\square$

**Reduction-closed Statement 2.23** (Endpoint-release feedback horizon). *The endpoint feedback loss has an equivalent phase-horizon form in the current sweep. A support-motion representative can be absorbed by endpoint self-feedback only if its distance from the current support window is at most*

$$\text{support width} + |\text{generator side depth} - \text{fill side depth}|.$$

*For all 11 support-motion rows the representative lies outside this horizon. The narrowest row is 19:12: its representative is at distance 40, while the support width is 20 and the side-depth skew is 10, so the feedback horizon is 30 and the phase-horizon surplus is 10.*

*Status and reference.* The endpoint-release feedback-horizon audit is generated by

`experiments/prime_matrix_affine_twin_endpoint_release_feedback_horizon_audit.py`.

It proves, for the current sweep, the identities

$$\text{feedback loss} = \text{window distance} - \text{side-depth skew},$$

and

$$\text{post-credit overcritical units} = \text{window distance} - (\text{support width} + \text{side-depth skew}).$$

Thus the residual critical error is exactly the distance by which the CRT representative crosses the feedback horizon. All rows have positive horizon surplus, so the current self-feedback absorption route is closed and the residual is routed to `EndpointReleaseFeedbackHorizon-PDEC`. The global obligation remains to exclude persistent horizon surplus, or to route it to orientation-change, source-rematerialization, or SAE exits. This does not close the row/column theorem.  $\square$

**Reduction-closed Statement 2.24** (Endpoint-release feedback-horizon slack). *In the same current sweep, the feedback-horizon surplus is exactly the missing side-depth skew needed for absorption. If a representative has distance  $d$  from the current support window and the support width is  $W$ , then absorption requires total side-depth skew at least*

$$\max(0, d - W).$$

*Hence the required extra skew is*

$$\max(0, d - W) - |\text{generator side depth} - \text{fill side depth}|,$$

*which equals the phase-horizon surplus row by row. The total required skew is 2292, while the current skew supplies only 95; the missing extra skew is 2197. The narrowest row is again 19:12, where distance 40 and support width 20 require total skew 20, but the current skew is 10.*

*Status and reference.* The endpoint-release feedback-horizon slack audit is generated by

`experiments/prime_matrix_affine_twin_endpoint_release_feedback_horizon_slack_audit.py`

It verifies that all 11 rows have positive required extra skew, that this extra skew equals the phase-horizon surplus, and that a pure orientation flip does not change the absolute skew. It also inherits from the moving-key source-rematerialization ledger that same-orientation exact rematerialization is absent. Thus the current anonymous slack absorption route is closed and the remaining obstruction is routed to `EndpointSkewGrowth-PDEC/OrientationChangingPrimitiveKey`-SAE. The global row/column theorem remains open until those named exits are excluded or absorbed.  $\square$

**Reduction-closed Statement 2.25** (Endpoint-release bidirectional skew hull). *The current slack rows also have a simultaneous CRT-hull form. Each formal generator/fill residue pair gives a local alignment condition in the variable  $P$ ,*

$$P \equiv r_g \pmod{q-2}, \quad P \equiv r_f - \Delta \pmod{q},$$

where  $\Delta$  is the affine delay. For the current  $q = 31$  atom, the combined modulus is 899 and the primitive support interval is  $[2669, 2688]$ , of width 20. The nearest representatives of all 12 formal pairs lie in the hull  $[2304, 3122]$ , of width 819. Thus a single support hull would need left and right extensions 365 and 434; after the feedback horizons are credited, positive two-sided deficits 335 and 409 remain. Moreover

$$899 - 819 = 80,$$

which equals the affine delay.

*Status and reference.* The bidirectional skew-hull audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
bidirectional_skew_hull_audit.py
```

It imports the inverse-alignment equivalence only as a local CRT-alignment discipline: the global statement that the minimal zero-row alignment  $x$  exceeds  $P$  is not used as a theorem. The audit proves that the current formal-pair system cannot be absorbed by one-sided skew growth, because below and above representatives both remain outside the credited feedback horizons. The remaining obstruction is routed to `BidirectionalSkewHull-PDEC/ColumnCRT` or to the moving-family multiplicity exits. This is a current-sweep closure, not a final unconditional row/column theorem.  $\square$

**Reduction-closed Statement 2.26** (Endpoint-release circular aperture). *The bidirectional hull also has to be read on the circle modulo  $q(q-2)$ , not only in the current linear cut. For the current  $q = 31$  atom, the previous interval  $[2304, 3122]$  has width 819 in one chosen cut of the modulus 899. Removing the largest circular open gap instead gives the minimal circular alignment arc*

$$[3029, 3586],$$

of width 558. The largest open gap runs from the actual pair 19:8 to the pair 13:9 and has open width 341. The affine delay gap of width 80 is present, but it is only the third largest circular gap, so the previous equality  $899 - 819 = 80$  is a linear-cut rigidity, not the complement of the circular minimal arc.

*Status and reference.* The circular-aperture audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
circular_aperture_audit.py
```

It verifies that even after the optimal circular cut, the minimal arc has width 558, whereas the primitive support has width 20. After optimally shifting the support interval to  $[3568, 3587]$ , the total extension still required is 539; after crediting the most favorable one-sided feedback horizon, a conservative deficit 509 remains. Thus the current single-sided circular-aperture absorption route is closed for this sweep. The remaining obstruction is routed to `CircularAperture-PDEC/ColumnCRT`, `SAE`, or the already named moving-family multiplicity exits. This refinement corrects the linear-hull wording but does not close the global row/column theorem.  $\square$

**Reduction-closed Statement 2.27** (Endpoint-release anchored circular depth). *The circular aperture can be sharpened by keeping the unique actual packet as an anchor. In the current sweep the minimal circular arc is  $[3029, 3586]$ , and its right endpoint 3586 is the actual pair 19:8 after one period of the modulus 899. Hence an actual-anchored absorption of the whole arc requires the generator phase and the shifted-fill phase to both reach the left endpoint 3029. The required common left depth is*

$$D = 3586 - 3029 = 557.$$

*The current generator and fill left depths are 18 and 28, so the required endpoint releases are 539 and 529, for a total of 1068. This is 53.4 times the primitive support width 20.*

*Status and reference.* The anchored circular-depth audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
anchored_circular_depth_audit.py
```

It shows that the previous one-sided circular support extension 539 hides a second endpoint release 529: the intersection support moves only when both underlying phase intervals move. The same-orientation moving-key explanation also fails at this anchored depth. The lower-side AffineTwin depth formulas would require

$$q = 2D - 5 = 1109, \quad q = D + 3 = 560,$$

so there is no common odd-prime key. Thus the current same-orientation anchored circular-depth absorption route is closed for this sweep. The remaining obstruction is routed to anchored circular-depth PDEC, ColumnCRT, SAE, orientation-changing keys, or the moving-family multiplicity exits; the global row/column theorem remains open.  $\square$

**Reduction-closed Statement 2.28** (Anchored parity no-go). *For a same-orientation lower-side AffineTwin key to absorb an anchored common left depth  $D$ , the generator and fill depth formulas must give the same prime parameter:*

$$q_g = 2D - 5, \quad q_f = D + 3.$$

*Thus a common key forces  $2D - 5 = D + 3$ , hence  $D = 8$  and  $q = 11$ . The actual-anchored circular arc in the current sweep instead forces  $D = 557$ . It gives*

$$q_g = 1109, \quad q_f = 560.$$

*The fill-side candidate is an even integer larger than 2, so it cannot be an odd-prime AffineTwin key.*

*Status and reference.* The anchored parity no-go audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
anchored_parity_nogo_audit.py
```

It records both obstructions: the forced depth is 549 away from the only common-depth solution  $D = 8$ , and its odd parity makes the fill-side candidate  $D + 3$  even. Therefore the current same-orientation anchored parity absorption route is closed for this sweep. The remaining obstruction is a global family version of this no-go, or one of the already named exits: orientation-changing keys, ColumnCRT/PDEC, SAE, or moving-family multiplicity. The global row/column theorem remains open.  $\square$

**Reduction-closed Statement 2.29** (Endpoint-release cut-anchor sweep). *The anchored parity obstruction can be checked against every circular cut, not only the optimal one. In the current  $q = 31$  atom there are 12 adjacent CRT cuts on the circle modulo 899. The unique actual pair 19:8 is retained in every lifted arc: it is the left endpoint for one cut, the right endpoint for one cut, and an interior point for the remaining ten cuts. For each cut we anchor the actual representative in the lifted arc and charge both the generator interval and the shifted-fill interval for the left and right depths needed to cover that arc.*

*The smallest arc is still the previous cut 19:8  $\rightarrow$  13:9, with width 558 and actual pair at the right endpoint. Its total endpoint release is  $1068 = 53.4 \cdot 20$ , where 20 is the primitive support width. Across all cuts, the endpoint release ranges from 1068 to 1737, so no cut fits inside the support budget.*

*Status and reference.* The cut-anchor sweep audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
cut_anchor_sweep_audit.py
```

It also checks the same-orientation moving-key formulas for every cut. On the left lower side a common key can occur only at  $D = 8$ ; the closest positive left depth in the sweep is  $D = 58$ . On the right above side the shifted-fill right depth is fixed at  $D = 1$ ; the closest positive right depth in the sweep is  $D = 342$ . Hence changing the circular cut or moving the actual endpoint does not reopen the same-orientation absorption channel. Endpoint cuts fall into the left anchored parity no-go or the right fixed-fill no-go; interior cuts require both side releases simultaneously. The current sweep is closed, but the global theorem still requires a family version of this cut-anchor no-go, or one of the named exits: orientation-changing keys, ColumnCRT/PDEC, SAE, or moving-family multiplicity.  $\square$

**Reduction-closed Statement 2.30** (Cut-anchor ColumnCRT compression). *The circular cut is a gauge choice, not a new actual residue. In the current cut-anchor sweep all 12 cuts retain the same actual pair 19:8. The actual representative is either 2687 or  $3586 = 2687 + 899$ , and in both cases*

$$P \equiv 889 \pmod{899}.$$

*Thus the fixed  $q = 31$  actual slot contributes one ColumnCRT atom of mass  $1/899$ , not twelve independent atoms of total mass  $12/899$ . The apparent extra mass*

$$\frac{11}{899}$$

*is only cut multiplicity.*

*Status and reference.* The cut-anchor ColumnCRT compression audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
cut_anchor_columncrt_compression_audit.py
```

It registers the persistent fixed-residue escape as a `CutAnchorColumnCRT-PDEC` object with phase  $P \equiv 889 \pmod{899}$ , equivalently  $P \equiv 19 \pmod{29}$  and  $P \equiv 21 \pmod{31}$  after the fill shift. Therefore, after same-orientation cut-anchor absorption is closed, changing the cut cannot supply additional capacity. If  $q$  or the residues move, the case leaves this fixed ledger and returns to the existing AffineTwin moving-family SAE/ColumnCRT ledger. The current sweep closes cut multiplicity as an independent capacity source; it does not yet exclude the global ColumnCRT/PDEC or orientation-changing exits.  $\square$

**Reduction-closed Statement 2.31** (Actual-anchor replacement). *After cut-anchor compression, keeping the actual pair 19:8 gives the single fixed residue*

$$P \equiv 889 \pmod{899}.$$

*If the counterexample chain tries to abandon this anchor, then some other formal pair must become actual, or else an unused target residue must arrive. Both alternatives require a CRT phase jump larger than the primitive support width 20. Among existing formal pairs, the closest replacement is 19:12, at CRT distance 58, and it requires total endpoint release 70. Among unused targets, the closest replacement is 20:9, at CRT jump 59, and it already needs a new side residue.*

*Status and reference.* The actual-anchor replacement audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
actual_anchor_replacement_audit.py
```

It imports the support-motion and unused-target ledgers only as audited local interfaces. The first interface covers the 11 unsupported formal pairs: every replacement CRT jump exceeds the support width, every replacement requires both endpoint releases, and the minimum release is  $70 = 3.5 \cdot 20$ . The second interface covers the unused target residues: every CRT jump again exceeds the support width, and every target requires a new side residue. Thus actual-anchor replacement is closed for the current sweep. The global theorem still requires a family version of this no-go, or exclusion of the named support-motion, unused-target, ColumnCRT/PDEC, and moving-family exits.  $\square$

**Reduction-closed Statement 2.32** (Near-miss 61/59 phase fracture). *The closest apparent bridge between actual-anchor replacement and moving-key rematerialization is not a bridge in the current sweep. The closest existing formal replacement is still 19:12: its CRT jump is 58, its required common side depth is 58, and its endpoint release is 70. The corresponding one-sided moving-key formulas produce  $q = 61$  and  $q = 111$ . Meanwhile the closest unused-target replacement is  $19:12 \rightarrow 20:9$ , with CRT jump 59, and the nearest available gap source to the candidate  $q = 61$  also has gap 59.*

*Status and reference.* The near-miss phase-fracture audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
near_miss_61_59_phase_fracture_audit.py
```

It records that  $q = 61$  is prime and  $q - 2 = 59$  is prime, but  $61 \equiv 1 \pmod{4}$ , so the AffineTwin delay  $(11q - 21)/4$  is not integral, and there is no matching gap-61 source. The nearest actual source has signature

$$\text{gap} = 59, \quad \text{generator} = 61, \quad \text{fill} = 59, \quad \text{sides} = \text{minus} \rightarrow \text{minus}, \quad p \text{ delay} = 70.$$

This differs from the  $q = 61$  AffineTwin source signature

$$\text{gap} = 61, \quad \text{generator} = 59, \quad \text{fill} = 61, \quad \text{sides} = \text{minus} \rightarrow \text{plus}.$$

The other moving-key candidate,  $q = 111$ , is composite. Thus the integer 59 appears in two nearby roles but not in the same phase role: as the nearest available gap source to  $q = 61$ , and as the unused-target CRT jump. The latter still exceeds the support width 20 and introduces a new side residue. Hence the current-sweep 61/59 near miss cannot carry hidden actual load. The global theorem still requires a family version of this no-go, or routing to source-rematerialization, unused-target, ColumnCRT/PDEC, or moving-family exits.  $\square$

**Reduction-closed Statement 2.33** (Phase-scale bridge exhaustion). *In the current sweep there is no other exact phase-scale bridge among the moving-key candidates, the available gap sources, and the unused-target CRT jumps. Across all one-sided moving-key candidates, no candidate  $q$  is equal to an available gap-source scale, and no candidate  $q$  is equal to an unused-target jump scale. The only exact offset bridge of the form  $q - 2 = \text{gap source} = \text{unused-target jump}$  is the already fractured case  $q = 61$ ,  $q - 2 = 59$ .*

*Status and reference.* The phase-scale bridge exhaustion audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
phase_scale_bridge_exhaustion_audit.py
```

It compares the 19 moving-key candidates

61, 65, 96, 111, 119, 123, 154, 181, 235, 293, 297, 355, 386, 575, 699, 761, 1375, 1499, 1747

with the available source scales

31, 43, 59

and the unused-target jump scales

59, 60, 90, 150, 281, 343, 345, 374, 375.

The audit records

$$\{q : q = \text{gap source}\} = \emptyset, \quad \{q : q = \text{unused jump}\} = \emptyset,$$

and

$$\{q : q - 2 = \text{gap source} = \text{unused jump}\} = \{61\}.$$

The previous near-miss audit already closes this unique offset bridge: the gap-59 source has the wrong role and orientation for a  $q = 61$  AffineTwin source, and the unused-target jump 59 still introduces a new side residue. Therefore the current sweep has no anonymous exact or offset scale bridge. The global theorem still requires a family version of this exhaustion, or a route to source-rematerialization, unused-target, ColumnCRT/PDEC, or moving-family exits.  $\square$

**Reduction-closed Statement 2.34** (Support-width near-scale fracture). *Even after exact phase-scale bridges are exhausted, the current sweep has no support-width near-scale bridge. With support width 20, the only moving-key candidates that are simultaneously within distance 20 of an available source scale and an unused-target jump scale are  $q = 61$  and  $q = 65$ . The first is the already fractured prime-but-not-AffineTwin case; the second is composite.*

*Status and reference.* The support-width near-scale audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
support_width_nearscale_fracture_audit.py
```

It enumerates all events satisfying

$$|s - q| \leq 20, \quad |s - (q - 2)| \leq 20, \quad |j - q| \leq 20, \quad |j - (q - 2)| \leq 20,$$

where  $s$  is a current gap-source scale and  $j$  is an unused-target CRT jump. It records

$$\{q : \text{near source}\} = \{61, 65\},$$



$$\{q : \text{near unused jump}\} = \{61, 65, 96, 111, 154, 293, 297, 355, 386\},$$

and

$$\{q : \text{near both source and unused jump}\} = \{61, 65\}.$$

There is no viable near-scale bridge. For  $q = 61$ , the AffineTwin source gate fails by the same delay, congruence, and missing-source invariants as in the near-miss audit. For  $q = 65$ ,  $q$  is composite. The remaining near-jump candidates have no near source and cannot form a three-way phase bridge. Thus support-width proximity cannot replace exact CRT/source signatures in the current sweep. A global proof must still show the corresponding family near-scale no-go, or route failures to source-rematerialization, unused-target, ColumnCRT/PDEC, or moving-family exits.  $\square$

**Reduction-closed Statement 2.35** (Orphan near-jump source deficit). *The remaining support-width near-scale candidates that are near an unused-target jump but not near a source scale are source-deficient in the current sweep. The orphan near-jump candidates are*

$$96, 111, 154, 293, 297, 355, 386.$$

*For all of them, the nearest source scale is farther than the support width 20. The smallest source distance is 35, attained at  $q = 96$ , leaving a positive source deficit  $35 - 20 = 15$ .*

*Status and reference.* The orphan near-jump source-deficit audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
orphan_nearjump_source_deficit_audit.py
```

It records that the seven orphan near-jump candidates split as six composite candidates and one prime-but-not-AffineTwin candidate. Every corresponding unused-target near-jump event introduces a new side residue, and no moving source is materialized. Thus target-side proximity does not create a three-way bridge: the source side remains outside the support-width tolerance. Any attempt to repair the orphan near-jump would have to move or create a source family, which is precisely the source-rematerialization or moving-family exit, not an anonymous primitive-support absorption. A global proof still requires the corresponding family no-go or routing to named PDEC/SAE exits.  $\square$

**Reduction-closed Statement 2.36** (Near-jump carrier exhaustion). *All near-jump carriers are exhausted in the current sweep. The carrier set is*

$$61, 65, 96, 111, 154, 293, 297, 355, 386.$$

*It splits into two disjoint parts:*

$$\{61, 65\}$$

*near both a source scale and a target jump, and*

$$\{96, 111, 154, 293, 297, 355, 386\}$$

*near only a target jump. The first part fails the source gate; the second part has source distance greater than the support width. In every case the target-side near-jump event requires a new side residue.*

*Status and reference.* The near-jump carrier exhaustion audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
nearjump_carrier_exhaustion_audit.py
```

It records the source-status histogram

$$\{\text{near-source gate fracture} : 2, \quad \text{orphan source deficit} : 7\}$$

and the moving-route histogram

$$\{\text{CompositeQ} : 7, \quad \text{PrimeButNotTwinAffine} : 2\}.$$

The nearest orphan source distance is 35, still larger than the support width 20, and all carrier jump events need new side residues. Hence a near-target phase cannot by itself carry actual load. Repairing it forces one of the named exits: source rematerialization, unused-target residue arrival, ColumnCRT/PDEC, or moving-family migration. This closes the current-sweep near-jump carrier but not the global row/column theorem.  $\square$

**Reduction-closed Statement 2.37** (Carrier-arrival routing). *Every target-side near-jump event from the exhausted carrier set routes to an already registered unused-target arrival atom. There are 27 such events, using all 9 arrival atoms and 5 target pairs. The missing-match count is 0. The event-level new-side-residue demand is 50, compressed to generator residues*

$$10, 16, 17, 18, 20$$

and fill residues

$$5, 6, 7, 30.$$

*The only exact zero-phase event is  $q = 61$ ,  $q - 2 = 59$ , jump 59; it still requires the new generator residue 20.*

*Status and reference.* The carrier-arrival routing audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
carrier_arrival_routing_audit.py
```

It matches each carrier event against the unused-target arrival ledger by source pair, target pair, and CRT jump. All matches are present, and every matched target is outside both the current formal product and the current actual support. Therefore near-target proximity is only a repeated projection of the unused-target arrival packet, not a new actual-load carrier. A persistent version must prove a global new-residue arrival bound or route to new-generator, new-fill, ColumnCRT/PDEC, or moving-family exits. This closes the current-sweep anonymous target-side carrier but not the global row/column theorem.  $\square$

**Reduction-closed Statement 2.38** (Arrival pressure product). *If the carrier-arrival packet is materialized as side residues, then the current sweep crosses the square-root pressure gate. The base side counts are*

$$(A_g, A_f) = (3, 4), \quad A_g A_f = 12 < \sqrt{29 \cdot 31}.$$

*The carrier packet adds generator residues*

$$10, 16, 17, 18, 20$$

and fill residues

$$5, 6, 7, 30.$$

*Thus full materialization gives side counts (8, 8) and*

$$(8 \cdot 8)^2 - 29 \cdot 31 = 3197 > 0.$$

Moreover the minimal crossing is exact to one unit: any two two-sided target atoms give counts (5, 6) and

$$(5 \cdot 6)^2 - 29 \cdot 31 = 1.$$

*Status and reference.* The carrier-arrival pressure-product audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
carrier_arrival_pressure_product_audit.py
```

It starts from the already routed carrier-arrival packet and updates the formal side-residue counts exactly. Since the updated product is above the square-root gate, the packet cannot remain an anonymous unused-target arrival. It is a named **SuperSqrt/PressureProduct-PDEC** atom. Therefore, after excluding anonymous target-side carriers, the remaining global obligation is to exclude this PDEC family or keep it as the explicit named exit; the row/column theorem is not yet unconditionally closed.  $\square$

**Reduction-closed Statement 2.39** (Arrival projection deficit). *The preceding square-root crossing is not an actual overload in the current sweep. Project the formal side product back to the common support window. For each minimal crossing row the formal product is 30, while the actual projection consists of exactly three points:*

$$\{\text{actual anchor } 19:8\} \cup \{\text{the two selected target atoms}\}.$$

*Thus the projection deficit is 27, and the actual square-root slack is  $29 - 3 = 26$ . For the full carrier packet the formal product is 64, but the actual projection has only 6 hits, with deficit 58 and slack 23.*

*Status and reference.* The carrier-arrival projection-deficit audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
carrier_arrival_projection_deficit_audit.py
```

It enumerates the side-residue Cartesian product for each minimal crossing and intersects it with the 20 residue pairs actually present in the common support window. The excess  $30 > 29$  is therefore a formal-product excess, not an actual-load excess. Any global continuation must either tighten product accounting to actual projection, or produce a named projection-collision or support-escape PDEC. The current sweep has no anonymous actual overload.  $\square$

**Reduction-closed Statement 2.40** (Support-graph cap). *In the current fixed AffineTwin slot the common support is a graph, not a two-dimensional product set. The target window consists of the 20 pairs*

$$(1, 21), (2, 22), \dots, (10, 30), (11, 0), \dots, (20, 9),$$

*so each generator residue has at most one fill partner and conversely. Hence any actual projection from side sets is bounded by the graph size 20. Since*

$$20 \leq \lfloor \sqrt{29 \cdot 31} \rfloor = 29,$$

*the fixed slot cannot produce an actual square-root overload.*

*Status and reference.* The support-graph cap audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
support_graph_cap_audit.py
```

It verifies that the common window is functional in both directions and has constant fill offset modulo 31. The general fixed-slot inequality is also recorded: for an AffineTwin modulus  $q \geq 13$ , the support width  $W = (q+9)/2$  satisfies  $W^2 \leq q(q-2)$ , equivalently  $3q^2 - 26q - 81 \geq 0$ . Therefore the current ProductAccounting problem is closed for fixed slots. Any remaining escape must move or change the support graph, which is the named primitive-slot support-escape exit, not an anonymous load surplus.  $\square$

**Reduction-closed Statement 2.41** (Moving-slot graph-cap route). *In the current endpoint-release sweep there is no anonymous moving-slot actual overload. A fixed or same-orientation AffineTwin slot remains bounded by the graph cap; the candidate fixed-family moduli are  $q = 31, 43, 103$ , and each satisfies  $W^2 \leq q(q-2)$ . If the support is truly moved while keeping the current primitive identity, the narrowest atom already requires endpoint release 70 against support width 20.*

*Status and reference.* The moving-slot route audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
moving_slot_graph_cap_route_audit.py
```

It records that all eleven support-motion candidates require both endpoint releases and that the fixed primitive depth identities break on both sides. Allowing a same-orientation moving key does not repair this: at the narrowest atom the depth formulas give  $q_g = 111$  and  $q_f = 61$ . Taking all one-sided candidate  $q$  values through the source-rematerialization gate gives no exact rematerialized source. Thus the present sweep routes every moving-slot escape to a named exit: orientation-changing primitive key, source rematerialization, ColumnCRT/PDEC, SAE, or unused-target arrival control. This is a reduction of the remaining interface, not a final unconditional closure of the row/column theorem.  $\square$

**Reduction-closed Statement 2.42** (Moving-family persistence pressure). *The current moving-family escape is not an anonymous capacity reservoir. Among the candidate moduli  $q = 31, 43, 103$ , only  $q = 31$  is source-materialized. The candidate  $q = 43$  has a same-gap source but with the wrong orientation and delay, and  $q = 103$  has no gap source. The non-realized candidates have one-sided generator pressure, but their paired pressure products remain below the square-root gate.*

*Status and reference.* The moving-family persistence pressure audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
moving_family_persistence_pressure_audit.py
```

It combines the source-materialization gate, epoch-pair sparse ledger, paired pressure ledger, threshold route classifier, and fill catchup budget. The source gate blocks 28 formal pairs: 16 from  $q = 43$  and 12 from  $q = 103$ . All minimal threshold-crossing routes require fill-side residue arrival; if all three candidates crossed, the minimum fill-arrival demand would be 11 new fill residues, with Rankin mass 23339/137299. Repeated fill arrival is routed to reset or ColumnCRT-PDEC. Hence the present sweep routes moving-family persistence to named exits: fill-arrival control, high-density epoch-pair PDEC/ColumnCRT, pressure-product PDEC, or source rematerialization. This is still a reduction boundary, not a final global proof.  $\square$

**Reduction-closed Statement 2.43** (Fill-arrival projection gate). *In the current sweep, fill-side residue arrival alone cannot create an actual overload. The only source-materialized candidate is  $q = 31$ , and its minimal threshold crossing is not fill-only: it requires at least two new generator residues. When such co-arrivals are counted formally, the minimal side product is 30, but its actual projection has only 3 hits and remains below  $\lfloor \sqrt{29 \cdot 31} \rfloor$ .*

*Status and reference.* The fill-arrival projection gate audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
fill_arrival_projection_gate_audit.py
```

For  $q = 31$ , every fill-arrival threshold crossing requires generator co-arrival and then returns to the support-graph projection ledger. The unused-target ledger shows that the needed target residues also require new generator residues and have CRT jumps beyond the current support width. For  $q = 43$  and  $q = 103$ , formal fill-only routes exist, but they are blocked before projection:  $q = 43$  has the wrong source signature and  $q = 103$  has no gap source. Thus the present sweep routes fill-arrival persistence to generator co-arrival, ProductAccounting tightening, source rematerialization, or ColumnCRT/PDEC. This remains a reduction boundary rather than a global exclusion theorem.  $\square$

**Reduction-closed Statement 2.44** (Generator-coarrival projection accounting). *In the current sweep, forcing fill arrival to co-arrive with generator residues still does not create an actual overload. For the realized candidate  $q = 31$ , all 31 nonempty subsets of the five unused-target atoms have been enumerated. There are 22 formally super-square-root subsets, but none projects above  $\lfloor \sqrt{29 \cdot 31} \rfloor$ .*

*Status and reference.* The generator-coarrival projection-accounting audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
generator_coarrival_projection_accounting_audit.py
```

The base packet has formal side product  $3 \cdot 4 = 12$  and one actual projection hit, the anchor  $19 : 8$ . The minimal coarrival crossing has formal product 30, but its actual projection has only 3 hits and slack 26 below  $\lfloor \sqrt{29 \cdot 31} \rfloor = 29$ . The full carrier packet has formal product 64, but only 6 actual hits and slack 23. Moreover every projection hit is exactly the anchor plus the selected target atoms; no hidden projection collision is being counted as free capacity. Thus the present sweep routes generator coarrival to ProductAccounting tightening, source rematerialization, ColumnCRT/PDEC, or moving-family persistence. This remains a local closed audit and not a global unconditional row/column proof.  $\square$

**Reduction-closed Statement 2.45** (Coarrival schema). *For every fixed AffineTwin support graph with  $q \geq 13$ , generator coarrival cannot by itself create an actual overload. The graph width is  $W = (q + 9)/2$ , and*

$$W^2 \leq q(q - 2) \iff 3q^2 - 26q - 81 \geq 0.$$

*The right hand side is already positive at  $q = 13$  and is increasing for  $q \geq 13$ .*

*Status and reference.* The family-schema audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
generator_coarrival_family_schema_audit.py
```

The current candidate moduli  $q = 31, 43, 103$  all pass this fixed-graph inequality. Hence the only way for coarrival to become an actual overload is to leave the fixed functional support graph. The present sweep routes those escapes to named ledgers: support motion requires endpoint release 70 against support width 20, source materialization blocks 28 formal pairs, the fixed actual anchor compresses to the single ColumnCRT atom  $P \equiv 889 \pmod{899}$ , and moving-family persistence remains a named pressure/source/fill exit. This closes the current-sweep schema, but the global row/column theorem still needs the schema promoted to all persistent AffineTwin families and the named exits excluded or absorbed.  $\square$

**Reduction-closed Statement 2.46** (Persistent-family promotion). *The current sweep promotes the coarrival schema through the active persistent AffineTwin family ledger. Fixed support graphs are bounded by the graph-width inequality; fixed  $q$  and fixed residue recurrence is routed to ColumnCRT/PDEC; moving  $q$  or moving residues are routed to epoch-pair SAE unless they become high-density.*

*Status and reference.* The promotion audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
persistent_family_promotion_audit.py
```

In the current ledger the candidate moduli are  $q = 31, 43, 103$ , with realized modulus 31. Their epoch-pair occupancy upper sum is 0.023577117628562343, below the threshold  $\eta = 0.025$ , and there is no high-density epoch pair. The fixed-slot recurrence count is 0; the 12 fixed-residue recurrences all drift in slot and are therefore the next ColumnCRT/PDEC input. Source materialization blocks 28 formal pairs, paired pressure does not cross, and fill arrival is routed to new-fill or reset PDEC. Thus the current sweep has no anonymous persistent-family capacity source. The remaining global obligations are an epoch-pair multiplicity bound and exclusion of fixed-residue slot drift by ColumnCRT/PDEC or SAE/Rankin.  $\square$

**Reduction-closed Statement 2.47** (Transport-frontier integration). *In the current AffineTwin endpoint-release sweep, fixed-residue slot drift is not an anonymous ColumnCRT capacity source. The 12 slot-drift pairs match 12 distinct transport cells, with no transport-cell recurrence and no exact phase translate. Chained recurrence has finite forward lifetime; non-chained recurrence must repeat the full transport-cell key, and the current reset-PDEC atom count is 0.*

*Status and reference.* The transport-frontier integration audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
transport_frontier_integration_audit.py
```

It imports the persistent-family promotion ledger and the fixed-residue transport-cell, iteration-drift, reset-PDEC, and persistence-frontier partition ledgers. The aggregate values are: 12 fixed-residue slot-drift pairs, 12 transport cells, 12 unique transport cells, 0 transport-cell recurrences, 0 exact phase translates, maximum forward transition count 8, and reset-PDEC atom count 0. The frontier partition has 324 physical records, split as 24 transport-cell records and 300 singleton-residue records, with no unclassified record. Thus the present sweep routes the fixed-residue drift branch to TransportResetPDECExclusion or SingletonResidue SAE/Rankin, while the global row/column theorem still requires those exits and the epoch-pair multiplicity bound to be proved globally.  $\square$

**Reduction-closed Statement 2.48** (Remaining-frontier bridge). *After transport-frontier integration, the computable current-sweep remainder bridges to three narrower ledgers. Transport reset has no current atom; singleton residue SAE is localized to the active prime-band endpoint problem; and AffineTwin epoch-pair multiplicity is currently below the  $\eta = 1/40$  sparse gate.*

*Status and reference.* The bridge audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
remaining_frontier_bridge_audit.py
```

It reads seven ledgers: transport integration, singleton profile, Rankin budget, one slot capacity, active source, active band, and epoch-pair multiplicity. The current singleton profile has 300 singleton packets and Rankin mass about 4.328474446, with one slot mass about 4.291749691 and two

slot share about 0.008484457. The one slot route has 40 active epochs, maximum occupancy ratio about 0.309859155, and minimum spare ratio about 0.690140845.

The active  $\ell$  source is exactly the prime band 23, ..., 109, with core 29, ..., 107 and minus-only endpoints 23, 109. The AffineTwin candidate epoch-pair product-mass upper sum is 0.023577118 < 0.025, with no high-density row.

Thus the current remainder is sharpened to endpoint growth or reset PDEC, transport reset PDEC exclusion, global epoch-pair multiplicity, and moving residue SAE/Rankin. These remain global obligations, not a completed unconditional proof.  $\square$

**Reduction-closed Statement 2.49** (Endpoint motion stencil). *In the current H-lower singleton route, endpoint motion of the active prime band has only named exits. The outward neighbor primes 19 and 113 have no current record; the endpoints 23 and 109 are minus-side singleton atoms; and the inward core edges 29 and 107 already have both-side support.*

*Status and reference.* The endpoint-motion stencil audit is generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_endpoint_motion_stencil_audit.py
```

The six-point stencil is

$$19 \rightarrow 23 \rightarrow 29, \quad 107 \rightarrow 109 \rightarrow 113.$$

The outer entries 19, 113 are empty in the current sweep. The endpoint entries 23, 109 each have one minus-side record, one residue, and one slot. The inward entries 29, 107 have both-side core support, with 4 and 6 records respectively. Hence current endpoint motion cannot be counted as anonymous capacity: outward motion requires a new endpoint arrival, while inward motion is absorbed into the core. The global obligations are endpoint outward arrival control, endpoint atom PDEC/SAE exclusion, and core-edge multiplicity control.  $\square$

**Reduction-closed Statement 2.50** (Endpoint motion gap). *In the current H-lower singleton route, endpoint expansion of the active prime band creates only three temporary internal prime gaps. The missing prime moduli are 31, 43, 59, and all are filled within the current sweep; the largest observed fill delay is 80 in the  $P$  parameter.*

*Status and reference.* The endpoint-motion gap router is generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_endpoint_motion_gap_router.py
```

It orders one-slot  $\ell$  values by their first activation. The three non-interval snapshots are: activation of 47 at  $P = 2063$  leaves 43 missing until delay 74; activation of 29 at  $P = 2687$  leaves 31 missing until delay 80; activation of 61 at  $P = 3187$  leaves 59 missing until delay 70. Every 1000-level prefix snapshot from 3000 to 10000 is already a complete prime interval, and the final active band is 23, ..., 109. Thus the current sweep routes endpoint motion to a gap-fill bound or to a Gap-PDEC/SAE if such a gap persists globally.  $\square$

**Reduction-closed Statement 2.51** (Gap-fill pair and corridor). *In the current H-lower singleton route, every endpoint-motion gap is repaired by the next first activation in  $\ell$ -activation order, and every repair lies inside the short corridor*

$$P_{\text{fill}} - P_{\text{gen}} \leq 3\ell_{\text{gap}}.$$

*The three gap-fill keys are distinct in the current sweep.*

*Status and reference.* The gap-fill pair and repair-corridor routers are generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_gap_fill_pair_router.py
```

and

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_gap_repair_corridor_router.py.
```

The three repairs are  $47 \rightarrow 43$  with delay 74,  $29 \rightarrow 31$  with delay 80, and  $61 \rightarrow 59$  with delay 70. Each has activation-rank delay 1, and each restores the exact prime interval after the fill. The corresponding full GapFillPair keys are pairwise distinct. The corridor ratios are bounded by 3: the tight row is  $\ell_{\text{gap}} = 31$ , where  $80 \leq 3 \cdot 31$  with margin 13; a  $2\ell_{\text{gap}}$  corridor is already false there. Thus the global remaining obligation is a short gap-repair corridor bound, or exclusion of a Corridor-PDEC/SAE when that bound fails.  $\square$

**Reduction-closed Statement 2.52** (Corridor phase budget). *In the current H-lower singleton route, the short repair corridor decomposes into the exact three-term identity*

$$P_{\text{fill}} - P_{\text{gen}} = d_{\text{gen},\text{right}} + g_{\text{bridge}} + d_{\text{fill},\text{left}}.$$

*All three current gap repairs satisfy this identity and keep positive  $3\ell_{\text{gap}}$  slack. The unique component excess is the phase bridge for  $\ell_{\text{gap}} = 31$ , but that excess is absorbed by the neighboring depth spares.*

*Status and reference.* The corridor phase-budget router is generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_corridor_phase_budget_router.py.
```

The three identities are

| $\ell_{\text{gap}}$ | $d_{\text{gen},\text{right}}$ | $g_{\text{bridge}}$ | $d_{\text{fill},\text{left}}$ | $P_{\text{fill}} - P_{\text{gen}}$ |
|---------------------|-------------------------------|---------------------|-------------------------------|------------------------------------|
| 43                  | 12                            | 40                  | 22                            | 74                                 |
| 31                  | 6                             | 46                  | 28                            | 80                                 |
| 59                  | 31                            | 31                  | 8                             | 70.                                |

The minimum remaining  $3\ell_{\text{gap}}$  slack is 13. The only component excess is  $46 - 31 = 15$  in the  $\ell_{\text{gap}} = 31$  phase bridge; the other two components provide total spare 28, leaving the same positive corridor slack 13. Hence the current short-corridor obstruction has been sharpened to a phase-budget obstruction: a global proof must bound the bridge component by neighboring depth spare, or route any persistent failure to a PhaseBridge-PDEC or SAE exit.  $\square$

**Reduction-closed Statement 2.53** (Phase-bridge excess return to AffineTwin). *The unique current phase-bridge excess is not a new anonymous capacity source. It collapses to the already registered AffineTwin fixed slot atom with*

$$q = 31, \quad q - 2 = 29.$$

*The excess is 15, the absorbing spare is 28, and the post-absorption slack is 13. The same atom satisfies the primitive affine identities and the coprime CRT lock*

$$P \equiv 19 \pmod{29}, \quad P \equiv 21 \pmod{31}.$$

*Hence the combined modulus  $29 \cdot 31 = 899$  exceeds the common support width 20 and isolates the fixed representative.*



*Status and reference.* This return chain is generated by the phase-bridge excess atom/source routers, the margin-slot normal-form and primitive-identity routers, the primitive affine-collapse router, and the AffineTwin slot-phase-lock router:

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_phase_bridge_excess_atom_router.py

experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_phase_bridge_excess_source_router.py

experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_margin_slot_primitive_identity_router.py
```

and the downstream AffineTwin phase-lock and moving-family routers. They record

$$\text{excess} = 15 = \text{generator margin} = 3\rho_{\text{jump}}, \quad \text{slack}_{\text{after}} = |\Delta b| = 13,$$

and the affine primitive identities

$$\begin{aligned} \ell_{\text{fill}} = \ell_{\text{gap}} = 31, \quad \ell_{\text{gen}} = \ell_{\text{gap}} - 2 = 29, \\ m_{\text{gen}} = (\ell_{\text{gap}} - 1)/2 = 15, \quad |\Delta b| = (\ell_{\text{gap}} - 5)/2 = 13, \quad |\Delta u| = (\ell_{\text{gap}} - 3)/4 = 7. \end{aligned}$$

Thus the bridge-excess atom is exactly the  $q = 31$  AffineTwin atom. Since 29 and 31 are coprime, the two slot congruences have modulus 899; the pair-support width is only 20, so the fixed slot contributes a single representative, already recorded as the fixed actual-anchor ColumnCRT atom. If the same  $q$  and residues recur, recurrence is a fixed-modulus ColumnCRT/PDEC object. If  $q$  or the residues move, the packet returns to the AffineTwin epoch-pair SAE/Rankin and moving-family exits. This closes the new current-sweep anonymous phase-bridge outlet, but the global row/column theorem still requires the moving-family multiplicity, endpoint-growth/reset, transport-reset, and moving-residue exits to be bounded or excluded.  $\square$

**Reduction-closed Statement 2.54** (AffineTwin sparse SAE envelope). *After the phase-bridge outlet returns to the AffineTwin moving-family ledger, the current candidate epoch-pairs pass the  $\eta = 1/40$  sparse gate. The candidate moduli are*

$$q = 31, 43, 103,$$

*the current product-mass upper sum is approximately*

$$0.023577117629 < 0.025,$$

*and there is no high-density epoch-pair row in the current sweep. Moreover, the single fixed-slot SAE mass has the telescoping envelope*

$$\frac{1}{q(q-2)} = \frac{1}{2} \left( \frac{1}{q-2} - \frac{1}{q} \right),$$

*so the odd tail from  $q \geq Q$  is at most  $1/(2(Q-2))$ .*

*Status and reference.* The epoch-pair multiplicity and sparse-SAE envelope routers are generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_affine_twin_epoch_pair_
multiplicity_router.py
```

and

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_affine_twin_sparse_sae_
global_envelope_router.py.
```

The current realized atom is the  $q = 31$  fixed slot with mass  $1/899$ , while the single-atom tail from  $q \geq 31$  is bounded by

$$\frac{1}{2(31-2)} = \frac{1}{58}.$$

This is a genuine global summability statement for one fixed AffineTwin atom per modulus. It does not, however, close the moving-family problem by itself: the  $\eta$  sparse gate alone would still permit  $O(\eta q(q-2))$  atoms at a fixed  $q$ , and each such atom has mass  $1/(q(q-2))$ . A saturated sparse row could therefore contribute about  $\eta$  for each  $q$ , which is not summable over infinitely many moduli. Hence the latest narrow obstruction is not the single-atom SAE mass; it is the per- $q$  multiplicity bound. A global proof must show  $O(1)$ , decay, or another summable bound for realized AffineTwin atoms per  $q$ , or route excess multiplicity to a HighDensityEpochPair PDEC/ColumnCRT exit.  $\square$

**Reduction-closed Statement 2.55** (AffineTwin square-root product gate). *The per- $q$  multiplicity target can be weakened to a square-root product gate. Let  $M_q$  be the candidate two-side residue product upper bound for one AffineTwin modulus  $q$ . If*

$$M_q^2 \leq q(q-2),$$

*then the corresponding SAE contribution is bounded by*

$$\frac{M_q}{q(q-2)} \leq \frac{1}{\sqrt{q(q-2)}} \leq \frac{1}{q-2}.$$

*In the current sweep the candidate moduli 31, 43, 103 all pass this gate. The largest product-to-capacity ratio is about 0.400222408.*

*Status and reference.* The square-root product and Brun-conditional router is generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_affine_twin_sqrt_product_
brun_router.py.
```

It records

| $q$ | $M_q$ | $q(q-2)$ | $M_q^2 \leq q(q-2)$ |
|-----|-------|----------|---------------------|
| 31  | 12    | 899      | yes                 |
| 43  | 16    | 1763     | yes                 |
| 103 | 12    | 10403    | yes.                |

The minimum squared slack is 755. Thus the current finite sweep has no SuperSqrtEpochPair row. If the square-root product gate is proved globally, then accepting the classical Brun convergence of reciprocal twin-prime sums would make the AffineTwin SAE tail summable, since AffineTwin moduli satisfy that both  $q$  and  $q-2$  are prime. This is recorded only as an external-input lane: the author-side self-contained route has not accepted Brun as an inline proof and has not proved the global square-root product bound. Therefore the latest self-contained obstruction is sharper: prove  $M_q^2 \leq q(q-2)$  for every persistent AffineTwin epoch-pair, or route every violation to a SuperSqrtEpochPair PDEC/ColumnCRT exit.  $\square$

**Reduction-closed Statement 2.56** (Return to actual-packet support). *The square-root product hard point is an actual-packet question, not merely a formal product question. Let  $M_q^{\text{form}} = A_g A_f$  be the side-residue Cartesian product upper bound, and let  $N_q = |\Pi_q|$  be the number of actual non-PDEC packets passing the two slots, CRT compatibility, common phase support, and non-reuse gates. If every such packet lies in the primitive AffineTwin two-slot support, then*

$$N_q \leq \frac{q+9}{2} \leq \sqrt{q(q-2)} \quad (q \geq 13).$$

*Status and reference.* The reduction is recorded in

`docs/monograph/prime-matrix-nonpdec-sqrt-phase-support-reduction.md`

and is compatible with the pressure-packet carrying ceiling file. The primitive support width is  $W_q = (q+9)/2$ . For  $q \geq 13$ ,

$$W_q \leq \sqrt{q(q-2)} \iff 3q^2 - 26q - 81 \geq 0,$$

and the right side is positive at  $q = 13$  and then increasing. A non-PDEC packet has a primitive phase projection into the common support set  $\Omega_q$ . If two distinct actual packets had the same primitive projection, the recurrence would be a repeated-residue/reset event or a fixed projection ColumnCRT/PDEC event. After those named exits are removed, the projection is injective:

$$\Pi_q \hookrightarrow \Omega_q, \quad N_q \leq |\Omega_q| \leq W_q.$$

Thus a formal SuperSqrt row has only three interpretations: product accounting used  $M_q^{\text{form}}$  where  $N_q$  is required; an actual projection collision gives a PDEC/ColumnCRT certificate; or an actual packet escaped the primitive two-slot support. The latest narrow task is therefore PrimitiveTwinSlotSupportExhaustion, together with replacing remaining SAE and Rankin ledgers by the actual-packet count  $N_q$  where they still use the formal product upper bound.  $\square$

**Reduction-closed Statement 2.57** (Current actual-packet contract). *In the current H-lower AffineTwin sweep, the formal product upper count is*

$$\sum_q M_q^{\text{form}} = 40,$$

*but the actual non-PDEC packet count is only*

$$\sum_q N_q = 1.$$

*The single actual packet occurs at  $q = 31$ ; the candidate rows  $q = 43$  and  $q = 103$  have no current actual packet.*

*Status and reference.* The actual-packet contract is generated by

`experiments/prime_matrix_affine_twin_actual_packet_contract.py`

and archived at

`docs/monograph/prime_matrix_affine_twin_actual_packet_contract.md`.

The current table is

| $q$ | $M_q^{\text{form}}$ | $N_q$ | $W_q$ | $q(q-2)$ |
|-----|---------------------|-------|-------|----------|
| 31  | 12                  | 1     | 20    | 899      |
| 43  | 16                  | 0     | 26    | 1763     |
| 103 | 12                  | 0     | 56    | 10403.   |

Every current actual packet is in primitive support and passes the actual square-root gate  $N_q \leq W_q \leq \sqrt{q(q-2)}$ . The formal-to-actual gap is 39, and there is no current projection-collision PDEC row. Therefore the current SuperSqrt pressure is product-accounting looseness, not actual load. The global obligations remain: prove the product-accounting tightening for all persistent AffineTwin families, prove primitive twin-slot support exhaustion, or route support escapes to their named PDEC/SAE exits.  $\square$

**Reduction-closed Statement 2.58** (Current formal-pair pruning). *In the current sweep, the whole formal-to-actual gap*

$$40 - 1 = 39$$

*is accounted for by named gates:*

$$39 = 11 + 28,$$

*where 11 formal pairs are CRT-window empty and 28 formal pairs fail source materialization. There is no unresolved current formal pair.*

*Status and reference.* The pruning certificate is generated by

`experiments/prime_matrix_affine_twin_formal_pair_pruning_audit.py`

and archived at

`docs/monograph/prime_matrix_affine_twin_formal_pair_pruning_audit.md.`

The current rows are

| $q$ | $M_q^{\text{form}}$ | $N_q$ | CRT empty | source unmaterialized |
|-----|---------------------|-------|-----------|-----------------------|
| 31  | 12                  | 1     | 11        | 0                     |
| 43  | 16                  | 0     | 0         | 16                    |
| 103 | 12                  | 0     | 0         | 12.                   |

For  $q = 31$ , the only supported pair is the residue pair (19,8); after the fill shift it gives the CRT class 889 mod 899, represented in the support window [2669, 2688] by 2687. The other eleven formal pairs have no support-window representative. For  $q = 43$ , the same gap exists only with the wrong generator/orientation/delay data; for  $q = 103$ , there is no current gap-fill source. Hence the present product-accounting gap is not hidden actual load. To globalize this step one must prove that every persistent AffineTwin formal pair either materializes as an actual primitive-supported packet, is CRT-window empty, fails source materialization as a named PDEC/SAE, or is explicitly routed as a primitive support escape.  $\square$

**Reduction-closed Statement 2.59** (Current formal-to-actual cutset). *In the current sweep, the formal-to-actual pipeline has no remaining anonymous capacity source. The formal product universe has 40 pairs, the actual packet count is 1, and the gap*

$$39 = 11 + 28$$

*is exactly the sum of CRT-window-empty pairs and source-unmaterialized pairs. The active CRT modulus is 899, the support width is 20, and the minimum empty-window distance is 40.*

*Status and reference.* The cutset router is generated by

`experiments/prime_matrix_formal_to_actual_global_cutset_router.py`

and archived at

`docs/monograph/prime_matrix_formal_to_actual_global_cutset_router.md.`

It imports the actual-packet, formal-pruning, source-gate, CRT-window, moving-slot, remaining-frontier, and endpoint-atom ledgers. The current support escape branch is already routed: the fixed support width is 20, the square-root floor is 29, exact rematerialized moving  $q$ -values are empty, transport-reset atoms are zero, the active singleton band is 23, ..., 109, and the endpoint band has two minus-only atoms. Thus a current counterexample cannot keep the formal product as hidden actual load. The global obligation is to promote this cutset to all persistent families and to exclude or summably absorb the named exits: source failure, CRT-window failure, primitive-support escape, endpoint/reset, transport reset, epoch-pair multiplicity, and moving-residue SAE/Rankin. This is still a reduction boundary, not a global unconditional row/column proof.  $\square$

**Reduction-closed Statement 2.60** (Cutset completeness). *The formal-to-actual cutset has a deterministic completeness law. For a fixed AffineTwin modulus  $q \geq 13$ , let*

$$F_q = G_q \times H_q$$

*be the formal generator/fill residue product. Then  $F_q$  is the disjoint union of source failures, CRT-window failures, primitive actual packets, and named primitive-support or projection escapes:*

$$F_q = S_q \sqcup C_q \sqcup P_q \sqcup E_q.$$

*Consequently*

$$M_q^{\text{form}} = |S_q| + |C_q| + |P_q| + |E_q|, \quad N_q = |P_q|.$$

*Status and reference.* The deterministic lemma is archived at

`docs/monograph/prime_matrix_formal_to_actual_cutset_completeness_lemma.md.`

For a formal pair  $a \in F_q$ , first test whether the matching gap-fill source exists with the required gap, generator, fill, orientation, delay, and slot-lock key. If not,  $a \in S_q$ . If the source exists, combine the generator and shifted-fill congruences modulo  $q(q-2)$ . If no representative lies in the primitive pair support,  $a \in C_q$ . If the representative exists and the primitive depth identities, non-reuse, and non-ColumnCRT conditions all hold, then  $a \in P_q$ . The remaining cases are precisely source-and-CRT hits that fail the primitive actual conditions; they form  $E_q$ , routed to ProjectionCollision/ColumnCRT/PDEC or PrimitiveTwinSlotSupportEscape-PDEC/SAE. The four alternatives are selected by the first failed gate, hence are disjoint and exhaustive. Thus formal overcount is never hidden actual load. What remains open is not classification, but global exclusion or summable absorption of the named exits.  $\square$

**Reduction-closed Statement 2.61** (After-cutset named-exit frontier). *After the deterministic cutset split, the current sweep has no anonymous post-cutset capacity or phase escape. The 40 formal pairs split as*

$$40 = 1 + 28 + 11,$$

*where 1 pair is an actual primitive packet, 28 pairs are source failures, and 11 pairs are CRT-window empty. The latter 11 pairs are routed to the named frontier consisting of window-edge or unused-target arrival, support motion and endpoint release, primitive identity shift, moving-slot families, transport reset, epoch-pair multiplicity, and moving-residue or singleton SAE.*

*Status and reference.* The frontier router is generated by

```
experiments/prime_matrix_after_cutset_named_exit_frontier_router.py
```

and archived at

```
docs/monograph/prime_matrix_after_cutset_named_exit_frontier_router.md.
```

The current numerical gates are: edge-collision candidates 11, support motion candidates 11, minimum endpoint release  $70 = 3.5 \cdot 20$ , minimum affine depth defect 70, no fixed highfactor slot isolation failure at the replay base, no transport reset atom, epoch-pair mass  $0.023577117628562343 < 1/40$ , 37 moving residue shapes, and 300 singleton residue packets. Hence the current after-cutset frontier is closed as a classification and routing statement. The next global obligation is still to exclude or summably absorb these named exits: SourceMaterializationFailure, CRTWindowEmpty, WindowEdge/UnusedTargetArrival, SupportMotion, PrimitiveIdentityShift, MovingSlotFamily, TransportReset, EpochPairMultiplicity, MovingResidueShape, and SingletonResidue. This is a sharper reduction boundary, not an unconditional row/column theorem.  $\square$

**Reduction-closed Statement 2.62** (Terminal named-exit choke set). *The ten after-cutset named exits in the current frontier compress to five terminal chokes: Source/CRT materialization, support-motion primitive identity, transport-singleton active- $\ell$ , epoch-pair paired pressure, and moving-residue SAE. All five are closed in the current sweep as routing and finite-capacity statements, but none is yet globally excluded.*

*Status and reference.* The terminal choke router is generated by

```
experiments/prime_matrix_global_named_exit_terminal_choke_router.py
```

and archived at

```
docs/monograph/prime_matrix_global_named_exit_terminal_choke_router.md.
```

Its current explicit margins are:

| quantity                                          | current value       |
|---------------------------------------------------|---------------------|
| unresolved formal pairs                           | 0                   |
| CRT phase margin                                  | 879                 |
| minimum extra endpoint release over support width | 50                  |
| transport reset atoms                             | 0                   |
| one-slot total spare ratio                        | 0.9038893044128646  |
| one-slot minimum epoch spare ratio                | 0.6901408450704225  |
| paired-pressure slack                             | 0.8398220244716351. |

Thus the latest obstruction is not a local capacity deficit. A persistent counterexample family must repeatedly realize one of the five terminal chokes; the task is to exclude those recurrences or summably absorb them through the registered PDEC/SAE/ColumnCRT ledgers. This is still a global reduction frontier, not an unconditional row/column proof.  $\square$

**Reduction-closed Statement 2.63** (Terminal choke amplification barrier). *In the current terminal choke set, the narrowest free phase jump is the CRT-window jump of size  $40 = 2 \cdot 20$ , but that jump is not a terminal escape: it is routed to unused-target arrival or window-edge PDEC/SAE. Among the terminal capacity barriers, the narrowest one-slot transport epoch is the minus  $\ell = 71$  epoch, which uses 22 of 71 residues and therefore needs 50 new residues to overflow. The support-motion endpoint release barrier has the same extra size  $70 - 20 = 50$ .*

*Status and reference.* The amplification-barrier router is generated by

`experiments/prime_matrix_terminal_choke_amplification_barrier_router.py`

and archived at

`docs/monograph/prime_matrix_terminal_choke_amplification_barrier_router.md.`

The current ordered barriers are

| barrier                  | factor  | additive units    |
|--------------------------|---------|-------------------|
| CRTWindowPhaseJump       | 2       | 40                |
| OneSlotTransportOverflow | 71/22   | 50                |
| SupportEndpointRelease   | 70/20   | 50                |
| EpochPairPairedPressure  | 899/144 | 1 to 8 fill units |
| FixedMovingSlotCRT       | 731/104 | 703.              |

Thus the current narrow terminal obstruction is a fifty-unit cross-lock: persistent recurrence must pay a common release/residue package of size 50, or else fall into reset, PDEC, SAE, or ColumnCRT. This identifies the next global interface, but does not yet exclude that interface for all persistent families.  $\square$

**Reduction-closed Statement 2.64** (Fifty-unit cross-lock carrier separation). *The fifty-unit cross-lock is not an immediate same-carrier contradiction in the current sweep. The support side is the  $q = 31$  atom with source pair  $(19, 12)$ , generator prime  $P = 2687$ , and endpoint release  $70 - 20 = 50$ . The one-slot side is the minus  $\ell = 71$  epoch, which uses 22 of 71 residues and needs 50 new residues to overflow. These two carriers have distinct moduli, distinct phase taxonomy, and disjoint current  $P$ -bands; therefore any persistent recurrence must pass through the cross-carrier modulus  $31 \cdot 71 = 2201$ .*

*Status and reference.* The carrier-separation router is generated by

`experiments/prime_matrix_fifty_unit_cross_lock_carrier_separation_router.py`

and archived at

`docs/monograph/prime_matrix_fifty_unit_cross_lock_carrier_separation_router.md.`

It records

`same_q_or_ell=false, same_side_taxonomy=false, same_p_band=false,`  
`direct_same_carrier_contradiction=false, combined_carrier_modulus=2201.`

The fixed delay of the support atom is 80. Modulo 71 this is 9, and  $\gcd(80, 71) = 1$ , so the first 50 transported residues are distinct. The first such block fitting the current minus-71 epoch runs from  $P = 4207$  to  $P = 8127$ , inside the recorded epoch range  $4177 \leq P \leq 9257$ . Thus a short repeated-residue PDEC is not automatic. If, however, these 50 transported arrivals are all new residues in the same minus-71 epoch, then  $22 + 50 > 71$ , so one-slot capacity fails. If they are not new, or if they cannot stay synchronized, the failure is routed to TransportReset-PDEC, SingletonResidue-SAE, support-motion/unused-target exits, or moving-carrier ColumnCRT. The next global interface is therefore

**CrossCarrierFiftyUnitSynchronizationPDECOrSAE.**

This closes the same-carrier shortcut and sharpens the remaining obstruction; it is not yet an unconditional row/column proof.  $\square$

**Reduction-closed Statement 2.65** (Cross-carrier fifty-unit residue saturation). *After reconstructing the complete singleton physical records, the current cross-carrier synchronization does not overflow the minus-71 epoch inside the recorded  $P$ -band. The already-used minus-71 residue set has 22 elements. The support lattice points admitted by the current epoch range contribute 64 distinct residues; 17 of them are already used, so the combined set has size 69, leaving exactly two missing residues, 0 and 62.*

*Status and reference.* The saturation router is generated by

```
experiments/prime_matrix_cross_carrier_fifty_unit_residue_saturation_router.py
```

and archived at

```
docs/monograph/prime_matrix_cross_carrier_fifty_unit_residue_saturation_router.md.
```

It records

```
used_residue_count_reconstructed=22, exact_used_matches_capacity_ledger=true,
admitted_step_range=[19,82], admitted_distinct_residue_count=64,
admitted_intersection_existing_count=17, admitted_new_residue_count=47,
union_size_after_admitted_band=69, spare_after_admitted_band=2.
```

Every contiguous 50-step block inside the current admitted band remains below capacity: the largest block union has size 62. Thus the branch “fifty arrivals are automatically fifty new residues” is false on the actual frontier. The precise remaining phase data are:

$$P = 9647 \mapsto 62, \quad P = 9727 \mapsto 0, \quad P = 9807 \mapsto 9.$$

The first two arrivals fill the two missing residues; the third is an old residue and hence must be routed to transport reset-PDEC if the same epoch persists. Therefore the next interface is

**TwoResidueSpareEndpointExtensionOrTransportResetPDEC.**

This is a sharper current-frontier obstruction, not a proof of the global row/column theorem.  $\square$

**Reduction-closed Statement 2.66** (Two-residue spare prime-anchor filter). *The two-residue near-fill route is not a valid actual-chain closure after the prime-anchor condition on  $P$  is imposed. In the same admitted lattice  $19 \leq t \leq 82$ , only 17 of the 64 values  $P = 2687 + 80t$  are prime anchors. They contribute only 13 new residues to the minus-71 epoch; together with the 22 existing residues this covers 35 of the 70 nonzero residue classes. The two near endpoint arrivals from the previous router are composite or structurally impossible as prime anchors.*

*Status and reference.* The prime-anchor filter router is generated by

```
experiments/prime_matrix_two_residue_spare_prime_anchor_filter_router.py
```

and archived at

```
docs/monograph/prime_matrix_two_residue_spare_prime_anchor_filter_router.md.
```

It records

```
admitted_lattice_step_count=64, admitted_prime_anchor_count=17,
admitted_prime_anchor_new_residue_count=13, prime_filtered_union_size=35,
prime_filtered_nonzero_spare=35, residue_zero_prime_anchor_impossible=true.
```



The previous near endpoints satisfy

$$9647 \mapsto 62, \quad 9727 \mapsto 0, \quad 9807 \mapsto 9,$$

but all three  $P$ -values fail the prime-anchor test. Moreover the residue 0 arithmetic progression has  $P \equiv 4047 \pmod{5680}$  and  $\gcd(4047, 5680) = 71$ , so for  $P > 71$  it cannot contain a prime anchor. The first prime anchor for residue 62 is delayed to  $P = 26687$ . Full nonzero-residue coverage in the scanned progression is delayed until  $P = 98047$ , and the first prime-anchor repeat after that is  $P = 98207$ . Thus the current interface is

`PrimeAnchorFilteredNonzeroResidueCoverageOrTransportResetPDEC.`

This closes the two-residue near-fill shortcut. It still leaves a global obligation: exclude persistent long arithmetic-progression prime-anchor coverage, or route the first repeat after coverage to transport reset-PDEC.  $\square$

**Reduction-closed Statement 2.67** (Prime-anchor post-band immediate repeat). *The long prime-anchor nonzero-residue coverage branch is preempted in the current primitive epoch. After the admitted band  $19 \leq t \leq 82$ , the first prime anchor in the progression  $P = 2687 + 80t$  is*

$$t = 90, \quad P = 9887, \quad \text{residue}_{71}(P) = 18.$$

*The residue 18 is already in the original minus-71 used-residue set. Thus no new missing nonzero residue is added before the first repeat.*

*Status and reference.* The immediate-repeat router is generated by

`experiments/prime_matrix_prime_anchor_postband_immediate_repeat_router.py`

and archived at

`docs/monograph/prime_matrix_prime_anchor_postband_immediate_repeat_router.md.`

It records

```
first_postband_prime_anchor=(90,9887,18), first_postband_prime_is_repeat=true,
first_postband_prime_is_original_used_repeat=true,
new_prime_anchor_count_before_first_repeat=0,
missing_nonzero_remaining_at_first_repeat=35,
coverage_before_reset_possible_in_same_epoch=false.
```

The intervening candidates  $t = 83, \dots, 89$ , namely

$$9327, 9407, 9487, 9567, 9647, 9727, 9807,$$

are composite. Therefore, if the same reset-free epoch persists past the admitted band, it hits a used residue before it hits any new missing nonzero class and must enter transport reset-PDEC. If the epoch does not persist to that anchor, the branch has already moved to endpoint motion/SAE. The latest interface is

`ImmediatePrimeAnchorRepeatTransportResetPDECOrEndpointMotionSAE.`

This closes the current long-coverage shortcut. It does not globally exclude transport reset or endpoint motion.  $\square$

**Reduction-closed Statement 2.68** (Prime-anchor repeat reset atom). *The first post-band repeat is an exact one-slot reset atom. The original minus-71 singleton record with residue 18 occurs at  $P = 7757$ . The first post-band prime anchor has  $P = 9887$  and the same residue. Their difference is*

$$9887 - 7757 = 2130 = 30 \cdot 71.$$

*Thus the first prime anchor after the admitted band is not a new coverage packet, but a lift by 30 full  $\ell$ -periods of an existing residue packet.*

*Status and reference.* The repeat-reset atom router is generated by

```
experiments/prime_matrix_prime_anchor_repeat_reset_atom_router.py
```

and archived at

```
docs/monograph/prime_matrix_prime_anchor_repeat_reset_atom_router.md.
```

It records

```
original_record_p=7757, repeat_prime_anchor_p=9887, repeat_residue=18,
p_delta=2130, p_delta_over_ell=30, exact_same_residue_ell_translate=true,
reset_atom_instantiated=true, missing_nonzero_remaining_at_reset_atom=35.
```

The associated original slot key is 644:128:71, and the formal unit is

```
one-slot-repeat-reset|side=minus|ell=71|residue=18|
lift_delta=30|p=7757->9887 .
```

Consequently, if the same reset-free epoch extends to  $P = 9887$ , the reset-PDEC atom is already present. To avoid it, the endpoint must cut before step 90, leaving only the composite buffer  $t = 83, \dots, 89$ . The next interface is

```
OneSlotPrimeAnchorRepeatResetPDECExclusionOrEndpointMotionSAE.
```

This instantiates the local reset atom. It does not yet globally exclude all one-slot reset atoms or endpoint-motion escapes.  $\square$

**Reduction-closed Statement 2.69** (Endpoint cut zero-gain). *If the endpoint is cut before the first post-band prime anchor in order to avoid the instantiated reset atom, the cut has no prime-anchor gain in the current primitive epoch. The only buffer before the reset is*

$$t = 83, \dots, 89, \quad P = 9327, 9407, 9487, 9567, 9647, 9727, 9807,$$

*and none of these  $P$ -values is prime.*

*Status and reference.* The zero-gain endpoint router is generated by

```
experiments/prime_matrix_endpoint_cut_zero_gain_router.py
```

and archived at

```
docs/monograph/prime_matrix_endpoint_cut_zero_gain_router.md.
```

It records

```
cut_buffer_step_count=7,  formal_gap_composite_row_count=2,
    formal_gap_composite_residues=[62,0],
    actual_prime_anchor_count_before_reset=0,
    actual_new_prime_anchor_count_before_reset=0,
    endpoint_cut_actual_gain_zero=true,
    reset_atom_instantiated_at_first_prime_anchor=true.
```

The apparent gap residues 62 and 0 occur at  $P = 9647$  and  $P = 9727$ , respectively, but 9647 has factor 11 and 9727 is divisible by 71. Thus they are formal gaps only, not actual prime-anchor arrivals. If the endpoint is not cut, the next value  $P = 9887$  is the one-slot repeat-reset atom. Hence the local dichotomy is

reset-PDEC   or   zero-gain endpoint cut SAE.

The latest interface is

ZeroGainEndpointCutSAEOrOneSlotResetPDECExclusion.

This closes the current endpoint-gain explanation, not the global SAE or reset-PDEC exclusion.  $\square$

**Reduction-closed Statement 2.70** (Endpoint cut no-payload SAE). *In the current primitive epoch, the endpoint-cut branch has empty actual payload. The cut buffer  $t = 83, \dots, 89$  contains no prime-anchor row, no new prime-anchor row, and no filtered repeat prime-anchor row. The apparent gap residues 62 and 0 do not survive prime-anchor filtering.*

*Status and reference.* The no-payload endpoint router is generated by

```
experiments/prime_matrix_endpoint_cut_no_payload_sae_router.py
```

and archived at

```
docs/monograph/prime_matrix_endpoint_cut_no_payload_sae_router.md.
```

It records

```
actual_prime_anchor_count_before_reset=0,
actual_new_prime_anchor_count_before_reset=0,
filtered_repeat_prime_anchor_count_before_reset=0,
actual_payload_empty=true,  actual_payload_mass=0,
missing_nonzero_before_cut=35,  missing_nonzero_after_cut=35,
missing_nonzero_set_preserved_by_cut=true.
```

Thus the endpoint cut cannot supply the capacity required by the counterexample chain; it is a no-payload SAE branch. If that branch is not taken, the first prime anchor is already the one-slot reset-PDEC atom. The current local dichotomy is

one-slot reset-PDEC   or   no-payload endpoint SAE.

The latest interface is

OneSlotResetPDECExclusionOrNoPayloadEndpointSAESummability.

This closes the current endpoint-cut actual-load exit, not the global reset or SAE summability problem.  $\square$

**Reduction-closed Statement 2.71** (One-slot reset prefix no-relief). *If the endpoint branch is not cut, the line first accepts the one-slot reset at  $t = 90$ ,  $P = 9887$ , residue 18. After that accepted reset, there is still no new missing nonzero residue until  $t = 115$ ,  $P = 11887$ , residue 30.*

*Status and reference.* The reset-prefix no-relief router is generated by

```
experiments/prime_matrix_one_slot_reset_prefix_no_relief_router.py
```

and archived at

```
docs/monograph/prime_matrix_one_slot_reset_prefix_no_relief_router.md.
```

It records

```
reset_step=90, reset_p=9887, reset_residue=18,
prefix_prime_anchor_count_before_first_relief=7,
prefix_new_missing_nonzero_count_before_first_relief=0,
prefix_repeat_prime_anchor_count_before_first_relief=7,
first_relief_step_gap_after_reset=25, first_relief_p_gap_after_reset=2000.
```

Thus a new actual-capacity path must first cross the accepted reset-PDEC atom; before the first relief all actual prime anchors remain old residues. If the reset is rejected, the endpoint branch has already been reduced to a no-payload SAE. The latest interface is

`AcceptedResetPDECExclusionOrDelayedReliefSupportMotionSAE.`

This closes the explanation that reset immediately supplies fresh capacity, not the global exclusion of accepted reset-PDEC atoms.  $\square$

**Reduction-closed Statement 2.72** (Accepted reset full relief horizon). *After accepting the one-slot reset at  $P = 9887$ , complete relief of the remaining 35 missing nonzero residues is not local. It first completes at  $t = 1192$ ,  $P = 98047$ , after 1102 synchronized steps.*

*Status and reference.* The accepted-reset full-relief horizon router is generated by

```
experiments/prime_matrix_accepted_reset_full_relief_horizon_router.py
```

and archived at

```
docs/monograph/prime_matrix_accepted_reset_full_relief_horizon_router.md.
```

It records

```
missing_nonzero_count_at_reset=35,
first_relief=(step=115,P=11887,residue=30),
full_relief=(step=1192,P=98047,residue=67),
full_relief_step_gap_after_reset=1102,
full_relief_p_gap_after_reset=88160,
new_relief_prime_anchor_count_until_full_relief=35,
repeat_prime_anchor_count_until_full_relief=225.
```

In addition, 101 formal missing-residue hits before full relief are composite and cannot be counted as actual load. Hence full relief is a long-horizon support-motion obligation, not a local repair inside the current primitive epoch. The latest interface is

`AcceptedResetPDECExclusionOrLongReliefHorizonSupportMotionSAE.`

This quantifies the delayed relief branch; it does not yet exclude the global accepted-reset family.  $\square$

**Reduction-closed Statement 2.73** (Long relief cycle debt). *The long-relief branch has an explicit 71-cycle phase debt. Among the remaining 35 nonzero residues, only 8 receive prime relief on the first post-reset phase hit; the other 27 require later 71-period returns.*

*Status and reference.* The long-relief cycle-debt router is generated by

```
experiments/prime_matrix_long_relief_cycle_debt_router.py
```

and archived at

```
docs/monograph/prime_matrix_long_relief_cycle_debt_router.md.
```

It records

```
ell=71, period_p=5680,
zero_cycle_relief_count=8, positive_cycle_debt_residue_count=27,
total_cycle_debt=101, total_composite_wait_count=101,
matches_full_horizon_composite_missing_count=true,
max_cycle_debt=15, max_cycle_debt_residue=67.
```

Thus the 101 composite formal missing-residue hits in the full-horizon ledger are exactly the sum of the per-residue cycle debts. The worst residue waits from  $P = 12847$  to  $P = 98047$ , a 15-period delay. The latest interface is

`LongReliefCycleDebtPDECEExclusionOrSupportMotionSAESummability.`

This records the phase cost of long relief; it does not yet prove global summability or global PDEC exclusion.  $\square$

**Reduction-closed Statement 2.74** (One-period relief deficit). *In the first complete 71-residue cycle after the accepted reset, every missing nonzero residue appears formally once, but only 8 of the 35 formal hits are actual prime relief.*

*Status and reference.* The one-period relief deficit router is generated by

```
experiments/prime_matrix_one_period_relief_deficit_router.py
```

and archived at

```
docs/monograph/prime_matrix_one_period_relief_deficit_router.md.
```

It records

```
period_is_complete_residue_cycle=true,
missing_nonzero_required=35, formal_missing_hit_count=35,
actual_relief_count_in_one_period=8,
composite_missing_count_in_one_period=27,
repeat_prime_anchor_count_in_one_period=10,
relief_deficit_after_one_period=27.
```

The 27 composite missing hits match exactly the positive cycle-debt residues. Thus one reset-local period is not enough actual load; it forces cycle debt or a support-motion SAE branch. The latest interface is

`OnePeriodReliefDeficitForcesCycleDebtPDECOrSupportMotionSAE.`

This is still a reduced hardpoint, not a global theorem closure.  $\square$

**Reduction-closed Statement 2.75** (Cycle-debt CRT cover pressure). *The 27 positive cycle-debt residues are not free local drift. Their composite waits form a CRT cover in the cycle coordinate, using 24 distinct blocking prime factors.*

*Status and reference.* The CRT cover pressure router is generated by

```
experiments/prime_matrix_cycle_debt_crt_cover_pressure_router.py
```

and archived at

```
docs/monograph/prime_matrix_cycle_debt_crt_cover_pressure_router.md.
```

It records

```
positive_cycle_debt_residue_count=27,
total_composite_waits=101,
global_unique_blocker_factor_count=24,
global_blocker_lcm=337212073559813724487421695331234639247,
max_row_residue=67, max_row_cycle_debt=15,
max_row_blocker_lcm=55140500775337593.
```

For a fixed missing residue the same-residue column is

$$P(k) = P_0 + 5680k.$$

Each composite wait imposes one congruence class in the cycle coordinate modulo its least prime blocker. Hence persistent reproduction of the debt pattern requires a large CRT phase package, not merely a local endpoint translation. The latest interface is

```
CycleDebtCRTCoverPressurePDECOrGlobalSupportMotionSAE.
```

This records the global-family pressure, while leaving the PDEC exclusion or SAE summability proof open.  $\square$

**Reduction-closed Statement 2.76** (Cycle-debt transverse CRT independence). *The cycle-debt CRT cover is transverse to the local period 5680. It cannot be absorbed by a local period translation or by a same-slot drift.*

*Status and reference.* The transverse CRT independence router is generated by

```
experiments/prime_matrix_cycle_debt_transverse_crt_independence_router.py
```

and archived at

```
docs/monograph/prime_matrix_cycle_debt_transverse_crt_independence_router.md.
```

It records

```
period_p=5680=2^4*5*71,
global_unique_blocker_factor_count=24, transverse_blocker_factor_count=24,
all_blocker_factors_coprime_to_period_p=true,
gcd_global_blocker_lcm_with_period_p=1,
combined_period_equals_product=true,
all_row_shift_replay_classes_zero=true,
all_row_lcm_exceeds_phase_support_width=true.
```

For a fixed residue column  $P(k) = P_0 + 5680k$ , every composite wait has a blocking prime  $q$  with  $\gcd(q, 5680) = 1$ . Replaying the same wait after a cycle shift  $K$  therefore forces  $K \equiv 0 \pmod{q}$ . Rowwise this gives  $K \equiv 0 \pmod{\text{lcm}_{\text{row}}}$ , and across all rows it gives

$$K \equiv 0 \pmod{337212073559813724487421695331234639247}.$$

The modulus is coprime to 5680, so this is an independent transverse CRT phase package, not a factor of the local period. The latest interface is

**TransverseCRTCoverPDECExclusionOrGlobalSupportMotionSAE.**

Thus local period absorption is closed in this certificate, while global PDEC exclusion or SAE summability remains open.  $\square$

**Reduction-closed Statement 2.77** (Cycle-debt sparse replay barrier). *If the transverse CRT cover keeps the same blocking graph, exact replay is extremely sparse. A non-sparse persistent branch must move the blocker graph and hence enters a moving-cover PDEC interface.*

*Status and reference.* The sparse replay barrier router is generated by

```
experiments/prime_matrix_cycle_debt_sparse_replay_barrier_router.py
```

and archived at

```
docs/monograph/prime_matrix_cycle_debt_sparse_replay_barrier_router.md.
```

It records

```
global_exact_replay_cycle_modulus= 337212073559813724487421695331234639247,
global_exact_replay_p_gap= 1915364577819741955088555229481412750922960,
    max_cycle_debt_support_width=15,
    full_relief_cycle_span_ceiling=16,
    single_full_debt_copy_per_full_relief_horizon=true,
exact_replay_branch_is_sae_sparse_current_certificate=true,
    non_sparse_persistence_forces_moving_blocker_map=true.
```

The full debt word can replay exactly only after a shift by the global transverse lcm. The observed full-relief horizon has length  $\lceil 1102/71 \rceil = 16$  residue cycles, whereas the replay modulus is the large integer displayed above. Thus any local or medium support-motion window shorter than that modulus contains at most one exact copy of the full word. The exact branch is therefore a sparse SAE branch. If a counterexample branch requires positive-density or short-window persistence, it must alter the blocking primes, phases, or row combination; this is the moving transverse cover PDEC branch. The latest interface is

**SparseReplaySAEOrMovingTransverseCoverPDEC.**

This is a sharper reduced hardpoint, not a global theorem closure.  $\square$

**Reduction-closed Statement 2.78** (Cycle-debt near-shift exit boundary). *The moving-cover branch cannot be a small shift of the old debt prefix. Any near shift either hits an exit prime inside the shifted word, or else has no old prefix left to reuse and must build a fresh cover.*

*Status and reference.* The near-shift exit-boundary router is generated by

`experiments/prime_matrix_cycle_debt_near_shift_exit_boundary_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_near_shift_exit_boundary_router.md`.

It records

```
near_shift_limit_cycles=16,  positive_cycle_debt_residue_count=27,
total_cycle_debt_mass=101,  max_cycle_debt=15,
all_near_shifts_close_old_prefix_reuse=true,
shift_1_exit_prime_collision_row_count=27,
shift_1_exit_prime_collision_debt_mass=101,
shift_max_cycle_debt_exit_prime_collision_row_count=1,
shift_near_limit_fresh_cover_required_row_count=27.
```

For every positive-debt residue, the old composite prefix is immediately followed by the first relief prime in that same residue column. A shift  $1 \leq K \leq d$  for a row of debt  $d$  pulls that exit prime into the shifted debt word. Hence  $K = 1$  breaks all 27 rows, while every  $1 \leq K \leq 15$  breaks all rows with  $d \geq K$ . At  $K = 16$  no row has old prefix support left, so all 27 rows require fresh cover material. The latest interface is

**FreshMovingCoverPDECOrGlobalSupportMotionSAE.**

Thus near-shift reuse of the old CRT cover is closed in this certificate; the remaining branch is fresh moving-cover PDEC or genuine global support-motion SAE.  $\square$

**Reduction-closed Statement 2.79** (Cycle-debt fresh-cover prime obstacle). *On the same 27 positive-debt support rows, every near shifted fresh window contains actual prime obstacles. Same-support fresh cover is therefore closed unless those actual primes are deleted by a PDEC.*

*Status and reference.* The fresh-cover prime-obstacle router is generated by

`experiments/prime_matrix_cycle_debt_fresh_cover_prime_obstacle_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_fresh_cover_prime_obstacle_router.md`.

It records

```
total_cycle_debt_mass_per_shift=101,  tested_shift_count=16,
total_tested_same_support_slots=1616,
total_prime_obstacles_all_near_shifts=364,
total_new_prime_obstacles_all_near_shifts=263,
all_near_shift_same_support_windows_have_prime_obstacles=true,
min_prime_obstacle_count_per_shift=17,
shift_near_limit_prime_obstacle_count=27,
same_support_fresh_cover_closed_current_certificate=true.
```



For each  $K = 1, \dots, 16$ , the shifted window on the same support rows has the same total length 101 as the original debt word. Direct primality checking of those actual  $P$ -values shows at least 17 prime obstacles in every such shifted word; the fully fresh  $K = 16$  window still has 27, all new. Hence a same-support fresh moving cover cannot be an all-composite word. The latest interface is

`FreshCoverPrimeObstaclePDECOrSupportRowReplacementSAE.`

The remaining branch must either delete actual prime obstacles, forming a prime-obstacle PDEC, or replace support rows and enter a support-row replacement SAE/PDEC interface.  $\square$

**Reduction-closed Statement 2.80** (Cycle-debt support-row replacement Hall barrier). *Support-row replacement is a Hall capacity problem over the 35 missing residue candidates. Fourteen of the sixteen near phases fail the Hall threshold test; only  $K = 13, 14$  survive.*

*Status and reference.* The support-row replacement Hall router is generated by

`experiments/prime_matrix_cycle_debt_support_row_replacement_hall_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_support_row_replacement_hall_router.md.`

It records

```
candidate_missing_residue_count=35,  needed_positive_debt_row_count=27,
total_demand_width=101,  max_required_width=15,
    hall_fail_shift_count=14,
    hall_survivor_shifts=[13,14],
    k13_assigned_immediate_relief_rows=6,
    k14_assigned_immediate_relief_rows=8.
```

For a phase  $K$ , a candidate row has capacity equal to the longest composite run starting at  $K$ , capped at the maximum required width 15. For every threshold  $t$ , Hall requires

$$\#\{\text{candidate rows with capacity} \geq t\} \geq \#\{\text{demand rows with debt} \geq t\}.$$

This condition fails for

$$K = 1, 2, \dots, 12, 15, 16.$$

Only  $K = 13$  and  $K = 14$  pass, and both are tight survivor phases requiring immediate-relief rows and large-scale row replacement. The latest interface is

`K13K14SupportReplacementSurvivorPDECOrGlobalSAE.`

This is a two-phase survivor hardpoint, not a global theorem closure.  $\square$

**Reduction-closed Statement 2.81** (K13/K14 tight-Hall survivor island). *The two support-row replacement survivors  $K = 13, 14$  are not free moving replacement phases. They form a tight Hall island whose zero-slack layers force row replacement, immediate-relief support, and transverse CRT pressure.*

*Status and reference.* The K13/K14 survivor rigidity router is generated by

`experiments/prime_matrix_cycle_debt_k13_k14_survivor_rigidity_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_k13_k14_survivor_rigidity_router.md.`

It records

```

k13_zero_slack_thresholds=[1,4,7,8],
k14_zero_slack_thresholds=[7,8,13,15],
k13_minimum_replacement_count=20, k14_minimum_replacement_count=19,
k13_minimum_immediate_relief_rows=6, k14_minimum_immediate_relief_rows=6,
k13_minimum_overstretch_units=61, k14_minimum_overstretch_units=65,
all_tight_layers_have_transverse_lcm_exceeding_period=true.

```

Here a zero-slack threshold  $t$  means that the number of candidate rows with capacity at least  $t$  is exactly the number of demand rows with debt at least  $t$ . Consequently the whole supply set of that layer is forced: deleting one such row, or moving one of its first  $t$  composite slots to a prime obstacle, breaks the Hall inequality at that threshold. For  $K = 13$ , the forced shells include the disjoint high shell

$$\{1, 13, 19, 30\} \rightarrow \{17, 23, 58, 67\}$$

at threshold 8, plus the singleton shell  $31 \rightarrow 15$  at threshold 7. For  $K = 14$ , the top shells force  $13 \rightarrow 67$ ,  $19 \rightarrow 23$ , and  $70 \rightarrow 15$  at the corresponding thresholds.

The router then solves exact minimum-cost matching problems rather than relying on the previous greedy witness. Even after optimization,  $K = 13$  requires at least 20 row replacements and 6 immediate-relief rows, while  $K = 14$  requires at least 19 row replacements and 6 immediate-relief rows. The forced composite slots in every tight layer have transverse CRT least common multiple larger than the local period 5680. The neighboring phases are not survivors:  $K = 12$  fails Hall at threshold 1, and  $K = 15$  fails at threshold 6. Thus the remaining support-row branch is compressed to

`K13K14TightHallCRTIslandPDECOrMovingSupportSAE.`

This still is not a global theorem closure; it replaces the previous free replacement interpretation with a named tight-Hall CRT island.  $\square$

**Reduction-closed Statement 2.82** (Shell phase graph obstruction). *The  $K = 13, 14$  tight-Hall island cannot be absorbed by a single residue translation of the support. Its forced shells split into multiple incompatible phase deltas.*

*Status and reference.* The shell phase graph router is generated by

`experiments/prime_matrix_cycle_debt_shell_phase_graph_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_shell_phase_graph_router.md.`

It records

```
k13_forced_shell_matching_product=252829237248000,
    k14_forced_shell_matching_product=2,
k13_forced_edge_count=1,  k14_forced_edge_count=3,
    k13_single_translation_support_possible=false,
    k14_single_translation_support_possible=false,
    k13_core_min_distinct_delta_lower_bound=7,
    k14_core_min_distinct_delta_lower_bound=4.
```

For each forced Hall shell the router builds the bipartite graph from supply residues to demand residues, keeping only edges whose capacity can cover the demand width. A single translation support motion would require one common  $\delta \pmod{71}$  across all forced shells. For  $K = 14$  three forced edges already have incompatible deltas:

$$13 \mapsto 67 \quad (\delta = 54), \quad 19 \mapsto 23 \quad (\delta = 4), \quad 70 \mapsto 15 \quad (\delta = 16).$$

The remaining 8..12 shell has only two possible matchings; the whole forced core still needs at least four distinct deltas. For  $K = 13$ , the singleton forced edge is  $31 \mapsto 15$  with  $\delta = 55$ , and the enumerable core shells  $\geq 8$ , 7..7, and 4..6 already require at least seven distinct deltas. The large 1..3 shell is not used to lower this count; it can only preserve or increase the number of deltas.

Thus the remaining branch is not an ordinary single-phase support translation. It is compressed to

```
NonAffineShellPhaseFragmentPDECOOrMultiDeltaSupportSAE.
```

This still is not a global theorem closure; it closes the single-translation support-motion explanation and leaves a non-affine, multi-delta support branch.  $\square$

**Reduction-closed Statement 2.83** (Multi-delta core CRT load). *The non-affine multi-delta support branch carries a large transverse CRT load on its forced core. It is not a lightweight phase-noise explanation.*

*Status and reference.* The multi-delta core CRT-load router is generated by

```
experiments/prime_matrix_cycle_debt_multi_delta_core_crt_load_router.py
```

and archived at

```
docs/monograph/prime_matrix_cycle_debt_multi_delta_core_crt_load_router.md.
```

It records

```
k13_enumerated_core_matching_count=96,
k13_minimum_delta_count=7,  k13_minimum_delta_matching_count=2,
    k13_best_min_delta_global_lcm_log10=36.165,
    k14_enumerated_core_matching_count=2,
k14_minimum_delta_count=4,  k14_minimum_delta_matching_count=1,
    k14_best_min_delta_global_lcm_log10=31.716.
```

For each core edge  $a \mapsto b$ , the router expands the source row from the given shift  $K$  for the full demand width of  $b$ , records the actual composite slots, and multiplies the distinct smallest blocking

factors. The resulting lanes are grouped by  $\delta = b - a \pmod{71}$ . For  $K = 14$ , the minimum multi-delta core is unique; its four lanes  $\delta = 4, 16, 28, 54$  carry total demand width 52 and have global least common multiple about  $10^{31.716}$ . For  $K = 13$ , the enumerable rigid core has 96 matchings, only two of which realize the minimum of seven deltas; the best such core has total demand width 71 and global least common multiple about  $10^{36.165}$ .

Both least common multiples are far larger than the local period 5680. Thus the latest branch is compressed to

`MultiDeltaCoreCRTLoadPDECOrLowShellFullResidueSAE.`

This still is not a global theorem closure: it closes the light multi-delta noise interpretation, while leaving a heavy CRT-load PDEC or the  $K = 13$  low-shell full-residue SAE as the next explicit interface.  $\square$

**Reduction-closed Statement 2.84** (K13 low-shell delta skeleton). *The  $K = 13$  low shell does not require a full-residue escape. Relative to the rigid core, it adds only two delta lanes and turns  $K = 13$  into a full-debt nine-lane CRT-load branch.*

*Status and reference.* The low-shell delta skeleton router is generated by

`experiments/prime_matrix_cycle_debt_low_shell_delta_skeleton_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_low_shell_delta_skeleton_router.md.`

It records

```
low_shell_possible_delta_count=71,
low_shell_minimum_delta_count=6,
low_shell_minimum_delta_witness=[23,35,38,58,68,70],
core_delta_count_before_low_shell=7,
low_shell_new_delta_count_over_core=2,
low_shell_new_deltas_over_core=[0,66],
full_k13_delta_count_after_low_shell=9,
full_k13_total_required_width=101,
full_k13_global_lcm_log10=42.095.
```

The router formulates the low shell as a binary matching problem. Edge variables choose the supply-to-demand matching, while delta variables record which residue lanes are used. First, with unit cost on every delta, the low shell alone needs only six deltas despite having all 71 deltas available. Second, after the seven core deltas from the  $K = 13$  rigid core are given zero cost, the low shell requires only two new lanes,  $\delta = 0$  and  $\delta = 66$ . Expanding the core and low-shell edges into their actual composite slots gives a full 101-width  $K = 13$  load with nine delta lanes and global CRT least common multiple about  $10^{42.095}$ .

Thus the latest branch is compressed to

`K13FullDebtNineLaneCRTLoadPDECOrResidualSlackTailSAE.`

This still is not a global theorem closure; it closes the low-shell full-residue free-escape interpretation and leaves a full-debt CRT-load PDEC or a residual slack-tail SAE.  $\square$

**Reduction-closed Statement 2.85** (K13 full-debt nine-lane residual tail). *The residual slack-tail in the  $K = 13$  full-debt nine-lane branch is not an artifact of the chosen matching. It is forced by the layer Hall ledger and already carries a transverse CRT load larger than the local period.*

*Status and reference.* The K13 full-debt nine-lane tail router is generated by

```
experiments/prime_matrix_cycle_debt_k13_full_debt_nine_lane_tail_router.py
```

and archived at

```
docs/monograph/prime_matrix_cycle_debt_k13_full_debt_nine_lane_tail_router.md.
```

It records

```
capacity_row_count=27, demand_row_count=27,
all_capacity_rows_mandatory=true,
total_capacity=119, total_demand_width=101,
unavoidable_tail_slot_count=18,
zero_slack_layers=[1,4,7,8],
allowed_deltas=[0,8,22,39,54,55,58,59,66],
min_tail_factor_count=8, min_tail_lcm_log10=11.488.
```

At threshold 1, the Hall count is 27 supply rows against 27 demand rows. Hence every positive-capacity row is mandatory in any full-debt matching. The total capacity is 119, while the full demand is 101; equivalently the sum of the layer slacks is 18. Thus the residual tail contains exactly 18 slots independently of the particular nine-lane matching witness.

The zero-slack layers 1, 4, 7, 8 are the local rigidity gates. Moving a slot through one of these layers without a compensating arrival would immediately break Hall capacity. Within the fixed nine-lane family, a MILP over all allowed 71 supply-demand edges optimizes the active tail blockers. Even under the minimizing objective, the tail activates at least eight distinct blocking factors,

3, 7, 11, 13, 19, 127, 151, 281,

whose least common multiple has logarithm about 11.488, already exceeding the local period 5680.

Thus the latest branch is compressed to

K13LayerSlackTailCRTInvariantPDECOrMovingFamilySAE.

This still is not a global theorem closure. It closes the interpretation that the residual slack-tail is a matching artifact or a cost-free tail; the remaining branch is a layer-invariant tail-CRT PDEC or a genuinely moving family SAE. □

**Reduction-closed Statement 2.86** (K13 tail gate drift). *The  $K = 13$  layer-tail profile has no near-shift replay. The only other Hall-surviving shift is  $K = 14$ , and it changes the zero-slack gates and increases the tail balance.*

*Status and reference.* The K13 tail gate-drift router is generated by

```
experiments/prime_matrix_cycle_debt_k13_tail_gate_drift_router.py
```

and archived at

```
docs/monograph/prime_matrix_cycle_debt_k13_tail_gate_drift_router.md.
```

It records

```

    hall_pass_shifts=[13,14],
    same_slack_vector_shifts=[13],  same_zero_gate_set_shifts=[13],
    k13_tail_balance=18,  k13_zero_slack_layers=[1,4,7,8],
    k14_tail_balance=31,  k14_tail_increase_over_k13=13,
    k14_zero_slack_layers=[7,8,13,15],
    k14_slack_l1_distance_from_k13=19.

```

The router compares every near shift  $1 \leq K \leq 16$  with the  $K = 13$  slack vector. The complete vector is repeated only at the base shift itself. The same is true for the zero-gate set. Since the Hall-surviving shifts are only 13 and 14, any near moving branch that leaves  $K = 13$  must enter  $K = 14$ .

But  $K = 14$  is not a same-profile replay. It preserves only the common gates 7, 8, loses the low gates 1, 4, gains the high gates 13, 15, and raises the tail balance from 18 to 31. Hence a moving family cannot carry the  $K = 13$  layer-tail invariant through the near-shift window without changing the gate structure.

Thus the latest branch is compressed to

**K13GateProfileNoNearReplayPDECOrK14HighGateDriftSAE.**

This still is not a global theorem closure. It closes the near same-profile moving replay interpretation; the remaining branch is either a fixed  $K = 13$  gate-profile PDEC or a  $K = 14$  high-gate drift SAE/PDEC.  $\square$

**Reduction-closed Statement 2.87** (K14 high-gate low-shell skeleton). *The  $K = 14$  high-gate drift branch is not an unregistered moving family. Its tight shells leave only two high-core matchings, and the remaining low shell compresses to a finite delta skeleton.*

*Status and reference.* The K14 high-gate low-shell skeleton router is generated by

```
experiments/prime_matrix_cycle_debt_k14_high_gate_low_shell_skeleton_router.py
```

and archived at

```
docs/monograph/prime_matrix_cycle_debt_k14_high_gate_low_shell_skeleton_router.md.
```

It records

```

    k14_zero_slack_layers=[7,8,13,15],
    k14_high_gate_core_alternative_count=2,
    k14_best_core_deltas=[4,16,28,54],
    k14_low_shell_minimum_delta_count=6,
    k14_low_shell_new_deltas_over_core=[2,48,58,70],
    k14_full_delta_count_after_low_shell=8,
    k14_full_total_required_width=101,
    k14_full_global_lcm_log10=48.508.

```

The high gates force the singleton shells  $13 \mapsto 67$ ,  $19 \mapsto 23$ , and  $70 \mapsto 15$ , together with one of two matchings on the 8.12 shell. For each high-core candidate, the router removes the already-used supply and demand residues and solves the remaining low-shell matching by a binary delta-skeleton

MILP. The best full branch has core deltas 4, 16, 28, 54, and the low shell adds only the four lanes 2, 48, 58, 70.

Expanding the resulting twenty-seven edges into actual composite slots covers all 101 demand-width units. The eight-lane load has global CRT least common multiple about  $10^{48.508}$ , far beyond the local period 5680. Thus the  $K = 14$  moving branch returns to a full-debt CRT-load PDEC; it is no longer an unnamed high-gate drift SAE.

Thus the latest branch is compressed to

`K13FixedGateProfilePDECOrK14FullDebtEightLaneCRTLoadPDEC.`

This still is not a global theorem closure. It closes the unregistered  $K = 14$  high-gate moving-family explanation and leaves a fixed  $K = 13$  gate-profile PDEC or the registered  $K = 14$  eight-lane CRT-load PDEC.  $\square$

**Reduction-closed Statement 2.88** (Two-survivor terminal CRT bifurcation). *The  $K = 13$  and  $K = 14$  terminal branches are not two views of the same support motion. They form a named terminal CRT bifurcation.*

*Status and reference.* The two-survivor terminal CRT bifurcation router is generated by

`experiments/prime_matrix_cycle_debt_two_survivor_terminal_crt_bifurcation_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_two_survivor_terminal_crt_bifurcation_router.md.`

It records

```
k13_delta_count=9,  k14_delta_count=8,
    common_deltas=[54,58],
delta_union_count=15,  delta_symmetric_difference_count=13,
source_intersection_count=19,  source_symmetric_difference_count=16,
target_intersection_count=27,  pair_intersection_count=1,
    union_lcm_log10=57.156,  tail_union_lcm_log10=24.632.
```

Both terminal branches cover the same 27 target demand residues. However, their source supports overlap in only 19 rows, and the actual source-to-target matching overlaps in only one edge,  $13 \mapsto 67$ . At the lane level, only the two deltas 54 and 58 are common; the symmetric difference contains 13 lanes. Thus  $K = 13$  and  $K = 14$  cannot be treated as a single support pattern with a small phase perturbation.

The combined blocker set has least common multiple about  $10^{57.156}$ , and the combined tail blocker set has least common multiple about  $10^{24.632}$ . Both are far larger than the local period 5680. Therefore the remaining branch is no longer an unnamed support-motion SAE. It is a terminal two-survivor CRT bifurcation PDEC that must be excluded directly.

Thus the latest branch is compressed to

`TwoSurvivorTerminalCRTBifurcationPDECExclusion.`

This still is not a global theorem closure. It closes the interpretation that the two survivor branches can absorb each other as one unnamed moving family.  $\square$

**Reduction-closed Statement 2.89** (Two-survivor switch-arrival pressure). *The terminal two-survivor bifurcation cannot be absorbed by the common anchor or by the two common delta lanes. Avoiding a fixed-branch PDEC forces a fresh-arrival ColumnCRT load or a branch-exclusive CRT-load exclusion.*

*Status and reference.* The two-survivor PDEC exclusion router is generated by

```
experiments/prime_matrix_cycle_debt_two_survivor_pdec_exclusion_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-two-survivor-pdec-exclusion-router.md.
```

It records

```
common_pair_count=1, common_pair_width=15
forced_rematched_target_count=26, forced_rematched_target_width=86
minimum_branch_exclusive_delta_width=70
minimum_branch_exclusive_delta_lcm_log10=32.582
k14_only_assigned_width=29, k14_only_capacity_at_k13=0
k14_only_capacity_at_k14=39, k14_arrival_lcm_log10=20.205
k13_same_support_deficit_at_k14=8, common_source_changed_count=18
```

The only common actual source-target edge is  $13 \mapsto 67$ , and it accounts for only 15 units of target width. Hence the remaining 26 targets, of total width 86, must be genuinely rematched. The common delta lanes 54, 58 do not carry this rematching load: each terminal branch still has at least 70 units of width on branch-exclusive deltas, with least common multiple at least  $10^{32.582}$  in the current certificate.

On the switching side, the  $K = 14$  branch uses eight source rows that have zero capacity at  $K = 13$ . These rows acquire 39 units of capacity at  $K = 14$  and carry 29 units of assigned demand, with arrival blocker lcm about  $10^{20.205}$ . Conversely, dragging the  $K = 13$  source set to  $K = 14$  leaves only 93 units of capacity against the 101-unit demand, a deficit of 8.

Thus the latest branch is compressed to

```
TerminalSwitchArrivalColumnCRTPDECOrBranchExclusiveCRTLoadExclusion.
```

This is still not a global theorem closure. It closes the interpretation that the terminal bifurcation can be repaired by the common anchor or the common delta lanes; the next obligation is to exclude the switch-arrival ColumnCRT load or the persistent branch-exclusive CRT loads.  $\square$

**Reduction-closed Statement 2.90** (Terminal switch arrival wall). *The  $K = 14$  fresh-arrival source is not a smoothly opening capacity source. It is an eight-prime entry wall followed by a post-wall CRT load.*

*Status and reference.* The terminal switch arrival wall router is generated by

```
experiments/prime_matrix_cycle_debt_terminal_switch_arrival_wall_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-terminal-switch-arrival-wall-router.md.
```

It records



```

arrival_source_count=8, all_arrival_sources_zero_capacity_at_k13=true
all_entry_wall_cycles_equal_13=true, all_entry_wall_positions_zero_at_k13=true
entry_wall_lcm_log10=39.482
postwall_capacity_total=39, postwall_assigned_width_total=29
postwall_assigned_lcm_log10=20.205
entry_plus_assigned_lcm_log10=59.687
entry_plus_postwall_lcm_log10=68.361

```

The eight  $K = 14$ -only source rows all have zero capacity at  $K = 13$ . More precisely, each of them is cut at  $K = 13$  by a prime obstacle at window-position 0. These eight entry primes have least common multiple about  $10^{39.482}$ , coprime to the local period 5680.

After crossing this entry wall, the  $K = 14$  branch obtains 39 post-wall capacity slots. It uses 29 of them for the actual assignment, with post-wall assigned blocker lcm about  $10^{20.205}$ . Combining the entry wall with the assigned post-wall load gives lcm about  $10^{59.687}$ , and combining the entry wall with all post-wall slots gives lcm about  $10^{68.361}$ , still coprime to 5680.

Thus the latest branch is compressed to

**EightPrimeEntryWallPostWallCRTExclusionOrBranchExclusiveCRTLoadExclusion.**

This is still not a global theorem closure. It closes the interpretation that the  $K = 14$  fresh-arrival capacity is an unnamed smooth deformation; the next obligation is to exclude the eight-prime entry-wall/post-wall CRT load, or the parallel branch-exclusive CRT loads.  $\square$

**Reduction-closed Statement 2.91** (Coupled branch and entry-wall load). *The entry-wall/post-wall load and the branch-exclusive load are not independent terminal exits. They couple on the  $K = 14$  branch; the  $K = 13$  branch leaves its own branch-exclusive CRT load.*

*Status and reference.* The coupled branch entry-wall router is generated by

```
experiments/prime_matrix_cycle_debt_coupled_branch_entry_wall_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-coupled-branch-entry-wall-router.md.
```

It records

```

k13_branch_exclusive_width=73, k13_branch_exclusive_lcm_log10=36.678
k14_branch_exclusive_width=70, k14_arrival_assigned_width=29
k14_arrival_branch_overlap_width=21
k14_branch_arrival_union_width=78
k14_branch_plus_entry_plus_postwall_lcm_log10=85.024
both_branches_plus_k14_entry_postwall_lcm_log10=99.349

```

Thus the  $K = 14$  terminal branch does not choose between entry-wall load and branch-exclusive load. It carries both. The 29-unit arrival load overlaps the 70-unit branch-exclusive load in 21 units, but their union still forces 78 units of actual load. After adding the entry wall and all post-wall slots, the  $K = 14$  coupled lcm is about  $10^{85.024}$ , coprime to the local period 5680.

The other terminal possibility is  $K = 13$ . There the remaining branch-exclusive load has width 73 and lcm about  $10^{36.678}$ . If both branch-exclusive loads are combined with the  $K = 14$  entry/post-wall load, the audit lcm is about  $10^{99.349}$ .

Thus the latest branch is compressed to

**K13BranchExclusiveCRTLoadExclusionOrK14CoupledEntryBranchCRTWallExclusion.**

This is still not a global theorem closure. It replaces the loose entry-wall/branch-exclusive pair by two branch-specific CRT-load exclusions.  $\square$

**Reduction-closed Statement 2.92** (Branch replay support gap). *The registered terminal branch loads cannot be reproduced by a local moving-slot replay. Any replay of the same blocker package has period at least its CRT lcm, which is already far larger than the phase support width.*

*Status and reference.* The branch replay support-gap router is generated by

```
experiments/prime_matrix_cycle_debt_branch_replay_support_gap_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-branch-replay-support-gap-router.md.
```

It records

```
period_p=5680, near_shift_limit_cycles=16
k13_branch_exclusive_width=73
k14_branch_arrival_union_width=78
k14_coupled_audit_slot_count_all_postwall=117
smallest_log10_margin_over_support_width=30.737
all_blocks_coprime_to_period=true
all_nonzero_replay_moduli_exceed_support_width=true
local_moving_slot_replay_excluded_for_registered_blocks=true.
```

The proof is a direct CRT replay calculation. Let  $M = 5680$ . If a registered blocker package  $B$  reappears after  $T$  local periods with the same blockers, then for every  $q \in B$

$$n_q + TM \equiv 0 \pmod{q}, \quad n_q \equiv 0 \pmod{q}.$$

All registered  $q$ 's are coprime to  $M$ . Hence  $T \equiv 0 \pmod{q}$  for every  $q \in B$ , and the least nonzero replay period is  $\text{lcm}(B)$ .

For the  $K = 13$  branch-exclusive block this replay modulus is about  $10^{36.678}$ , while the phase support width is 73. For the  $K = 14$  coupled branch-plus-entry-plus-postwall block the replay modulus is about  $10^{85.024}$ , while the actual support width is 78 and the full audit slot count is 117. Therefore the registered high-factor absorption slots cannot be carried by local moving-slot support motion.

Thus the latest branch is compressed to

```
BranchReplayColumnCRTPDECExclusionOrIsolatedTerminalAtomAbsorption.
```

This is still not a global theorem closure. It closes the local replay interpretation; any far replay is a named ColumnCRT/PDEC package, and any non-replaying survivor must be treated as an isolated finite atom.  $\square$

**Reduction-closed Statement 2.93** (Global dichotomy for branch replay). *The isolated-atom alternative is not a global structural escape. A persistent registered replay block is a Column-CRT/PDEC family; a nonpersistent registered block is only a finite atom requiring finite base checking.*

*Status and reference.* The branch replay global dichotomy router is generated by

```
experiments/prime_matrix_cycle_debt_branch_replay_global_dichotomy_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-branch-replay-global-dichotomy-router.md.
```

It records

```
registered_replay_block_count=6
minimum_cycle_replay_modulus_log10=32.582
minimum_p_space_columncrt_modulus_log10=36.337
maximum_p_space_columncrt_modulus_log10=103.103
persistent_registered_replay_routes_to_columncrt_pdec=true
isolated_atoms_cannot_form_infinite_registered_family=true
finite_atom_base_check_required=true.
```

There are only finitely many registered replay blocks in the current terminal certificate. If a global counterexample chain used one of them infinitely often with the same blocker package, the previous replay lemma gives a fixed cycle congruence  $T \equiv 0 \pmod{\text{lcm}(B)}$ . In the original  $P$ -coordinate this is a fixed ColumnCRT class modulo

$$5680 \cdot \text{lcm}(B).$$

The smallest such  $P$ -space modulus in the registered list is already about  $10^{36.337}$ .

If no registered replay block occurs infinitely often, then the registered atoms are finite and cannot be a global structural escape. They still require finite base checking; this router does not erase that obligation. If the blocker package changes, then the object is no longer the same replay block and must return to a PDEC/SAE route or to a new named router.

Thus the latest branch is compressed to

**BranchReplayColumnCRTPDECExclusionOrFiniteAtomBaseCheck.**

This remains below an unconditional theorem. The genuine structural hard point is now the exclusion of persistent branch-replay ColumnCRT/PDEC, plus the finite base check for any isolated atoms.  $\square$

**Reduction-closed Statement 2.94** (Finite atom boundary bridge). *The finite-atom check splits into a covered finite prefix and a post-100000 tail packet. The older direct grid verification up to  $P \leq 5000$  must not be used to absorb the present cycle-debt atoms.*

*Status and reference.* The finite atom boundary bridge router is generated by

```
experiments/prime_matrix_cycle_debt_finite_atom_boundary_bridge_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-finite-atom-boundary-bridge-router.md.
```

It records

```
direct_grid_verified_max_p=5000
dynamic_finite_bridge_range=[3001,99991]
dynamic_finite_bridge_closed=true
cover_pressure_atom_count=155
cover_pressure_all_atoms_in_dynamic_finite_bridge=true
arrival_wall_atom_count=55
arrival_wall_tail_atom_count=31
tail_atom_p_range=[101087,134047]
finite_atom_base_check_fully_closed=false.
```

The cycle-debt cover-pressure packet has 155 recorded  $P$ -coordinates, all lying in the already archived dynamic finite bridge  $3001 \leq P < 100000$ . These finite-prefix atoms are therefore absorbed by the existing bridge, not by the older direct grid verification up to  $P \leq 5000$ .

The terminal arrival-wall packet is different. It has 55 recorded  $P$ -coordinates. Only 24 lie below 100000; the remaining 31 are post-100000 tail atoms, with  $P$ -range 101087 to 134047. They consist of assigned slots, post-wall first primes, and tail slots. Hence the finite-atom alternative is not completely closed by the finite bridge.

Thus the latest branch is compressed to

**BranchReplayColumnCRTPDECExclusionOrPost100000TailAtomRunner.**

This is still not a global theorem closure. It closes the finite-prefix part of the finite-atom check and leaves the post-100000 tail packet to a tail-runner, tail lower-sieve, or ColumnCRT/PDEC route.  $\square$

**Reduction-closed Statement 2.95** (Post-100000 tail atom exact runner). *The post-100000 finite tail atoms in the registered terminal packet are exactly verified. The finite-atom branch is closed for the current registered objects, leaving the persistent branch-replay ColumnCRT/PDEC as the active structural obstruction.*

*Status and reference.* The post-100000 tail atom exact runner is generated by

`experiments/prime_matrix_cycle_debt_post100000_tail_atom_exact_runner.py`

and archived at

`docs/monograph/prime-matrix-cycle-debt-post100000-tail-atom-exact-runner.md.`

It records

```
tail_atom_count=31, tail_atom_p_range=[101087,134047]
kind_counts={assigned_slot:15,postwall_first_prime:8,tail_slot:8}
all_composite_atoms_divisible_by_recorded_factor=true
all_composite_recorded_factors_are_smallest=true
all_postwall_first_primes_verified=true
all_preprime_slots_composite_in_arrival_rows=true
finite_atom_branch_closed_for_registered_atoms=true.
```

The 31 tail atoms split into 23 composite slots and 8 post-wall first-prime boundaries. For every composite slot the recorded factor divides the slot and equals its smallest factor. For every post-wall first-prime boundary, direct trial division confirms primality in the recorded range. Moreover, in each arrival row all slots before the first-prime boundary are verified composite. Thus the finite post-100000 tail packet is no longer a parallel exit for the current registered terminal configuration.

Thus the latest branch is compressed to

**BranchReplayColumnCRTPDECExclusion.**

This still does not prove the row/column theorem. It removes the finite atom alternative and leaves the global persistent replay family as a named ColumnCRT/PDEC obstruction.  $\square$

**Reduction-closed Statement 2.96** (Fresh-modulus escalation). *A fixed finite ColumnCRT replay class is not a terminal structure for an infinite counterexample chain. New prime layers are coprime to the registered finite modulus and force unbounded modulus escalation, a PDEC/ColumnCRT defect, or a tail-sieve stability contradiction.*

*Status and reference.* The branch replay fresh-modulus escalation router is generated by

```
experiments/prime_matrix_cycle_debt_branch_replay_fresh_modulus_escalation_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-branch-replay-fresh-modulus-escalation-router.md.
```

It records

```
finite_atom_branch_closed_for_registered_atoms=true
registered_replay_block_count=6
fresh_prime_sample_count_per_block=8
all_registered_blocks_have_coprime_fresh_layers=true
minimum_first_fresh_log10_gain=2.400
minimum_sample_log10_gain=19.435
finite_crt_terminal_description_excluded=true.
```

Let a registered replay class have finite modulus  $Q$ . Every prime  $\ell \nmid Q$  is a new CRT coordinate independent of the old replay class. When such an  $\ell$  enters the active wheel, the old congruence modulo  $Q$  cannot decide the  $\ell$ -layer. A persistent counterexample chain must therefore absorb  $\ell$  into a larger modulus, fail at that layer as a named PDEC/ColumnCRT defect, or prove that the tail sieve absorbs the new coordinate.

The current six registered replay blocks make this explicit. Each has coprime fresh-prime layers. Adding only the first fresh prime increases the modulus by at least  $10^{2.400}$ , and adding the first eight fresh primes in the recorded sample increases it by at least  $10^{19.435}$ . Thus a finite registered ColumnCRT class cannot be a final stable object for an infinite counterexample chain.

Thus the latest branch is compressed to

```
UnboundedFreshModulusEscalationPDECOrTailSieveStabilityContradiction.
```

This is still not an unconditional closure. It identifies the remaining global obstruction as unbounded fresh-modulus escalation or a tail-sieve stability contradiction, rather than a fixed finite CRT atom.  $\square$

**Reduction-closed Statement 2.97** (Fresh-modulus tail-sieve bridge). *After finite ColumnCRT terminal classes have been excluded, a persistent branch replay family has only two remaining branch-replay exits. Either a fresh prime layer creates a PDEC/ColumnCRT defect, or the non-defective unbounded fresh-prime layers form the same one-residue-per-prime tail rough object used in the  $B = 3$  lower-sieve ledger.*

*Status and reference.* The fresh-modulus tail-sieve bridge router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_modulus_tail_sieve_bridge_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-modulus-tail-sieve-bridge-router.md.
```

It records

```
fresh_modulus_escalation_registered=true
finite_columncrt_terminal_excluded=true
non_pdec_unbounded_fresh_layers_force_tail_rough_object=true
tail_object_interface_closed=true
conditional_external_tail_sieve_closed=true
strict_self_contained_tail_sieve_closed=false
fresh_layer_pdec_excluded=false
row_column_unconditional_closed=false.
```

Let  $Q$  be the finite modulus of a registered replay class. Once the fixed finite terminal explanation is removed, every subsequent fresh prime  $\ell \nmid Q$  is an independent CRT coordinate. If the  $\ell$ -layer cannot be registered without a projection collision, residue repetition, moving-support escape, or ColumnCRT defect, the branch has entered the named fresh-layer PDEC exit. Otherwise the only information carried by that layer is one forbidden residue class modulo  $\ell$ . Repeating this over the unbounded fresh-prime layers gives the tail object

$$S(P) = \#\{1 \leq k < P : k \text{ avoids one prescribed residue class modulo each } q \leq P^{0.43}\}.$$

For each squarefree  $d < P$ , CRT counting gives

$$A_d(P) = \frac{P-1}{d} + r_d, \quad |r_d| \leq 1,$$

so this is exactly the same interface as the existing  $B = 3$  lower-sieve rough-object ledger. The imported tail ledger has

$$\alpha = 0.43, \quad \text{model main at } P = 100000 = 4896.256004,$$

and its ten-percent main term still exceeds 401 by 88.625600. Thus, if external explicit Mertens/Dusart estimates or the standard beta-sieve input are accepted, the tail-sieve stability branch is conditionally closed.

The strict self-contained branch is not closed by this router. It still requires the internal Mertens/PNT/Dusart reciprocal-prime tail proof and a separate exclusion of fresh-layer PDEC/ColumnCRT. The current branch-replay frontier is therefore

**FreshLayerPDECColumnCRTExclusionOrSelfContainedTailSieveStabilityClosure.**

This is a compression of the latest branch, not an unconditional proof of the row/column theorem.  $\square$

**Reduction-closed Statement 2.98** (Self-contained tail-sieve synchronization). *For the branch replay fresh-modulus branch, the old self-contained tail-sieve opening is no longer an active branch-replay obstruction after the later strict Mertens/PNT and  $B = 3$  total-variation synchronization certificates are imported. The remaining branch-replay obstruction is the fresh-layer PDEC/ColumnCRT exit.*

*Status and reference.* The synchronization router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_modulus_tail
_self_contained_sync_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-modulus-tail
-self-contained-sync-router.md.
```

It records

```
previous_tail_object_interface_closed=true
previous_non_pdec_unbounded_fresh_layers_force_tail_rough_object=true
strict_self_contained_mertens_tail_proved_latest=true
b3_tv_strict_self_contained_synchronized_latest=true
strict_self_contained_tail_sieve_closed_for_branch_replay=true
fresh_layer_pdec_excluded=false
row_column_unconditional_closed=false.
```

The previous bridge had reduced the non-PDEC unbounded fresh layers to the one-residue-per-prime  $B = 3$  tail rough object, but it kept the strict self-contained branch open under the coarse name

`SelfContainedDusartReciprocalPrimeProofAppendixXGe10372.`

Later strict synchronization certificates refine this coarse atom: the theta/PNT envelope and the Meissel–Mertens  $B_1$  interval are imported as self-contained inputs, and the  $B = 3$  total-variation ledger is synchronized with that Mertens tail closure. Hence the branch-replay strict remaining basis changes from

```
FreshLayerPDECColumnCRTExclusion
AND SelfContainedDusartReciprocalPrimeProofAppendixXGe10372
```

to

```
FreshLayerPDECColumnCRTExclusion.
```

This still does not close the row/column theorem. It only removes the old tail-sieve analytic opening from this branch; the fresh-layer PDEC/ColumnCRT obstruction remains active.  $\square$

**Reduction-closed Statement 2.99** (Local fresh-layer collision exclusion). *For the registered branch-replay blocks, a fresh-prime layer cannot create a local projection collision inside the recorded primitive support windows. Any remaining fresh-layer PDEC must therefore be a support-motion escape, a remote  $P$ -space ColumnCRT/PDEC recurrence, or a new moving-family block.*

*Status and reference.* The local collision router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_layer_local_collision_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-layer-local-collision-router.md.
```

It records

```
period_p=5680
registered_block_count=6
all_sample_fresh_primes_coprime_to_period_p=true
all_first_fresh_primes_exceed_support_width=true
all_first_fresh_primes_exceed_audit_slots=true
all_registered_samples_injective_on_local_windows=true
minimum_first_fresh_minus_support_width=178
minimum_first_fresh_minus_audit_slots=178
local_fresh_layer_projection_collision_excluded=true
row_column_unconditional_closed=false.
```

Let the local period step be  $M = 5680$ , and let  $\ell$  be a registered fresh prime. In every recorded sample,  $\gcd(M, \ell) = 1$ . If two slots  $j_1, j_2$  in a local window of length  $W$  collide modulo  $\ell$ , then

$$a + j_1 M \equiv a + j_2 M \pmod{\ell},$$

so  $\ell \mid (j_1 - j_2)M$ . Since  $M$  is invertible modulo  $\ell$ , this forces  $\ell \mid j_1 - j_2$ . But the recorded windows satisfy  $W < \ell$ , with minimum first-fresh surplus 178 over both support width and audit-slot count. Hence  $|j_1 - j_2| < \ell$ , and the only possible collision is  $j_1 = j_2$ .

Thus the local projection-collision subcase of fresh-layer PDEC is excluded for the registered blocks. The remaining branch-replay obstruction is

`FreshLayerSupportMotionEscapeOrRemoteColumnCRTPDECExclusion.`

This is still a named open exit, not a global unconditional proof.  $\square$

**Reduction-closed Statement 2.100** (Registered support-motion escape removal). *For the registered branch-replay blocks, the support-motion escape is not a remaining local mechanism. After local fresh-layer projection collisions and same-package local support drift are excluded, a persistent registered replay must be a remote  $P$ -space ColumnCRT/PDEC class; if the blocker package changes, it is an unregistered moving-family branch.*

*Status and reference.* The fresh support-motion global router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_support_motion_global_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-support-motion-global-router.md.
```

It records

```
period_p=5680
registered_block_count=6
local_fresh_layer_projection_collision_excluded=true
registered_local_support_motion_excluded=true
persistent_registered_replay_routes_to_columncrt_pdec=true
registered_support_motion_escape_closed=true
minimum_cycle_log10_margin_over_support_width=30.737
minimum_p_space_columncrt_modulus_log10=36.337
remote_pspace_columncrt_excluded=false
unregistered_moving_family_excluded=false
row_column_unconditional_closed=false.
```

The preceding local-collision certificate removes collisions inside the recorded primitive support windows. The earlier support-gap replay lemma also removes same-package local support drift: a nonzero replay by the same blocker package must move by a multiple of  $\text{lcm}(B)$ , and the least recorded margin over support width is  $10^{30.737}$ . Hence a registered block cannot persist through local support motion.

The global dichotomy then leaves only two possibilities. If a registered block recurs infinitely often, it is promoted to a fixed  $P$ -space ColumnCRT/PDEC class, whose smallest recorded modulus is about  $10^{36.337}$ . If the blocker package changes, the branch is not registered support motion but an unregistered moving-family branch requiring a new named router. The current branch-replay frontier is therefore

`RemotePspaceColumnCRTExclusionOrUnregisteredMovingFamilyRouter.`

This is a further compression of the branch-replay interface, not a proof of the row/column theorem. □

**Reduction-closed Statement 2.101** (Remote ColumnCRT feedback router). *Within the registered branch-replay chain, a remote  $P$ -space ColumnCRT class is no longer allowed to remain as a bare fixed-period terminal. Such a bare class is absorbed by the already recorded finite-atom runner, the fresh-modulus escalation router, and the strict tail-sieve synchronization. Therefore the current branch-replay remainder is narrowed to a materialized fresh-layer PDEC/ColumnCRT event or to an unregistered moving-family event.*

*Status and reference.* The feedback router is generated by

```
experiments/prime_matrix_cycle_debt_remote_columncrt_feedback_router.py
```



and archived at

`docs/monograph/prime-matrix-cycle-debt-remote-columncrt-feedback-router.md.`

It records

```
registered_remote_block_count=6
minimum_remote_pspace_columncrt_modulus_log10=36.337
maximum_remote_pspace_columncrt_modulus_log10=103.103
minimum_first_fresh_log10_gain=2.400
minimum_sample_fresh_log10_gain=19.435
registered_finite_atoms_absorbed=true
fixed_finite_remote_crt_not_terminal=true
non_pdec_fresh_tail_sieve_closed=true
bare_remote_pspace_columncrt_terminal_closed=true
materialized_fresh_layer_pdec_columncrt_excluded=false
unregistered_moving_family_excluded=false
row_column_unconditional_closed=false.
```

The point is only a feedback reduction. If the remote class is an isolated or finite registered atom, it has already been handled by the exact finite runner. If it is a fixed finite CRT replay class, the fresh-modulus escalation router shows that it cannot be a terminal infinite counterexample description. If fresh layers do not materialize a PDEC/ColumnCRT event, the non-PDEC unbounded fresh-layer branch has already been routed to and closed by the strict tail-sieve synchronization. Hence the broad remote ColumnCRT label can be removed as a bare terminal explanation.

The remaining branch-replay interface is consequently

**MaterializedFreshLayerPDECColumnCRTExclusionOrUnregisteredMovingFamilyRouter.**

This still leaves a real open obligation: one must exclude materialized fresh-layer PDEC/ColumnCRT events or unregistered moving-family recurrences. It is not a proof of the row/column theorem.  $\square$

**Reduction-closed Statement 2.102** (Fresh-layer PDEC admission firewall). *Within the current cycle-debt branch-replay corpus, the materialized fresh-layer PDEC/ColumnCRT label has no unnamed admitted instance. A future fresh-layer PDEC obstruction must be submitted as an explicit primitive same-formal-unit schema; otherwise the branch is an unregistered moving family.*

*Status and reference.* The firewall router is generated by

`experiments/prime_matrix_cycle_debt_fresh_layer_pdec_admission_firewall_router.py`

and archived at

`docs/monograph/prime-matrix-cycle-debt-fresh-layer-pdec-admission-firewall-router.md.`

It records

```
registered_block_count=6
minimum_first_fresh_minus_max_window=178
minimum_remote_plus_first_fresh_log10=38.803
local_registered_materialized_pdec_closed=true
current_materialized_pdec_frontier_closed=true
pdec_family_explicit_input_boundary_closed=true
newlayer_schema_admission_closed=true
newlayer_ranktwo_budget_independent_gate_removed=true
current_corpus_materialized_fresh_layer_pdec_closed=true
future_explicit_primitive_fresh_layer_pdec_schema_submitted=false
unregistered_moving_family_excluded=false
row_column_unconditional_closed=false.
```

For the registered blocks, every sampled first fresh prime is larger than the support and audit windows, and the local affine projection is injective. Thus local fresh-layer phase reuse cannot materialize a PDEC event inside the registered support. If a remote event is to be admitted as PDEC, it must pass the already registered PDEC boundary: the same formal unit, one fixed phase map, at least three physical primitive atoms after quotienting, non-tautology, rank at least two, and cap stability. ColumnCRT or displacement cases are absorbed before admission; sparse fallbacks require their own packet extractor.

Consequently the current branch-replay interface is narrowed to

`FutureExplicitPrimitiveFreshLayerPDECSchemaIfNewOrUnregisteredMovingFamilyRouter`.

This does not prove that no future primitive schema can exist, and it does not exclude unregistered moving families. It only removes the current unnamed materialized fresh-layer PDEC/ColumnCRT label.  $\square$

**Reduction-closed Statement 2.103** (Cycle-debt branch-replay current frontier zero). *In the present materialized cycle-debt branch-replay corpus, the remaining fresh-layer PDEC/ColumnCRT and unregistered moving-family labels are reduced to future admission firewalls rather than active terminal instances. The branch-replay subfrontier is therefore empty in the current corpus, while the global row/column theorem still depends on the separate final-input gates.*

*Status and reference.* The current-frontier-zero router is generated by

`experiments/prime_matrix_cycle_debt_branch_replay_current_frontier_zero_router.py`

and archived at

`docs/monograph/prime-matrix-cycle-debt-branch-replay-current-frontier-zero-router.md`.

It records

```
current_fresh_layer_pdec_frontier_closed=true
future_explicit_primitive_fresh_layer_pdec_schema_submitted=false
unregistered_moving_family_schema_submitted=false
future_pdec_schema_admission_discipline_closed=true
future_sparse_schema_admission_discipline_closed=true
final_input_firewall_boundary_closed=true
no_hidden_terminal_remaining=true
cycle_debt_branch_replay_current_materialized_frontier_zero=true
future_explicit_schema_global_nonexistence_proved=false
row_column_unconditional_closed=false.
```

The proof is a current-instance compression. A future primitive fresh-layer PDEC schema has not been submitted, so it is not an active mathematical object in the current branch-replay corpus; it is only an admission discipline. The same applies to an unregistered moving family unless it supplies source, shape, phase, persistence, and no-loss ledger fields. The PDEC and sparse schema firewalls specify how such a future object would have to re-enter.

Thus the current materialized cycle-debt branch-replay frontier is

`CycleDebtBranchReplayCurrentMaterializedFrontierZeroWithFutureSchemaFirewall`.

This is still not a proof of the row/column theorem. The global remainder is

`GlobalFinalInputsStillOpen`,

including the noncanonical legal-closure mode and the DStructure/Rankin promotion gate, unless future explicit schemas are submitted and separately excluded.  $\square$

## Chapter 3

# Prime-Matrix Contradiction Fields

Let  $P$  be an odd prime and write the  $P \times P$  prime matrix entries as

$$n(r, c) = (r - 1)P + c, \quad 1 \leq r, c \leq P.$$

The row and column programs attempt to show that every nontrivial row or column contains a prime. The current formal appendix reduces a row or non- $P$  column counterexample to a structured bad configuration after removing small-factor locks and tail anchors.

**Reduction-closed Statement 3.1** (A/B reduction, recorded form). *A nontrivial full-composite row or column, after small-factor and tail-anchor peeling, induces a Structured-EHPD bad configuration satisfying CRT balance, large-factor non-reuse, and anchor-capacity constraints.*

*Reference.* See `docs/row-column-reduction-formal-appendix.md` and the theorem list

`docs/monograph/prime-matrix-ab-entrance-theorem-list.md`.

This statement is a reduction interface; it does not by itself prove the Structured-EHPD exclusion theorem.  $\square$

### 3.1 Two-Point Rough-Pair Propositions

A natural next strengthening asks for two simultaneous nonzero-residue constraints in the back half of the prime matrix. Fix an even integer  $w > 0$  and ask whether there is an entry  $x$  in the back half of the  $P \times P$  matrix such that both  $x$  and  $x - w$  are nonzero modulo every prime  $p \leq P$ , or equivalently both  $x$  and  $w - x$  are nonzero modulo every prime  $p \leq P$ .

**Proposition 3.2** (Equivalence and strength of the two-point rough-pair problem). *For fixed even  $w > 0$ , the two conditions*

$$p \nmid x(x - w) \quad (p \leq P)$$

*and*

$$p \nmid x(w - x) \quad (p \leq P)$$

*are identical. In the region  $1 < x \leq P^2$ , any integer nonzero modulo every prime  $p \leq P$  is prime. Consequently, a proof of this two-point assertion for every sufficiently large  $P$  would imply infinitely many prime pairs of gap  $w$ ; in particular  $w = 2$  would imply the twin-prime conjecture.*

*Proof.* For each  $p \leq P$ , the condition  $p \nmid x - w$  is  $x \not\equiv w \pmod{p}$ , and the condition  $p \nmid w - x$  is the same congruence. Both formulations therefore exclude exactly the two residue classes 0 and  $w$  modulo each  $p$ , with possible collision when  $p \mid w$ .

If  $1 < x \leq P^2$  is nonzero modulo every prime  $p \leq P$  and is composite, then it has a prime factor  $q \leq \sqrt{x} \leq P$ , contradicting the nonzero-residue condition. Thus such an  $x$  is prime. If the two-point assertion holds for all sufficiently large  $P > w$ , then  $x$  and  $x - w$  are both prime in infinitely many growing back-half matrix intervals, giving infinitely many prime pairs of gap  $w$ .  $\square$

**Remark 3.3** (Status). This two-point problem is therefore not a routine corollary of the one-point row/column contradiction field. It is a prime-pair-level strengthening, naturally expressible as a two-point rough Jacobsthal problem. In this monograph it is recorded as a future contradiction-field direction, not as an unconditional theorem.

## 3.2 Two-Point Secondary-Sieve Contradiction Field

The two-point rough-pair problem admits a more detailed contradiction-field research program. The current working manuscript is recorded in

`docs/monograph/two-point-secondary-sieve-research.md.`

It studies the secondary sieve condition

$$x \not\equiv 0, w \pmod{p} \quad (p \leq P)$$

inside back-half matrix windows, and tries to exclude a two-point rough-pair counterexample by a hierarchy of local rigidity fields.

**Definition 3.4** (Two-point secondary-sieve field). Fix an even integer  $w > 0$ . A two-point secondary-sieve field is the structured ledger obtained from a hypothetical interval in which every candidate  $x$  fails at least one of the two congruence conditions  $p \nmid x$  or  $p \nmid x - w$  for some prime  $p \leq P$ . Its named channels are small-prime double locks, top-scale large-factor coverage, short-block zero-mass loss, multiplicative near-crossings, Fourier CRT defects, and smoothing boundary terms.

**Lemma 3.5** (Fixed-order local crossing complexity). *In the secondary-sieve manuscript, every fixed-order local crossing test is expanded only through truncated hard-skeleton weights*

$$d \leq R, \quad R \leq (\log P)^{C_R},$$

with coefficient mass  $\sum_{d \leq R} |\lambda_d| \leq (\log P)^{C_\lambda}$  and short-block shifts  $|h| < L \asymp (\log \log P)^2$ . For every fixed order  $k$ , the effective historical complexity satisfies

$$Q_{\text{eff},k} \ll_k \left( \sum_{d \leq R} |\lambda_d| \right)^k R^k L^{O(k)} \leq (\log P)^{C_k}.$$

In particular the required  $k = 2, 3$  local crossing tests have polylogarithmic effective complexity.

*Proof.* After expanding the  $k$  truncated weights, only moduli  $d_1, \dots, d_k \leq R$  occur. For fixed new-line primes  $p_i$ , phases, and block shifts, all conditions are one-variable linear congruences. By the Chinese remainder theorem the local system is either inconsistent or occupies one residue class modulo a divisor of

$$\text{lcm}(d_1, \dots, d_k) \text{lcm}(p_1, \dots, p_k).$$

The primes  $p_i$  are the explicit variables being summed in the crossing estimate; they contribute the usual  $Y^k$  scale, not hidden historical complexity. The historical part is bounded by  $\text{lcm}(d_1, \dots, d_k) \leq R^k$ , the block shifts contribute  $L^{O(k)}$  patterns, and the coefficient expansion contributes  $(\sum |\lambda_d|)^k$ . This proves the displayed bound. The true residual set  $U_Y$  is not expanded into a full primorial period.  $\square$

**Lemma 3.6** (Zero-mass and smoothing discharge). *Assume the SC2/diagonal package supplies the moment bounds*

$$M_1 := \frac{\sum_B X_B R_B}{\sum_B X_B} \leq 0.45, \quad M_2 := \frac{\sum_B X_B \binom{R_B}{2}}{\sum_B X_B} \geq 0.05,$$

and

$$M_3 := \frac{\sum_B X_B \binom{R_B}{3}_+}{\sum_B X_B} \leq 0.03.$$

Then the weighted zero-hit mass satisfies

$$\frac{\sum_B X_B 1_{R_B=0}}{\sum_B X_B} \geq 0.57.$$

Moreover, if the directional-balance endpoint estimate gives

$$N_{\text{endpoint}}/N_{\text{total}} \leq C_{\text{end}}\Delta + o(1),$$

then choosing  $\Delta \leq 0.01/C_{\text{end}}$  gives a smooth-to-sharp loss  $< 0.03$  for all sufficiently large  $P$ .

*Proof.* For every integer  $R \geq 0$ ,

$$1_{R=0} \geq 1 - R + \binom{R}{2} - \binom{R}{3}_+.$$

Multiplying by  $X_B \geq 0$  and summing over blocks gives

$$\frac{\sum_B X_B 1_{R_B=0}}{\sum_B X_B} \geq 1 - M_1 + M_2 - M_3 \geq 1 - 0.45 + 0.05 - 0.03 = 0.57.$$

For smoothing, choose even smooth cutoffs  $W_- \leq 1_{|t|<1} \leq W_+$  whose difference is supported in  $1 - \Delta \leq |t| \leq 1 + \Delta$ . The sharp and smoothed counts differ only on this endpoint layer. The displayed directional-balance bound therefore gives loss at most  $C_{\text{end}}\Delta + o(1)$ , which is  $< 0.03$  after the stated choice of  $\Delta$  and then taking  $P$  large.  $\square$

**Lemma 3.7** (Adaptive true-hit layering dichotomy). *Let  $U_Y$  be the current true residual set and define, for each new prime  $p > Y$ ,*

$$a_p = \frac{1}{|U_Y|} \#\{x \in U_Y : x \equiv 0 \text{ or } w \pmod{p}\}.$$

*Fix  $\eta = 0.05$ . Either some  $p$  has  $a_p > \eta$ , in which case the configuration is a Single-Prime CRTDefect, or the remaining primes can be greedily partitioned into consecutive layers  $\mathcal{P}_j$  satisfying*

$$\nu_j^{\text{real}} := \sum_{p \in \mathcal{P}_j} a_p \leq 0.4,$$

*and, except for the final layer,  $\nu_j^{\text{real}} > 0.4 - \eta \geq 0.35$ .*

*Proof.* If  $a_p > \eta$  for some  $p$ , then a fixed positive proportion of  $U_Y$  lies in the two residue classes 0,  $w$  modulo  $p$ . This is exactly the Single-Prime CRTDefect exit, to be excluded by the CRTDefect/Directional Balance part of the SC2 package. Otherwise all  $a_p \leq \eta$ . Starting from the smallest remaining prime, add primes to the current layer until adding the next one would make the partial sum exceed 0.4. The completed layer has sum at most 0.4, and because the next summand is at most  $\eta$ , every non-final completed layer has sum greater than  $0.4 - \eta$ . This gives the claimed adaptive true-hit decomposition.  $\square$

**Proposition 3.8** (Recorded conditional closure chain for the two-point secondary sieve). *Using the three lemmas above, assume the following remaining input holds with the constants recorded in the secondary-sieve research note:*

1. *the I3-Core SC2/CRTDefect package controls the true-residual two-line near-crossing error, the odd-prime  $45^\circ$  main-synchronization peak, Single-Prime CRTDefect exits, the normalized  $q = 2$  parity skeleton, the moment bounds  $M_1, M_2, M_3$ , and the endpoint Directional Balance used in the zero-mass and smoothing discharge lemma.*

*Then the two-point secondary-sieve contradiction field has no remaining top-scale escape channel under the parameter choice*

$$\alpha_0 = 0.75, \quad c_0 = 1/2, \quad \nu_j^{\text{real}} \leq 0.4$$

*for adaptive true-hit layers, subject to the same fixed- $w$  convention used in the research manuscript.*

*Proof sketch and status.* The argument is a conditional synthesis, not a final unconditional theorem. A top-scale covering excess is first decomposed into adaptive true-hit layers; any single-prime concentration is routed to the Single-Prime CRTDefect exit. The remaining excess is converted by a TCA linear bridge into a positive short-block load. The block load is forced into the SMC/SC2 near-crossing ledger. The  $45^\circ$  main synchronization is split into large-factor reuse, shared-complement factors, and good synchronized pairs; reuse and shared factors are excluded by short-window rigidity, while good pairs are sent to an auxiliary-prime Fourier CRT-defect. Directional balance excludes the Fourier defect unless the top-scale energy is already below the capacity threshold. Finally, the weighted Bonferroni zero-mass estimate supplies enough zero-hit block mass to make the capacity comparison incompatible.

The current audit is archived under `docs/archive/monograph-reviews/` and classifies this as a research proposition conditional on the single I3-Core package above. The fixed-order complexity input, zero-mass/smoothing discharge, and adaptive layering dichotomy have been inlined as the preceding lemmas. The fact that an unconditional proof would imply fixed-gap prime-pair consequences, including the twin-prime case when  $w = 2$ , is not a logical obstruction. It only marks the strength of the claim. The reason this chapter remains conditional is that I3-Core has not yet been inlined as an independently verified unconditional proof.  $\square$

**Proposition 3.9** (Window-compressed conditional variant). *Keep the same fixed even integer  $w > 0$  and the same remaining I3-Core input as in the preceding proposition. Let  $I$  be any consecutive back-half window of  $H(P)$  rows in the  $P \times P$  matrix, so that  $|I| = H(P)P + O(P)$ . By the fixed-order complexity lemma, write*

$$Q_{\text{eff}} \leq C_Q (\log P)^C.$$

*For a prescribed middle-scale error budget  $\varepsilon_{\text{SC2}} > 0$ , the conditional row threshold supplied by the current ledger is*

$$H_{\min}^{\text{cond}}(P; \varepsilon_{\text{SC2}}) = \left\lceil \frac{C_Q}{\varepsilon_{\text{SC2}}} P^{1/2} (\log P)^C \right\rceil.$$

Equivalently, one may use the asymptotic form

$$H(P) \geq H_{\min}^{\text{asym}}(P) := \left\lceil P^{1/2}(\log P)^{C_*} \right\rceil, \quad C_* > C.$$

Under either threshold, the same conditional contradiction-field chain gives an element

$$x \in I, \quad p \nmid x(x-w) \quad (p \leq P).$$

When  $x - w > 1$ , both  $x$  and  $x - w$  are primes.

*Recorded scale check.* The only scale-sensitive term in the recorded chain is the middle/high-scale  $K = 2$  near-crossing error. With top layer  $Y = P^{3/4}$  and  $Q_{\text{eff}} \leq (\log P)^C$ , replacing the original back-half length  $\asymp P^2$  by  $|I| \asymp H(P)P$  changes the normalized error to

$$\frac{Q_{\text{eff}} Y^2}{|I|} \ll \frac{P^{1/2}(\log P)^C}{H(P)}.$$

The displayed lower bound on  $H(P)$  makes this at most  $\varepsilon_{\text{SC2}}$  in the explicit form, or  $o(1)$  in the asymptotic form. Hence the constants in the TCA, SC2, zero-mass, finite-package, and smoothing ledgers are unchanged. The conclusion is therefore a conditional window-compressed version of the previous proposition. It becomes an unconditional prime-pair theorem only if I3-Core is independently proved and accepted.  $\square$

**Remark 3.10** (Referee boundary). The two-point secondary-sieve chapter is included to document the contradiction-field architecture and its parameter ledger. The window-compression audit is recorded in `docs/monograph/two-point-window-compression-and-unconditionality.md`. It must not be cited as an unconditional proof of prime pairs, the twin-prime conjecture, or Goldbach-type assertions unless I3-Core is independently proved and accepted.

**Remark 3.11** (Current obstruction). The current hard attack reduces I3-Core to a true-residual correlation problem. The local CRT and matrix rigidities do not by themselves exclude a model residual set concentrated on a new prime's bad residue class while still satisfying all previously imposed two-residue exclusions and short-window non-reuse constraints. Thus the remaining task is not another bookkeeping step but a genuine lower-bound/correlation theorem for the true residual set in new prime moduli.

**Remark 3.12** (Latest TRC-1G reduction). The single-prime part of the remaining TRC input has been sharpened to a heavy-atom prohibition. If more than 40% of the true residual mass lies in the two bad classes  $0, w \pmod p$  for some new prime  $p > Y$ , then Parseval gives a non-zero Fourier coefficient modulo  $p$  of size  $> 1/5$  after normalization. Equivalently, the three long-direction difference families  $0, \pm w \pmod p$  carry a fixed positive proportion of the true-residual pair energy. The existing matrix rigidities remove the short-direction, 45°-locked, reuse/shared-complement, and endpoint pieces. The remaining unproved input is therefore a new-modulus long-direction balance theorem for the true residual set, not a finite-template or local CRT bookkeeping issue.

**Remark 3.13** (Total large-incidence reduction). The pointwise new-modulus balance can be weakened further. It is enough to prove the total large-incidence inequality

$$\sum_{P^\alpha < p \leq P} \#\{x \in U_{P^\alpha}(I) : p \mid x(x-w)\} < |U_{P^\alpha}(I)|.$$

Indeed, a point not counted on the left is unsieved by every prime up to  $P$ , and in the  $P^2$  matrix range this forces both  $x$  and  $x - w$  to be prime. The CRT model predicts the normalized left side to be  $2 \log(1/\alpha) + o(1)$ ; for  $\alpha = 3/4$  this is  $2 \log(4/3) = 0.57536 \dots$ , leaving margin  $0.42463 \dots$ . The unproved part is the true-residual total-incidence estimate with an error below that margin.

**Remark 3.14** (RB–TLI audit boundary). The direct attempt to prove the total large-incidence estimate separates cleanly into a numerator and a denominator. The numerator is an upper-sieve problem: after writing  $x = pm$  or  $x = w + pm$ , the variable  $m$  again avoids two residue classes modulo each old prime, giving an upper bound of shape

$$A_p \leq 2C_+ |I| V_2(P^\alpha) / p + R_p.$$

The denominator requires a model-level lower bound  $|U_{P^\alpha}(I)| \geq c_- |I| V_2(P^\alpha)$ . For  $\alpha = 3/4$ , this route would need  $C_+/c_- < 1/(2 \log(4/3)) = 1.738 \dots$ . This relative Buchstab total-incidence input is the remaining unproved prime-pair-strength statement.

**Remark 3.15** (Local collapse when  $q \mid w$ ). If an old odd prime  $q$  divides the fixed even difference  $w$ , then the two forbidden classes  $0, w \pmod{q}$  coalesce into a single forbidden class. The local density is therefore  $1 - 1/q$  instead of  $1 - 2/q$ , giving the singular local gain  $(q-1)/(q-2)$ . This increases the absolute number of admissible residual points and weakens the small-prime synchronization peaks. In the relative RB–TLI ratio, however, the same local factor multiplies both  $|U_Y|$  and the model numerator  $A_p \sim 2|I|V_w(Y)/p$  for all new primes  $p > Y > |w|$ . Thus the main relative constant remains  $2 \log(1/\alpha)$ ; the collapse improves finite constants but does not by itself close the hardest case  $w = 2$ .

**Remark 3.16** (Buchstab semiprime transition for  $w = 2$ ). The latest numerical audit of the hardest case  $w = 2$  shows that the naive constant  $2 \log(1/\alpha)$  is not the correct conditional main term after the old primes have been sieved. For  $\alpha > 2/3$ , any new-prime hit  $p \mid x$  or  $p \mid x - 2$  with  $Y = P^\alpha < p \leq P$  forces the complementary quotient to be prime, because it is  $Y$ -rough and smaller than  $Y^2$ . Thus the large-incidence layer is a Buchstab semiprime transition. The corresponding two-sided main constant is

$$K(\alpha) = \frac{2 \log((2 - \alpha)/\alpha)}{1 + \log((2 - \alpha)/\alpha)}.$$

Exact scans recorded in `docs/rb-tli-w2-scan-large.md` give, for example,  $E_U D = 0.691696$  at  $P = 10007, \alpha = 3/4$ , compared with  $K(3/4) = 0.676220$ . The remaining target is therefore a two-point stability theorem for this Buchstab semiprime transition, not the earlier naive incidence model.

**Remark 3.17** (BST error decomposition). The refined scan decomposes the remaining error. At  $P = 10007, \alpha = 3/4$ , the shell contributions in the five exponent intervals between 0.75 and 1 track the Buchstab shell constants, and the covariance between the  $x$ -side and  $(x-2)$ -side large-incidence counts is approximately  $-9.3 \cdot 10^{-5}$ . The zero-incidence set is independent of the intermediate cutoff  $\alpha$ , as it is exactly the set unsieved by all primes up to  $P$ . Thus the remaining hard input has been reduced to a shifted biprime roughness estimate of the form

$$\#\{p, m : pm \in I, pm - 2 \in U_Y\} \leq (B(\alpha) + o(1)) |U_Y|.$$

This is the current BST–2 obstruction.

**Remark 3.18** (BMD reduction). The BST–2 obstruction can be written as a one-dimensional multiplicative sieve on biprime variables. After setting  $x = pm$ , the condition  $pm - 2$  is unsieved by old primes is, for each  $q \leq Y$ ,

$$pm \not\equiv 2 \pmod{q}.$$

The prime  $q = 2$  is peeled off deterministically: for odd primes  $p, m$ , the integer  $pm - 2$  is odd. Thus the multiplicative sieve is taken over odd  $q$ . For each odd  $q \leq Y$ , since  $p, m > Y \geq q$ , the forbidden



curve has  $(q-1)$  points inside  $(\mathbb{F}_q^\times)^2$ , out of  $(q-1)^2$  possible non-zero pairs. The local forbidden density is therefore  $1/(q-1)$ . The remaining unproved estimate is a biprime multiplicative dispersion bound: weighted sums of the errors in the congruences  $pm \equiv 2 \pmod{d}$ , with odd squarefree  $d$ , must be  $o(|U_Y|)$  for the Buchstab/Rosser weights used in the transition. This is the current BMD input.

**Remark 3.19** (Row-column input for BMD). If the prime-matrix row/column theorem is accepted as a conditional input, it can be used only to remove zero-section degeneracies in complete CRT blocks: entire empty rows, columns, or unit residue cells are then not allowed to masquerade as the main BMD error. It does not by itself prove BMD, because BMD is a signed distribution statement. For odd  $d$ , expanding

$$1_{pm \equiv 2 \pmod{d}} = \frac{1}{\varphi(d)} + \frac{1}{\varphi(d)} \sum_{\chi \neq \chi_0} \bar{\chi}(2) \chi(p) \chi(m)$$

shows that the remaining obstruction is a bilinear character-dispersion estimate

$$\sum_{d \leq D} \frac{|\lambda_d|}{\varphi(d)} \sum_{\chi \neq \chi_0} \left| \sum_{Y < p \leq P} \chi(p) \sum_{\substack{m \in J_p \\ m \text{ prime}}} \chi(m) \right| = o(|U_Y|).$$

Thus the row/column input gives a useful BMD-Zero exclusion, but the genuine remaining hard point is BMD-Char.

**Remark 3.20** (Direct BMD attack via an  $E_2$  Bombieri–Vinogradov input). Set  $N \asymp P^2$  and

$$a_n = \#\{(p, m) : Y < p \leq P, m \text{ prime}, n = pm, n \in I\}.$$

For  $\alpha > 2/3$  the multiplicity of  $a_n$  is bounded. After the deterministic modulus 2 has been peeled off, the BMD remainder for odd squarefree  $d$  is exactly the arithmetic-progression discrepancy of this restricted two-prime convolution:

$$R_d = \sum_{\substack{n \in I \\ n \equiv 2 \pmod{d}}} a_n - \frac{1}{\varphi(d)} \sum_{\substack{n \in I \\ (n, d) = 1}} a_n.$$

Thus BMD follows from the standard Bombieri–Vinogradov theorem for restricted  $E_2$  sequences at level

$$d \leq P(\log P)^{-B} = N^{1/2}(\log N)^{-B+O(1)}.$$

Indeed, if

$$\sum_{d \leq P/\log^B P} \max_{(a, d) = 1} |R_d(a)| \ll_A N(\log N)^{-A},$$

then every Rosser/Buchstab weighted remainder is  $o(|U_Y|)$  since  $|U_Y| \asymp N/(\log P)^2$ . The condition  $\alpha < 1$  ensures  $Y < P/\log^B P$  for large  $P$ , so all one-prime forbidden curves  $q \leq Y$  are included in the available distribution range. Therefore BMD is closed if this standard  $E_2$  BV input is cited; a fully self-contained manuscript must instead prove that input in an appendix.

**Remark 3.21** (Self-contained BV- $E_2$  obligation). The local BV- $E_2$  appendix records the exact self-contained obligation. A direct multiplicative large-sieve estimate is insufficient: in the balanced dyadic block  $p \sim P_1$ ,  $m \sim M_1$ , with  $P_1 \asymp M_1 \asymp P$  and  $Q \asymp P/\log^B P$ , Cauchy plus the large sieve gives size roughly  $P^3/\log^{2B} P$ , while the required Bombieri–Vinogradov error is  $P^2/\log^A P$ . Thus the only genuinely deep remaining self-contained input is a Type-II dispersion/Kloosterman-cancellation estimate for the balanced convolution blocks. If that standard dispersion theorem is cited, the BMD step is closed as an external-input theorem; if not, this is the next appendix that must be proved line by line.

**Remark 3.22** (Weighted BE2-3K core). BMD does not require the full  $\max_a$  Bombieri–Vinogradov statement. It only needs the fixed class  $2 \bmod d$  against the well-factorable Rosser/Buchstab weights. After Cauchy, the self-contained problem is a variance bound for

$$T_r = \sum_{d \leq Q} \lambda_d \left( \sum_{s \equiv 2r^{-1} \bmod d} \beta_s - \frac{1}{\varphi(d)} \sum_{(s,d)=1} \beta_s \right).$$

Opening  $\sum_r |T_r|^2$ , the off-diagonal conditions

$$rs_1 \equiv 2 \pmod{d_1}, \quad rs_2 \equiv 2 \pmod{d_2}$$

give, by CRT and Fourier expansion of the interval condition, the Kloosterman phase

$$e\left(-\frac{2h\overline{s_1}\overline{d_2}}{d_1} - \frac{2h\overline{s_2}\overline{d_1}}{d_2}\right).$$

Thus the true no-black-box core is a weighted bilinear Kloosterman dispersion estimate, denoted BE2-3K in the notes. Proving BE2-3K gives BE2-3, then weighted BV- $E_2$ , then BMD.

**Remark 3.23** (Referee status of the no-black-box claim). At the current stage the manuscript is not yet fully no-black-box. The BMD obstruction has been reduced to the single deep BE2-3K Kloosterman-dispersion estimate. If a BFI/Deshouillers–Iwaniec/Kuznetsov type mean-value theorem is cited, BMD is closed as an external-input theorem. If no external input is allowed, BE2-3K remains the unique missing line-by-line proof obligation.

**Remark 3.24** (Further reduction of BE2-3K). The BE2-3K core can be narrowed further. Point-wise Weil bounds for individual incomplete reciprocal sums give only single-modulus square-root cancellation and do not provide the arbitrary logarithmic saving required after summing over the  $d_1, d_2, h$  families. The common-divisor strata  $(d_1, d_2) > 1$  are not the main obstruction: compatibility forces  $s_1 \equiv s_2$  modulo the common divisor and yields only polylogarithmic loss. The remaining genuine input is a windowed Kloosterman spectral large-sieve estimate, denoted KLS-window in the notes. Its role is precisely

$$\text{KLS-window} \Rightarrow \text{BE2-3K} \Rightarrow \text{BE2-3} \Rightarrow \text{weighted BV-}E_2 \Rightarrow \text{BMD}.$$

**Remark 3.25** (External-theorem closure of KLS-window). The KLS-window input is tied to two classical external sources: the Deshouillers–Iwaniec spectral Kloosterman large-sieve technology, as developed in *Kloosterman sums and Fourier coefficients of cusp forms*, Invent. Math. 70(2), 219–288 (1982), and the Bombieri–Friedlander–Iwaniec dispersion framework with well-factorable weights, as in *Primes in Arithmetic Progressions to Large Moduli. II*, Math. Ann. 277, 361–394 (1987). With these external theorems allowed, the analytic chain closes as

$$\text{DI+BFI} \Rightarrow \text{KLS-window} \Rightarrow \text{BE2-3K} \Rightarrow \text{BE2-3} \Rightarrow \text{weighted BV-}E_2 \Rightarrow \text{BMD}.$$

This is an external-deep-theorem closure, not a fully self-contained reproof of Kuznetsov/dispersion theory.

## Chapter 4

# Zeta-Zero Contradiction Fields

Assume an off-critical zero  $\rho = \beta + i\gamma$  with  $\beta > 1/2$ . The RH contradiction-field manuscript transfers the resulting smooth prime anomaly into a CRT rough-composite ledger, decomposes it into sparse/dense/tail/internal/global exits, and proves that no named escape channel remains under the recorded local theorems and restricted external inputs.

**Theorem 4.1** (RH contradiction-field synthesis, recorded form). *Under the numbered local theorems, controlled-exit lemmas, threshold hierarchy, and restricted external inputs recorded in `paper/rh-proof/rh-contradiction-field.tex`, an off-critical zero has no free escape channel in the contradiction-field event graph.*

*Reference.* See the RH manuscript in `paper/rh-proof/` and the final audit in `docs/`. The submission warning in the RH manuscript remains in force until independent line-by-line referee verification and final monograph page-number checks are completed.  $\square$



## Chapter 5

# Unified Dependency Map

The two programs share the same logic:

counterexample  $\Rightarrow$  structured covering field  $\Rightarrow$  named exits  $\Rightarrow$  capacity/no-cycle contradiction.

The prime-matrix program is more geometric and finite-CRT in nature; the zeta-zero program is analytic and scale-asymptotic. The monograph treats both as projections of the same contradiction-field method.

| Layer           | Prime-matrix projection                            | RH projection                                          |
|-----------------|----------------------------------------------------|--------------------------------------------------------|
| Baseline        | CRT nonzero class balance                          | smoothed CRT candidate baseline                        |
| Local rigidity  | large-factor non-reuse, diagonal locks             | sparse/dense/tail routing                              |
| Capacity        | anchor coverage capacity                           | projection and square-function capacity                |
| No-cycle        | recursive peeling and shell migration              | GEE transfer/no-cycle ledger                           |
| External inputs | finite verification and Structured-EHPD            | explicit formula, KL, Vaaler, BG/Baker, sieve, Vaughan |
| Final status    | reduction package, not final unconditional theorem | verification package with submission warning           |



# Chapter 6

## Directory and Theory System

The detailed working directory and theory-system map is maintained in

`docs/monograph/combined-monograph-directory-and-theory-system.md`.

This chapter records the concise version used by the merged monograph.

### 6.1 Proposed Book Structure

| Part | Contents                                                                                                | Status                                        |
|------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| I    | Unified contradiction-field<br>method: entrance, local rigidity,<br>exits, capacity, no-cycle potential | Methodological framework                      |
| II   | Prime-matrix row/column<br>program: CRT skeleton, A/B<br>reduction, Structured-EHPD<br>interface        | Reduction package; not final<br>theorem       |
| III  | Two-point secondary sieve: TLI,<br>BST, BMD, WBE2, BE2-3K,<br>KLS-window                                | External-deep-theorem closure for<br>BMD      |
| IV   | Zeta-zero contradiction field:<br>explicit-formula entrance and<br>controlled exits                     | Verification package; not final RH<br>theorem |
| V    | Dependency map, referee<br>checklist, optimization routes                                               | Audit and future work                         |

### 6.2 Latest Two-Point Chain

For the  $w = 2$  hardest two-point branch, the current analytic chain is

$$\text{DI+BFI} \Rightarrow \text{KLS-window} \Rightarrow \text{BE2-3K} \Rightarrow \text{BE2-3} \Rightarrow \text{WBE2} \Rightarrow \text{BMD} \Rightarrow \text{BST-2} \Rightarrow \text{BST} \Rightarrow \text{TLI}.$$

The chain is closed only in the external-deep-theorem sense: DI and BFI are cited as classical analytic inputs. A fully self-contained version would require reproving the spectral Kloosterman large-sieve and dispersion machinery.

### 6.3 Prime-Matrix Low-Hole Closure Package

The latest prime-matrix update closes the low-hole bucket submodule as a finite-certificate plus explicit-external-theorem package. The verification is split into five non-overlapping ranges:

|                         |                                                 |
|-------------------------|-------------------------------------------------|
| $13 \leq P < 61$        | : low-range final certificate,                  |
| $61 \leq P \leq 103$    | : narrow-band collision-energy certificate,     |
| $107 \leq P \leq 229$   | : exact $P/5$ splitting recursion,              |
| $233 \leq P \leq 13207$ | : integer cross-multiplied product certificate, |
| $P \geq 13208$          | : Rosser–Schoenfeld explicit constants.         |

The low-range certificate checks 15414 zero buckets: 15282 close by whole-set Hall deficit, 108 close by bridge-critical deletion, and the remaining 24 exceptional phases at  $P = 41$  close by exact finite set-cover dynamic programming. The narrow band  $61 \leq P \leq 103$  covers 23100 low phases by fixed ascending high-prime residue blocks. The two finite tail certificates cover 23 and 1520 prime rows respectively, the latter by exact integer cross multiplication rather than floating-point comparison.

For the infinite tail, Rosser–Schoenfeld Corollary 1, formulas (3.5)–(3.6), supplies the prime-counting bounds, while Theorem 7, formula (3.26), supplies the Mertens product bound. Since  $P \geq 13208$  implies  $\lfloor P/5 \rfloor \geq 2641$ , the factor  $1 + 1/(2 \log^2 \lfloor P/5 \rfloor)$  is below 1.009, hence the manuscript’s conservative 1.03 Mertens constant is admissible.

This closes the BPN low-hole bucket interface in the stated finite-certificate/external-theorem sense. It does not by itself promote the whole prime-matrix row/column program to a final theorem, because the remaining global exits still require the separately named PDEC/SAE and Rankin-ledger acceptance obligations.

### 6.4 Prime-Matrix WSH-Hall Phase Certificate

The latest row-program compression keeps the endpoint status honest. The terminal RCI branch has been reduced to the comparison between no-tail primes and balanced two-tail semiprimes. The next local interface is

$$\text{RCI/PDEC} \Rightarrow \text{Distributed-RCI} \Rightarrow \text{WSH-Hall/PDEC}.$$

Here WSH-Hall means that every balanced two-tail semiprime in a row can be injectively matched to a same-row prime through an offset allowed by the small-prime wheel. If such a Hall matching fails, the failing interval must be routed to a named exit: persistent endpoint CRT defect, tail-anchor concentration, or endpoint prime deficit.

A finite phase certificate is archived in

`docs/monograph/prime-matrix-wsh-hall-phase-certificate.md`.

For  $17 \leq p \leq 2000$ ,  $y = \lfloor p/e \rfloor$ , and wheel primes 2, 3, 5, 7, 11, 13, it scans 215074 rows containing balanced two-tail semiprimes. It finds no Hall matching failure, no failure under the candidate radius  $3 \log^2 q$ , maximum minimal matching radius 132, maximum ratio  $R/\log^2 q = 2.3771082468335174$ , and minimum margin  $\#\text{primes} - \#\text{semiprimes} = 1$ . The certificate rows list semiprime labels, matched primes, offsets, fixed-offset loads, and wheel-phase loads; each matched offset is checked against the congruence condition  $d \not\equiv -b \pmod{r}$ .

This is a computational-certificate update, not a global theorem. The remaining proof obligation is either a uniform WSH-Hall theorem or a proof that every Hall defect necessarily triggers PDEC, Tail-anchor, or Endpoint deficit.



## 6.5 WSH-Hall Defect Trichotomy

The corresponding defect audit is archived in

`docs/monograph/prime-matrix-wsh-hall-defect-trichotomy.md`.

It converts a possible Hall failure into a consecutive block  $B_0$  of balanced two-tail semiprimes with

$$|N_R(B_0)| < |B_0|.$$

The same block is then examined through five projections: endpoint prime deficit, tail-factor load, small-wheel residue load, fixed-offset capacity, and the terminal mirror  $n \mapsto q^2 - n$ . The resulting target is the named interface

WSH-Expansion-or-Defect.

If Tail-anchor and fixed-offset/PDEC thresholds do not fire, one must prove the expansion inequality

$$|N_R(B_0)| \geq |B_0|$$

for  $R = C \log^2 q$ ; otherwise the failing block must be absorbed by Endpoint/PDEC. A forced-radius audit on the 44 tight certificate rows produces 124 explicit Hall defects and records their tail, phase, offset, and mirror data. This audit fixes the certificate fields and exit geometry; it is not a substitute for the uniform expansion theorem.

## 6.6 WSH Expansion Margin Ledger

The expansion term itself is audited in

`docs/monograph/prime-matrix-wsh-expansion-margin-ledger.md`.

For  $17 \leq p \leq 2000$ ,  $R = \lceil 3 \log^2 q \rceil$ , and the same  $y = \lfloor p/e \rfloor$ , the audit scans every consecutive balanced-semiprime block in every relevant row. It checks 8440419 blocks across 215074 rows and finds

$$\min_B (|N_R(B)| - |B|) = 1,$$

with no zero-surplus block. Only seven blocks have surplus at most 1.

This strengthens the finite certificate from “a matching exists” to “every interval-Hall block has positive margin” in the audited range. It still does not prove the uniform theorem. The remaining target is:

WSH Positive Expansion or Named Defect.

After Tail-anchor, fixed-offset/PDEC, and endpoint-mirror deficit exits are removed, one must prove the positive expansion inequality for all formal blocks.

## 6.7 Short Critical Block Reduction

The short-critical-block audit is archived in

`docs/monograph/prime-matrix-wsh-short-critical-block-reduction.md`.

It reruns the expansion-margin audit with tight surplus threshold 2. Among the same 8440419 consecutive semiprime blocks, only 45 have surplus at most 2, and every such block has at most three semiprimes. The distribution is:

| surplus | $ B $ | count |
|---------|-------|-------|
| 1       | 1     | 4     |
| 1       | 2     | 3     |
| 2       | 1     | 24    |
| 2       | 2     | 10    |
| 2       | 3     | 4     |

This suggests the next reduction:

SCB-1:  $|B| \geq 4 \Rightarrow$  positive expansion or named defect,

SCB-2:  $|B| \leq 3 \Rightarrow$  local exclusion or named defect.

This is still finite evidence and a proof target, not a theorem. It is useful because SCB-2 is a small local problem involving only one, two, or three two-tail semiprimes, where tail labels, fixed offsets, wheel residues, and endpoint mirrors can be written explicitly.

## 6.8 SCB-2 Local Routing Certificate

The SCB-2 local certificate is archived in

`docs/monograph/prime-matrix-wsh-scb2-local-exclusion-template.md`.

It enumerates all consecutive blocks with  $|B| \leq 3$  for  $17 \leq p \leq 2000$ . The block counts are

| $ B $ | count   | $\min( N_R(B)  -  B )$ |
|-------|---------|------------------------|
| 1     | 1496400 | 1                      |
| 2     | 1281326 | 1                      |
| 3     | 1088870 | 2                      |

There are no negative-surplus or zero-surplus short blocks. The local routing statement is:

$$|B| \leq 3, \quad |N_R(B)| < |B| \Rightarrow \text{Endpoint/PDEC deficit},$$

with additional Tail-repeat and fixed-offset-full-load checks on tight two- and three-point blocks. Thus SCB-2 is closed as a routing lemma modulo the still-open Endpoint/PDEC exclusion.

## 6.9 SCB-1 Long-Block Certificate

The complementary long-block certificate is archived in

`docs/monograph/prime-matrix-wsh-scb1-long-block-expansion-template.md`.

It enumerates all consecutive balanced two-tail semiprime blocks with  $|B| \geq 4$  for  $17 \leq p \leq 2000$  and the same radius  $R = \lceil 3 \log^2 q \rceil$ . The audit checks 4573823 long blocks and finds

$$\min_{|B| \geq 4} (|N_R(B)| - |B|) = 3,$$

with no negative-surplus or zero-surplus long block. The four tight long blocks have sizes 4 or 5, and every tight block carries both Endpoint-margin and fixed-offset-full-load labels.

This is a finite certificate, not the uniform SCB-1 theorem. Its proof-theoretic role is to isolate the next target:

long-block positive expansion   or   fixed-offset/PDEC absorption.

Together with SCB-2, the present WSH chain is therefore:

SCB-2 routing modulo Endpoint/PDEC,      SCB-1 finite long-block margin,

The remaining global obstructions are fixed-offset/PDEC absorption and Endpoint/PDEC exclusion.

## 6.10 Fixed-Offset/PDEC Absorption

The fixed-offset absorption interface is archived in

`docs/monograph/prime-matrix-wsh-fixed-offset-pdec-absorption.md`.

It proves the no-third-escape lemma behind the fixed-offset-full-load label. If  $p < q$  are consecutive primes and  $1 < n < q^2$  is composite, then  $n$  has a prime factor at most  $p$ . Hence, if a fixed offset  $d$  is allowed by the wheel 2, 3, 5, 7, 11, 13 for every  $b \in B$ , every missing candidate  $b + d$  is explained by some factor in  $(13, p]$ .

The finite ledger

`docs/monograph/prime-matrix-wsh-fixed-offset-pdec-ledger.md`

checks the tight SCB-1 long blocks. It finds 11 full-load offset rows and 47 candidates, among which 15 are prime and 32 are missing. Every missing candidate has an explaining factor at most  $p$ ; there is no unexplained candidate.

Thus fixed-offset-full-load is no longer an unnamed escape. It routes to expansion, Tail-anchor, PDEC/low-mod CRTDefect, or SAE/Endpoint. The still-open global target is

FO-PDEC: distributed explaining factors  $\Rightarrow$  persistent PDEC or sparse SAE/Endpoint.

## 6.11 FO-PDEC Low-Mod Equations

The low-mod equation audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-hard-attack.md`.

For every missing candidate  $n = b + d = (r - 1)q + c$  and every explaining factor  $\ell$ , the exact CRT equation is

$$r \equiv 1 - cq^{-1} \pmod{\ell}.$$

If  $b = uv$  is the corresponding two-tail semiprime, the same missing candidate also satisfies

$$uv + d \equiv 0 \pmod{\ell}.$$

The finite audit checks 43 such low-mod equations in the tight fixed-offset ledger. There are no CRT-equation failures and no bilinear-equation failures.

This closes the equation layer of FO-PDEC. It does not close the global theorem. The remaining target is the quantitative inequality

$$\text{low-mod defect energy} \geq \text{PDEC threshold},$$

unless the non-persistent part is absorbed by SAE/Endpoint.

## 6.12 FO-PDEC Energy Lower Bound

The energy ledger is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-energy-lemma.md`.

For a multiset  $E$  of explaining equations, let  $t_\ell$  be the number of equations using  $\ell$  and let  $s_\ell$  be the number of occupied row residues modulo  $\ell$ . Define

$$\mathcal{E}_\ell(E) = t_\ell - \frac{s_\ell t_\ell}{\ell}, \quad \mathcal{E}(E) = \sum_\ell \mathcal{E}_\ell(E).$$

For every  $0 < \theta < 1$ , either some explaining factor has high load  $t_\ell > \theta\ell$ , which is a Tail/PDEC exit, or

$$\mathcal{E}(E) \geq (1 - \theta)|E|.$$

Thus FO-PDEC now produces low-mod energy unconditionally, modulo the named high-load exit.

The finite ledger with  $\theta = 0.25$  gives global energy per equation

$$0.9547817296210457$$

and no high-load factor. The remaining obstruction is not energy production; it is the final threshold comparison on the same bad-window set:

$$U_{\text{CRT}}(E) < \mathcal{E}(E) \quad \text{or} \quad L_{\text{PDEC}}(E) \leq (1 - \theta)|E|.$$

## 6.13 FO-PDEC Explicit Threshold

The explicit threshold ledger is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-threshold-comparison.md`.

For each explaining factor  $\ell$ , it computes the nonzero Fourier threshold

$$M_\ell(E) = \max_{1 \leq h < \ell} \left| \sum_{\rho \bmod \ell} g_\ell(\rho) e^{2\pi i h \rho / \ell} \right|.$$

The current best projection is

$$\ell = 199, \quad h = 95, \quad M_\ell(E) = 3.959247567099438.$$

Thus the remaining task is now concrete:

$$U_{\text{CRT},199} < 3.959247567099438$$

for the same vector  $g_{199}$ . Otherwise this projection must be absorbed by SAE/Endpoint.

## 6.14 FO-PDEC Dual-Cluster Obstruction

The dual-cluster audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-dual-cluster-route.md`.

For the best projection, the mass is 4 and the Fourier threshold is 3.959247567099438, so the trivial mass bound misses closure by only

0.04075243290056196.

The best frequency  $h = 95$  sends the support to a short dual arc of length 11 modulo 199:

$40, 126, 61 \mapsto 19, 30, 24$ .

Thus the last obstruction is a concrete dual-cluster exclusion problem: this short dual arc must be ruled out by a PDEC dual certificate, or else absorbed as a non-persistent SAE/Endpoint event.

## 6.15 FO-PDEC Multiplicity Legitimacy

The primitive-cluster audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-multiplicity-legitimacy.md`.

It checks whether the best projection counts independent bad-window equations or repeated physical candidates. Under the equation or block-local convention, the projection has mass 4 and Fourier value

3.959247567099438.

Under physical deduplication, it has mass 2 and Fourier value

1.9699193446802263.

Thus the final PDEC comparison must first fix the multiplicity convention. The lower bound and the upper bound must be computed on the same multiset. Otherwise the argument must switch to the primitive threshold or absorb the repeated events by SAE/Endpoint.

## 6.16 FO-PDEC Formal Unit Consistency

The formal-unit audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-formal-unit-route.md`.

It checks a stricter requirement: the PDEC vector  $g(t)$  must come from one formal bad-window set or multiset with one phase map. The current strong value

3.959247567099438

is obtained by pooling the finite certificate library. After splitting by formal unit, the audit gives

| convention                                 | best Fourier value |
|--------------------------------------------|--------------------|
| global library                             | 3.959247567099438  |
| layer deduplication                        | 2.9698366905785227 |
| physical deduplication                     | 1.9997507790353146 |
| single coordinate-deduplicated formal unit | 1.0.               |

Thus the strong threshold cannot yet be used as a single-branch PDEC lower bound. One must prove FormalUnit-Stitching for the cross- $q$  events and NestedBlock-Independence for exact nested-coordinate duplicates, or route the repeated events to SAE/Endpoint. This is now the narrowest FO-PDEC obstruction before any global unconditional conclusion can be claimed.

## 6.17 FO-PDEC Branch Separation

The branch-separation theorem and stitching audit are archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-branch-separation-theorem.md.`

They separate two issues. First, events from different matrix widths  $q$  cannot be pooled into one PDEC vector without a cross- $q$  persistence theorem. Second, two occurrences with the same

$$(q, row, candidate\_row, column, offset, n, \ell, \rho)$$

give the same CRT equation and must be quotiented unless a weighted Hall-dual independence row is supplied. In the present finite ledger there are 7 exact nested-coordinate duplicates and 9 cross-level physical reuses. After coordinate deduplication inside a single formal unit, the best Fourier value recorded by the audit is only 1.0. Therefore the next narrow obstruction is Weighted Hall Dual Independence, or else SAE/Endpoint absorption of the exact duplicates.

The subsequent dominance audits discharge these finite FO-PDEC subgates without promoting the full theorem. The nested duplicates are laminar repeated coordinates; the fractional weighted Hall differences add one semiprime and one neighboring prime, with zero surplus, so they cannot pay for a second independent unit of mass. The cross- $q$  reuses are coordinate-chart overlaps of the same physical candidate, not independent persistent events. Thus the raw library value 3.959247567099438 and the coordinate-cap value 2.9698366905785227 cannot be used as single-formal-unit lower bounds in the audited branch.

## 6.18 FO-PDEC Physical Primitive Tautology and SAE Absorption

The physical primitive audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-physical-primitive-tautology.md.`

After physical deduplication the current 199-factor frontier has only two physical atoms. For a prime modulus  $\ell$  and two distinct residues  $a, b$ , choosing

$$h \equiv (b - a)^{-1} \pmod{\ell}$$

makes the dual residues adjacent. Hence every two-point support satisfies

$$\max_{h \neq 0} |e(ha/\ell) + e(hb/\ell)| = 2 \cos(\pi/\ell).$$

For  $\ell = 199$  this is exactly

$$2 \cos(\pi/199) = 1.9997507790353146.$$

Therefore the remaining physical primitive value is a two-point Fourier tautology, not a usable PDEC defect threshold.

The two atoms are then routed to the SAE/Endpoint interface and audited in

`docs/monograph/prime-matrix-wsh-fo-pdec-sae-endpoint-absorption.md`.

Each associated fixed-offset fiber has a local prime survivor: the 250541 atom has witness 250543, and the 1664237 atom has witness 1664227. In all four source rows the 199-factor load in the same fiber is 1. Consequently the audited two-atom primitive branch is absorbed by LocalSurvivor witnesses.

This closes only the current finite  $\ell = 199$  FO-PDEC sample chain:

raw library signal  $\rightarrow$  nested/weighted rejection,  
coordinate-cap signal  $\rightarrow$  cross- $q$  chart-overlap rejection,  
physical two-point signal  $\rightarrow$  Fourier tautology,  
two physical atoms  $\rightarrow$  LocalSurvivor witness absorption.

It does not close the global prime-matrix theorem. The next honest frontier is the global LocalSurvivor certificate family, or a future non-tautological primitive PDEC family with at least three physical atoms or an extra fixed-frequency constraint. The non-tautological PDEC admission audit records that the current materialized frontier has no such candidate: the raw three-point signal is not admissible as one primitive formal unit, nested and weighted duplicate recovery are blocked, cross- $q$  reuse is only chart overlap, and the physical primitive remnant has two atoms. Those two atoms are already absorbed by LocalSurvivor/SAE witnesses. Thus any future PDEC certificate must introduce a genuinely new same-formal-unit primitive family with at least three physical atoms, or an additional constraint defeating the two-point tautology.

## 6.19 Materialized LocalSurvivor Packet Ledger

The current LocalSurvivor packet ledger is archived in

`docs/monograph/prime-matrix-local-survivor-materialized-packet-ledger.md`.

It combines the Triad-A1 SparseCap ledger, the FO-PDEC two-atom SAE/Endpoint absorption ledger, and the RPZ Endpoint-SAE finite ledger. The combined current ledger records

| quantity                      | value |
|-------------------------------|-------|
| materialized packets          | 9     |
| materialized phase atoms      | 32    |
| local survivor witnesses      | 5     |
| finite PDEC atoms             | 25    |
| open materialized obligations | 0.    |

Thus every LocalSurvivor/SAE packet currently materialized by the proof frontier is either witnessed, vacuous in the current ledger, or routed to a finite PDEC packet beyond the early  $P \times P$  rows.

The packet-generation contract is archived in

`docs/monograph/prime-matrix-local-survivor-packet-generation-contract.md`.

It records the structural dichotomy for any future sparse escape. If the window data can be extracted, it is a finite LocalSurvivor packet and must be checked by a witness or blocker-deficit

certificate. If the same finite signature persists along an infinite counterexample family, it is no longer a sparse terminal: according to the repeated signature it returns to PDEC, ColumnCRT, endpoint PDEC, TailAnchor, or CofactorAnchor. If the finite signatures do not persist and the layers keep escaping, the branch enters CleanKLS/DLS admission.

Consequently the next narrow frontier is

LocalSurvivorPacketGenerationOrNonTautologicalPDEC.

This is still not a final unconditional theorem. It says that the current materialized LocalSurvivor obligations are exhausted and that no unnamed LocalSurvivor terminal remains; the remaining work is packet-extractor completeness, a genuinely non-tautological primitive PDEC family, or the CleanKLS/DLS and referee inputs.

The known-entry extractor audit is archived in

`docs/monograph/prime-matrix-local-survivor-packet-extractor-coverage.md`.

It checks eight currently named LocalSurvivor entry types. Four are covered by machine extractors: Triad-A1 SparseCap, FO-PDEC two-atom SAE/Endpoint, RPZ Endpoint-SAE, and the materialized packet aggregator. Four are covered by contract routes: generic SAE, persistent finite signatures, layer escape to CleanKLS/DLS, and descent/seam return. The audit has no missing or open known entry.

The new-entry admission audit is archived in

`docs/monograph/prime-matrix-new-sparse-entry-admission-audit.md`.

It checks the possible sources of a future sparse entrance: short-window or endpoint sparse packets, PDEC dual sparse caps, endpoint seams, Bohr caps, column-tail/cofactor sparse packets, descent seams, persistent signatures, and layer escape. Each is already admitted to a named route: SAE/LocalSurvivor, PDEC, ColumnCRT, TailAnchor/CofactorAnchor, or CleanKLS/DLS. Hence there is no additional unnamed LocalSurvivor entry route in the current contract system.

After these two audits the LocalSurvivor branch is conditionally exhausted: any future explicitly introduced sparse route must bring its own extractor schema and finite ledger. The current narrow frontier is therefore

NonTautologicalPDECOrCleanKLS.

This remains a reduction boundary, not the final prime-matrix theorem.

The clean KLS side is no longer an unnamed black-box exit in the current routing table. The A1 clean-admission router checks that K1-K9 failures return to PDEC/SAE/Multiplicity/Promotion, while a fully admitted clean residual enters the Kuznetsov-LS atom SC-9. The SC-9 frontier router expands this atom into KZ-A-KZ-E: KZ-A-KZ-D are routed by smoothing, trace specialization, Bessel decay, and the spectral-large-sieve/pretrace chain; KZ-E remains the actual WFD block non-concentration input NC-BLK, or else a precisely matched external DI/BFI original dispersion theorem. Thus the updated narrow frontier is

NonTautologicalPDECOrNCBLK.

This also remains a reduction boundary, not a final unconditional row/column theorem. The NC-BLK boundary reconciliation audit records the exact reading of this label. On the canonical RIW/Buchstab source branch, the NC-BLK clean-KLS chain is already absorbed by the existing same-set capacity boundary. On the broader full-S non-AP generic branch, NC-BLK remains



an exact-source-entropy or external DI/BFI/Kuznetsov route and is not imported into the self-contained theorem. Thus the label no longer denotes an unnamed CleanKLS exit, but it also does not promote the whole row/column theorem.

The subsequent global-terminal-family boundary router updates the frontier one more time. After the NC-BLK reconciliation and the non-tautological PDEC admission audit, the current materialized frontier has no local terminal instance left to eliminate: the non-tautological primitive PDEC candidate count is zero, the materialized LocalSurvivor ledger has no open packet, all known sparse entry routes are admitted, and NC-BLK is reconciled with the canonical boundary. The router records the new narrow label

**CurrentMaterializedFrontierExhausted\_GlobalTerminalFamiliesOpen.**

Equivalently, the next required theorem is no longer a fixed numerical improvement for the current  $\ell = 199$  sample or a new reading of NC-BLK. It is the global terminal-family exclusion package: every terminal object emitted by a minimal counterexample must be certified by a PDEC-family exclusion, a LocalSurvivor witness/deficit certificate, a CleanKLS/DLS internal large-sieve certificate, or an explicitly accepted external/referee input. This is again a boundary closure, not a final unconditional row/column theorem.

The split router for this global terminal family then removes the remaining ambiguity in the word “family”. The non-final reductions are closed: there is no fourth terminal route, the current LocalSurvivor and NC-BLK ledgers are not independent blockers, and the finite-projection terminal dichotomy sends positive-limsup mass to PDEC-CAP and diffuse mass to CleanKLS/DLS. Thus the self-contained mathematical remainder is

**PDEC\_CAP\_OR\_INTERNAL\_CleanKLS\_LargeSieve,**

while final promotion also keeps the D-structure/Rankin referee gate. In particular, the frontier is not an unnamed local sample problem; it is the pair of named analytic terminal estimates plus the referee interface.

The self-contained terminal bottleneck router then removes one more apparent branch. Inside the current canonical-source boundary, InternalCleanKLS is not an independent minimal blocker: a clean large-sieve failure returns a dual concentration object to PDEC/SAE, the canonical NC-BLK/CleanKLS branch is already absorbed by the same-set capacity boundary, and the unrestricted generic full- $S$  WFD self-contained theorem is refuted and not claimed. Hence the independent self-contained mathematical bottleneck is

**PDEC\_CAP\_SameSetGlobalDualCertificate,**

namely a same-bad-window global dual capacity certificate  $U_{\text{CRT}} < L_{\text{PDEC}}$  for the PDEC families. The D-structure/Rankin referee gate remains a separate final-promotion condition, so this is still not a final unconditional row/column theorem.

The next PDEC-CAP router then opens this named bottleneck rather than treating it as a black box. All currently materialized PDEC intermediate gates are routed: the DualCap families have same- $M_Q$  mass sources and closed early  $P$ -row exits; the current PersistentCaps route through promotion deletion or NoDeletion-KL/PDEC/CleanKLS; the current ForcedCaps route into multi-bucket ActualPaymentStitching; and LHB projection-stitching is removed by the common full-period completion set. The profinite ActualPaymentStitching router then closes the remaining APS logic by the finite-projection pigeonhole dichotomy. The PDEC-CAP theorem is still not proved. Its current narrow internal frontier is now the single terminal estimate

**PrimitiveMultiAtomSameFormalUnitPDECCertificate.**

Persistent real payment  $\Gamma$  and fixed-shell low-mod persistence are now treated as the same finite-signature formal-unit problem: the former is a phase-bucket/tail-column signature, and the latter is a shell/displacement signature. ColumnCRT displacement is absorbed as displacement PDEC, PDEC dual failure is absorbed as cap refinement, and formal-unit mismatch is absorbed by the multiplicity-stitching normalization. Thus the single persistent terminal is PersistentFiniteSignaturePDECColumnCRT. The flat-clean SC-9 branch has been reconciled with the canonical-source boundary: clean failure returns to PDEC/SAE, canonical NC-BLK is absorbed by the same-set boundary, and generic WFD is not imported into the self-contained claim. Finally, the persistent terminal admission router removes raw ColumnCRT, dual-failure, multiplicity, and two-point-tautology pseudo-terminals. Thus the remaining admitted object must be a primitive multi-atom same-formal-unit PDEC certificate. The intermediate diffuse split is now routed: persistent KL or phase-residue mutual information returns to refined/new-layer PDEC, while only the KL-flat branch may enter CleanKLS/DLS; divergent deletion already implies support exhaustion or a named sparse/PDEC return, and HRO reduces deletion non-divergence to occupancy saturation or TailIndependence. The latter is already NoDeletion/CleanKLS. The occupancy-kernel router then applies the HRO injection bound  $|\text{Occ}_t| \leq \min(|H_Q(t)|, r)$ : sparse old-hole or residue-deficient phases cannot saturate, and any remaining saturation must contain a near-full old-hole selector kernel. The common-variable router writes  $c_b = \rho_b + rk_b$ , turning every old-prime ban into one forbidden residue for the same shell variable  $k_b$ . Hence no legal shell is a capacity/Hall deletion, fixed-shell mass is PDEC/ColumnCRT persistence, and shell-dispersed flat mass is the SC-9 atom. External KLS input closes the flat branch only as an external-theorem version.

## 6.20 Triad-A1 Self-Contained Theorem-Boundary Closure

The final Triad-A1 review separates the exact theorem that is now closed from a stronger statement that is not true under the recorded hypotheses. In plain terms, the closed result is the following. Once the source of the A1 payment measure is fixed to the canonical RIW/Buchstab decision-tree source, every remaining same-set capacity gate in the full- $S$  terminal chain is internally accounted for. The proof no longer needs to borrow the external FullS-KLS theorem for this canonical-source branch.

**Proved-in-text Statement 6.1** (Canonical-source self-contained boundary). *The Triad-A1 same-set capacity / full- $S$  terminal is self-contained on the canonical RIW/Buchstab source branch.*

*Recorded proof certificate.* The certificate is recorded in

`docs/monograph/prime-matrix-triad-a1-self-contained-theorem-boundary-review.md`.

It checks seven gates: exact theorem statement boundary, final closure certificate, absorption by the same-set frontier, canonical-source provenance, branch coverage without silent generic upgrade, no false generic claim, and separation of the external FullS-KLS contract. The review verdict is

APPROVE\_CANONICAL\_SOURCE\_SELF\_CONTAINED\_BOUNDARY,

with no open review gate. □

**Not-claimed Statement 6.2** (Unrestricted generic WFD self-contained theorem). The manuscript does not claim an unrestricted generic full- $S$  well-factorable WFD self-contained theorem.

*Reason for non-claim.* The generic anti-atom input needed for that broader statement is refuted by the moving-delta model recorded in the A1 no-go audit. Hence the generic WFD version is not an open missing lemma of the canonical-source proof; it is a different and stronger statement that is false under the current formal hypotheses. The external FullS–KLS contract remains available as an external-theorem route, but it is kept separate from the self-contained canonical-source claim.  $\square$

**Remark 6.3** (Boundary in everyday language). The row-program terminal proof has reached a closed theorem-boundary for the canonical source: the bookkeeping, provenance, and branch split now agree on what is proved. What remains outside this boundary is not a hidden gap in that theorem. It is either a broader generic WFD theorem that the current model refutes, or one of the separate global prime-matrix terminal certificate obligations listed below for promoting the whole row/column program to a final unconditional theorem.

## 6.21 Claim Status Legend

The monograph uses the following status labels.

| Label                     | Meaning                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------|
| Proved-in-text            | The statement is proved inside the manuscript under the stated hypotheses.                                       |
| Reduction-closed          | The statement has been reduced to a smaller named interface.                                                     |
| External-theorem closed   | The statement is closed after citing explicitly named external theorems.                                         |
| Computational-certificate | The statement is closed over a finite range or certified submodule by reproducible scripts and archived outputs. |
| Referee-block             | A proof package exists, but it still requires line-by-line referee verification.                                 |
| Not claimed               | The statement is not asserted as a theorem of the manuscript.                                                    |

The status discipline and promotion rules are maintained in

`docs/monograph/claim-status-discipline.md`.

The manuscript provides theorem-like environments with matching names for future editing. A statement may be promoted only when its required proof, reduction endpoint, external theorem template, or certificate package is present.

## 6.22 External Theorem Index

The external-input index is maintained in

`docs/monograph/external-theorem-index.md`.

For the prime-matrix low-hole tail, the current explicit package is Rosser–Schoenfeld 1962: Corollary 1, formulas (3.5)–(3.6), supplies the required  $\pi(x)$  bounds, and Theorem 7, formula (3.26), supplies the required Mertens product bound. These constants close the  $P \geq 13208$  tail in the external-theorem sense once the finite certificates below that threshold are accepted.

For the two-point sieve, the critical external package is DI plus BFI. The index records the required checks for the KLS-window input: Kloosterman phase normalization, modulus range, frequency range, inverse-variable support, well-factorable weights, gcd strata, smoothing and sawtooth losses, and the final choice of the logarithmic-loss parameter  $B(A)$ . The standalone adaptation template is maintained in

`docs/monograph/kls-window-di-bfi-adaptation-template.md`.

It closes the BMD distribution input only in the external-deep-theorem sense; the separate BMD-to-TLI transfer remains a hard obligation.

## 6.23 Critical Chain Audit

The detailed chain audit is maintained in

`docs/monograph/key-proof-chain-optimization-audit.md`.

Its main purpose is to prevent status mixing. Prime-matrix existence rigidity may support zero-section exclusion, but it does not replace signed spectral distribution estimates such as BMD-Char. Numerical experiments may guide constants and structures, but they are not proof inputs. The RH branch remains a verification package and is not claimed as an unconditional proof.

## 6.24 Optimization Priorities

1. Separate theorem environments for proved-in-text, reduction-closed, external-theorem closed, and referee-block statements.
2. Replace the overly strong  $\max_a \text{BV-}E_2$  formulation by the precise WBE2 well-factorable weighted statement in the main line.
3. Rewrite KLS-window directly in the notation of the cited DI/BFI theorems to reduce translation risk.
4. Compress the prime-matrix program around the single Structured-EHPD exclusion interface.
5. Preserve the RH warning language until every controlled exit has independent referee verification.

## 6.25 Pre-External-Referee Hard Obligations

The current hard-obligation ledger is maintained in

`docs/monograph/pre-external-referee-remaining-obligations.md`.

It records the author-side status of every terminal interface that cannot honestly be promoted to a final theorem without further proof or independent verification. The decisive interfaces are:

| Program | Hard interface | Required closure before theorem promotion |
|---------|----------------|-------------------------------------------|
| <hr/>   |                |                                           |

|                 |                                            |                                                                                                                                                     |
|-----------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Prime matrix    | Structured-EHPD /<br>D-structure exclusion | Prove the exclusion theorem directly, or replace it by a fully proved RHI-type large-prime incidence inequality.                                    |
| Prime matrix    | PDEC / SAE / Rankin exits                  | Prove persistent endpoint CRT defect exclusion, local isolated-escape exclusion, and complete the formal Rankin certificate ledger.                 |
| Prime matrix    | RRD / OSPC /<br>Selberg-uniform constants  | Put all constants in one convention and prove each budget item over the full asymptotic range, not only in sampled windows.                         |
| Two-point sieve | I3-Core true-residual package              | Prove true-residual new-modulus balance, TCA, SC2, 45° synchronization peak control, Directional Balance, Zero Mass, and endpoint smoothing.        |
| Two-point sieve | DI/BFI to KLS-window adaptation            | State the external theorems in their standard variables and verify phase, modulus, frequency, weights, gcd strata, and logarithmic-loss absorption. |
| Two-point sieve | BMD to TLI transfer                        | Prove the Buchstab transfer and total-large-incidence comparison without using a hidden prime-pair lower bound.                                     |
| RH branch       | controlled exits                           | Prove every sparse, dense, tail, internal, analytic, and global exit under a single load convention.                                                |

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For the RRD/OSPC constants, the current acceptance matrix is maintained in

`docs/monograph/h5-rrd-ospc-proof-obligation-matrix.md`.

It splits the remaining constant obligation into six independently checkable inequalities and does not claim that those inequalities have already been proved. The first item, RRD-low routing, is recorded in

`docs/monograph/h5-1-rrd-low-exit-theorem.md`.

It proves that an RRD-low excess enters either OSPC\* or a weighted CRT-defect exit; excluding those exits remains part of the PDEC/SAE and tail-anchor obligations. The absorption of these two exits into the unified PDEC-or-SAE interface is recorded in

`docs/monograph/h5-4-ospc-weighted-crtdefect-absorption.md`.

This absorption is not an exclusion theorem; the required PDEC and SAE certificates remain open hard obligations. The PDEC certificate format itself is now recorded in

`docs/monograph/h4-pdec-certificate-template.md`.

It fixes the data, Fourier lower bound, admissible CRT upper-bound certificates, and failure-routing rules for persistent defects. This is a certificate-format closure only; the actual PDEC certificates for the formal bad-window families remain to be supplied. The first source lemmas for admissible PDEC constraint rows are recorded in

`docs/monograph/h4-pdec-constraint-source-lemmas.md.`

They justify mass, phase-capacity, mirror-pair, low-hole-bucket, and conditional-routing rows when their stated hypotheses hold. They do not by themselves provide the full constraint table or the final dual upper bound. The first admissibility table for those rows is recorded in

`docs/monograph/h4-pdec-admissible-constraint-table.md.`

It separates tautological, finite-certificate, symbolic, conditional-routing, still-unproved, and rejected rows. In particular, full-cycle CRT balance is not allowed to control an arbitrary bad-window subset. The column-cap source rule is recorded in

`docs/monograph/h4-pdec-column-cap-source-lemma.md.`

It allows column information to enter only through finite column-projection capacity, symbolic column-capacity theorems, or conditional routing to ColumnRadius/ColumnCRT/Tail-anchor exits. The coefficient ledger for those rows remains open. The first coefficient ledger is recorded in

`docs/monograph/h4-pdec-column-cap-coefficient-ledger.md.`

It registers finite column-witness, RCI/CDB, LHB column-residue, and conditional ColumnDefect rows. Most rows still require machine-readable phase blocks before they can become actual rows of the dual certificate matrix. The first materialized phase-block file is

`docs/monograph/h4-pdec-lhb-column-phase-blocks.json.`

It gives finite  $Q = 2310$  LHB column rows for  $13 \leq p \leq 47$ . Empty anomaly blocks are machine-readable finite rows; diagnostic support blocks still require a projection-capacity proof before entering the final dual system. The support-to-capacity transfer conditions are recorded in

`docs/monograph/h4-pdec-lhb-support-to-capacity-transfer.md.`

They make explicit that support size is not a multiplicity bound for the persistent count vector  $g(t)$ ; a phase-indicator reduction, a multiplicity cap, or an allowed-set projection bound is still required. The multiplicity route is made explicit in

`docs/monograph/h4-pdec-lhb-multiplicity-cap-route.md.`

It states the precise acceptance contract: for the same  $(p, Q, S, \tau)$  one must prove  $g(t) \leq M(t)$ , and only then may a diagnostic block  $C$  be promoted to the capacity row  $\sum_{t \in C} g(t) \leq \sum_{t \in C} M(t)$ . The first finite multiplicity-cap certificate is

`docs/monograph/h4-pdec-lhb-multiplicity-cap-certificate.json.`

It takes  $M(t)$  to be the high-layer CRT completion count  $C_P(t; Q)$  and verifies, for  $Q = 2310$  and  $p = 13, 17, 19, 23, 29, 31, 37, 43, 47$ , that the WHOLEDEF and BRIDGED support blocks have  $\sum_{t \in C} M(t) = 0$ . This is an allowed-set projection certificate: it can be used in the LHB branch after

proving  $S \subset Z_{\text{LHB}}(p, Q)$ , but it is not by itself a global PDEC exclusion theorem. The attachment lemma for that branch is

`docs/monograph/h4-pdec-lhb-attachment-lemma.md.`

It proves that an LHB-typed full-row bad window is indeed an element of  $Z_{\text{LHB}}(p, Q)$ . Thus the remaining global task is no longer this attachment step, but the classification step: every formal PDEC bad window must either be LHB-typed or be routed to a named SAE, ColumnCRT/ColumnRadius, TailAnchor, or Rankin exit. The classification step is recorded in

`docs/monograph/h4-pdec-bad-window-classification-lemma.md.`

It combines the unified PDEC-or-SAE dichotomy, column-conflict routing, tail-anchor persistence, and Rankin access rules to split each named low-mod bad-window family into SAE, LHB-PDEC, or Routed-PDEC. This is still a routing theorem: the non-LHB exits and the final dual  $U_{\text{CRT}} < L_{\text{PDEC}}$  certificate remain open. The homogeneous-splitting rule is recorded in

`docs/monograph/h4-pdec-homogeneous-splitting-lemma.md.`

It closes the bookkeeping case in which one attempted certificate mixes different  $(Q, \tau, F, \kappa, p)$  types: such a family must be partitioned into homogeneous subfamilies before applying UPS-1 or the bad-window classification. Thus mixed type is not an additional mathematical exit; the remaining exits are SAE, ColumnCRT/ColumnRadius, TailAnchor, and Rankin. The column-defect routing contract is recorded in

`docs/monograph/h4-pdec-column-defect-routing-contract.md.`

It defines the conditional certificate objects for ColumnRadius and ColumnCRT rows: radius rows use a threshold  $D_0$ , a column-witness selector, and phase-compatible weights  $W_D(t)$ , while displacement rows use a label  $\ell$ , a non-zero displacement residue  $a$ , a threshold  $L_D$ , and phase-compatible weights  $W_{\ell,a}(t)$ . Violating such a row routes to the named ColumnRadius or ColumnCRT exit after the zero displacement class has been separated by the same-column prime-witness rigidity. This is a routing-contract closure only; the exits themselves still require materialized weights, thresholds, and exclusion certificates. The first finite materialization of those weights is recorded in

`docs/monograph/h4-pdec-column-defect-weight-certificate.json.`

It uses the refined finite phase  $\tau_{\text{fin}} = (p, q, \text{row})$  for  $p \leq 1000$  and the five tightest RCI rows for each  $p$ . In this finite domain, the blocks  $D_{\text{col}} > 81$  and displacement-residue load  $> 2$  are both empty, over 835 phases. This is a finite refined-phase certificate; it does not provide a global  $D_0, L_D$  theorem or exclude the ColumnRadius/ColumnCRT exits. The recursive zero-row peeling route is isolated in

`docs/monograph/prime-matrix-recursive-peeling-zero-row-hardpoint.md.`

It proves the exact layer-peeling identity: after a  $p$ -sieve zero window is peeled to the previous prime level, any resurrected survivor must be a multiple of the peeled prime with rough cofactor. Thus the recursive object is a punctured zero window, not automatically a lower-level zero row. The accompanying finite audit finds no forced consecutive lower zero rows in the known first-zero-row samples. The useful remaining target is therefore an absorption theorem routing those

resurrected points to ColumnCRT, ColumnRadius, or TailAnchor, rather than a direct short-period contradiction. The adjacent-shell refinement is recorded in

`docs/monograph/prime-matrix-adjacent-shell-recursive-descent-route.md.`

For adjacent primes  $p < q$ , it proves that the only composite old  $p$ -sieve survivor in  $[1, q^2]$  is  $q^2$ . Hence a non-first  $q$ -zero row first descends to an old  $p$ -sieve  $q$ -window, with only the terminal  $q^2$  puncture in the last row. That window either contains a full  $p$ -aligned zero row or is a seam window whose two guard intervals have lengths  $a$  and  $p - (q - p) - a$ . The resulting narrow target is SeamGuard-Elimination: guards cannot absorb all forced survivors without entering SAE, PDEC, or ColumnCRT. The multi-level seam descent audit is recorded in

`docs/monograph/prime-matrix-seam-multilevel-descent-route.md.`

In this conditional model, if a seam interval is already an old  $p$ -sieve zero window, then after descending to a lower prime  $h < p$  the only resurrected points are those with  $P^-(n) \in (h, p]$ , together with the terminal  $q^2$  puncture in the last  $q$ -row. Any full  $h$ -aligned row avoiding these resurrected points is therefore a forced  $h$ -zero row. The finite ledger checks all 21339 seam windows for  $p \leq 500$  and a deterministic  $p \leq 2000$  sample of 11488 seam windows, with no persistent blocking cases. This is not a global theorem; it sharpens the remaining gate to the SMD-Global Inequality, or else a proof that persistent revival blocking enters SAE, PDEC, or ColumnCRT. The tail-mirror refinement is recorded in

`docs/monograph/prime-matrix-seam-tail-mirror-descent-route.md.`

For a forced lower  $h$ -zero row with row number  $R$ , set  $N_h = M_h/h$  and  $\rho = ((R - 1) \bmod N_h) + 1$ . If  $\rho \leq h$  or  $N_h - \rho + 1 \leq h$ , then periodicity or CRT reflection places a conditional zero row inside the lower  $h \times h$  matrix. The finite ledgers above again show full success in the same domains. This motivates the sharper TailMirror-SMD gate: every genuine seam counterexample must hit such a lower head/tail phase, or its persistent non-hit phases must be routed to SAE, PDEC, or ColumnCRT. The all-branch version is recorded in

`docs/monograph/prime-matrix-total-zero-row-recursive-descent-route.md.`

It treats aligned containment, seam windows, and the terminal  $q^2$  puncture by the same descent rule. In the  $p \leq 500$  ledger all 21936 non-first  $q$ -rows, including 597 aligned rows and 21339 seam rows, force a lower matrix head/tail phase. In the deterministic  $p \leq 2000$  sample all 11583 checked rows do the same. The remaining theorem-level statement is TotalDescent-TM: every genuine higher zero row must force a lower  $h \times h$  zero row by periodicity or tail reflection, or else its non-hit phase set must enter SAE, PDEC, or ColumnCRT. The fixed  $h = 3$  refinement is recorded in

`docs/monograph/prime-matrix-total-descent-h3-margin-route.md.`

Each complete 3-row contains a unique candidate  $c_R$  coprime to 6. If  $5 \leq P^-(c_R) \leq p$ , that row is blocked by a resurrected point when descending from the  $p$ -sieve to the 3-sieve; if  $P^-(c_R) > p$ , then  $c_R$  is already a survivor of the old  $p$ -sieve and contradicts the assumed  $p$ -sieve zero window. Thus the narrowest remaining form of TotalDescent is the six-wheel inequality

$$\#\{R : J_R \subset I_s\} > \#\{R : J_R \subset I_s, 5 \leq P^-(c_R) \leq p\}.$$

The audit

`docs/monograph/prime-matrix-total-descent-h3-margin-audit.md`



checks this for all 1,552,462 non-first  $q$ -rows with  $p \leq 5000$ , with minimum margin 1. This is a finite certificate and a sharpened interface, not a global theorem: a final proof must establish the six-wheel inequality uniformly, or route full blocking to SAE/PDEC/ColumnCRT. The hard-attack reduction

`docs/monograph/prime-matrix-h3-sixwheel-hard-attack.md`

clarifies the remaining obstruction. The direct six-wheel inequality sits at the Buchstab/linear-sieve boundary  $u = 2$ , i.e. at square-root interval length near  $x = q^2$ . Therefore a proof cannot be obtained by a generic short-interval theorem or a finite template. If the six-wheel candidates are all blocked, they admit the first-factor partition  $n = \ell m$  with  $5 \leq \ell \leq p$ ,  $m$  in an interval of length  $q/\ell$ , and  $P^-(m) \geq \ell$ . The next theorem-level gate is consequently an H3 full-blocking defect theorem: high first-factor load enters Tail/PDEC, distributed low-load coverage produces low-mod PDEC energy, and endpoint residue concentration is SAE or ColumnCRT. The near-miss audit

`docs/monograph/prime-matrix-h3-full-blocking-defect-audit.md`

checks the same ledger for all  $p \leq 5000$ . Among 1,552,462 non-first  $q$ -rows, only 4015 have margin at most 20, and these occur only up to  $p = 317$  and  $q = 331$ . Their aggregate first-factor load is led by the small-prime skeleton  $5, 7, 11, 13, \dots$ . This data does not prove the theorem, but it identifies the correct next split: small-first-factor overload should be absorbed by Tail/PDEC, while low-load many-label blocking should produce low-mod PDEC/ColumnCRT energy. The small-factor envelope route

`docs/monograph/prime-matrix-h3-small-factor-envelope-route.md`

makes this split deterministic. For a cutoff  $y$ , let  $C_y$  be the load carried by first factors at most  $y$ , let  $R_y = A - C_y$ , and let  $\ell_+(y)$  be the next prime. If full blocking holds, then either  $C_y$  is already a small-skeleton overload, or at least

$$K_y = \left\lceil \frac{R_y}{\lfloor q/\ell_+(y) \rfloor + 1} \right\rceil$$

middle-tail labels are active. The finite ledger for the near-miss rows gives maximum forced values  $K_y = 7$  for  $y = 31$  and  $K_y = 9$  for  $y = 43$ . The remaining proof obligation is to convert small-skeleton overload into Tail/PDEC, or many active labels into H3-PDEC energy, with endpoint losses routed to SAE/ColumnCRT. The global scaling audit

`docs/monograph/prime-matrix-h3-global-scaling-law-route.md`

records the underlying scale. The natural six-wheel rough count is

$$\#A_s \prod_{5 \leq \ell \leq p} \left(1 - \frac{1}{\ell}\right) \asymp \frac{q}{\log q}.$$

In the finite ledger, from  $p = 331$  onward every prime layer has minimum H3 margin at least 21, and bucket averages track this Mertens scale. Thus a formal counterexample must annihilate a growing  $q/\log q$  rough-residue scale. The current proof task is to show that such annihilation forces small-skeleton Tail/PDEC overload, many-label H3-PDEC energy, or endpoint SAE/ColumnCRT. The constant-level formula is recorded in

`docs/monograph/prime-matrix-h3-universal-scaling-inequality.md`.

In the finite ledger,  $M_{H3}(p, s) \geq 0.30 q / \log q$  holds from  $p = 113$  onward, and  $0.40 q / \log q$  holds from  $p = 2011$  onward. These are not asserted as unconditional short-interval prime theorems. They are target constants for a defect inequality: if the H3 margin falls below  $cq / \log q$ , then SmallSkeletonOverload, ManyLabel-PDEC, or Endpoint-SAE/ColumnCRT must fire. The global tail-energy lemma

`docs/monograph/prime-matrix-h3-global-tail-energy-lemma.md`

proves the deterministic core of this defect inequality. For any cutoff  $y$ , target  $B$ , threshold  $L$ , and modulus

$$d \geq 2 \left( \left\lfloor \frac{q}{\ell_+(y)} \right\rfloor + 1 \right),$$

where  $\ell_+(y)$  is the next prime after  $y$ , the implication

$$M_{H3}(p, s) < B \implies C_y > \#A_s - B - 2L \quad \text{or} \quad E_y(d) > L$$

holds for every adjacent pair  $p < q$  and every row window  $2 \leq s \leq q$ . Substituting  $B = cq / \log q$  and  $L = \lambda q / \log q$  gives a uniform infinite-scale defect formula. This is still not the final H3 lower bound: the remaining theorem-level exits are SmallSkeletonOverload  $\Rightarrow$  Tail/PDEC and TailEnergy  $\Rightarrow$  H3-PDEC/ColumnCRT. The first exit is bridged in

`docs/monograph/prime-matrix-h3-small-skeleton-pdec-bridge.md`.

Let

$$V_y = \#A_s \prod_{5 \leq r \leq y} \left( 1 - \frac{1}{r} \right), \quad D_y = V_y - R_y,$$

where  $R_y = \#\{n \in A_s : P^-(n) > y\}$ . The overload inequality  $C_y > \#A_s - B - 2L$  gives  $R_y < B + 2L$ , hence

$$D_y > V_y - B - 2L.$$

Thus, whenever  $V_y \geq (c + 2\lambda + \eta)q / \log q$ , small-skeleton overload forces a PDEC defect of size  $D_y > \eta q / \log q$ . The remaining issue is not this bridge, but the exclusion of such persistent PDEC defects and of the tail-energy alternative. The tail alternative is put in Fourier form in

`docs/monograph/prime-matrix-h3-tail-energy-fourier-bridge.md`,

where Parseval gives  $\mathcal{F}_y(d) = dE_y(d)$  for the non-zero residue frequencies of the tail first-factor labels. The synthesis

`docs/monograph/prime-matrix-h3-unified-defect-criterion.md`

therefore proves the conditional closure criterion

$$D_y \leq V_y - B - 2L \quad \text{and} \quad \mathcal{F}_y(d) \leq dL \implies M_{H3}(p, s) \geq B.$$

With  $B = cq / \log q$  and  $L = \lambda q / \log q$ , this is the current global scale theorem: the empirical  $q / \log q$  law is explained by two explicit low-mod defect exclusions. The unconditional row theorem still requires proving those exclusions uniformly. The source of the  $q / \log q$  scale is clarified in

`docs/monograph/prime-matrix-h3-first-row-scale-bridge.md`.

Let  $\Pi_1(q) = \pi(q)$  be the number of primes in the first row up to  $q$ . By the prime number theorem,  $\Pi_1(q) \sim q/\log q$ . On the other hand, adjacent-shell singleton rigidity says that below  $q^2$  every old  $p$ -sieve survivor, apart from the endpoint  $q^2$ , is prime. Hence the average H3 margin over the non-first  $q$ -rows satisfies

$$\frac{1}{q-1} \sum_{s=2}^q M_{H3}(p, s) \sim \frac{q}{2 \log q} \sim \frac{1}{2} \Pi_1(q).$$

The factor  $1/2$  is simply  $\log q / \log(q^2)$ . Thus H3 is the square-shell projection of the first-row prime scale; a zero H3 row would have to remove a full first-row-scale mass from one row and therefore must be accounted for by the low-mod defects above. The pointwise boundary is recorded in

`docs/monograph/prime-matrix-h3-pointwise-closure-boundary.md`.

The average identity above is rigorous, but it does not imply pointwise positivity. The missing statement is a pointwise scale-transfer theorem, for example

$$M_{H3}(p, s) \geq \kappa \pi(q) \quad (2 \leq s \leq q)$$

for some fixed  $\kappa > 0$ , or equivalently the uniform exclusion of the  $D_y$  and  $\mathcal{F}_y(d)$  defects in the H3 unified criterion. The analytic boundary is recorded in

`docs/monograph/prime-matrix-h3-square-root-short-interval-barrier.md`.

It identifies this pointwise transfer with a prime lower bound in aligned intervals of length  $q = \sqrt{q^2}$  near  $x = q^2$ . This is exactly the square-root short-interval scale, where ordinary average PNT input and the linear sieve at  $u = 2$  do not supply a pointwise positive lower bound. The remaining theorem is therefore a genuine square-root defect exclusion, not a bookkeeping consequence of the first-row average. The direct hard attack is archived in

`docs/monograph/prime-matrix-h3-square-root-defect-exclusion-hard-attack.md`.

It shows that the  $D_y$  branch can be controlled at a small sieve level by the standard one-dimensional interval-sieve fundamental lemma. The obstruction is the tail branch: for a natural low modulus  $d$ , the tail Fourier energy satisfies

$$\frac{1}{2} T_y \leq E_y(d) \leq T_y,$$

so it is essentially the tail-label count. The final hard core is therefore to exclude tail labels and double-rough points as an exact full-row filler, or to force them into ColumnCRT, endpoint phase, or cofactor-structure defects. The local rigidity of such a filler is recorded in

`docs/monograph/prime-matrix-h3-tail-filler-rigidity-hardcore.md`.

Adjacent H3 candidates differ by 2 or 4, and candidates two steps apart differ by 6. If both endpoints are tail-filled, all their large factor families are disjoint. A repeated tail label must be separated by at least  $y/4$  candidate steps, and any block of diameter less than  $y$  consumes at least two fresh large prime factors per candidate. Thus the remaining obstruction is the global concatenation of these local  $2/4/6$  rough short-difference units into a whole row. The global concatenation interface is sharpened in

`docs/monograph/prime-matrix-h3-tail-filler-global-chain-capacity.md`.

For any continuous tail-filled block  $B$ , full concatenation forces

$$\text{EdgeCap}(B; y) = |B| - 1, \quad \text{TriCap}(B; y) = |B| - 2.$$

Here  $\text{EdgeCap}$  counts compatible adjacent 2/4 tail edges and  $\text{TriCap}$  counts compatible 2/4/6 triples. If  $y > \sqrt{q+6}$ , a fixed ordered tail-label pair contributes at most once in the row; if  $y > q^{2/3}$ , every tail-filled point is a double-rough semiprime. Hence the final H3 hard input can now be stated as an explicit capacity exclusion: prove that these edge or triple capacities are strictly below the full-chain requirements, or that full capacity forces PDEC, ColumnCRT, endpoint phase, or cofactor-structure defects. This is a necessary-capacity reduction, not yet an unconditional proof of H3. The macroscopic full-tail branch is then attacked in

`docs/monograph/prime-matrix-h3-tail-edge-selberg-exclusion.md`.

For a fixed even shift  $\delta \in \{2, 4, 6\}$  and an interval  $I$  of length  $H$ , the classical two-dimensional Selberg upper-bound sieve gives

$$\#\{n \in I : (n(n+\delta), P(y)) = 1\} \leq A_\delta(\theta) \frac{H}{(\log y)^2} \quad (3 \leq y \leq H^\theta).$$

Every filled adjacent edge gives a  $y$ -rough pair with shift 2 or 4, and every filled triple gives a  $y$ -rough endpoint pair with shift 6. Hence a continuous tail-filled block  $B$  must satisfy

$$|B| \leq 2 + A_*(\theta) \frac{q+6}{(\log y)^2}.$$

Thus a macroscopic block of length comparable with  $q$ , in particular a whole H3 row of about  $q/3$  candidates, cannot be filled only by tail labels at sufficiently large scale. The remaining H3 obstruction is consequently narrower: any surviving bad row must split tail points into many short blocks, and the small-factor cuts or short-block endpoint phases must be routed to PDEC, ColumnCRT, endpoint, or cofactor defects. The short-block routing is made exact in

`docs/monograph/prime-matrix-h3-shortblock-singleton-barrier.md`.

If the tail points are decomposed into maximal consecutive tail blocks, with total tail mass  $T$ , number of blocks  $J$ , and internal adjacent-tail edges  $E$ , then

$$T = J + E.$$

The two-dimensional sieve controls  $E$ , so a bad row with  $T$  still of order  $q/\log y$  would have almost all tail mass in singleton tail blocks. The small-factor cuts themselves are not yet a PDEC contradiction, because the small skeleton has natural size comparable with  $q$ . The final local obstruction is therefore the singleton-tail system: a point  $a_j = \ell_j m_j$  with  $\ell_j > y$  is isolated between two small-skeleton neighbours, so  $a_j \equiv 0 \pmod{\ell_j}$  and simultaneously satisfies two non-zero congruences modulo small labels  $r_-, r_+ \leq y$ . One must prove that many such squeezed singleton phases force PDEC, ColumnCRT, endpoint, or cofactor defects. The squeezed singleton phase is converted into a CRT capacity criterion in

`docs/monograph/prime-matrix-h3-singleton-clamp-defect-criterion.md`.

For a fixed clamp datum  $c = (\delta_-, \delta_+, r_-, r_+)$ , the two small-skeleton congruences are either incompatible or define one class  $\rho(c) \pmod{\text{lcm}(r_-, r_+)}$ . Adding the tail label  $\ell > y$  gives one class modulo  $\ell \text{lcm}(r_-, r_+)$ , hence one row contains at most

$$1 + \left\lfloor \frac{q + O(1)}{\ell \text{lcm}(r_-, r_+)} \right\rfloor$$

points from that unit. Consequently many singleton tails must enter one of three branches: tail-label concentration, clamp low-mod concentration, or distributed singleton capacity. The first two are named defect entrances. The final unresolved branch is the distributed signed excess of double-rough semiprimes over the prime survivor channel. This distributed branch is made bilinear in

`docs/monograph/prime-matrix-h3-distributed-singleton-bilinear-obstruction.md.`

For  $y > q^{2/3}$ , every singleton tail is a product  $\ell m$  with  $y < \ell \leq p$  and  $m$  prime. The clamp class  $\rho(c) \pmod{R(c)}$  is equivalent to the moving cofactor congruence

$$m \equiv \ell^{-1} \rho(c) \pmod{R(c)}.$$

Thus the final distributed obstruction is a signed bilinear prime-counting defect over the short cofactor intervals  $I/\ell$ . Without tail-label concentration, clamp concentration, endpoint accumulation, or ColumnCRT/cofactor bias, one must prove that this signed bilinear error is only  $O(q/\log^2 y)$ ; this is the current last H3 input. The non-zero-frequency bridge is recorded in

`docs/monograph/prime-matrix-h3-bilinear-large-sieve-defect-bridge.md.`

Expanding the cofactor residue condition modulo  $R(c)$  into non-principal characters gives

$$\chi(\ell^{-1} \rho(c)) = \chi(\rho(c)) \overline{\chi(\ell)}.$$

Hence a large distributed singleton error forces a non-principal bilinear character sum over the short cofactor intervals. A plain large-sieve bound is still too weak at this short window scale; the final input is a dispersion/Kloosterman-type estimate for this family, or a proof that the corresponding non-zero frequency is already a PDEC, ColumnCRT, or cofactor defect. The additive Kloosterman reduction is

`docs/monograph/prime-matrix-h3-dsb-kloosterman-window-reduction.md.`

The congruence  $m \equiv \rho(c) \bar{\ell} \pmod{R(c)}$  gives, after finite additive Fourier expansion, the inverse phase

$$e\left(-\frac{h\rho(c)\bar{\ell}}{R(c)}\right).$$

Thus the final H3 frequency input is a short-window Kloosterman sum with variable  $\ell$ , cofactor window  $I/\ell$ , modulus  $R(c)$ , and frequency  $h$ . The remaining audit splits into four branches: KLS-window coverage of the active parameters, high-lcm clamp escape, high-frequency endpoint tail, and coefficient concentration. The high-lcm branch is routed further in

`docs/monograph/prime-matrix-h3-dsb-high-lcm-clamp-routing.md.`

If  $R(c) > R_0$ , then one row contributes at most

$$1 + \left\lfloor \frac{q + O(1)}{R_0} \right\rfloor$$

points from a fixed clamp unit. Hence any high-lcm mass of order  $q/\log y$  must activate many almost-singleton high-lcm units. Across a persistent family of bad rows this gives a non-zero CRT/Fourier defect, hence a PDEC/ColumnCRT exit; if it is not persistent, it is a sparse single-window escape. This does not yet exclude the branch, but it removes high-lcm mass from the

low-modulus KLS-window estimate and routes it to the existing final exits. The common energy source of the two high-lcm exits is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-fourier-energy-clamp.md`.

For a homogeneous high-lcm modulus block, with residue-counting measure  $\mu$  and total mass  $U$ , finite Plancherel gives

$$\sum_{1 \leq h < R} |\widehat{\mu}(h)|^2 = R \sum_{a \bmod R} \mu(a)^2 - U^2.$$

In particular, when  $U \leq R/2$ , this non-zero energy is at least  $RU/2$ . Hence high-lcm escape necessarily injects Fourier energy: persistent energy must be excluded by PDEC/ColumnCRT, and isolated energy by SAE/endpoint analysis. The persistent branch is converted to a PDEC threshold in

`docs/monograph/prime-matrix-h3-dsb-hlc-pdec-threshold-bridge.md`.

For one homogeneous formal unit  $B$  with phase-counting vector  $g_B$ ,

$$L_{\text{HLC}}(B) = \left( \frac{R \sum_a g_B(a)^2 - U_B^2}{R - 1} \right)^{1/2}$$

is a forced non-zero-frequency lower bound. Thus a persistent high-lcm unit is excluded by a same-unit certificate  $U_{\text{CRT}}(B) < L_{\text{HLC}}(B)$ . The case  $U_B > R/2$  is not a new escape; it is dense effective-modulus concentration and routes back to KLS-window, PDEC/ColumnCRT, or SAE/endpoint. If this PDEC upper certificate fails, the failure is localized in

`docs/monograph/prime-matrix-h3-dsb-hlc-pdec-failure-localization.md`.

For a failing non-zero frequency  $h$  and direction  $\zeta$ , every cap

$$\mathcal{C}_\alpha = \{a : \Re(\zeta e(ha/R)) \geq \alpha\}, \quad 0 \leq \alpha < L/U,$$

has mass at least  $(L - \alpha U)/(1 - \alpha)$ . Hence failure is a concrete Bohr-cap concentration, which must become a same-unit PDEC constraint or a sparse SAE/endpoint certificate. The cap is further decomposed in

`docs/monograph/prime-matrix-h3-dsb-hlc-bohr-cap-component-route.md`.

With  $d = (h, R)$  and  $R' = R/d$ , the cap is the  $d$ -fold lift of one arc in  $\mathbb{Z}/R'\mathbb{Z}$ . Each lifted component has length at most  $d + \omega(\alpha)R$ , where  $\omega(\alpha) = \arccos(\alpha)/\pi$ , and one component carries mass at least  $(L - \alpha U)/(d(1 - \alpha))$ . Thus the remaining cap obstruction is either a large-gcd low-effective-modulus concentration or a short-arc component concentration. The large-gcd branch is descended in

`docs/monograph/prime-matrix-h3-dsb-hlc-high-gcd-descent.md`.

If  $d = (h, R) > 1$  and  $R' = R/d$ , the projected vector  $g'(b) = \sum_{a \equiv b \pmod{R'}} g(a)$  satisfies

$$\widehat{g}_R(h) = \widehat{g}_{R'}(h/d),$$

and the Bohr-cap mass is preserved. Hence each high-gcd step strictly lowers the effective modulus and must terminate in a low-effective-modulus PDEC/KLS exit or in the short-arc case. The short-arc case is quantified in

`docs/monograph/prime-matrix-h3-dsb-hlc-short-arc-density-pressure.md`.

If a short arc has length at most  $D_0 + \omega(\alpha)R$  and mass forced by the Bohr-cap lower bound, then its density relative to the global density  $U/R$  is at least

$$\frac{R(L - \alpha U)}{D_0(1 - \alpha)U(D_0 + \omega(\alpha)R)}.$$

If this pressure exceeds  $1 + \epsilon$ , a local-density PDEC row or SAE/endpoint certificate is forced; otherwise only the explicit parameter residual remains. The low-pressure residual is optimized in

`docs/monograph/prime-matrix-h3-dsb-hlc-short-arc-pressure-optimizer.md.`

Writing  $t = L/U$  and

$$\Psi_{D_0, R}(t) = \sup_{0 \leq \alpha < t} \frac{t - \alpha}{D_0(1 - \alpha)(D_0/R + \omega(\alpha))},$$

the residual is the one-dimensional condition  $\Psi_{D_0, R}(t) \leq 1 + \epsilon$ . This implies the same-unit flatness estimate

$$\sum_a g(a)^2 \leq \frac{1 + (R - 1)\tau^2}{R} U^2$$

and the corresponding support lower bound. Thus low-pressure short-arc mass is routed back to the KLS-window coefficient-norm check or to coefficient concentration. The KLS admission step is separated in

`docs/monograph/prime-matrix-h3-dsb-hlc-l2-flat-cls-admission.md.`

It lists six required checks for an HLC L2-flat residual: effective modulus, effective frequency, endpoint smoothing, coefficient  $L^2$  norm, gcd/unit strata, and dyadic or tail-label splitting. The L2-flat estimate supplies the coefficient-norm check; any other failed item is a named exit, and only if all six pass may the KLS-window input be invoked. Clean admission is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-clean-cls-reduction.md.`

A clean HLC formal unit is one in which the six named exits do not occur; then the K1–K6 admission checks pass. The remaining object is the clean HLC Kloosterman window

$$\mathcal{K}_{\text{HLC}}(B),$$

and the remaining external/self-contained input is an estimate  $\mathcal{K}_{\text{HLC}}(B) = O(q/\log^2 y)$ , denoted HLC–KLS-ext in the notes. The external-theorem adaptation is now separated in

`docs/monograph/prime-matrix-h3-dsb-hlc-cls-external-adaptation.md.`

It matches the clean HLC object to a DI/BFI/Kuznetsov-type windowed Kloosterman input: the inverse phase is  $e(-h\rho(c)\bar{\ell}/R(c))$ , the modulus and frequency checks are supplied by K1–K2, endpoint smoothing by K3, coefficient  $L^2$  control by K4, gcd/unit strata by K5, and dyadic or tail-label losses by K6. Thus the clean HLC branch is closed in the external-deep-theorem version. A fully self-contained version would still require reproving the corresponding spectral/dispersion Kloosterman theorem inside this manuscript. The self-contained obligation is further compressed in

`docs/monograph/prime-matrix-h3-dsb-hlc-cls-core-reduction.md.`

There the clean HLC window is decomposed into dyadic Type-I/II blocks with standard kernel  $\mathfrak{S}(\mathcal{D})$ , and the proof of HLC-KLS-ext is reduced to one core estimate  $|\mathfrak{S}(\mathcal{D})| \leq \mathcal{N}(\mathcal{D})/\log^A y$ . Thus the remaining no-black-box line is this single windowed Kloosterman spectral mean estimate. The no-black-box spine is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kls-core-self-contained-spine.md.`

It completes the smooth completion, Kloosterman quadratic-form reduction, coefficient  $L^2$  book-keeping, and the implication from the Kuznetsov large-sieve atom to the core estimate. It also records why pointwise Weil bounds alone do not yield the required arbitrary logarithmic saving. Hence a fully self-contained version now has one precise remaining analytic line: prove the specialized Kuznetsov large-sieve atom for this window family. That line is expanded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kuznetsov-ls-atom-expansion.md.`

The atom is decomposed into smoothing, the specialized Kuznetsov trace formula, Bessel transform decay, the spectral large sieve including oldforms and Eisenstein spectrum, and the well-factorable dispersion step that supplies the logarithmic saving. The manuscript therefore records the exact no-black-box obligations rather than hiding them under the single phrase “spectral theory”. The Bessel-transform part is now treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-c-bessel-transform-decay.md.`

It uses the Bessel differential equation, the self-adjoint Bessel operator, and repeated integration by parts to obtain the required rapid transform decay. The remaining no-black-box analytic atoms at this point are the specialized Kuznetsov formula, the spectral large sieve, and the well-factorable dispersion logarithmic saving. The spectral-large-sieve atom is further reduced in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-d-spectral-large-sieve-spine.md.`

This dualizes the spectral large sieve to a spectral projection kernel, accounts for oldforms, Eisenstein spectrum and holomorphic forms as polylogarithmic losses, and reduces KZ-D to a single pre-trace kernel bound with diagonal size  $T^2$  and off-diagonal Schur size  $N_0$ . The pre-trace kernel is further decomposed in

`docs/monograph/prime-matrix-h3-dsb-hlc-ptk-d-pretrace-kernel-bound.md.`

There the diagonal term is recorded as the local Weyl volume, and the remaining off-diagonal part is reduced to a spatial lattice-point/correlation row-column bound, denoted LPC-D. The spatial kernel is split in

`docs/monograph/prime-matrix-h3-dsb-hlc-lpc-d-spatial-correlation-split.md.`

The far tail is absorbed by kernel decay, the identity-near sector is empty or parabolic, and the parabolic cusp sector is controlled by divisor bookkeeping. The remaining spatial line is the generic hyperbolic local-correlation bound GHLC-D. That last spatial line is now treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-ghlc-d-local-schur-closure.md.`

The key correction is to estimate the pre-trace object as an integral kernel, not by pointwise hyperbolic neighbour counting. On the local radius  $r_* = T^{-1} \log^B y$  one has

$$T^2 \int_0^{r_*} (1 + Tr)^{-A} \sinh r \, dr = O_A(1),$$



so the local ball area cancels the  $T^2$  height of the kernel. Combining this local  $L^1$  kernel mass with the Poincare-packet Schur normalization gives GHLC-D, hence LPC-D, PTK-D, and KZ-D. At that point the remaining analytic atoms were KZ-B, the specialized Kuznetsov trace formula, and KZ-E, the well-factorable dispersion logarithmic saving. The KZ-B specialization is treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-b-kuznetsov-trace-specialization.md`.

It derives the needed trace formula from the automorphic kernel, Poincare-packet unfolding, the double-coset Kloosterman geometric side, spectral Plancherel, and Bessel-transform normalization. This closes the formula-conversion part; it does not supply the final logarithmic saving. Hence the remaining no-black-box analytic atom in the HLC branch is KZ-E, the well-factorable dispersion logarithmic-saving step. The current KZ-E attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-e-well-factorable-dispersion-spine.md`.

It internalizes the well-factorable convolution algebra, the dispersion variance identity, the CRT reduction to a standard Kloosterman phase, and the gcd/endpoint polylogarithmic ledger. This proves that KZ-E would follow from one remaining estimate, denoted WFD-core: a windowed well-factorable Kloosterman dispersion mean estimate. That core is further balanced in

`docs/monograph/prime-matrix-h3-dsb-hlc-wfd-core-balanced-factor-reduction.md`.

Using the square-root well-factorable decomposition  $\lambda_c = \sum_{c=uv} \alpha_u \delta_v$ , the condition  $c \asymp C$  forces  $u, v \asymp C^{1/2}$  up to polylogarithmic boundary blocks. The gcd strata are polylogarithmic, and CRT factors the phase modulo  $uv$  into coupled phases modulo  $u$  and  $v$ . At this intermediate stage the remaining atom is BWFD-core, a balanced two-modulus well-factorable Kloosterman dispersion mean estimate. The subsequent completion attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bwfd-core-spectral-completion-attack.md`.

It performs exact finite Fourier completion in the  $s$  variable and uses complete Kloosterman multiplicativity modulo  $uv$ . This proves that BWFD-core follows from one still narrower atom, BSC-core: balanced complete Kloosterman bilinear correlation with arbitrary logarithmic saving. Ordinary KZ-D spectral large sieve recovers only the raw quadratic norm scale, and pointwise Weil bounds give only single-sum square-root cancellation; neither supplies the required  $\log^{-A}$  saving. The current direct attack on BSC-core is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bsc-core-kloosterman-fraction-attack.md`.

It expands the complete Kloosterman sums term by term and exposes the reciprocal-fraction phase  $e(\bar{v}R/u + \bar{u}T/v)$ . One-sided degeneracy is localized to quadratic congruences such as  $(a_h + \ell)x^2 + b_h \equiv 0 \pmod{u}$ ; the remaining no-black-box atom is KFLS-core, a balanced Kloosterman-fraction large-sieve estimate with arbitrary logarithmic saving. The next square-kernel attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kfls-core-square-kernel-attack.md`.

It expands the square of the reciprocal-fraction sum into a four-modulus phase kernel. This also rules out an absolute Schur proof of the full kernel: the exact diagonal recovers only the raw quadratic norm scale. The remaining atom is therefore CFQK-core, a centered four-modulus Kloosterman-fraction correlation estimate covering the semi-diagonal and true off-diagonal layers. The block-centering correction is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-cfqk-core-block-centering-attack.md`.

It shows that the entire same- $(u, v)$  block diagonal is local variance and cannot be forced to be logarithmically small for arbitrary admissible coefficients. The remaining obligations are therefore BD-CEN, the block-centering identity, OSQK-core for one shared modulus, and TFQK-core for the true four-modulus layer. The BD-CEN audit is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bd-cen-dispersion-centering-audit.md.`

It checks the current KZ-E dispersion spine and finds only the  $h = 0$  main-term subtraction, not the same- $(u, v)$  block projection required by BD-CEN. Thus the first blocking obligation is the identity (BDC-5); OSQK-core and TFQK-core are conditional downstream obligations until that identity is proved. The subsequent no-go note is

`docs/monograph/prime-matrix-h3-dsb-hlc-bd-cen-no-go-route-fork.md.`

It shows that, for the current unblocked WFD/KZ-E object, BD-CEN is false rather than merely unproved: a single nonzero block with nonzero frequency survives the  $h = 0$  subtraction but has no cross-block counterpart. The remaining choices are therefore SOURCE-CEN, BLK-energy-core, or an explicit external DI/BFI input. The SOURCE-CEN no-go note

`docs/monograph/prime-matrix-h3-dsb-hlc-source-cen-no-go.md`

checks the first of these choices. It shows that source block-centering would replace the unblocked linear WFD target in (KE-13) by a different centered object and would leave a block mean term which still has to be estimated. Thus SOURCE-CEN is not an identity for the present target. The subsequent BLK-energy audit

`docs/monograph/prime-matrix-h3-dsb-hlc-blk-energy-core-obstruction.md`

checks the second choice. As an arbitrary coefficient-array theorem at the square-kernel level, a one-block one-atom test makes  $\sum_b |S_b|^2$  comparable with the raw norm, so a bare arbitrary logarithmic saving for BLK-energy-core is false. The remaining self-contained obligation is therefore the sharper NC-BLK input: prove block non-concentration for the actual WFD coefficients produced by the Type-I/II, Fourier, and well-factorable structure. Without that new input, this branch is only closed in an external-theorem version using a precisely matched DI/BFI dispersion theorem. The CRT lift-parity variant is audited in

`docs/monograph/prime-matrix-crt-lift-parity-audit.md.`

For adjacent primes  $p < q$ , the non-trivial  $q$ -columns are not covered by the new prime  $q$ ; therefore a hypothetical  $q$ -row zero does descend to an old  $p$ -sieve zero window of length  $q$ . However, the old  $p$ -sieve row period already has an even number of zero rows by reflection symmetry, so multiplying the period by the odd factor  $q$  preserves evenness and does not create a parity contradiction. The retained conclusion is only the early-window dichotomy: an aligned lower zero row or a seam window routed to PDEC/SAE. The attempted recursive continuation is audited in

`docs/monograph/prime-matrix-recursive-mirror-descent-audit.md.`

It separates three objects: zero rows in a full CRT period at fixed width  $q$ , zero rows after peeling the largest base prime, and zero rows in an actual smaller-width matrix. These objects do not imply one another. In the finite audit, full-period zero rows occur for several adjacent pairs, but no

early  $q^2$  zero row and no actual smaller matrix zero row is forced. The next usable gate remains seam-window exclusion, not recursive parity. The annulus-mirror localization audit

`docs/monograph/prime-matrix-annulus-mirror-localization-audit.md`

separates two reflections. The CRT reflection preserving zero residue classes is  $n \mapsto -n \pmod{M_p}$  and sends a small row to the tail of the CRT period. The terminal reflection  $n \mapsto q^2 - n$  does land in an early interval, but it changes the forbidden classes from 0 to  $q^2 \pmod{\ell}$ . It is therefore a terminal non-zero-class block, not a smaller prime-matrix zero row. The scaled-row and half-width version is audited in

`docs/monograph/prime-matrix-scaled-peeling-halfwidth-audit.md`.

It shows that the scaled height  $nP/p$  is only an approximate locator: a lower aligned zero row appears only when the upper window fully contains a lower row. In the same samples, passing to a prime near half the upper width gives no automatic pair of consecutive lower zero rows, because the half-level sieve resurrects points whose least factor lies above the half-width level. The resurrected-point absorption route is now recorded in

`docs/monograph/prime-matrix-rpz-absorption-defect-route.md`

with the finite audit

`docs/monograph/prime-matrix-rpz-absorption-defect-audit.md`.

It routes complete absorption into five alternatives: a survivor remains, TailAnchor load, ColumnCRT displacement load, ColumnRadius, or a residual Distributed-RPZ branch. In the known first-zero-row samples the half-level has 15 resurrected points, no missing absorber, and maximum one-window label and top-column loads equal to 1. Thus the route does not justify a single-window TailAnchor exclusion; the next theorem-level gate is the Distributed-RPZ capacity barrier, or a proof that distributed absorption necessarily returns to one of the named column or tail exits. The first attack on this gate is recorded in

`docs/monograph/prime-matrix-rpz-sliding-plateau-barrier.md`.

It proves a sliding-platform multiplicity formula: if a zero window remains zero under a consecutive block of shifts, then a resurrected source in the common core repeats with the full platform multiplicity. Hence distributed absorption on such a platform is either a persistent TailAnchor load or, after source deletion, a boundary-compressed branch. The audit in

`docs/monograph/prime-matrix-rpz-sliding-plateau-audit.md`

finds platform length 5 in the known samples and TailAnchor triggering at  $T_0 = 4$  in all five cases. The remaining gate is therefore the boundary-compressed RPZ branch, not unrestricted distributed absorption. The boundary-compressed branch is further routed in

`docs/monograph/prime-matrix-rpz-boundary-compressed-core-route.md`.

It proves that, if TailAnchor is excluded, the center interval  $J_{T_0} = [L + A + T_0, R + B - T_0]$  is an  $h$ -sieve zero interval. The audit

`docs/monograph/prime-matrix-rpz-bcb-core-audit.md`

finds that all 11 center survivors in the known samples are TailAnchor core sources; after conditional TailAnchor deletion, all five center intervals are clean and contain an aligned half-width row. The next gate is therefore the BCB grid/endpoint exclusion: either the center contains an aligned lower zero row, or the endpoint seam enters SAE, PDEC, or ColumnCRT. The exact grid criterion is recorded in

`docs/monograph/prime-matrix-rpz-bcb-grid-endpoint-criterion.md.`

For  $J = [u, v]$ ,  $N = v - u + 1$ , and  $\delta_h(u) \equiv 1 - u \pmod{h}$ , the interval contains an aligned  $h$ -row if and only if  $N \geq \delta_h(u) + h$ . Failure is therefore a finite endpoint seam phase. The audit in

`docs/monograph/prime-matrix-rpz-bcb-grid-endpoint-audit.md`

matches this criterion in all five samples and finds no endpoint-defect sample. The remaining gate is endpoint persistence exclusion: a failed grid phase must be isolated as SAE or persistent as PDEC/ColumnCRT. This endpoint gate is routed in

`docs/monograph/prime-matrix-rpz-bcb-endpoint-persistence-route.md`

and audited in

`docs/monograph/prime-matrix-rpz-bcb-endpoint-phase-ledger.md.`

For fixed  $(h, N)$ , endpoint failure is a finite residue ledger in  $u \bmod h$ . In the known samples there are no actual endpoint failures and only two possible failure phases. Sparse loads are SAE obligations; persistent same-phase loads are PDEC/ColumnCRT objects. Thus the RPZ endpoint branch is now named, not closed: the remaining work is either certificate closure for SAE/PDEC/ColumnCRT or recursive descent from the lower  $h$ -zero row. The two remaining RPZ attacks are now advanced together in

`docs/monograph/prime-matrix-rpz-dual-track-closure-route.md.`

Track A fixes the certificate interfaces for RPZ-SAE, RPZ-PDEC, and RPZ-ColumnCRT. Track B uses the lower-zero descent audit

`docs/monograph/prime-matrix-rpz-lower-zero-descent-audit.md`

which shows that, in the known BCB samples, all six conditional lower zero rows descend to a  $p = 2$  impossible zero row with no endpoint/grid obstruction. This is still a sample-level descent audit; the global task is to prove descent-grid persistence or route every obstruction to the named certificate interfaces. The obstruction side is now made explicit in

`docs/monograph/prime-matrix-rpz-lower-descent-obstruction-ledger.md`

and

`docs/monograph/prime-matrix-rpz-certificate-materialization-interface.md.`

For each adjacent descent  $p \rightarrow r$ , obstruction is a finite phase condition modulo  $\prod_{\ell \leq r} \ell$  and is classified as either grid failure or endpoint-puncture blocking. The current descent tree has twenty actual transition nodes, all successful, with no actual blocked node. This does not close the global

theorem; it reduces any future obstruction to the SAE finite package, PDEC endpoint phase row, or ColumnCRT displacement row. The certificate skeleton builder

`docs/monograph/prime-matrix-rpz-certificate-skeleton-package.md`

materializes these rows: two endpoint SAE candidates, two endpoint PDEC/ColumnCRT rows, and three lower-descent grid-failure PDEC/ColumnCRT rows. These are certificate obligations, not completed exclusions. The endpoint SAE finite certificate

`docs/monograph/prime-matrix-rpz-endpoint-sae-finite-certificate.md`

checks that the two endpoint candidates have possible load two but actual load zero in the current BCB ledger. Hence the current finite endpoint SAE line is vacuously closed; global SAE exclusion is still not proved. The lower grid-failure certificate

`docs/monograph/prime-matrix-rpz-lower-grid-fail-avoidance-certificate.md`

reduces every adjacent descent  $p \rightarrow r$  to the exact test  $\delta = -(a-1)(p-r) \bmod r$  and grid success iff  $\delta \leq p-r$ . The current descent tree has twenty actual transition nodes and none hits grid failure. The remaining global task is to prove this phase inequality for formal descent paths, or route its failure to PDEC/ColumnCRT. The formal path note

`docs/monograph/prime-matrix-rpz-formal-descent-phase-inequality.md`

sharpens this statement: once a formal descent path exists, the phase inequality is automatic from interval containment. The true remaining task is path existence for every formal counterexample branch, or exclusion of the first obstruction phase through SAE/PDEC/ColumnCRT. The first-obstruction dichotomy

`docs/monograph/prime-matrix-rpz-first-obstruction-dichotomy.md`

removes endpoint puncture as a genuine obstruction: a contained lower row containing the endpoint  $ap$  would force  $r \mid a$ , contradicting the roughness condition needed for the endpoint to revive. Hence every non-descending branch has a first grid-failure seam phase, which must be discharged by SAE/PDEC/ColumnCRT certificates. The first-grid-failure seam certificate

`docs/monograph/prime-matrix-rpz-first-grid-fail-seam-certificate.md`

puts this obstruction in normal form. If  $g = p-r$  and  $g < \delta < r$ , the bad row is the union of two adjacent lower-row caps of lengths  $\delta$  and  $r+g-\delta$ , with conserved missing mass  $r-g$  and at most the right endpoint  $ap$  as an  $r$ -rough survivor. The certificate produces twelve seam phase rows covering all 1752 grid-failure phases in the current ledger. It now also records the exact PDEC support row for each seam:

$$S_\tau = \{a \bmod Q : a \equiv \rho \pmod{r}\}, \quad F_\tau = \mathbf{1}_{a \equiv \rho \pmod{r}} - 1/r,$$

whose non-zero Fourier support is contained in the frequencies  $jQ/r$ ,  $1 \leq j < r$ . The same certificate splits the endpoint  $ap$  into 1348 phases already killed by lower labels and 404  $Q$ -unit endpoint phases. Thus the next narrow gate is no longer support extraction, but either an endpoint-PDEC upper bound  $U_{\text{CRT}} < L_{\text{PDEC}}$  for these same bad-window families or a ColumnCRT displacement certificate with explicit selectors  $\Pi, \lambda$  and threshold  $L_D$ . The unit-endpoint gate certificate

`docs/monograph/prime-matrix-rpz-unit-endpoint-columncrt-gate.md`

pushes this one step further. For a unit endpoint seam,

$$c \equiv ap \equiv p\rho \equiv r + (p - r) - \delta \pmod{r},$$

so the endpoint lies in a fixed non-trivial lower column. If  $\pi = hr + c$  is a same-column prime witness, then the endpoint lower row satisfies  $H \equiv -cr^{-1} \pmod{p}$ , and the displacement  $d = h - H \pmod{p}$  is independent of the row phase and non-zero. In the current ledger all twelve gate rows have explicit witnesses and non-zero displacements, covering all 404 unit endpoint phases. This routes persistent unit endpoint seams to fixed non-zero ColumnCRT candidates; it still does not exclude the ColumnCRT threshold  $L_D$  or prove that formal counterexample families avoid these gates. The threshold obstruction audit

`docs/monograph/prime-matrix-rpz-columncrt-threshold-obstruction.md`

then shows why this cannot be closed by merely lowering  $L_D$ . Inside a fixed unit gate, all unit endpoint residues already have the same label  $p$  and the same non-zero displacement class  $d \pmod{p}$ , so the intrinsic load of that class is

$$R_{p,d} \geq \prod_{\ell < r} (\ell - 1).$$

In the present ledger the maximum intrinsic load is 48, and the test value  $L_D = 2$  would send ten gate rows, covering 400 phases, into ColumnCRTDefect rather than exclude them. Thus the remaining legitimate closures are a formal-family avoidance theorem, an endpoint-PDEC upper bound, or an independent exclusion theorem for the resulting ColumnCRTDefect. The three-route audit

`docs/monograph/prime-matrix-rpz-three-route-closure-audit.md`

compares these options under the same ledger. It finds no closed route yet, but ranks formal-family avoidance first: the present lower-descent ledger has twenty actual transition nodes and no grid-failure node, while the endpoint-PDEC route still lacks an admissible  $U_{\text{CRT}}$  upper bound and the ColumnCRT route has been reduced to an independent defect-exclusion theorem. The formal phase automaton

`docs/monograph/prime-matrix-rpz-formal-phase-automaton.md`

turns the first route into an explicit start-phase admission problem. For each prime level  $p$  it defines an accepted set  $A_p \pmod{P(p)}$ : phases in  $A_p$  descend canonically through successive prime levels to  $p = 2$ , while phases outside  $A_p$  hit a first-grid-fail seam and return to the already materialized exits. In the current ledger all six BCB-Core start rows belong to their accepted sets. The global missing theorem is that every formal counterexample-family start phase lies in  $A_p$ , or else is captured by the seam exits. The rejected-phase absorption certificate

`docs/monograph/prime-matrix-rpz-rejected-phase-absorption.md`

removes the second ambiguity within the present automaton range. It checks all 27924 rejected phases, finds twelve distinct first-fail seam rows, and verifies that every one of them is already materialized by the first-grid-fail seam certificate. Hence the RPZ formal route is now a strict two-way routing statement: an accepted start phase descends to  $p = 2$ , while a rejected start phase enters the named seam/PDEC/ColumnCRT certificate chain. This is an absorption theorem, not an exclusion theorem; the remaining hard point is the actual discharge of those seam/PDEC/ColumnCRT exits, or a proof that formal counterexample starts always lie in  $A_p$ . The seam exit pressure ledger

`docs/monograph/prime-matrix-rpz-seam-exit-pressure-ledger.md`

then compresses this remaining exit. The 27924 rejected phases reduce to twelve seam rows; within those rows 1348 endpoint phases are already killed by lower labels, while the remaining 404 unit endpoint phases fall into ten fixed non-zero ColumnCRT displacement classes, with maximum aggregated load 96. Thus the next gate is no longer a phase search: one must either prove formal-family avoidance of the twelve seam rows, provide  $U_{\text{CRT}} < L_{\text{PDEC}}$  for their single-residue PDEC supports, or prove an independent exclusion theorem for the ten ColumnCRT displacement classes. The symbolic ladder certificate

`docs/monograph/prime-matrix-rpz-symbolic-ladder-certificate.md`

rewrites the preferred avoidance route as local adjacent-prime digits. For each adjacent pair  $p > r$ , with  $g = p - r$ , it uses

$$\delta_p(a) = -(a - 1)g \pmod{r}, \quad \delta_p(a) \leq g$$

as the exact success digit. Failure of this inequality is precisely the first-grid-fail seam already materialized above. The certificate verifies the resulting counting formula by full enumeration through modulus  $P(19) = 9699690$  and symbolically records the same ladder through  $p = 97$ . The remaining theorem is now sharply stated: derive these digit inequalities from the formal counterexample construction itself, not from a finite phase enumeration. The BCB start-digit ledger

`docs/monograph/prime-matrix-rpz-bcb-start-digit-ledger.md`

extracts these digits for the current BCB-Core starts. All twenty digit nodes are safe, but ten of them have zero margin  $\delta_p(a) = p - r$ . This is a useful negative result for the proof strategy: the desired avoidance theorem cannot be closed by a coarse positive margin estimate. It must use exact congruence information for the start row modulo each lower prime  $r$ . The accepted lower-row selector ledger

`docs/monograph/prime-matrix-rpz-bcb-accepted-row-selector.md`

weakens the required positive statement to the exact selector condition

$$C_h(J) \cap A_h \neq \emptyset, \quad C_h(J) = \{m : J \supset [(m - 1)h + 1, mh]\}.$$

For the current BCB-Core samples all five core intervals have such a selector, and all six candidate lower rows are accepted. This is still only a current-ledger certificate; the global theorem must prove selector existence for every formal BCB core interval, or else route the all-rejected selector failure back to the seam/PDEC/ColumnCRT exits. Finally, the selector gap ledger

`docs/monograph/prime-matrix-rpz-selector-gap-threshold.md`

splits selector existence into two verifiable mechanisms. If the number of complete candidate rows in  $C_h(J)$  exceeds the maximum cyclic rejected gap of  $A_h$ , length alone forces a selector. In the current samples this explains two of the five BCB cores; the remaining three are short-candidate cases and require exact endpoint phase information. Thus the next proof obligation is not merely to enlarge  $J$ , but to control the endpoint phase of short cores relative to the rejected gaps of  $A_h$ . The short-candidate endpoint phase ledger

`docs/monograph/prime-matrix-rpz-short-candidate-phase-ledger.md`

performs this refinement at modulus  $hP(h)$  rather than just modulo  $h$ . In the current ledger the three short-candidate BCB records all hit an accepted selector. However, the full  $(h, \ell)$  phase

families still contain all-rejected endpoint phases: 588 for  $(h, \ell) = (7, 13)$ , 11880 for  $(11, 19)$ , and 344760 for  $(13, 25)$ . These rejected phases are not unnamed escapes; each carries a first-failure seam key and therefore returns to the seam/PDEC/ColumnCRT exits. The remaining theorem-level gate is consequently exact: prove that the formal BCB construction avoids these all-rejected  $hP(h)$  endpoint phases, or close the corresponding seam/PDEC/ColumnCRT certificates. The follow-up forbidden-block ledger

`docs/monograph/prime-matrix-rpz-short-phase-block-formula.md`

removes another layer of enumeration. Since the relevant short cores have length  $\ell < 2h$ , a starting endpoint  $u$  contains at most one complete  $h$ -row. For a lower row  $m$ , the endpoint interval that contains it is exactly

$$mh - \ell + 1 \leq u \leq (m - 1)h + 1,$$

of width  $\ell - h + 1$ . Hence the all-rejected endpoint set is the disjoint union of such blocks over rejected row phases  $m \bmod P(h)$ . The formula matches enumeration in all three short families, and the current endpoints have positive distance from the rejected endpoint union. The remaining non-enumerative input is therefore narrower still: derive from the formal BCB construction that the induced candidate row phase belongs to  $A_h$ , or else close its named first-failure exit. The BCB candidate phase identity ledger

`docs/monograph/prime-matrix-rpz-bcb-candidate-phase-identity.md`

then makes the induced row phase explicit. If the top zero row is  $R$ , the top prime is  $P$ , the half-prime is  $h$ , the plateau extremes are  $s_-$ ,  $s_+$ , and the tail threshold is  $T$ , the forced core is

$$L = (R - 1)P + 1 + s_- + T, \quad U = RP + s_+ - T.$$

The complete lower rows inside this core are exactly

$$m_{\min} = \left\lfloor \frac{L + h - 2}{h} \right\rfloor + 1, \quad m_{\max} = \left\lfloor \frac{U}{h} \right\rfloor.$$

Equivalently, after writing  $P = Qh + d$ , these phases depend only on  $R \bmod hP(h)$  and the plateau parameters. In the current ledger the formula matches all five BCB samples and all six induced lower row phases lie in  $A_h$ . This is the sharp remaining input: prove the same residue implication for every formal BCB counterexample, or send any rejected induced phase to its first-failure exit. The accepted-preimage ledger

`docs/monograph/prime-matrix-rpz-bcb-accepted-preimage-ledger.md`

enumerates this final residue condition for the current BCB parameter families. The  $P = 13, h = 5$  family has no bad residue. The other four families still have bad residues, split between no-candidate endpoint failures and all-rejected candidate phases. The actual top-row residues in the ledger all lie in the selector preimage, with positive distance from the bad set where the bad set is nonempty. Thus the residue preimage computation is now explicit, but the proof is still not global: the formal counterexample construction must be shown to land in the selector preimage, or the bad residue must be discharged through its named BCB endpoint or first-failure seam/PDEC/ColumnCRT exit. The core-run obstruction ledger

`docs/monograph/prime-matrix-rpz-bcb-core-run-obstruction.md`



adds a stronger upstream test for the no-TailAnchor branch. In that branch BCB-Core forces  $J_T$  itself to be an  $h$ -sieve zero interval, so its length cannot exceed the maximum cyclic run of integers divisible by primes  $\leq h$  modulo  $P(h)$ . For the current layers these exact maxima are

$$h = 5 : 5, \quad h = 7 : 9, \quad h = 11 : 13, \quad h = 13 : 21.$$

The five current BCB cores have lengths 9, 13, 15, 19, 25, respectively, so every one of them exceeds the corresponding low-sieve run bound, with minimum margin 4. Thus the current finite BCB no-TailAnchor samples are obstructed before the accepted-preimage question is reached. The global theorem-level input is now a Jacobsthal-type low-sieve run bound for the formal layers; without such a bound, the remaining bad cases still route to the named endpoint and first-failure exits. This input is isolated in

`docs/monograph/prime-matrix-rpz-bcb-jacobsthal-closure-interface.md`.

Writing  $G(h)$  for the maximum length of a consecutive interval all of whose points are divisible by some prime  $\leq h$ , the exact BCB closure condition is

$$G(h) < P + m - 1 - 2T,$$

where  $m$  is the sliding platform length. Under this inequality the no-TailAnchor BCB branch closes immediately; if it fails, the proof must keep the TailAnchor, endpoint, first-failure, PDEC, and ColumnCRT exits active. The risk scan

`docs/monograph/prime-matrix-rpz-bcb-jacobsthal-risk-scan.md`

checks this linear closure route against the Ziller–Morack primordial Jacobsthal data. It shows that the current finite samples are safe, but the naive global specialization  $m = 5$ ,  $T = 4$ , and only  $P > 2h$  fails already at  $h = 43$ : there  $G(h) = 89$ , the minimal top prime is  $P = 89$ , and the core length is only 85. Thus the finite core-run obstruction cannot be promoted to a global theorem by a blanket linear bound. A global proof must either force longer platforms  $m$ , prove stronger restrictions on the formal BCB parameter set, or close the TailAnchor/endpoint/first-failure exits. The platform-threshold ledger

`docs/monograph/prime-matrix-rpz-bcb-platform-length-threshold.md`

rewrites this requirement as the exact integer condition

$$m \geq G(h) - P + 2 + 2T.$$

For  $T = 4$ , the first substantive unsafe layer with  $h \geq 5$  is again  $h = 43$ , where the required platform length is 10 rather than 5. The next attack must therefore prove platform-length growth or prove that short platforms trigger one of the named exits. The short-platform embedding route

`docs/monograph/prime-matrix-rpz-bcb-short-platform-embedding-route.md`

handles the second alternative. If  $N = P + m - 1 - 2T \leq G(h)$  and no TailAnchor is present, then the forced core  $J_T$  is an  $h$ -sieve zero interval of length  $N$  and hence must be embedded inside an extremal low-sieve covered block modulo  $P(h)$ . Therefore its left endpoint lies in a finite Jacobsthal endpoint set  $E_{h,N}$ . Persistent occurrence of such an endpoint phase is no longer an unnamed escape: it is routed to SAE if sparse, or to PDEC/ColumnCRT if persistent. The accompanying certificate

`docs/monograph/prime-matrix-rpz-bcb-short-platform-embedding-certificate.md`

materializes the max-block subfamily  $E_{h,N}^{\max}$  by reconstructing CRT left endpoints from the Ziller–Morack modulus representation. For the current  $m = 5, T = 4$  interface, the first non-trivial short platform is  $h = 43, P = 89, N = 85$ , with 240 distinct max-block endpoint phases; across the scanned table, 1197 reconstructed longest blocks pass the coverage and boundary checks. This remains a certificate for the longest-block subfamily, not a proof that the full endpoint set  $E_{h,N}$  has been exhausted.

These obligations are not typographical tasks. They are the remaining theorem-level gates. Until they are closed, the monograph may be submitted only as a synthesis, dependency, and verification manuscript, not as a final unconditional proof of the prime-matrix theorem, prime-pair assertions, or RH.

## Chapter 7

# Referee Closure Audit

This chapter records the strict closure standard used for the whole monograph. A contradiction-field chain is considered closed only if its entrance, decomposition, exits, and global contradiction are all verified under the same load convention.

**Definition 7.1** (Closure certificate). A closure certificate for a contradiction field consists of four data:

1. an entrance theorem sending every counterexample into the field;
2. a disjoint or source-deleting decomposition into named exits;
3. for each exit, an absorption, descent, transfer, external theorem, or finite certificate;
4. a global comparison showing that final load is simultaneously bounded below and above in incompatible ways.

If any item is replaced by an unverified structural input, the conclusion is conditional on that input.

**Theorem 7.2** (Current monograph closure). *The present manuscript closes the audit and dependency structure of the two contradiction-field programs: all known entrances, decompositions, exits, external inputs, finite certificates, and referee obligations are explicitly named. It does not close the final unconditional theorem status of either program.*

*Proof.* For the prime-matrix program, the A/B appendix gives the entrance reduction from a row or column counterexample to a Structured-EHPD bad configuration. Inside the BPN branch, the low-hole bucket submodule is now closed by the low-range certificate, the narrow-band collision certificate, the two finite tail certificates, and the Rosser–Schoenfeld explicit tail package recorded above. Inside Triad-A1, the same-set capacity / full- $S$  terminal is now closed as a canonical-source self-contained theorem-boundary: the source provenance is canonical RIW/Buchstab, the final closure certificate has no open gates, and the unrestricted generic WFD self-contained statement is explicitly refuted rather than claimed. These are genuine submodule and theorem-boundary closures. The D-structure exclusion and the remaining global PDEC/SAE/Rankin or LocalSurvivor/CleanKLS certificates remain decisive inputs requiring independent line-by-line review. Therefore the whole chain is closed as a reduction, certificate, and boundary package but not as a final unconditional prime-matrix theorem.

For the RH program, the RH manuscript has completed U1–U5: statement wording, AEX acceptance, EXT localization, manuscript wording, BibTeX/PDF compilation, and warning-free LaTeX logs. The final audit still requires independent referee verification of every local theorem

and controlled exit. Therefore the chain is closed as a verification package but not as a final unconditional RH theorem.

The status table and the referee audit in `docs/monograph/` record exactly which pieces are proved, reduced, external, computational, or awaiting review. Thus the monograph's logical boundary is explicit and closed, while promotion to final theorem status remains conditional on external verification.  $\square$

**Theorem 7.3** (Prime-matrix frontier). *The current prime-matrix row/column unconditional theorem is routed to named terminal gates but is not closed. The canonical-source Triad-A1 self-contained boundary is closed, and the unrestricted generic WFD self-contained theorem is refuted and not claimed. After the global terminal-family split and the self-contained bottleneck router, the independent self-contained mathematical bottleneck is the same-set PDEC capacity certificate*

*PDEC\_CAP\_SameSetGlobalDualCertificate.*

*Inside that bottleneck, the transverse formal-unit embedding and the downstream canonical layer admission/nonzero-transfer/thin-return gate are now closed on the canonical-source self-contained branch. The remaining materialized open gate is the generic/external original DI/BFI route*

*DIBFIQuantifiedNoProjection  
WindowCertificate* ,

*after the APS profinite dichotomy, the persistent finite-signature unification, and the SC-9 boundary reconciliation are closed as routing laws, and after the persistent terminal admission and finite-arc routers remove raw, low-rank, continuous-cap, and unnamed finite-arc pseudo-terminals. The previous transverse clean large-sieve atom is now routed to the A1 CleanKLS/SC-9 frontier, and its self-contained source-support branch is further embedded into the canonical A1/KZ-E source; the canonical layer closure then returns to the already closed A1 selector/decision-tree/source-provenance boundary. The PDEC-CAP theorem itself remains open in the broader global/external sense. Final theorem promotion still separately requires the D-structure/Tail-log4/finite Rankin referee interface. The boundary-lift router records the canonical-source branch as*

*NoFurtherCanonicalSourceSelfContainedPDECCapGap,*

*while keeping the generic/external DI/BFI route and the final D-structure/Rankin referee interface outside that self-contained closure. The subsequent terminal-promotion closure router sharpens the canonical-source self-contained boundary to*

*NoFurtherCanonicalSourceTerminalPromotionGap.*

*Proof.* The frontier router

`docs/monograph/prime-matrix-row-column-unconditional-frontier-router.md`

checks the merged boundary audit, the Triad-A1 terminal confluence router, the terminal triad contract, the formal-unit audit, the stitching feasibility audit, the nested-duplicate dominance audit, the line-by-line referee matrix, and the claim-status table. It records

`row_column_unconditional_closed=false,`

while preserving the closed canonical-source boundary and the refuted generic branch. The terminal triad has no fourth exit, so any final counterexample must be absorbed by PDEC, LocalSurvivor, or CleanKLS/DLS certificates, together with the still guarded D-structure/Tail-log4/Rankin interfaces.

The global terminal-family boundary router and its split router then show that the current materialized frontier is exhausted and that all non-final reductions are closed. The open terminal family gates reduce to PDEC-CAP, KLS/external-or-internal large sieve, and the D-structure/Rankin referee interface. For the fully self-contained route, the remaining choice was

PDEC\_CAP\_OR\_INTERNAL\_CleanKLS\_LargeSieve.

The self-contained bottleneck router closes the non-final CleanKLS ambiguity. CleanKLS failure returns to PDEC/SAE, the canonical clean branch is absorbed by the closed same-set boundary, the generic full- $S$  self-contained strengthening is refuted and not claimed, and the SC-9 route has already been accounted for. Therefore InternalCleanKLS is no longer an independent self-contained blocker. What remains is exactly the global same-set PDEC dual capacity certificate  $U_{\text{CRT}} < L_{\text{PDEC}}$ , plus the independent referee-promotion gate. This proves the stated frontier and also explains why it is not yet a final unconditional theorem.

The PDEC-CAP same-set global-dual router refines that frontier once more. It records

closed\_current\_materialized\_pdec\_gates=true

and

pdec\_cap\_same\_set\_global\_dual\_closed=false.

Thus the current local PDEC materialization is exhausted, but the global proof is not. The profinite APS router closes the missing dichotomy: along the inverse tower of signatures, the real payment graph  $\Gamma_n$  either has a positive-limsup persistent finite signature, which becomes a multi-bucket PDEC formal unit, or avoids every finite signature, which is a distributed CleanKLS/DLS input unless FiberDeletion or NoDeletion-KL has already routed it back to PDEC. The first persistent narrow hardpoint is therefore the terminal estimate

PrimitiveMultiAtomSameFormalUnitPDECCertificate.

The diffuse terminal split router first sharpens the nonpersistent side. No unnamed FiberDeletion, NoDeletion-KL, or CleanKLS exit remains; the side is reduced to either SC-9 or a finite-shell low-mod persistence object. The latter is a PDEC/ColumnCRT input. The persistent-signature unification router then identifies persistent MFU and fixed-shell persistence as the same finite-signature formal-unit grammar; ColumnCRT displacement, PDEC cap failure, and formal-unit mismatch all route back into the PDEC/ColumnCRT family rather than creating a parallel terminal. The SC-9 boundary reconciliation then removes the flat-clean SC-9 branch as an independent canonical-source blocker: clean failure returns to PDEC/SAE, canonical NC-BLK is absorbed by the same-set boundary, and generic WFD is not claimed as self-contained. The persistent terminal admission router then forces any remaining PDEC object to be primitive, multi-atom, same-formal-unit, non-tautological, and not already absorbed by SAE/Endpoint. The deletion support-exhaustion bridge closes the implication from divergent deletion to exhausted support or named sparse/PDEC return; the HRO router then shows that if deletion does not diverge, either TailIndependence returns to NoDeletion/CleanKLS or old-hole residue occupancy saturates. The occupancy-kernel router uses  $|\text{Occ}_t| \leq \min(|H_Q(t)|, r)$  to remove the sparse-old-hole subcase. Therefore the deletion-side hardpoint first becomes the dense old-hole selector kernel. The common-variable router then rewrites each selected column as  $c_b = \rho_b + rk_b$ ; every old-prime condition forbids exactly one residue of the same shell variable  $k_b$ . The resulting alternatives are named: no legal shell gives capacity/Hall deletion, positive mass on a fixed shell or finite shell packet gives PDEC/ColumnCRT persistence,

and genuine multishell flatness enters the already reconciled SC-9/NC-BLK boundary. An external KLS/DI/BFI/Kuznetsov citation would close only the external-input version.

The primitive terminal is then reduced further. The primitive multi-atom rank router removes rank-zero and rank-one pseudo-terminals: after physical deduplication, such remnants are reuse defects, two-point Fourier tautologies, fixed-shell PDEC/ColumnCRT objects, or SAE objects. What remains is a rank-at-least-two cap-stable primitive kernel. The cap-stable kernel router rewrites its inequality by the contrapositive of cap localization: if a direction reaches  $L_{\text{PDEC}}$ , then a direction cap has mass at least  $(L_{\text{PDEC}} - \alpha M)/(1 - \alpha)$  and must return to SAE, refined PDEC, ColumnCRT, or multiplicity normalization. Hence a genuinely cap-stable kernel only has to certify that all legal direction caps stay below their thresholds. The uniform-cap finite-basis router then removes the continuum in  $(\zeta, \alpha)$ : for a finite formal unit, every such cap is the preimage of a finite cyclic arc under a nontrivial character. Finally, the finite-arc transverse router splits a high arc into low transverse support, persistent transverse bias, or transverse-flat dispersion. These route respectively to SAE/ColumnCRT/fixed-shell PDEC, refined PDEC, or CleanKLS/DLS or an external large-sieve input. The current narrow PDEC-CAP hardpoint is thereby reduced to transverse clean analysis. The transverse clean reduction router observes that a finite arc fixes only one character direction; a rank-at-least-two primitive kernel still leaves a transverse quotient. Nonflat transverse defects are named returns to SAE, refined PDEC, or ColumnCRT. If none occurs, the residual is exactly a transverse  $L^2$ -flat clean large-sieve atom. The current narrow PDEC-CAP hardpoint is therefore first reduced to

**TransverseQuotientCleanLargeSieveAtom.**

The transverse clean-atom frontier router then imports the already registered A1 CleanKLS/SC-9 grammar. K1-K9 admission failure returns to PDEC, SAE, ColumnCRT, or multiplicity normalization. If admission succeeds, the object enters the SC-9 frontier: self-contained actual-coefficient NC-BLK or external DI/BFI. The canonical NC-BLK absorption cannot be used for a generic transverse quotient until the quotient coefficients are shown to have the canonical RIW/Buchstab source support, or until actual transverse block non-concentration is proved directly. The naive factor-residue incidence bridge is blocked by the internal-fiber obstruction. Thus the current self-contained hardpoint is

**TransverseSourceSupportNonconcentrationCertificate.**

This certificate is then split once more. The A1/KZ-E actual-source provenance ledger is closed for the canonical RIW/Buchstab pre-Cauchy source, the canonical RIW support route is reduced to Buchstab-layer support, and the raw thick squarefree support count is closed. The transverse embedding router then proves that the transverse quotient formal unit is a legal restriction, quotient, or conditioning of that canonical A1/KZ-E source without changing its coefficients: actual payment is a deterministic pushforward, finite signatures are finite projections, direction arcs are preimage restrictions, and the transverse quotient is a finite factor. The canonical layer closure router then observes that the downstream layer-admission/nonzero-transfer/thin-return gate is exactly the already routed A1 chain: selector retention, finite signatures, RIW/Buchstab decision tree, actual source provenance, and canonical branch boundary. Hence the canonical-source self-contained PDEC-CAP branch has no remaining layer-transfer gate. The original external DI/BFI route is still separate and requires

**DIBFIQuantifiedNoProjectionWindowCertificate,**

unless one explicitly accepts a windowed KLS external theorem as an external input. This does not promote the full row/column statement to an unconditional theorem.

Finally, the boundary-lift router

```
prime-matrix-self-contained-pdec-cap-boundary-lift-router
```

propagates this conclusion back to the upper self-contained bottleneck. It records

```
canonical_source_self_contained_pdec_bottleneck_closed=true
```

and

```
narrowest_self_contained_boundary=NoFurtherCanonicalSourceSelfContainedPDECCapGap.
```

The same audit keeps

```
pdec_cap_same_set_global_dual_closed=false
```

and

```
row_column_unconditional_closed=false.
```

Thus the PDEC-CAP gap has been removed only from the canonical-source self-contained branch. The broader generic/external branch still needs the quantified no-projection DI/BFI certificate, and the whole prime-matrix theorem still needs the global terminal family promotion review and the D-structure/Rankin referee interface.

The canonical terminal-promotion closure router then reconciles the older global split with the boundary lift. It observes that the current materialized terminal frontier is exhausted, the terminal triad has no fourth exit, PDEC-CAP promotion is closed on the canonical-source branch, canonical CleanKLS/NC-BLK is absorbed or returns to PDEC/SAE, and all known LocalSurvivor entrances are covered by witness, extractor, or admission contracts. It records

```
latest_self_contained_hardpoint_closed=true
```

and

```
open_self_contained_gates=[].
```

Thus the latest self-contained boundary is

```
NoFurtherCanonicalSourceTerminalPromotionGap.
```

This is still a canonical-source theorem-boundary closure, not an unrestricted global terminal-family exclusion theorem. The external DI/BFI route and the D-structure/Rankin referee gate remain outside the closure.

The final self-contained theorem router then packages the exact statement boundary. The approved self-contained statement is the Prime Matrix canonical-source terminal theorem boundary: the Triad-A1 same-set/full- $S$  terminal together with canonical terminal promotion on the canonical RIW/Buchstab source branch. Under that exact boundary it records

```
canonical_source_self_contained_theorem_closed=true
```

and

```
open_self_contained_gates=[].
```

Its terminal is

```
NoFurtherCanonicalSourceSelfContainedTheoremBoundaryGap.
```

The router also records the non-claims: the unrestricted generic full- $S$  WFD self-contained theorem is refuted and not claimed, and the unrestricted/global Prime Matrix row-column unconditional theorem is not claimed. The external DI/BFI route and the D-structure/Rankin referee promotion gate remain outside this self-contained theorem boundary.

The global-unconditional obstruction router makes the last point explicit. It records

```
current_corpus_global_unconditional_self_contained_closure_possible=false
```

and

```
row_column_unconditional_closed=false.
```

The named blockers are the refuted unrestricted generic WFD route, the still open actual full- $S$  source bridge or strengthened source anti-atom theorem, the DI/BFI no-projection window certificate, and the D-structure/Tail-log4/finite Rankin promotion gate. Thus the current manuscript closes the exact canonical-source self-contained theorem boundary, but does not close the unrestricted global row-column theorem without new inputs.

The actual-source bridge reconciliation router sharpens this last obstruction list. It checks the canonical provenance ledger, the source-lock branch split, and the final canonical-source theorem certificate together. On the canonical RIW/Buchstab branch it records

```
actual_source_bridge_closed_for_canonical_branch=true.
```

Hence the former broad blocker

```
ActualFullSSourceBridge
```

is superseded inside the canonical self-contained theorem boundary. The global complement is still open, however:

```
actual_source_bridge_closes_global_unrestricted=false.
```

The updated global blockers are

```
UnrestrictedGenericWFD,
NoncanonicalFullSComplementAntiAtomOrExternalDIBFI,
FullSNonAPStrengthenedSourceAntiAtom,
DIBFIQuantifiedNoProjectionWindowCertificate,
DStructureTailLog4FiniteRankinPromotion.
```

Thus the actual-source bridge is closed only for the canonical branch. Full global unrestricted closure still requires a strengthened anti-atom for the noncanonical full- $S$  complement, or an external quantified DI/BFI no-projection certificate, together with the final D-structure/Rankin promotion gate.

The noncanonical-complement input-contract router then fixes the remaining input boundary. It records

```
contract_boundary_closed=true,    canonical_branch_removed_from_remainder=true,
```

but also

```
generic_wfd_template_available=false,
noncanonical_complement_closed=false  for the current corpus.
```



Thus the remaining full- $S$  complement is not an unrestricted generic WFD lemma. The moving-delta separation blocks that template, while the canonical branch has already been subtracted. Under the current terminal map there is no hidden fourth route: one must prove actual source identity with the canonical RIW/Buchstab source, prove a strengthened anti-atom for the actual noncanonical source, or provide the quantified no-projection DI/BFI route. Final unconditional row-column promotion still additionally requires the D-structure/Tail-log4/Rankin acceptance gate.

The noncanonical-complement trilemma router sharpens this boundary. It records

trilemma\_boundary\_closed=true,

but also

self\_contained\_noncanonical\_package\_closed=false,  
external\_contract\_package\_closed\_if\_fulls\_kls\_ext\_accepted=true.

The router separates the only legal closure modes after the canonical branch has been subtracted:

actual source identity,  
strengthened actual-source anti-atom,  
or acceptance/proof of FullS-KLS-ext.

Its negative content is important: the generic full- $S$  self-contained anti-atom route is refuted by the moving-delta capacity model under the recorded formal WFD, Type/Fourier, K4/K6, and naive-incidence hypotheses. Hence the noncanonical package cannot be closed by returning to the generic WFD template. Without the external FullS-KLS-ext input, one must prove actual source identity or a strengthened actual-source anti-atom; with the external input, the remaining primary-source issue is the line-by-line `DIBFIPrimarySourceSpecializationProof`.

The closure-input atlas finally reviews the explored models together. The CRT/MinRep equivalence, cylindrical line covering, the  $P$ -column anchor wheel, the layered  $P^2 \pm k$  wheel, far-tail cofactor inversion, SN recursive peeling, named-exit absorption, and PDEC cap refinement no longer point to independent unnamed branches. They all feed the same terminal certificate package:

SAE-Cert, PDEC-Cert, ColumnCRT-Cert,  
CleanMultishellKLS, TotalDescent.

Together with the noncanonical full- $S$  complement package and the D-structure/Rankin promotion package, these are the minimal remaining input basis. Thus future work should prove these packages directly rather than search for another generic WFD template or a fixed numerical constant.

The D-structure/Rankin promotion package is now fixed as an acceptance boundary rather than a hidden terminal. The corresponding router records

promotion\_package\_boundary\_closed=true,  
promotion\_package\_independently\_accepted=false,  
rankin\_sample\_pass=true.

It checks that the package is listed in the minimal input basis, that it is separated from the canonical-source self-contained boundary, that PM-16 remains a BLOCK-REFEREE item in the line-by-line matrix, and that the Rankin ledger has a checkable certificate format. The current sample has exact smooth-core count 39, allowed budget 40.0, and both the Rankin and exact budget tests passing.

This is a format and sample verification, not a global acceptance of the whole promotion package. Final promotion still requires independent acceptance of the D-structure/Structured-EHPD entrance reductions, Tail-log4 BG/RKS theorem-number and adaptation checks, reproducible finite-verification hashes, and Rankin certificates for all formal colored corridors, with failures routed to PDEC/SAE.

The terminal-certificate package has a further compression. The no-unnamed-escape machine and the named-exit absorption contract are already closed at the routing level. The canonical-source clean KLS branch is absorbed by the final canonical boundary, and the noncanonical clean branch belongs to the noncanonical complement or to an external KLS route. TotalDescent is also not a separate terminal: a formal descent path reaches  $p = 2$ , while a first failure is a first-grid-fail seam already routed to PDEC, ColumnCRT, or SAE. Therefore the independent terminal certificate work is reduced to

**SAE-Cert, PDEC-Cert, ColumnCRT-Cert.**

This is a structural compression of the first package, not yet a proof of those three certificate families.

The ColumnCRT branch is then absorbed once more. A persistent nonzero column displacement overload is a displacement PDEC after adjoining the displacement signature to the finite phase space; a sparse displacement overload is an SAE/endpoint event; and balanced displacement loads become admissible constraints in the PDEC dual problem. RPZ unit endpoint gates show that tuning the displacement threshold alone cannot prove exclusion, but they do not create a third terminal family. Hence the independent terminal remainder is reduced to

**SAE-Cert, PDEC-Cert** including endpoint, displacement, cofactor, and primitive variants.

This is still not a proof of the terminal certificate package; it is the removal of ColumnCRT as an independent terminal.

The SAE branch is also removed as an independent terminal by the SAE absorption router. The router checks the SAE local-reduction lemma, the materialized LocalSurvivor packet ledger, the known extractor coverage audit, the new-sparse-entry admission audit, the packet-generation dichotomy, and the persistent-terminal admission boundary. It records

**sae\_independent\_terminal\_removed=true.**

The absorption law is the following. A sparse or single-window escape must either emit a finite LocalSurvivor packet with a witness or blocker-deficit certificate, or else a finite signature persists and returns to PDEC, ColumnCRT, TailAnchor, or CofactorAnchor; if the signature layers keep escaping, the branch enters CleanKLS/DLS admission or an external input package. The currently materialized LocalSurvivor/SAE packets are exhausted, the known sparse entries have extractors or contract fallbacks, and no additional unnamed sparse entry is admitted by the current contract system. Hence SAE is not a third terminal family. The first terminal package is now reduced to

**PDEC-Cert** including endpoint, displacement, cofactor, primitive, and  
non-tautological variants,

together with the hygiene rule that any genuinely new sparse route must bring an explicit packet-extractor schema. This still does not prove the terminal package or the unrestricted row-column theorem; it deletes SAE as an independent terminal and pushes the remaining structural burden back to PDEC or to a future explicit sparse schema.

The broad PDEC family is then turned into an explicit input boundary. The PDEC-family router records

```
pdec_family_explicit_input_boundary_closed=true,
current_materialized_pdec_frontier_closed=true,
canonical_source_pdec_cap_closed=true,
global_pdec_family_unconditional_closed=false.
```

It checks the persistent-terminal admission gate, the non-tautological PDEC admission audit, the primitive-rank boundary, the rank-two cap-stable inverse step, the finite cyclic-arc reduction, the transverse clean reduction, the clean-frontier route, and the canonical-layer closure. Thus the currently materialized non-tautological primitive PDEC candidate count is zero, and the canonical-source PDEC-CAP route has returned to the final self-contained boundary. A future PDEC obstruction is not admitted as a label: it must come with a same-formal-unit phase map, at least three physical primitive atoms after deduplication, no two-point tautology, rank at least two after shell and column quotients, cap stability under finite cyclic-arc localization, and no sparse, persistent-biased, or clean-externalized transverse return. Hence the first package is now reduced to future explicit primitive PDEC schemas and future explicit sparse packet schemas, while generic branches remain in the noncanonical trilemma and final theorem promotion remains in the D-structure/Rankin acceptance package.

The future sparse branch is also turned from a hygiene rule into an explicit admission boundary. The sparse-schema router records

```
future_sparse_packet_schema_boundary_closed=true,
current_materialized_sparse_frontier_closed=true,
global_sparse_family_unconditional_closed=false.
```

It uses the materialized LocalSurvivor ledger, the known packet-extractor coverage audit, the new-sparse-entry admission audit, the SAE absorption router, the packet-generation dichotomy, and the PDEC-family boundary. Thus a future sparse obstruction is not admitted as a bare terminal label. It must specify the source class, a finite window or fixed-offset fiber, the candidate set, the blocker families and their incidence projection, a witness or a strict blocker-deficit inequality, phase and formal-unit normalization with deduplication, a finite-signature persistence test, a layer-escape test, and a reproducible ledger with no open local obligation. If the finite signature persists, the branch returns to explicit PDEC/ColumnCRT/Tail/Cofactor schemas; if the packet layers escape, it enters CleanKLS/DLS or an explicit external large-sieve input. Hence the current materialized sparse frontier is zero, but the global sparse family is not unconditionally excluded.

Combining these items gives the final input firewall. The firewall router records

```
final_input_firewall_boundary_closed=true,
current_materialized_terminal_frontier_closed=true,
no_hidden_terminal_remaining=true,
all_final_inputs_independently_accepted=false,
row_column_unconditional_closed=false.
```

The remaining named inputs are exactly future explicit primitive PDEC schemas, future explicit sparse packet-extractor schemas, one legal branch of the noncanonical full-*S* trilemma, and independent D-structure/Rankin promotion acceptance. Thus the author-side boundary and naming problem is closed: there is no hidden terminal left in the current material. The unrestricted row-column

theorem is still not promoted, because the named inputs have not all been proved or independently accepted.

The four named inputs are then attacked one by one. The attack router records

```
current_materialized_frontier_zero=true,    no_hidden_terminal_remaining=true,
conditional_closure_chain_complete=true,    all_current_obligations_closed=false,
unconditional_closure_possible_from_current_corpus=false.
```

The future primitive PDEC and future sparse packet-extractor inputs have no present materialized obligation: they are admission firewalls for future routes. The real current obligations are the noncanonical full- $S$  complement and D-structure/Rankin promotion. The canonical actual-source branch is closed, but the unrestricted noncanonical complement is not; the recorded primary-source audit rejects deriving the required full- $S$  non-AP WFD Kloosterman large-sieve estimate from the existing DI/BFI sources. The D-structure/Rankin boundary is named and the sample Rankin certificate passes, but the full ledger and independent acceptance are not complete. Therefore the logical chain is conditionally complete: if future PDEC/sparse routes are discharged by their schemas, one legal noncanonical branch is proved or accepted, and the D-structure/Rankin package is independently accepted, then the no-hidden-terminal chain promotes the theorem. Without those new inputs, the present corpus cannot honestly claim the unrestricted unconditional row-column theorem.

The noncanonical input is finally narrowed further. The noncanonical-narrowing router records

```
noncanonical_narrowing_boundary_closed=true,    ap_source_lift_rejected=true,
generic_self_contained_antiatom_refuted=true,    exact_source_entropy_closed=false,
external_full_s_contract_closed_if_accepted=true,
self_contained_noncanonical_closed=false.
```

The AP-source lift is rejected by the AP/non-AP branch definition and by the object ledger; the generic self-contained anti-atom is refuted by the moving-delta capacity model; the canonical branch has already been removed. Therefore the noncanonical complement has only two real remaining inputs: an internal theorem proving exact source entropy, equivalently a strengthened source anti-atom for the actual full- $S$  non-AP source, or an external/new proof of the full- $S$  non-AP WFD Kloosterman large-sieve input.

The last-remaining-atoms router packages this boundary into the final open atoms. It records

```
last_remaining_atom_boundaries_closed=true,
conditional_logic_chain_complete=true,
all_last_atoms_proved_or_accepted=false,
row_column_unconditional_closed=false.
```

The final atoms are

```
ActualFullSNonAPExactSupportAtom,
ModulusDependentCompletedFullSKLSInput,
DStructureRankinIndependentAcceptance.
```

The first two are alternative closures for the noncanonical full- $S$  complement: either an internal exact support/source-entropy theorem for the actual full- $S$  non-AP source, or an external/new modulus-dependent completed full- $S$  Kloosterman large-sieve input. The third is the independent promotion

acceptance for the D-structure/Tail-log4/finite Rankin package. Thus the no-hidden-terminal logic chain is conditionally complete: one of the first two atoms, together with the D-structure/Rankin acceptance atom, promotes the row-column theorem. Since the present corpus has not proved or independently accepted these atoms, the unrestricted unconditional row-column theorem remains open.

The three-final-atoms hard-attack router then pushes this boundary one step further. It records

```
three_atom_attack_boundary_closed=true,
conditional_logic_chain_complete=true,
all_three_atoms_proved_or_accepted=false,
row_column_unconditional_closed=false.
```

The final mathematical choice is:

```
ActualFullSNonAPSourceCapacityAntiAtomForActualSource
or
CDependentResidueWeightSpectralCancellationInput
```

The first input is a theorem that the actual full- $S$  non-AP source capacity measure has no moving same- $(u, v)$  atom. The generic version is refuted by the moving-delta model, and it is not forced by K4/K6 or naive incidence. The second input is a spectral/dispersion cancellation theorem for the completed  $c$ -dependent residue weights  $B_{c,x} = \sum_k \beta_{x+kc}$ ; pointwise Weil,  $L^2$  budgeting, the ordinary large sieve, and flat-residue shortcuts do not supply it. The final promotion input is:

```
DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
```

Thus the minimal unconditional input basis is

```
(ActualFullSNonAPSourceCapacityAntiAtomForActualSource
or CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
```

The current corpus has not proved this mathematical alternative and has not completed the independent promotion acceptance, so the unrestricted unconditional row-column theorem is not yet a proved theorem of the manuscript.

The final unconditional-closure attempt router then checks whether the mathematical alternative can be closed from the current corpus. It records

```
final_attempt_boundary_closed=true,
math_lanes_collapsed_to_common_core=true,
internal_math_proof_found_in_current_corpus=false,
external_math_match_found_in_current_corpus=false,
independent_promotion_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The  $c$ -dependent completed-residue route is not an independent terminal: finite Fourier inversion connects it to the existing BWFD-BSC-KFLS chain, whose self-contained remaining duty is actual same- $(u, v)$  block non-concentration, unless a precisely matched external DI/BFI/Kuznetsov dispersion theorem is supplied. The actual-source anti-atom is the same moving-block spread/source-entropy duty. Therefore the irreducible final input basis is

```
(MovingBlockSpreadNCBLKForActualFullSNonAPWFDCoefficients
OR PreciselyMatchedExternalDIBFIKuznetsovDispersionTheorem)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
```

The current corpus does not prove moving-block spread, does not contain a line-by-line external theorem match for the current full- $S$ , non-AP, uncentered, no-projection object, and has not completed independent D-structure/Rankin promotion acceptance. Thus the final input boundary is closed, but the unrestricted unconditional theorem is still not proved.

The irreducible-input refinement router then sharpens the external label. The phrase “precisely matched external DI/BFI/Kuznetsov dispersion theorem” cannot be read as a claim that the existing primary DI/BFI sources have already been matched to this object: the BFI AP theorem, the DI/Maynard  $J$ -scale, and the AP-source lift route are all blocked for the present full- $S$ , non-AP, uncentered, no-projection WFD window. The refined basis is

```
(MovingBlockSpreadNCBLKForActualFullSNonAPWFDCoefficients
OR FullSNonAPWFDKLSTheoremInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Here FullSNonAPWFDKLSTheoremInput means a new proof or an independently cited theorem which directly estimates the current full- $S$  non-AP WFD Kloosterman/dispersion object. The router records

```
refinement_boundary_closed=true,
row_column_unconditional_closed=false.
```

The subsequent joint attack compares the two remaining mathematical lanes directly. The canonical RIW/Buchstab source branch has already been closed and subtracted from the noncanonical full- $S$  complement, so its support theorem cannot be imported as a proof of the noncanonical moving-block anti-atom. The internal lane is therefore narrowed to an exact support package for the actual noncanonical source:

```
FullSNonAPExactFactorSupportPackageForActualNoncanonicalSource.
```

The package has four subclauses:

```
exact factor-support lower bound,
balanced-range thresholds,
Type/Fourier capacity compatibility,
factor-residue incidence bridge or direct support proof.
```

The external lane remains

```
FullSNonAPWFDKLSTheoremInput.
```

The BFI, DI, and Maynard sources provide the AP/dispersion/Kuznetsov technology class, but the current audit finds no ready theorem covering the present  $c$ -dependent completed residue weights, uncentered no-projection full- $S$  non-AP object, with arbitrary logarithmic saving. Thus the latest basis is

```
(FullSNonAPExactFactorSupportPackageForActualNoncanonicalSource
OR FullSNonAPWFDKLSTheoremInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The two lanes can be narrowed once more. On the internal side, the full- $S$  balanced-range threshold is closed, since  $C \asymp P/\log^{O(1)} P$  and the balanced factors of  $c = uv$  dominate every fixed logarithmic threshold. The remaining exact-support and Type/Fourier-capacity clauses are therefore a single source anti-atom contract:

`FullSNonAPStrengthenedSourceAntiAtomContractForActualNoncanonicalSource.`

On the external side, the full- $S$  window may be completed modulo  $c$ ; the real residue is the  $c$ -dependent completed weight

$$B_{c,x} = \sum_k \beta_{x+kc}.$$

Thus the external theorem input is narrowed to

`CDependentResidueWeightSpectralCancellationInput.`

The refined terminal basis is

`(FullSNonAPStrengthenedSourceAntiAtomContractForActualNoncanonicalSource  
OR CDependentResidueWeightSpectralCancellationInput)  
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.`

The dual-lane common-core reconciliation router then checks whether the internal and external lanes are still independent self-contained obstacles. It records

`common_core_reconciliation_closed=true,  
self_contained_common_core_proved=false,  
external_contract_accepted_as_final_input=false,  
dstructure_rankin_independent_acceptance_completed=false,  
row_column_unconditional_closed=false.`

The reconciliation law is this: if `CDependentResidueWeightSpectralCancellationInput` is accepted as a FullS-KLS or  $c$ -dependent spectral theorem, it is an external theorem input. If it is proved internally from the present corpus instead, finite Fourier completion sends it back to the BWFD-BSC-KFLS chain, and then to actual same- $(u,v)$  block non-concentration. That is the same moving-block source-entropy core as the internal source anti-atom lane. Consequently the reconciled basis is

`((ActualA1FullSSourceLockOrNewFullSNonAPSourceAntiAtomTheoremInput)  
OR AcceptedFullSKLSExtOrCDependentResidueWeightSpectralCancellationInput)  
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.`

The fully self-contained basis, without an external black box, is

`ActualA1FullSSourceLockOrNewFullSNonAPSourceAntiAtomTheoremInput  
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.`

Thus the independence gap between the two mathematical lanes is closed. What remains is not two separate self-contained obstacles, but one actual-source anti-atom/source-lock input plus the independent D-structure/Rankin promotion acceptance. The unrestricted unconditional row-column theorem is therefore still not proved.

The self-contained narrowest-core router then removes one more ambiguity from the no-external-black-box basis. It records

```

        narrowest_core_reduction_closed=true,
    canonical_source_lock_absorbed_for_canonical_branch=true,
        source_lock_option_removed_from_global_remainder=true,
            external_black_box_used=false,
        generic_self_contained_antiatom_available=false,
            noncanonical_actual_source_core_proved=false,
    dstructure_rankin_independent_acceptance_completed=false,
        row_column_unconditional_closed=false.

```

The source-lock option has already been absorbed by the canonical RIW/Buchstab branch and the actual-source provenance ledger. It closes the canonical-source self-contained theorem, but it does not close the unrestricted global noncanonical full- $S$  complement. The generic self-contained anti-atom is also unavailable, since the moving-delta model refutes it under the recorded formal WFD/Type/Fourier hypotheses. Therefore the latest fully self-contained global basis is

```

    ActualNoncanonicalFullSSourceEntropyOrStrengthenedAntiAtomTheoremInput
    AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

The first input means exact source entropy, equivalently no moving same- $(u, v)$  atom, for the actual noncanonical full- $S$  non-AP WFD coefficients. It is not an unrestricted generic WFD template. This is a sharper boundary, not a proof of the remaining source core or of the independent D-structure/Rankin promotion gate.

The noncanonical source-core atomization router then removes the remaining naming duality between “source entropy” and “strengthened anti-atom.” It records

```

        source_core_atomization_closed=true,
    entropy_antiatom_duality_removed=true,
        balanced_range_threshold_closed=true,
            k4_k6_or_naive_incidence_suffices=false,
        canonical_import_allowed=false,
    actual_support_capacity_core_proved=false,
    dstructure_rankin_independent_acceptance_completed=false,
        row_column_unconditional_closed=false.

```

Exact source entropy and the strengthened anti-atom are two formulations of the same actual noncanonical source-capacity core. The balanced range threshold is already closed in the full- $S$  regime. K4/K6 residue and dyadic controls do not imply moving factor-pair support, naive factor-residue incidence is blocked by the large internal fiber inside one  $(u, v)$  block, and canonical RIW/Buchstab support cannot be imported into the noncanonical complement. Hence the latest self-contained basis is

```

    ActualNoncanonicalFullSFactorSupportCapacityTheoremInput
    AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

The first input asks for log-power lower absolute support of the exact  $u$ - and  $v$ -factors in every surviving actual noncanonical full- $S$  non-AP balanced block, together with Type/Fourier capacity



compatibility so that no moving  $(u, v)$  pair receives an unrecorded capacity multiplier. This closes a naming and shortcut boundary, but it is not yet a proof of that support-capacity theorem.

The actual capacity-ledger microatom router then checks one last possible shortcut inside that support-capacity statement. It records

```

        microatom_boundary_closed=true,
        support_only_suffices=false,
    registered_multiplier_discipline_would_suffice_with_support=true,
        exact_uv_support_proved=false,
    registered_multiplier_discipline_proved=false,
        actual_final_capacity_antiatom_proved=false,
    dstructure_rankin_independent_acceptance_completed=false,
        row_column_unconditional_closed=false.

```

The correction is that raw support width is not itself a source anti-atom. A model with many raw  $(u, v)$  pairs but one unrecorded Type/Fourier/fiber multiplier can still concentrate the final capacity measure on a single moving pair. Conversely, if all such multipliers are registered in the same formal unit and bounded by a fixed logarithmic power, then exact support does imply the final anti-atom: with  $L = \log y$ ,  $|\alpha_u|, |\delta_v| \leq L^C$ , registered multipliers  $W_{u,v} \leq L^E$ , and  $S_u S_v \geq L^{2A+4C+E}$ , one obtains

$$\max_{u,v} M_{u,v} / \sum M_{u,v} \leq L^{-2A}.$$

Thus the sharp self-contained input is best stated as

```

ActualFinalCapacityAntiAtomLedgerForNoncanonicalFullIS
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

A proof may supply the package

```

ActualNoncanonicalExactUVSupportLowerBound
AND ActualTypeFourierRegisteredCapacityMultiplierDiscipline.

```

This closes the support-only shortcut and fixes the conditional implication, but it still does not prove the final actual capacity anti-atom ledger or the independent D-structure/Rankin acceptance.

The registered capacity-multiplier discipline router then attacks the second micro-input in that package. It records

```

        registered_capacity_multiplier_discipline_closed=true,
        exact_uv_support_proved=false,
        actual_final_capacity_antiatom_proved=false,
    dstructure_rankin_independent_acceptance_completed=false,
        row_column_unconditional_closed=false.

```

This is a bookkeeping closure, not a new cancellation theorem. The Type decomposition, dyadic summation, CRT phase normalization, the Fourier  $h$ -window and tail, the coefficient, gcd, endpoint and smoothing losses, the full- $S$  completion fiber, and tail-label bookkeeping are all registered inside the same actual formal unit and cost only fixed logarithmic powers. These costs are absorbed into the final capacity measure  $M_{u,v}$ , or equivalently into a registered multiplier  $W_{u,v} \leq L^E$ . The router does not use the external DI/BFI no-projection theorem-match branch and does not prove factor support.

Consequently the fully self-contained source-side basis is reduced from

ActualNoncanonicalExactUVSupportLowerBound  
 AND ActualTypeFourierRegisteredCapacityMultiplierDiscipline

to the single source micro-input

ActualNoncanonicalExactUVSupportLowerBound.

Together with the independent promotion gate, the latest basis is

ActualNoncanonicalExactUVSupportLowerBound  
 AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

This closes the multiplier-discipline item, but the exact  $u, v$ -support theorem and the independent D-structure/Rankin acceptance remain open.

The ExactUVSupport terminal-attack router then audits the remaining source micro-input. It records

```
exact_uv_support_terminal_boundary_closed=true,
registered_capacity_multiplier_discipline_closed=true,
exact_uv_support_proved=false,
actual_final_capacity_antiatom_proved=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The audit removes the remaining shortcut routes. Formal WFD hypotheses allow point-supported factors, so they do not force actual  $u, v$ -support. K4/K6 and naive factor-residue incidence still live on the wrong projections. Raw Buchstab/Mertens counting gives enough squarefree products only after the canonical layer is admitted, and the canonical RIW/Buchstab support chain closes only the canonical-source branch. It cannot be imported into the noncanonical full- $S$  complement.

Thus the current fully self-contained basis is still

ActualNoncanonicalExactUVSupportLowerBound  
 AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

If the exact-support theorem is supplied, the already closed multiplier discipline gives the final capacity anti-atom by the recorded inequality. Without that theorem, and without independent D-structure/Rankin acceptance, the unrestricted row-column theorem cannot yet be stated as unconditional.

The ExactUVSupport failure-packetization router then rewrites the remaining source input in its contrapositive finite form. It records

```
failure_packetization_closed=true,
exact_uv_support_proved=false,
actual_support_failure_packet_exclusion_proved=false,
actual_final_capacity_antiatom_proved=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

After the actual noncanonical clean block, exact  $u, v$ -support notion, and registered multiplier threshold are fixed, the positive lower-bound input

ActualNoncanonicalExactUVSupportLowerBound

is equivalent to the absence of a positive-mass support-failure packet:

**ActualNoncanonicalSupportFailurePacketExclusion.**

Such a packet must carry the source class, formal-unit id, block key, exact  $u$ - and  $v$ -support sets, the  $L^{2A+4C+E}$  threshold, the registered capacity profile, return tests to the named exits, and a reproducible certificate. Hence support failure can no longer remain an unnamed obstruction. This packetization is an equivalent boundary reformulation, not a proof of packet exclusion. The latest self-contained input basis can therefore be written as

**ActualNoncanonicalSupportFailurePacketExclusion**  
**AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.**

The manuscript still lacks a proof that all such packets are absent or must return to the PDEC, SAE, ColumnCRT, CleanKLS, or external KLS exits, and it still lacks independent D-structure/Rankin acceptance.

The support-failure packet return-dichotomy router then closes the return alphabet for these packets. It records

**support\_failure\_packet\_return\_dichotomy\_closed=true,**  
**clean\_core\_packet\_exclusion\_proved=false,**  
**actual\_final\_capacity\_antiatom\_proved=false,**  
**dstructure\_rankin\_independent\_acceptance\_completed=false,**  
**row\_column\_unconditional\_closed=false.**

An actual noncanonical support-failure packet cannot be a fifth terminal family. If it is not a clean-core packet, then it must return through one of the existing named interfaces: an isolated finite packet gives a LocalSurvivor/SAE certificate; a persistent finite signature gives an explicit primitive PDEC schema; a column, displacement, endpoint or cofactor load returns through the ColumnCRT-to-PDEC/SAE absorption; and a drifting moving block enters CleanKLS/DLS, exact source entropy, or an explicit external KLS input. The canonical and generic-template escapes have already been blocked by the ExactUVSupport audit.

Thus the latest source-side micro-input is

**ActualNoncanonicalCleanCoreSupportFailurePacketExclusion.**

A clean-core packet is a positive-mass actual noncanonical full- $S$  non-AP balanced block, in the same formal unit, below the exact  $u, v$ -support threshold, with no canonical import, no finite sparse witness, no persistent PDEC signature, no column or displacement defect, and no external or generic CleanKLS exit. The latest self-contained input basis is therefore

**ActualNoncanonicalCleanCoreSupportFailurePacketExclusion**  
**AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.**

This closes return completeness, not clean-core exclusion. The manuscript still lacks a proof that such clean-core packets are impossible, or a direct proof of the final capacity anti-atom.

The clean-core moving-atom sharp-input router then corrects one final over-strong formulation. It records

**clean\_core\_moving\_atom\_sharp\_boundary\_closed=true,**  
**clean\_core\_moving\_atom\_exclusion\_proved=false,**  
**actual\_final\_capacity\_antiatom\_proved=false,**  
**dstructure\_rankin\_independent\_acceptance\_completed=false,**  
**row\_column\_unconditional\_closed=false.**

Excluding every clean-core low-support packet is sufficient, but it is stronger than the final capacity anti-atom requires. Once the registered multiplier discipline is closed, the contrapositive of the recorded capacity inequality says that any final  $M_{u,v}$ -atom produces a low-support packet. The converse is not needed: a low-support packet that does not concentrate final capacity is not a terminal obstruction. Therefore the sharp source-side input is

**ActualNoncanonicalCleanCoreMovingAtomExclusion.**

Here a clean-core moving atom is a positive-mass actual noncanonical full- $S$  non-AP balanced block, after all return tests, with some pair  $(u, v)$  satisfying

$$M_{u,v} / \sum_{u,v} M_{u,v} > \log^{-2A}.$$

The latest self-contained input basis is

**ActualNoncanonicalCleanCoreMovingAtomExclusion**  
**AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.**

This is only a sharp reformulation. It is not implied by formal WFD, Type decomposition, Fourier smoothing, or fixed-projection diffuse; the moving-delta obstruction remains.

The clean-core terminal normal-form router then removes the last naming ambiguity around that sharp input. It records

```
clean_core_terminal_normal_form_closed=true,
internal_exact_entropy_proved=false,
external_completed_kls_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The internal normal form of **ActualNoncanonicalCleanCoreMovingAtomExclusion** is exact clean-core source entropy: for every post-return clean-core moving block  $b = (u, v)$ ,

$$\max_b M_b / \sum_b M_b \leq \log^{-2A}.$$

The external alternative is not a generic DI/BFI or Kuznetsov citation; it is the completed, modulus-dependent full- $S$  input **ModulusDependentCompletedFullSKLSInput**. Thus the latest conditional terminal input basis is

**(ExactCleanCoreFullSNonAPWFDSourceEntropy**  
**OR ModulusDependentCompletedFullSKLSInput)**  
**AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.**

The fully self-contained basis is

**ExactCleanCoreFullSNonAPWFDSourceEntropy**  
**AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.**

This closes the terminal input normal form only. It does not prove exact entropy, accept the completed KLS input, or complete the independent D-structure/Rankin gate.

Finally, the clean-core exact-entropy atom router rewrites the failure mode of that self-contained input. It records

```

clean_core_exact_entropy_atom_boundary_closed=true,
clean_core_terminal_support_incidence_proved=false,
    exact_clean_core_entropy_proved=false,
dstructure_rankin_independent_acceptance_completed=false,
    row_column_unconditional_closed=false.

```

If exact clean-core entropy fails, there is a clean-core moving atom. The registered multiplier discipline removes unrecorded Type/Fourier/fiber capacity as an explanation; packetization forces any support failure to be reproducible; and the return dichotomy routes every non-clean-core packet back to the named exits. The only remaining self-contained failure mode is a clean-core terminal support atom. A sufficient actionable proof package is therefore

```

CleanCoreTerminalSupportIncidenceTheorem
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

Here `CleanCoreTerminalSupportIncidenceTheorem` means that every positive-mass post-return actual clean-core full- $S$  non-AP WFD block has enough exact  $u/v$  support, inside the same formal unit, to absorb the divisor bound and all registered capacity multipliers. This theorem is not proved in the current corpus.

The clean-core support-incidence attack router then audits whether this theorem follows from the already closed ingredients. It records

```

clean_core_support_incidence_attack_boundary_closed=true,
    clean_core_exact_layer_transfer_proved=false,
clean_core_terminal_support_incidence_proved=false,
    external_completed_kls_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
    row_column_unconditional_closed=false.

```

The range threshold, registered capacity budget, and raw squarefree/Buchstab counting in thick blocks are no longer the obstruction. The naive factor-residue incidence bridge is still blocked by the internal  $h, \ell, x, z$  fiber, and canonical RIW/Buchstab support cannot be imported into the noncanonical clean-core branch. The remaining internal input is

```

CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn.

```

The latest conditional basis is therefore

```

(CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn
OR ModulusDependentCompletedFullSKLSInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

The fully self-contained basis replaces the parenthesized alternative by

```

CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn.

```

This input requires exact clean-core layer admission, nonzero transfer to the actual  $\alpha/\delta$  coefficients, and return of thin or rejected blocks to the named exits. It is still open.

The clean-core layer-transfer path router then separates the available canonical template from what is needed for the noncanonical clean-core residual. It records

```

clean_core_layer_transfer_path_boundary_closed=true,
clean_core_path_partition_proved=false,
clean_core_exact_layer_transfer_proved=false,
external_completed_kls_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.

```

The canonical RIW/Buchstab decision tree and source-provenance ledger are closed only on the canonical-source branch. The noncanonical clean-core residual must supply its own actual coefficient path partition, or leave through the completed KLS or named-return interfaces. The newest internal input is

```
CleanCoreExactCoefficientPathPartitionNoCancellationAndThinReturn.
```

It asks for a polylogarithmic partition of the actual clean-core  $\alpha/\delta$  coefficients into exact path signatures, nonzero same-path contribution without cancellation, and return of thin, over-budget, cancelling, or unidentified-source blocks to the named exits.

The clean-core path-source firewall router then moves this input to the coefficient source level. It records

```

clean_core_path_source_firewall_boundary_closed=true,
clean_core_precauchy_source_law_proved=false,
clean_core_path_partition_proved=false,
external_spectral_atom_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.

```

The path partition must be built on the pre-Cauchy actual clean-core  $\alpha/\delta$  coefficient formula. The canonical RIW/Buchstab decision tree is only a canonical-source template, and generic WFD well-factorability is blocked by the moving-delta model. The latest conditional basis is

```

(CleanCorePreCauchyCoefficientSourceLawAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

The self-contained source-side input is

```
CleanCorePreCauchyCoefficientSourceLawAndReturn.
```

It requires an exact source formula before Cauchy, Type/Fourier and completion, plus polylog path signatures, same-path nonzero contribution, and named return for source-law failure, over-budget paths, or thin/rejected blocks.

The clean-core pre-Cauchy source-law atom router then compresses this source-law input to an

origin-generation ledger. It records

```

    clean_core_precauchy_source_law_atom_boundary_closed=true,
    origin_generation_ledger_implication_closed=true,
clean_core_original_coefficient_generation_ledger_proved=false,
    clean_core_precauchy_source_law_proved=false,
    external_spectral_atom_accepted=false,
    dstructure_rankin_independent_acceptance_completed=false,
    row_column_unconditional_closed=false.

```

The compression theorem is that a pre-Cauchy source law is exactly an origin ledger for the actual clean-core  $\alpha/\delta$  coefficients in one formal unit before Cauchy, dispersion, Type/Fourier and completion. If that ledger lists every summand, branch key,  $u/v$  map, sign and local factor, then the branch keys give the exact path signatures, the registered K6/tail-label costs give the polylog path budget, and same-key nonzero or sign-split refinement gives no cancellation. Missing source entries, over-budget paths, thin/rejected blocks or unresolved cancellation must return to the named exits or to the external spectral input. The latest conditional basis is

```

(CleanCoreOriginalCoefficientGenerationLedgerAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

The self-contained source-side input is

```

CleanCoreOriginalCoefficientGenerationLedgerAndReturn.

```

This input is not proved here. The router only shows that the former source-law input has no smaller honest self-contained evidence than the actual origin ledger; the canonical RIW/Buchstab ledger remains scoped to the canonical-source branch, and generic WFD well-factorability cannot replace it.

The clean-core origin-source admission router then compresses the origin ledger to a primitive source-constructor admission gate. It records

```

    clean_core_origin_source_admission_boundary_closed=true,
    constructor_admission_implies_origin_ledger=true,
    unregistered_source_return_absorbed=false,
clean_core_primitive_source_constructor_admission_proved=false,
clean_core_original_coefficient_generation_ledger_proved=false,
    external_spectral_atom_accepted=false,
    dstructure_rankin_independent_acceptance_completed=false,
    row_column_unconditional_closed=false.

```

An origin ledger is not an estimate; it first needs a source constructor. If the actual clean-core  $\alpha/\delta$  coefficients are emitted by a pre-Cauchy primitive constructor inside one formal unit, then the emitted-summand schema expands into the origin ledger. If no such constructor is registered, the object must return as an unregistered-source failure rather than remain a clean-core terminal. The latest conditional basis is

```

(CleanCorePrimitiveSourceConstructorAdmissionAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

The self-contained source-side input is

```
CleanCorePrimitiveSourceConstructorAdmissionAndReturn.
```

This input is still open: the current material proves the canonical constructor only on the canonical-source branch and blocks generic WFD as a constructor, but it does not prove noncanonical clean-core constructor admission or absorption of every unregistered source.

The clean-core constructor source-class firewall then separates this admission gate by source class. It records

```
constructor_source_class_firewall_boundary_closed=true,
source_class_partition_closed=true,
unregistered_source_return_absorbed=true,
actual_noncanonical_primitive_constructor_formula_proved=false,
clean_core_primitive_source_constructor_admission_proved=false,
external_spectral_atom_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The firewall has five classes. The canonical constructor is already closed but scoped to the canonical-source branch. A generic well-factorable template is not a constructor. Unregistered or mixed formal-unit sources return through Multiplicity/Stitching or the K7 formal-unit failure. The external spectral class remains the completed  $c$ -dependent residue-weight input. Thus the only self-contained source-side atom is

```
ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn.
```

Equivalently, the latest conditional basis is

```
(ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The source-side input asks for a pre-Cauchy definition of the actual noncanonical clean-core coefficients, a deterministic summand emitter for  $\alpha/\delta$ , branch keys,  $u/v$  maps, signs and local factors, and compatibility with the same formal unit used by the support, capacity and return ledgers. This formula is not proved here.

Finally, the external-lemma parameter-match router compares this remaining self-contained atom with the available DI/BFI/Kuznetsov tools. It records

```
external_lemma_parameter_match_boundary_closed=true,
external_lemmas_match_constructor_formula=false,
external_lemmas_close_self_contained_remainder=false,
external_spectral_atom_accepted=false,
actual_noncanonical_primitive_constructor_formula_proved=false,
row_column_unconditional_closed=false.
```

The comparison is structural. DI/Kuznetsov and BFI act after coefficients, residue weights, or Kloosterman phases have already been produced: they give spectral or AP-distribution control. The current self-contained atom is earlier in the chain: it asks for the pre-Cauchy actual noncanonical



source constructor and summand emitter. Therefore those external lemmas cannot replace the constructor formula. They remain relevant only to the external spectral branch, whose narrow target is

```
CDependentResidueWeightSpectralCancellationInput.
```

The conditional basis is unchanged:

```
(ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis remains

```
ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The clean-core reverse-provenance functor router then audits whether the older geometric and payment models can be used in the opposite direction. It records

```
reverse_provenance_functor_boundary_closed=true,
payment_pushforward_functoriality_available=true,
pushforward_reverse_uniqueness_rejected=true,
finite_projection_recovers_gamma_not_source=true,
constructor_formula_equivalent_to_registered_fiber_emitter=true,
registered_primitive_prepushforward_fiber_emitter_proved=false,
actual_noncanonical_primitive_constructor_formula_proved=false,
row_column_unconditional_closed=false.
```

The point is functorial rather than numerical. The actual payment graph  $\Gamma$ , its finite projections, cyclic arc restrictions, and transverse quotients are forward pushforwards or finite factors of a pre-Cauchy source. They preserve the registered formal unit and prevent source replacement, but they do not invert the pushforward: several primitive source disintegrations may have the same  $\Gamma$ . Thus one cannot recover a primitive constructor merely from the downstream payment graph.

Modulo the already closed return discipline, however, the constructor formula is equivalent to a more auditable input:

```
RegisteredPrimitivePrePushforwardFiberEmitterAndReturn.
```

Such an emitter must list, before Cauchy/dispersion, finite or polylogarithmically many primitive preimage summands over each positive-mass clean-core payment atom, with  $\alpha/\delta$ , branch keys,  $u/v$  maps, signs, local factors, formal-unit registration, and coefficient identities. Empty fibers, incompatible registrations, over-budget fibers, thin blocks, or unregistered sources must return through named exits. The latest conditional basis is therefore

```
(RegisteredPrimitivePrePushforwardFiberEmitterAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis becomes

```
RegisteredPrimitivePrePushforwardFiberEmitterAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The clean-core fiber-emitter field audit then separates the already available payment skeleton from the still missing signed coefficient lift. It records

```

fiber_emitter_field_audit_boundary_closed=true,
    payment_fiber_skeleton_closed=true,
    first_cover_payment_map_closed=true,
    payment_count_identity_closed=true,
    alpha_delta_coefficient_lift_proved=false,
    polylog_branch_schema_proved=false,
registered_primitive_prepushforward_fiber_emitter_proved=false,
    row_column_unconditional_closed=false.

```

The payment-level skeleton is now explicit: a completion  $y$  and a low hole  $c$  are sent by the first-cover rule  $\text{pay}(c, y)$  to a payment atom in  $\Gamma$ , and the count identity

$$\text{payment\_count} = \sum_{\text{phase}} M(\text{phase}) |H_{\text{low}}(\text{phase})|$$

has been checked. ActualPaymentStitching also leaves no fourth payment-layer exit. What is not supplied by this skeleton is the signed clean-core primitive coefficient table. Thus the newest self-contained input is

```

ExactAlphaDeltaLiftForRegisteredPaymentFiberSkeletonAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

The conditional basis is

```

(ExactAlphaDeltaLiftForRegisteredPaymentFiberSkeletonAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

The alpha/delta disintegration router then rewrites this lift in its intrinsic form. It records

```

alpha_delta_disintegration_boundary_closed=true,
    payment_base_map_closed=true,
lift_equivalent_to_signed_disintegration_dictionary=true,
    canonical_decision_tree_template_scoped=true,
    generic_or_unregistered_dictionary_blocked=true,
registered_alpha_delta_disintegration_dictionary_proved=false,
    exact_alpha_delta_lift_proved=false,
    row_column_unconditional_closed=false.

```

Let  $\Phi$  denote the first-cover payment map from primitive source summands to the completion-hole payment skeleton. An exact alpha/delta lift is equivalent to a registered signed disintegration dictionary for the actual noncanonical source measure over  $\Phi$ : one must list the signed preimage summands, branch keys,  $u/v$  maps, signs, local factors, variation/support budgets, and the push-forward identity. The canonical RIW/Buchstab decision tree is a model for such a dictionary only on the canonical branch; generic well-factorability or unregistered sources do not provide it. The latest self-contained basis is therefore

```
RegisteredAlphaDeltaDisintegrationDictionaryAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
(RegisteredAlphaDeltaDisintegrationDictionaryAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Finally, the disintegration automaticity router removes a false hard point. It records

```
disintegration_automaticity_boundary_closed=true,
discrete_payment_base_map_closed=true,
signed_fiber_disintegration_formal=true,
pushforward_identity_is_real_gate=true,
actual_signed_source_measure_phi_compatibility_budget_proved=false,
row_column_unconditional_closed=false.
```

Once a signed source measure  $\nu$  and the first-cover map  $\Phi$  are given, the fiber decomposition

$$\nu = \sum_a \nu|_{\Phi^{-1}(a)}$$

is formal on the discrete payment skeleton. The real compatibility gate is not disintegration itself, but the proof that  $\nu$  is the actual noncanonical alpha/delta source measure in the same formal unit, that  $\Phi_*\nu$  equals the target payment-side coefficient, and that variation, absolute support, and branch-key budgets are registered. The latest self-contained basis is

```
ActualSignedSourceMeasurePhiCompatibilityBudgetAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
(ActualSignedSourceMeasurePhiCompatibilityBudgetAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This is a boundary refinement, not a proof of the signed source or budget identity.

The geometric Phi/budget bridge router then returns the line-covering, cylindrical,  $P$ -column anchor, layered wheel, and dynamic-capacity models to this same interface. It records

```
geometric_phi_budget_bridge_boundary_closed=true,
geometric_payment_base_available=true,
geometry_defines_signed_source_measure=false,
geometry_proves_phi_pushforward_identity=false,
geometry_supplies_budget_return_shape=true,
actual_signed_source_measure_phi_compatibility_budget_proved=false,
row_column_unconditional_closed=false.
```

The geometric models provide the unsigned payment base map, candidate variation/support budgets, and named returns such as PDEC, SAE, ColumnCRT, and CleanKLS. They do not define the pre-Cauchy signed  $\alpha/\delta$  source measure, nor do they prove the coefficient identity  $\Phi_*\nu = \text{target payment-side coefficient}$ . Hence the previous single input splits into two atoms:

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND GeometricVariationBranchBudgetCertificateOrNamedReturn.
```

The latest conditional basis is therefore

```
((ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND GeometricVariationBranchBudgetCertificateOrNamedReturn)
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND GeometricVariationBranchBudgetCertificateOrNamedReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This again refines the boundary rather than proving the two remaining atoms.

The geometric variation/branch budget router then attacks the second atom. It records

```
geometric_budget_attack_boundary_closed=true,
geometry_ledger_alphabet_closed=true,
dprc_analytic_capacity_bound_proved=false,
signed_variation_branch_lift_proved=false,
geometric_variation_branch_budget_certificate_proved=false,
row_column_unconditional_closed=false.
```

The line-cylinder, bottom-deficit,  $P$ -column anchor, dynamic promoted wheel, layered wheel, and input-atlas ledgers close the unsigned support and named-return alphabet. They do not control the total variation of the actual signed source, since mass may cancel after pushforward along  $\Phi$ . The budget atom therefore splits as

```
GeometricVariationBranchBudgetCertificateOrNamedReturn
=> DPRCAAlpha043CenteredDiscrepancyOrNamedLayerReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND DPRCAAlpha043CenteredDiscrepancyOrNamedLayerReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
((ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND DPRCAAlpha043CenteredDiscrepancyOrNamedLayerReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn)
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the budget-side interface, not the two new inputs.

The DPRC centered-discrepancy router then attacks the DPRC atom itself. It records

```
dprc_centered_discrepancy_boundary_closed=true,
rsm_identity_closed=true,
bes_compression_closed=true,
dprc_centered_discrepancy_input_proved=false,
row_column_unconditional_closed=false.
```

The relative-sieve margin identity reduces  $T_Y < S_Y$  to an explicit model-gap/finite ledger and a centered positive-discrepancy bound. The BES compression reduces that bound to the exclusion of simultaneous high  $L^1$  and high  $L^2$  positive beta-bucket synchronization. If such a danger intersection is not directly excluded, the DLS route requires a named return:

```
PointLoad/ColumnCRT, ShortWindow/SAE, LowPhase/PDEC.
```

Thus

```
DPRCAIpha043CenteredDiscrepancyOrNamedLayerReturn
=> ExplicitModelGapAndFiniteDPRCLedger
AND BESDangerIntersectionExclusionOrDLSNamedReturn.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND BESDangerIntersectionExclusionOrDLSNamedReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
((ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND BESDangerIntersectionExclusionOrDLSNamedReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn)
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the DPRC interface, not the finite/model-gap or BES-DLS proofs.

The BES-DLS named-return router then attacks the BES-DLS atom. It records

```
bes_dls_named_return_boundary_closed=true,
dls_kernel_and_danger_algebra_closed=true,
bes_dls_named_return_input_proved=false,
row_column_unconditional_closed=false.
```

The centered kernel, danger intersection, spike-bucket pigeonhole, and quadratic energy expansion are now registered. If the BES danger intersection occurs, it may no longer remain an unnamed synchronization failure. It must enter one of three exits:

```
PointLoad -> ColumnCRT/tail-anchor/displacement PDEC,
ShortWindow -> SAE/LocalSurvivor/persistent sparse PDEC,
LowPhase -> W-unit PDEC/new-layer PDEC/flat DLS.
```

Thus

```
BESDangerIntersectionExclusionOrDLSNamedReturn
=> DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSLowPhasePDECNewLayerOrFlatDLSBound.
```

The latest self-contained basis is

```

ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSLowPhasePDECNewLayerOrFlatDLSBound
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

This closes the named-return boundary, not the three microinputs.

The DLS LowPhase PDEC/flat-DLS router then attacks the LowPhase atom. It records

```

dls_lowphase_boundary_closed=true,
lowphase_input_proved=false,
row_column_unconditional_closed=false.

```

LowPhase is not a single fixed-modulus law. If a fixed-wheel unit-class peak persists, it must return a  $W$ -unit PDEC. If a new Fourier layer concentrates in low dimension, it must return a new-layer PDEC. If both fixed-wheel and new-layer peaks dilute, the remaining mass is genuinely high-modulus and flat, so it must be absorbed by flat DLS/KLS. Thus

```

DLSLowPhasePDECNewLayerOrFlatDLSBound
=> DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND DLSNewLayerFourierConcentrationPDECReturn
AND DLSFlatHighModLargeSieveAbsorption.

```

The latest self-contained basis is

```

ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND DLSNewLayerFourierConcentrationPDECReturn
AND DLSFlatHighModLargeSieveAbsorption
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

This closes the LowPhase boundary, not the three new microinputs.

The new-layer external-lemma matching router then compares the most structural LowPhase microinput with the already registered external FullS-KLS and CleanKLS/DLS interfaces. It records

```

direct_external_lemma_closes_newlayer=false,
self_contained_newlayer_closed=false,
row_column_unconditional_closed=false.

```

The comparison is not a new numerical experiment. It is a parameter-level boundary check: external KLS/FullS lemmas estimate a prepared, completed, coefficient-given flat spectral block. The current new-layer atom sits one layer earlier. It must first prove that low-dimensional concentration on the added wheel fibre is an actual PDEC certificate, or that after deleting all such fibres the remaining object satisfies the clean flat-admission hypotheses. Hence

```

DLSNewLayerFourierConcentrationPDECReturn
=> ExactNewLayerFiberPDECProjectionMorphism
AND NewLayerNoConcentrationImpliesFlatAdmission.

```

The latest self-contained basis becomes

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND ExactNewLayerFiberPDECProjectionMorphism
AND NewLayerNoConcentrationImpliesFlatAdmission
AND DLSFlatHighModLargeSieveAbsorption
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Thus the external theorem can attach only after a new-layer projection/admission morphism; it cannot be used directly to erase the self-contained new-layer obligation.

The new-layer PDEC projection router then attacks the first of these two atoms. It records

```
newlayer_projection_slicer_closed_schema_admission_open,
exact_newlayer_projection_morphism_closed=false,
row_column_unconditional_closed=false.
```

Two deterministic sublemmas are closed. First, on the same full-period completion set  $C_P$ , lifting  $Q$  to  $rQ$  cannot create support outside the old projection. Second, a large  $r \nmid h$  Fourier coefficient on the new fibre can be rotated and sliced by a layer-cake argument to produce a cyclic-arc or half-plane cap with comparable signed mass discrepancy. Hence

```
ExactNewLayerFiberPDECProjectionMorphism
=> RegisteredNewLayerPDECFormalUnitAndCapStableSchema.
```

The latest self-contained basis is therefore

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND RegisteredNewLayerPDECFormalUnitAndCapStableSchema
AND NewLayerNoConcentrationImpliesFlatAdmission
AND DLSFlatHighModLargeSieveAbsorption
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The remaining obstruction is formal PDEC admission for the sliced cap: the same  $\Omega, \tau, w$  and phase map must be fixed, the cap must be non-tautological and cap-stable, and its lower and upper budgets must be computed in that same formal unit.  $\square$

**Theorem 7.4** (Early-zero gap carrier and CRT-asymmetry router). *The early-zero gap-carrier observation is a valid entrance reduction but not by itself a final contradiction. If the  $k$ -th  $P$ -row*

$$I_{P,k} = [(k-1)P+1, kP]$$

*contains no prime, and if  $a$  and  $b$  are the nearest primes respectively to the left and to the right of  $I_{P,k}$ , then  $a, b$  are adjacent primes and  $b - a > P$ . The CRT period  $M_P$  repeats the small-factor covering pattern, but it does not repeat the primality of the carrier endpoints. Consequently a persistent endpoint phase is a PDEC/ColumnCRT object, an isolated endpoint phase is an SAE object, and the non-periodic prime endpoint case returns to the H3-DSB/KLS branch, whose current self-contained hard core is NC-BLK unless an external DI/BFI dispersion input is accepted.*

*Proof.* The finite certificate is recorded in

`docs/monograph/prime-matrix-early-zero-gap-crt-asymmetry-router.md.`

If  $I_{P,k}$  contains no prime, then every prime between  $a$  and  $b$  would have to lie inside  $I_{P,k}$ , contradicting the hypothesis. Hence  $a, b$  are adjacent. Since  $a \leq (k-1)P$  and  $b \geq kP+1$ ,

$$b - a \geq (kP+1) - (k-1)P = P+1 > P.$$

This proves the carrier-gap lemma.

The second assertion is a boundary statement. Small-factor coverage of a row is periodic modulo  $M_P$ ; primality of the carrier endpoints is not periodic modulo  $M_P$ . Therefore the carrier cannot be used as a naked CRT-period contradiction. If its endpoint phase recurs as a fixed formal unit, it is a PDEC/ColumnCRT candidate. If it is isolated, it is SAE. If the endpoint primality cannot be made periodic while the covering pressure persists, the already established H3 singleton-tail reduction sends the obstruction to the H3-DSB/KLS chain. The present document has not closed NC-BLK internally and has not replaced it by an accepted external DI/BFI theorem, so the global row/column theorem remains open.  $\square$

**Theorem 7.5** (Period-lift carrier drift router). *Let*

$$L_P = \prod_{\substack{q < P \\ q \text{ prime}}} q.$$

*If  $x$  is a  $P$ -zero-row multiplier, then  $x + tL_P$  is again a  $P$ -zero-row multiplier for every  $t \geq 0$ . The corresponding intervals are shifted by  $PL_P$ . However, the adjacent prime carriers around these intervals are not forced to translate by  $PL_P$ . Thus the period-lifted counterexample cover and the true adjacent-prime endpoint chain have an intrinsic asymmetry: persistent endpoint drift is a PDEC/ColumnCRT object, sparse drift is SAE, and non-periodic drift with persistent covering pressure returns to the H3-DSB/KLS NC-BLK or external DI/BFI branch.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-early-zero-period-lift-carrier-drift-router.md.`

For each non-trivial column  $1 \leq c < P$ , the zero-row hypothesis gives a prime  $q < P$  with  $q \mid Px + c$ . Since  $q \mid L_P$ ,

$$P(x + tL_P) + c \equiv Px + c \pmod{q},$$

so the same small prime  $q$  covers the lifted entry. This proves exact period lift of the covering chain.

Let  $a_t, b_t$  be the adjacent primes immediately outside the lifted interval  $[P(x + tL_P) + 1, P(x + tL_P) + P]$ . The congruence calculation above contains no condition forcing  $a_t = a_0 + tPL_P$  or  $b_t = b_0 + tPL_P$ . It only proves absence of primes inside the lifted interval. Hence endpoint translation is not a CRT consequence. Any infinite structured use of the endpoint drift must therefore be named: fixed recurring drift gives a PDEC/ColumnCRT formal unit; isolated drift gives SAE; and non-periodic endpoint drift leaves the tail filler/short-window dispersion problem already registered as H3-DSB/KLS with NC-BLK or external DI/BFI as the remaining input. This proves the router but not the final row/column theorem.  $\square$

**Theorem 7.6** (Q2-stage endpoint inversion router). *Let an early zero row be straddled by adjacent primes  $Q_1 < Q_2$ . If  $Q_2 < P^2$ , then every composite in the open carrier gap  $(Q_1, Q_2)$  already has*



a prime factor below  $P$ ; otherwise the branch enters the square-anchor endpoint problem. At the  $Q_2$ -stage, the full wheel

$$M_{\leq Q_2} = \prod_{\substack{\ell \leq Q_2 \\ \ell \text{ prime}}} \ell$$

does not reproduce the two prime endpoints. For every  $t \geq 1$ ,

$$Q_1 + tM_{\leq Q_2} \equiv 0 \pmod{Q_1}, \quad Q_2 + tM_{\leq Q_2} \equiv 0 \pmod{Q_2},$$

so the endpoint-stable full-wheel replay is impossible. The surviving branches are: a persistent closed carrier block, which is a ColumnCRT/PDEC object; a sparse carrier block, which is SAE; or moving endpoints/support, which returns to the moving-family H3-DSB/KLS branch.

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-q2-carrier-stage-crt-asymmetry-router.md`.

The below-square assertion follows because a composite  $n < Q_2 < P^2$  has a least prime factor at most  $\sqrt{n} < P$ , while adjacency of  $Q_1, Q_2$  excludes primes in the open gap. The endpoint inversion is the displayed congruence: both endpoints divide the full  $Q_2$ -wheel modulus, so any positive full-wheel copy of either endpoint is composite. If one uses only  $M_{< Q_2}$ , the left endpoint  $Q_1$  is still killed by its own zero class, and the right endpoint  $Q_2$  is not forced to remain prime by CRT data. Thus the full  $Q_2$ -stage gives a genuine local contradiction only to endpoint-stable replay. Without endpoint-stable replay, the obstruction is precisely one of the named ColumnCRT/PDEC, SAE, or moving-family exits. Hence the router is closed, but the global row/column theorem is not.  $\square$

**Theorem 7.7** ( $Q_2$  endpoint replacement and aperture growth). *In the  $Q_2$ -stage of the preceding theorem, full-wheel replay creates a replacement debt for the true adjacent-prime endpoints. More precisely, in the below-square branch the full  $Q_2$ -wheel copy of  $[Q_1, Q_2]$  is a closed composite block. Hence if  $A_t < B_t$  are the adjacent primes straddling that copied block, then*

$$B_t - A_t \geq Q_2 - Q_1 + 2.$$

*Consequently a fixed bounded carrier aperture cannot support infinitely many endpoint-stable replays. A fixed aperture of width  $W$ , starting from closed carrier width  $W_0 = Q_2 - Q_1 + 1$ , permits at most*

$$\left\lfloor \frac{W - W_0}{2} \right\rfloor$$

*such replacements. Any infinite continuation must move or enlarge the aperture and is therefore routed to moving-aperture ColumnCRT/PDEC, SAE, or the H3-DSB/KLS moving-family branch.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-q2-endpoint-replacement-aperture-growth-router.md`.

For an early zero row  $[(k-1)P+1, kP]$ , the left endpoint satisfies  $Q_1 \leq (k-1)P = kP - P$ , with equality possible only in the  $k=2$  boundary case  $Q_1 = P$ ; the right endpoint satisfies  $Q_2 > kP$ . In the  $Q_2 < P^2$  branch, every composite in the open gap has least prime factor below  $P$ , and the full  $Q_2$ -wheel kills  $Q_1$  and  $Q_2$  themselves. Thus every point of the copied closed interval

$$[Q_1 + tM_{\leq Q_2}, Q_2 + tM_{\leq Q_2}]$$

is composite for  $t \geq 1$ . The nearest true primes around this copied block must lie outside it, giving the displayed +2 lower bound. Iterating this lower bound inside a fixed aperture gives the finite bound  $\lfloor (W - W_0)/2 \rfloor$ . Therefore the bounded-aperture branch is closed; the remaining branches are exactly the named moving, PDEC, SAE, and H3 exits.  $\square$

**Theorem 7.8** (Fresh endpoint layer cascade). *After the bounded-aperture branch is removed, any continued full-wheel endpoint replay must introduce an unbounded sequence of fresh endpoint prime layers. Let  $M_j$  be the full-wheel modulus at stage  $j$ , and let  $B_{j+1}$  be the new right adjacent-prime endpoint forced after the copied closed composite block. Then*

$$B_{j+1} > M_j.$$

*If  $L_j = \log M_j$ , the next full wheel satisfies*

$$L_{j+1} \geq L_j + \log B_{j+1} > 2L_j.$$

*Thus the endpoint cascade cannot terminate in a fixed finite CRT period. A persistent fresh endpoint phase is a ColumnCRT/PDEC object; a sparse cascade is SAE; and a non-PDEC unbounded fresh-layer cascade is routed to the one-residue per fresh prime tail-sieve/H3-DSB/KLS branch.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-q2-endpoint-fresh-layer-cascade-router.md.`

At stage  $j$ , full-wheel replay moves the current closed carrier by at least  $M_j$  and turns it into a closed composite block. The true right adjacent prime around that copied block must therefore lie to the right of the copied block, hence beyond  $M_j$ . This prime is not contained in any fixed finite endpoint layer below the old bound. When the next wheel includes it, its next copy is divisible by itself, so the same replacement mechanism repeats. The displayed logarithmic inequality follows immediately. Since a finite CRT period contains only finitely many prime layers, it cannot support this infinite cascade. The only remaining possibilities are the named PDEC, SAE, or tail-sieve/H3 exits.  $\square$

**Theorem 7.9** (Fresh-layer tail-mass dichotomy). *In the non-PDEC branch of the fresh endpoint layer cascade, assume the local aperture at stage  $j$  is  $W_j$  and the fresh endpoint prime is  $B_j$ . If the fresh layer does not create a persistent ColumnCRT/PDEC correlation, then its local one-residue contribution is bounded by*

$$\frac{W_j}{B_j}.$$

*Under the controlled-aperture condition  $\log W_j = o(\log B_j)$ , and in particular under the endpoint-replacement model  $W_j = W_0 + 2j$ , the total fresh-tail mass is summable:*

$$\sum_j \frac{W_j}{B_j} < \infty.$$

*Thus the controlled non-PDEC fresh-tail branch is SAE. If instead the aperture repeatedly grows fast enough to track the fresh modulus, the branch is no longer local endpoint replacement; it is aperture explosion or global support motion and must be routed to H3-DSB or moving-support PDEC.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-q2-fresh-layer-tail-mass-dichotomy-router.md.`

The previous cascade theorem gives  $L_{j+1} > 2L_j$ , where  $L_j = \log M_j$  is the logarithm of the full-wheel modulus, and also gives  $B_{j+1} > M_j$ . Hence the denominators  $B_{j+1}$  have lower bounds  $\exp(L_j)$

whose logarithms grow at least double-exponentially in the stage index. If  $\log W_j = o(\log B_j)$ , then for all sufficiently large  $j$  one has  $W_j \leq B_j^{1/2}$ , so

$$\frac{W_j}{B_j} \leq B_j^{-1/2},$$

and the right-hand series converges by the same double-exponential lower growth. The linear endpoint-replacement aperture  $W_j = W_0 + 2j$  is a special case.

The only ways to avoid this SAE absorption are exactly the two named failures of the hypotheses. A persistent concentration of fresh-layer hits is, by definition, a ColumnCRT/PDEC formal unit. A failure of controlled aperture is an aperture explosion or support-motion event, not the local endpoint replay branch. Therefore the router closes the controlled fresh-tail outlet but not the global row/column theorem.  $\square$

**Theorem 7.10** (Q2 aperture-explosion schema firewall). *After the controlled fresh-tail branch has been absorbed into SAE, any remaining aperture-explosion branch must present an explicit persistent structure: support motion, a blocker-package change, a fresh-layer ColumnCRT/PDEC unit, or a moving-family schema. In the current materialized corpus, unnamed aperture explosion is not an admissible terminal. Registered support motion has already been routed away, current materialized fresh-layer PDEC/ColumnCRT has already been blocked by the admission firewall, and the branch-replay current frontier is zero. Hence the only surviving form is a future explicit schema, which must reopen under the stated firewall; the global row/column theorem is still not proved.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-q2-aperture-explosion-schema-firewall-router.md.`

The previous theorem leaves only two ways out of SAE: persistent fresh-layer correlation, which is PDEC/ColumnCRT, or failure of controlled aperture, which is support motion or aperture explosion. The current registered support-motion certificate eliminates local support drift for the registered blocks and routes persistent registered replay to named ColumnCRT/PDEC objects. The fresh-layer admission firewall eliminates unnamed materialized PDEC/ColumnCRT in the current corpus. The current-frontier-zero certificate then states that no active branch-replay terminal instance remains in the materialized ledger.

Therefore an aperture-explosion claim cannot be kept as an unnamed terminal. It must specify its source family, blocker package, shape, phase map and persistence test, with a reproducible ledger. Such a future schema may reopen the branch, but it is not a present closed counterexample chain. This proves the schema-firewall synchronization only, not the final theorem.  $\square$

**Theorem 7.11** (Q2 CRT ladder to final exact-source alignment). *The Q2-stage CRT ladder has no remaining current unnamed local terminal: the controlled fresh tail is SAE, and aperture explosion must enter an explicit schema firewall. However, this does not by itself prove the row/column theorem. The CRT ladder controls positions and phases of allowed and forbidden classes; it does not produce a mass-dispersion estimate for the actual pre-Cauchy source over exact  $(u, v)$ -fibers. Therefore the Q2 branch can promote only through one of the final source inputs:*

*NonterminalExactUVFiberAperiodicityEstimateForActualPreCauchySource  
or ExternalDIBFIKuznetsovDispersionTheoremMatch,*

*together with independent DStructure/Rankin promotion acceptance.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-q2-to-final-exact-source-alignment-router.md.`

The preceding Q2 routers prove that a full  $Q_2$ -wheel repeats the composite cover but destroys endpoint-prime replay, that continued endpoint replacement forces fresh endpoint layers, that controlled fresh tails are summable, and that current unnamed aperture explosion is barred by the schema firewall. Thus there is no current Q2-local terminal left to carry an infinite counterexample chain.

The remaining obstruction is of a different type. The final source anti-atom concerns the distribution of the actual signed pre-Cauchy source among exact  $(u, v)$ -fibers before Cauchy, pair energy and terminal extraction. CRT wheel rigidity gives congruence positions, not a bound on that source mass. Hence the strict self-contained branch must prove the nonterminal exact-UV fiber aperiodicity estimate, equivalently the already named source-domain rank/no-collapse package. The alternative is an accepted external DI/BFI/Kuznetsov dispersion theorem. In either case the separate DStructure/Rankin promotion gate remains open. This proves alignment of the remaining Q2 branch, not unconditional closure.  $\square$

**Theorem 7.12** (Global CRT homogeneity frontier). *The  $Q_1/Q_2$  adjacent-prime carrier gives a genuine endpoint obstruction: at the full  $Q_2$ -wheel the two endpoint primes cannot replay as primes. Nevertheless this obstruction is not, by itself, a global CRT phase contradiction. For any squarefree wheel modulus  $M_Y$  and any fresh prime  $r \nmid M_Y$ , the lifts  $a + tM_Y$ ,  $0 \leq t < r$ , of a fixed old residue  $a$  run through all residue classes modulo  $r$ . Hence the fresh layer deletes exactly one class in each old fiber. Finite CRT growth is therefore a homogeneous one-class deletion mechanism, not a deterministic short-interval occupancy theorem, unless the additional phase equations form a registered PDEC/ColumnCRT collision.*

*Consequently the pure global-CRT phase-contradiction outlet is closed as a final proof route. The remaining strict self-contained branch must pass through actual-source exact- $(u, v)$  fiber dispersion and then the pointwise primitive alpha/delta kernel table. Its first active atom is*

*AlphaRowAnchorPhaseEmissionFormulaLedger,*

*together with the independent pre-Cauchy arithmetic identity, the same-unit rank/multiplicity certificate, and independent DStructure/Rankin promotion acceptance.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-global-crt-homogeneity-frontier-router.md.`

The previous Q2 routers already prove endpoint inversion, fixed-period removal, controlled fresh-tail SAE absorption, and the aperture-explosion schema firewall. These remove the local endpoint-replay and unnamed moving outlets. The remaining question is whether the infinite CRT wheel itself forces a global phase contradiction.

The lift calculation above is exact because  $M_Y$  is invertible modulo the fresh prime  $r$ . As  $t$  varies modulo  $r$ , the class  $a + tM_Y$  also varies through all classes modulo  $r$ ; precisely one value is 0. Thus each finite wheel extension applies the same relative deletion to every old fiber. The Euler product explains the critical density scale, but it does not supply a lower bound for the actual mass in a moving interval of length  $P$  near  $P^2$ .

Therefore CRT position and phase rigidity must be supplemented by actual source information. The already archived strict descent identifies that source information with exact- $(u, v)$  fiber aperiodicity and then with the pointwise primitive alpha/delta kernel table. Since the alpha-row emission

formula, the signed pre-Cauchy identity, the rank/multiplicity certificate, and DStructure/Rankin promotion acceptance remain open, this theorem is a frontier synchronization, not a proof of the row/column theorem.  $\square$

**Theorem 7.13** (Global CRT route to terminal saturation). *The global CRT/Q1-Q2 route does not stop at the coarse alpha-row atom. After the pure CRT homogeneity firewall is imported, the local alpha-row anchor/phase formula has already been synchronized to the terminal PDEC/CleanKLS gate. The PDEC/CleanKLS gate has then been attacked to the current internal saturation frontier: the CleanKLS/Kuznetsov-DLS arm returns to the strict acyclic terminal family, the nonrecursive constructor/signed-lift breaker returns to the signed coordinate-source cycle, and the seed-cycle-cut branch is saturated. Therefore the current non-cyclic strict inputs are*

*AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate  
or NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact,*

*together with the model-gap ledger, rate preservation for the moving-atom packet, and independent DStructure/Rankin promotion acceptance.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-global-crt-terminal-saturation-sync-router.md.`

The previous theorem leaves the strict route at the pointwise alpha/delta kernel table. The archived alpha-row terminal synchronization shows that the unsigned carry-shell, source-tuple and phase-wheel parts of the alpha formula are local geometric data, while the signed-lift and anchor-overload parts return to the global PDEC/CleanKLS terminal gate and the model-gap ledger.

The archived terminal split then separates the terminal gate into a direct same-set PDEC scope arm and a clean Kloosterman large-sieve arm. Continuing the clean arm through the Kuznetsov/DLS files returns to the strict acyclic terminal family rather than producing a self-contained log-saving theorem. Continuing the nonrecursive breaker returns to the signed source-coordinate cycle; continuing seed-cycle-cut returns to the same row-level fixed point unless a new joint emitter formula is supplied.

Thus the CRT route, the alpha-row route and the current terminal-family route all meet at the same saturated frontier. A proof must either supply the same-set PDEC scope match or a new explicit joint alpha/delta constructor formula, while also paying the model, rate and DStructure/Rankin gates. Since none of these has been proved here, this is a sharpening of the frontier, not unconditional closure.  $\square$

**Theorem 7.14** (Global CRT route to branch-trace frontier). *The non-PDEC side of the current global CRT frontier can be sharpened further. The alternative “new explicit joint alpha/delta formula” is not a single undifferentiated atom. If it is split through the existing alpha-side, same-row, row-level and signed-source routes, it returns to the already registered source-coordinate fixed point. Therefore a genuinely new formula must be anti-split: it must produce atomic joint rows before alpha-side projection. The ordinary anti-split declaration still degrades to the old split cycle unless it contains built-in word/coefficient pairing, and that built-in pairing further requires an exact atomic joint branch trace. Thus the current strict frontier is*

*AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate  
or ExactAtomicJointBranchTraceSignedCoefficientFormulaOrReturn,*

*with the ExactUV, model-gap, rate-preservation and DStructure/Rankin gates still present.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-global-crt-branch-trace-frontier-router.md.`

The same-set PDEC scope arm remains an independent possible new certificate, but it is not proved in the present strict corpus. On the other arm, the archived joint-formula obligation says that the old joint declaration, alpha-side primitive rule and same-row origin identity return to the signed-source fixed point. The anti-split router therefore requires a non-split joint primitive word/coefficient formula. The declaration firewall then sharpens this to atomic pre-Cauchy joint rows with built-in pairing, and the atomic pairing router sharpens it to a built-in signed coefficient pairing closed form.

The signed coefficient is odd branch data: it depends on orientation and local factors, not only on the unsigned CRT skeleton. Hence the closed-form frontier is an exact atomic branch trace that, before Cauchy, Phi and payment pushforward, gives the basis word, signed coefficient, alpha/delta payload, orientation/local factor, exact  $(u, v)$ , and named returns in the same formal unit. The current corpus has not supplied that trace formula. This proves the frontier sharpening only, not the row/column theorem.  $\square$

**Theorem 7.15** (Global CRT route to signed-payload frontier). *After importing the strict atomic-payload audit, the branch-trace side of the current global CRT frontier sharpens from*

*ExactAtomicJointBranchTraceSignedCoefficientFormulaOrReturn*

*to*

*AtomicSignedPayloadTraceConstructorBeforeAssignmentOrReturn.*

*Moreover, in the present self-contained internal corpus the PDEC same-set arm is saturated: unless a new acyclic same-set scope certificate, or an external no-projection DI/BFI input, is supplied, that arm returns to the new-joint formula line, and the new-joint line has already been routed through branch trace to signed payload. Thus the internal self-contained frontier is*

*AtomicSignedPayloadTraceConstructorBeforeAssignmentOrReturn  
and ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem  
and ExplicitModelGapAndFiniteDPRCLedger  
and RatePreservationLedger\_FOR\_moving\_atom\_packet  
and DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.*

*If a new PDEC scope input is retained, the global activity basis is instead*

*AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate  
or AtomicSignedPayloadTraceConstructorBeforeAssignmentOrReturn,*

*with the same ExactUV, model, rate and DStructure/Rankin gates.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-global-crt-signed-payload-sync-router.md.`

The preceding theorem left the non-PDEC side at an exact atomic branch trace. The archived signed-payload audit separates such a trace into its visible coordinate part and its signed payload part. The coordinate part follows the CRT anchor, phase and word-coordinate data. The payload part consists of the orientation bit, local-factor product, signed coefficient, same-row word/coefficient

identity, alpha/delta prepushforward identity and exact  $(u, v)$  key. These are not consequences of the unsigned CRT skeleton.

The PDEC arm is treated separately. The archived scope-saturation audit says that current internal PDEC arguments either require a new same-set scope certificate, require an external no-projection DI/BFI input, or return to the new-joint formula line. The branch-trace synchronization has already routed that new-joint line to exact atomic branch trace, and the payload audit routes exact atomic branch trace to signed payload. Therefore the self-contained internal route has the stated payload frontier, while the new PDEC scope route is retained as an independent unproved input. No signed payload constructor, PDEC scope certificate, ExactUV incidence theorem, model-gap ledger, rate preservation ledger, or DStructure/Rankin promotion is proved here, so this is a frontier sharpening rather than unconditional closure.  $\square$

**Theorem 7.16** (Predecessor-gap P-CRT uniformity route). *Let  $p^- < P$  be consecutive primes. A large gap  $P - p^-$  creates a local first-row actual-prime deficit in the columns  $p^- + 1, \dots, P - 1$ , but it does not create an imbalance in the complete  $P$ -level CRT wheel. In the full period  $M_{\leq P} = PM_{<P}$ , before imposing the  $P$ -zero class, each residue class modulo  $P$  contains exactly  $\varphi(M_{<P})$  residues coprime to  $M_{<P}$ . After imposing the  $P$ -zero class, the  $P$ -column is deleted and every nonzero column  $c \in \mathbb{F}_P^*$  still contains exactly  $\varphi(M_{<P})$  surviving CRT classes. The reflection  $n \mapsto -n \pmod{M_{\leq P}}$  also pairs  $c$  with  $P - c$  exactly. Therefore a predecessor-gap argument can close the row/column theorem only if it supplies a localization transfer from this complete-wheel identity to the initial  $P \times P$  square, or else a new PDEC scope or signed-payload input.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-predecessor-gap-pcrt-uniformity-router.md.`

The complete-wheel count is a direct CRT bijection. For every unit  $a \pmod{M_{<P}}$  and every  $c \pmod{P}$ , there is a unique residue  $n \pmod{PM_{<P}}$  satisfying  $n \equiv a \pmod{M_{<P}}$  and  $n \equiv c \pmod{P}$ . Hence each  $P$ -column has  $\varphi(M_{<P})$  such classes before the  $P$ -sieve. The  $P$ -sieve removes exactly the class  $c = 0$ , leaving all nonzero columns equal. Negation gives the stated reflection symmetry.

The gap  $P - p^-$  is different in kind. It says that several first-row columns contain no prime atom, not that the full wheel has unequal nonzero columns. Passing from the full-period identity to the initial  $P \times P$  arc is precisely a short localization problem. The archived scan for  $P \leq 5000$  records the largest predecessor gaps and observes that the first-row missing columns are repaired later in those finite squares, but this empirical repair is not used as proof. Without a localization transfer, the route returns to the current frontier: same-set PDEC scope, signed payload, ExactUV, model, rate and DStructure/Rankin gates. Thus this theorem closes a false shortcut, not the row/column theorem.  $\square$

**Theorem 7.17** (Localized P-CRT transfer and the Linnik-2 barrier). *The column side of*

*LocalizedPCRTCColumnUniformityTransferToInitialPxPSquare*

*is equivalent to the following pointwise least-prime-in-progressions assertion for prime moduli:*

$$\forall c \in \mathbb{F}_P^* \quad \exists \ell \leq P^2 \quad \text{prime with} \quad \ell \equiv c \pmod{P}.$$

*Thus any proof that localizes complete  $P$ -wheel nonzero-column uniformity to the initial  $P \times P$  square must either prove this Linnik-2 type pointwise AP assertion, or else introduce a genuinely new structural input such as a same-set PDEC scope certificate or a signed-payload constructor.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-localized-pcrt-transfer-linnik2-barrier-router.md.`

The  $c$ -th non- $P$  column of the initial square is exactly the finite progression

$$c, c + P, c + 2P, \dots, c + (P - 1)P,$$

intersected with the prime numbers. Since  $1 \leq c < P$ , every term is congruent to  $c \pmod{P}$  and is at most  $P^2$ . Therefore the column has a prime if and only if the displayed pointwise AP assertion holds for that residue  $c$ . Requiring this for every nonzero column gives the asserted equivalence.

The complete CRT wheel does not contain this pointwise first-prime information. It says that the nonzero columns have equal total mass over a full period  $M_{\leq P}$ , while the column theorem asks for a hit in the first  $P$  terms of each residue class. Mean AP estimates or complete-period symmetry can at best control averaged exceptions; they do not remove a single bad residue class. The archived finite scan up to  $P \leq 3000$  records no missing class and a largest first-prime row 108, but this is only diagnostic. In the absence of a new pointwise AP theorem, the route returns to the active frontier consisting of PDEC scope, signed payload, ExactUV, model, rate and DStructure/Rankin gates. Hence this is a barrier identification, not an unconditional proof.  $\square$

**Theorem 7.18** (Terminal-row CRT atoms and the square-root gate). *For  $P = 5$ , the terminal-row and next-row atoms 23 and 29 are units modulo  $M_{\leq 5} = 2 \cdot 3 \cdot 5$ . For  $P = 7$ , the atoms 43, 47 and 53 are units modulo  $M_{\leq 7} = 2 \cdot 3 \cdot 5 \cdot 7$ . Hence none of these atoms can be absorbed by a prime factor  $\leq P$ . More generally, if  $p_+$  is the next prime after  $P$  and*

$$1 < n < p_+^2, \quad (n, M_{\leq P}) = 1,$$

*then  $n$  is prime.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-terminal-row-crt-atom-router.md.`

For  $P = 5$ ,

$$23 \equiv (1, 2, 3) \pmod{(2, 3, 5)}, \quad 29 \equiv (1, 2, 4) \pmod{(2, 3, 5)}.$$

For  $P = 7$ ,

$$43 \equiv (1, 1, 3, 1) \pmod{(2, 3, 5, 7)},$$

$$47 \equiv (1, 2, 2, 5) \pmod{(2, 3, 5, 7)},$$

$$53 \equiv (1, 2, 3, 4) \pmod{(2, 3, 5, 7)}.$$

Every displayed coordinate is nonzero, so the hypothesis that any of these numbers is divisible by a prime  $\leq P$  contradicts its CRT vector.

For the general square-root gate, suppose  $n$  is composite and  $(n, M_{\leq P}) = 1$ . Then the least prime factor of  $n$  is strictly larger than  $P$ , hence at least  $p_+$ . A composite integer all of whose prime factors are at least  $p_+$  is at least  $p_+^2$ , contradicting  $n < p_+^2$ . Thus  $n$  is prime.

This proves an atom-level principle: a reduced CRT atom in the terminal row or the immediate post-square row is automatically a true prime. It does not prove that every prime  $P$  has such an atom in a prescribed short row. Complete CRT-period symmetry supplies full-period uniformity and reflection, but not the local pointwise existence needed in the initial square. The remaining global input is therefore a terminal-row reduced-residue existence theorem, the localized  $P$ -CRT/Linnik-2 route, or the already registered AP zero-packet, Page-sparsity, signed-payload, or PDEC frontiers.  $\square$



**Theorem 7.19** (Terminal rows as square-phase Jacobsthal special phases). *For  $1 \leq r < P$ , the next-row and last-row reduced atoms are exactly*

$$(P^2 + r, M_{<P}) = 1, \quad (P^2 - r, M_{<P}) = 1.$$

*Equivalently, the next-row full-cover failure is the covering of  $r = 1, \dots, P - 1$  by the forbidden classes*

$$r \equiv -P^2 \pmod{q}, \quad q < P,$$

*and the last-row full-cover failure is the covering by*

$$r \equiv P^2 \pmod{q}, \quad q < P.$$

*Thus either full cover forces the square phase  $P^2$  to start a covered block of length  $P - 1$  in the primordial period  $M_{<P}$ .*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-terminal-row-square-phase-bridge-router.md.`

For every prime  $q < P$ , since  $P$  is a unit modulo  $q$ ,

$$q \mid P^2 + r \iff r \equiv -P^2 \pmod{q},$$

and

$$q \mid P^2 - r \iff r \equiv P^2 \pmod{q}.$$

Therefore the absence of a reduced atom on one side is exactly the statement that these one-residue-per- $q$  forbidden classes cover the whole interval  $[1, P - 1]$ . In the primordial period this is precisely a consecutive covered block of length  $P - 1$  beginning at the corresponding square phase.

The full-period Jacobsthal shortcut is not available in the present corpus: known and archived finite data already show primordial periods with covered blocks of length at least  $P - 1$ . The useful remaining target is not a uniform period-wide maximum-run bound, but the special-phase assertion that the particular phase  $P^2$  cannot align with such a long block, or that such alignment is a PDEC/SAE/ColumnCRT defect. Hence this is a route sharpening, not an unconditional row/column proof.  $\square$

**Theorem 7.20** (Linnik-2 defects and nonprincipal character obstruction). *For a prime modulus  $P$ , put*

$$\vartheta_a(P) = \sum_{\substack{\ell \leq P^2 \\ \ell \text{ prime} \\ \ell \equiv a \pmod{P}}} \log \ell, \quad a \in \mathbb{F}_P^*,$$

*and for each Dirichlet character  $\chi \bmod P$  put*

$$T_\chi(P) = \sum_{a \in \mathbb{F}_P^*} \chi(a) \vartheta_a(P).$$

*Let  $T_0(P)$  denote the principal-character mass and let*

$$N_a(P) = \sum_{\chi \neq \chi_0} \overline{\chi(a)} T_\chi(P)$$

be the nonprincipal projection at the class  $a$ . Then

$$\vartheta_a(P) = \frac{T_0(P) + N_a(P)}{P - 1}.$$

Consequently a Linnik-2 zero-column defect, namely  $\vartheta_a(P) = 0$  for some  $a \in \mathbb{F}_P^*$ , forces the exact negative phase alignment

$$N_a(P) = -T_0(P)$$

and hence

$$\sum_{\chi \neq \chi_0} |T_\chi(P)|^2 \geq \frac{T_0(P)^2}{P - 2}.$$

Thus the remaining AP route is equivalent to excluding this pointwise nonprincipal negative projection at  $x = P^2$ , or else returning to the same-set PDEC scope or signed-payload structural frontier.

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-linnik2-nonprincipal-character-obstruction-router.md`.

All identities are finite orthogonality identities on the group  $\mathbb{F}_P^*$ . Since

$$T_\chi(P) = \sum_{b \in \mathbb{F}_P^*} \chi(b) \vartheta_b(P),$$

character inversion gives the displayed formula for  $\vartheta_a(P)$ . If a nonzero column has no prime below  $P^2$ , then  $\vartheta_a(P) = 0$ , so the nonprincipal projection must cancel the entire principal mass. Applying Cauchy's inequality to

$$T_0(P) = \left| \sum_{\chi \neq \chi_0} \overline{\chi(a)} T_\chi(P) \right|$$

and using  $\sum_{\chi \neq \chi_0} |\chi(a)|^2 = P - 2$  gives the stated energy lower bound.

The archived finite scan up to  $P \leq 3000$  finds no theta-zero residue class, with largest normalized negative theta deficit about 0.686688 at  $P = 73$ , but this empirical fact is not used in the proof. The theorem only sharpens the hard point: complete CRT symmetry and averaged AP estimates do not by themselves exclude the required negative nonprincipal projection. No pointwise projection bound, PDEC scope certificate, signed-payload constructor, ExactUV incidence theorem, model-gap ledger, rate-preservation ledger, or DStructure/Rankin promotion is proved here.  $\square$

**Theorem 7.21** (Energy capacity and rank-one phase for the Linnik-2 defect). *In the notation of the preceding theorem, set*

$$\rho_a(P) = -\frac{N_a(P)}{T_0(P)}.$$

Then

$$\rho_a(P) = 1 - \frac{\vartheta_a(P)}{T_0(P)/(P - 1)}.$$

Thus a zero-column defect is exactly the rank-one condition  $\rho_a(P) = 1$ . The energy condition

$$\sum_{\chi \neq \chi_0} |T_\chi(P)|^2 \geq \frac{T_0(P)^2}{P - 2}$$

is only a necessary condition for such a defect. Therefore an  $L^2$  capacity argument can close this route only if it proves the strict reverse inequality. Otherwise the remaining obligation is a pointwise phase statement: exclude a negative projection of size  $T_0(P)$  in one evaluation direction.

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-linnik2-rankone-phase-capacity-router.md.`

The first identity follows directly from  $\vartheta_a(P) = (T_0(P) + N_a(P))/(P - 1)$ . Hence  $\vartheta_a(P) = 0$  if and only if  $\rho_a(P) = 1$ . The displayed energy lower bound is exactly the Cauchy consequence proved in the preceding theorem. Its converse is false as a matter of logic: total nonprincipal energy above the floor does not force the energy to point in the single evaluation direction attached to  $a$ .

The evaluation vectors  $v_a = (\chi(a))_{\chi \neq \chi_0}$  satisfy

$$\langle v_a, v_b \rangle = \begin{cases} P - 2, & a = b, \\ -1, & a \neq b, \end{cases}$$

so the residue tests are simplex directions in the nonprincipal character space. The archived scan up to  $P \leq 3000$  records 423 prime moduli for which the total-energy floor is already exceeded while no zero residue class exists. This diagnostic is not a proof input; it identifies why the capacity route is too coarse. The remaining named input is the rank-one negative evaluation projection exclusion at  $x = P^2$ , or else a return to same-set PDEC scope or signed payload.  $\square$

**Theorem 7.22** (Three-claim frontier and the sharp AP zero-packet barrier). *The three main claims currently have distinct frontiers. The row/column chain has been sharpened to the rank-one condition*

$$\rho_a(P) < 1 \quad (a \in \mathbb{F}_P^*),$$

which is equivalent to

$$\vartheta_a(P) > 0 \quad (a \in \mathbb{F}_P^*).$$

*The two-point secondary-sieve chain remains conditional on the I3-Core true residual total-incidence input, or on the external DI/BFI-Kloosterman large-sieve route. The RH contradiction-field chain remains a verification package awaiting independent acceptance of its controlled exits.*

*For the row/column AP branch, the rank-one condition is therefore precisely the sharp positivity statement*

$$\theta(P^2; P, a) > 0 \quad \text{for every } a \in \mathbb{F}_P^*.$$

*In explicit-formula language this requires the signed nonprincipal zero packet in each residue class to be strictly smaller than the main term, which is of order  $P$ . Complete CRT averages, Bombieri-Vinogradov type mean estimates, standard Linnik bounds with exponent  $> 2$ , and GRH-shape errors  $O(P \log^2 P)$  do not by themselves imply this  $x = P^2$  pointwise positivity.*

*Proof.* The certificate is recorded in

`docs/monograph/three-claims-frontier-rankone-explicit-formula-router.md.`

The equivalence  $\rho_a(P) < 1 \iff \vartheta_a(P) > 0$  follows from the identity

$$\rho_a(P) = 1 - \frac{\vartheta_a(P)}{T_0(P)/(P - 1)}$$

proved in the preceding theorem. Since  $\vartheta_a(P)$  is exactly the Chebyshev prime weight in the class  $a \bmod P$  up to  $P^2$ , positivity is equivalent to a prime in that nonzero class below the square barrier.

The explicit formula for  $\theta(P^2; P, a)$  has a principal term of size about  $P$ , plus signed nonprincipal zero packets and controlled trivial terms. Therefore the sharp remaining analytic input is a pointwise bound on those signed packets for every  $a$ . Mean estimates and full-period CRT identities only

control averages, not a single exceptional simplex direction. GRH-shape errors at  $x = P^2$  are of size  $P \log^2 P$ , too large to beat a main term of size  $P$  without additional sharp constants or cancellation. Thus this theorem is a frontier separation and barrier identification, not a promotion of any of the three claims to final unconditional theorem status.  $\square$

**Theorem 7.23** (Explicit AP zero-packet Siegel split). *The remaining row/column AP input*

*ExplicitAPZeroPacketBoundBeatingMainTermAtXEqualsP2*

*splits into two genuine analytic branches. First, any exceptional real zero  $\beta$  of a real nonprincipal character modulo  $P$  contributes at  $x = P^2$  on the scale*

$$P^{2\beta-1}/\beta.$$

*If  $\beta = 1 - \lambda/\log P$ , this is  $P \exp(-2\lambda)/\beta$ , hence still of the same order as the principal term for bounded  $\lambda$ . This is the square-scale Siegel-bias branch. Second, after extracting any real exceptional packet, the nonreal zeros leave a signed residue packet; closing it requires pointwise phase cancellation in every evaluation direction a mod  $P$ , not just average energy or complete CRT uniformity.*

*Consequently the current AP branch is not a single unnamed zero-packet error. It is the conjunction*

*SiegelExceptionalBiasExclusionAtSquareScale  
AND NonrealZeroPacketResiduePhaseCancellationAtXEqualsP2,*

*with the trivial and finite explicit-formula terms kept as a downstream constant ledger once a positive main-term margin is available.*

*Proof.* The certificate is recorded in

docs/monograph/prime-matrix-explicit-ap-zero-packet-siegel-split-router.md.

The character explicit formula for  $\theta(P^2; P, a)$  has a principal term of size  $P$  and a signed sum of nonprincipal zero terms. A real zero close to 1 cannot be dismissed by a mean theorem, because its square-scale contribution remains comparable to the main term on one of the two real character half-classes. Deuring–Heilbronn repulsion organizes the remaining zeros but does not itself prove that this biased half-class is positive before  $P^2$ .

After this real branch is separated, the nonreal packet is still pointwise in the residue  $a$ . Bombieri–Vinogradov, complete CRT identities, total nonprincipal energy bounds, and GRH-shape estimates do not exclude a single rank-one evaluation direction from carrying too much signed mass at the square barrier. Therefore the latest analytic AP route remains open and must either prove the two displayed branches or return to the already named structural inputs: same-set PDEC scope, atomic signed payload, exact- $(u, v)$  multiplicity, model/rate preservation, and the D-structure Rankin promotion gate. No unconditional row/column theorem is claimed here.  $\square$

**Theorem 7.24** (Siegel branch as a quadratic half-class margin). *For prime modulus  $P$ , let  $\chi_P$  be the quadratic character and define*

$$T_\chi(P) = \sum_{\substack{\ell \leq P^2 \\ \ell \text{ prime}, \ell \neq P}} \chi_P(\ell) \log \ell, \quad T_0(P) = \sum_{\substack{\ell \leq P^2 \\ \ell \text{ prime}, \ell \neq P}} \log \ell.$$

*Then the real-character part of the Siegel branch is measured by*

$$r_{\text{quad}}(P) = \frac{|T_\chi(P)|}{T_0(P)}.$$

*The dangerous quadratic half-class retains an average main-term margin proportional to  $1 - r_{\text{quad}}(P)$ . Thus the square-scale Siegel branch is reduced to either proving the margin theorem*

*QuadraticHalfClassSquareScaleBiasMarginTheorem,*

*or accepting an independently certified effective no-Siegel-zero or  $\beta$ -gap input strong enough at  $x = P^2$ .*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-siegel-quadratic-halfclass-margin-router.md.`

The identity is the specialization of the nonprincipal character expansion to the unique real quadratic character modulo  $P$ . The sum  $T_\chi$  is the difference between the Chebyshev weights falling in the quadratic-residue and quadratic-nonresidue half-classes. If the real projection almost equals the principal mass in magnitude, one half-class may have only a small average margin left before the nonreal packet and finite terms are charged. Conversely, any explicit bound  $r_{\text{quad}}(P) \leq 1 - \eta(P)$  supplies exactly that half-class margin.

The archived finite scan  $7 \leq P \leq 1000$  records a maximum diagnostic ratio about 0.13027, attained at  $P = 7$ . This scan is not used as an infinite proof. The theorem only identifies the exact remaining self-contained target for the real branch. The nonreal zero-packet phase-cancellation branch and the structural PDEC/signed-payload/ExactUV/D-structure inputs remain open.  $\square$

**Theorem 7.25** (Nonreal packet as two half-class simplexes). *After removing the principal character and the quadratic character, define*

$$R_a = (P - 1)\theta(P^2; P, a) - T_0(P) - \chi_P(a)T_\chi(P).$$

*Let  $w_a$  be the evaluation vector over the remaining nonreal characters. Then*

$$\langle w_a, w_b \rangle = \begin{cases} P - 3, & a = b, \\ -2, & a \neq b \text{ and } \chi_P(a) = \chi_P(b), \\ 0, & \chi_P(a) \neq \chi_P(b). \end{cases}$$

*Thus the nonreal packet splits into two orthogonal regular simplexes indexed by the quadratic-residue and quadratic-nonresidue half-classes. A zero column in class  $a$  would force*

$$R_a = -T_0(P) - \chi_P(a)T_\chi(P),$$

*so the nonreal packet must supply a rank-one negative projection of size at least*

$$(1 - |T_\chi(P)|/T_0(P))T_0(P).$$

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-nonreal-halfclass-simplex-phase-router.md.`

The character orthogonality relation over  $\mathbb{F}_P^*$  gives  $\sum_\chi \overline{\chi(a)}\chi(b) = P - 1$  if  $a = b$ , and 0 otherwise. Removing the principal character subtracts 1, and removing the quadratic character subtracts  $\chi_P(a)\chi_P(b)$ . This gives the displayed inner products. The identity for  $R_a$  is the inverse character expansion with those two components removed. Setting  $\theta(P^2; P, a) = 0$  gives the displayed zero-column requirement.

Cauchy's inequality gives a necessary energy floor

$$E_{\text{nonreal}} \geq \frac{T_0(P)^2(1 - |T_\chi(P)|/T_0(P))^2}{P - 3},$$

but this is still only a capacity condition. The archived scan  $7 \leq P \leq 1000$  records many cases where the nonreal energy is above this floor while no zero column occurs. Hence the remaining analytic AP input is not an energy statement, but

`NonrealHalfClassSimplexRankOneProjectionExclusionAtP2.`

No unconditional row/column theorem is claimed.  $\square$

**Theorem 7.26** (Half-class compensation variance for the nonreal packet). *Keep the notation of the preceding theorem and fix  $s \in \{+1, -1\}$ . Let*

$$H_s = \{a \in \mathbb{F}_P^* : \chi_P(a) = s\}, \quad h = |H_s| = \frac{P-1}{2},$$

and define the half-class mean

$$\mu_s = \frac{T_0(P) + sT_\chi(P)}{P-1}.$$

For  $a \in H_s$  the nonreal residual is exactly

$$R_a = (P-1)(\theta(P^2; P, a) - \mu_s),$$

and therefore

$$\sum_{a \in H_s} R_a = 0.$$

If a zero column occurs at  $a_0 \in H_s$ , then

$$R_{a_0} = -(P-1)\mu_s, \quad \sum_{\substack{a \in H_s \\ a \neq a_0}} R_a = (P-1)\mu_s.$$

Consequently the sharp one-hole variance floor is

$$\sum_{a \in H_s} R_a^2 \geq ((P-1)\mu_s)^2 \frac{h}{h-1}.$$

Equality is attained, at the level of the centered half-class geometry, by the flat compensator in which the non-hole residues all have

$$\theta(P^2; P, a) = \mu_s \frac{h}{h-1}.$$

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-nonreal-halfclass-compensation-variance-router.md.`

Since  $T_0 + sT_\chi = 2 \sum_{a \in H_s} \theta(P^2; P, a)$ , the mean over  $H_s$  is  $\mu_s$ . Substituting this identity into the previous residual formula gives  $R_a = (P-1)(\theta_a - \mu_s)$ , and summing over  $H_s$  gives the centering identity.

If  $\theta_{a_0} = 0$ , then the displayed value of  $R_{a_0}$  follows immediately, and the compensation identity follows from centering. Cauchy's inequality applied to the remaining  $h - 1$  residues gives

$$\sum_{\substack{a \in H_s \\ a \neq a_0}} R_a^2 \geq \frac{((P-1)\mu_s)^2}{h-1},$$

so adding  $R_{a_0}^2$  gives the stated floor. Equality requires all remaining residuals to be equal, i.e. the flat compensator above.

Thus the previous energy floor is not only necessary but convexly sharp. It cannot by itself exclude a zero column, because the flat compensator is a nonnegative centered half-class vector. The remaining self-contained obligation is therefore the actual prime-phase input

`HalfClassCompensationMassNonconcentrationOrColumnCRTPDEC,`

which must either rule out such flat or near-flat compensation for the true prime-induced vector at  $x = P^2$ , or route persistent compensation phases to a registered ColumnCRT/PDEC or moving-family return. No unconditional row/column theorem is claimed here.  $\square$

**Theorem 7.27** (Ratio-Fourier lock for half-class compensation). *Let  $Q \subset \mathbb{F}_P^*$  be the subgroup of quadratic residues and fix a candidate puncture  $a_0 \in H_s$ . Put*

$$f_{a_0}(u) = \theta(P^2; P, a_0 u), \quad u \in Q,$$

*so that  $u = 1$  is the punctured coordinate. Let*

$$z = f_{a_0}(1), \quad c_{a_0} = \frac{\sum_{u \in Q, u \neq 1} f_{a_0}(u)}{h-1}, \quad \mu_s = \frac{z + (h-1)c_{a_0}}{h}.$$

*Then the excess over the one-point variance floor is exactly*

$$\sum_{a \in H_s} R_a^2 - (P-1)^2(z - \mu_s)^2 \frac{h}{h-1} = (P-1)^2 \sum_{\substack{u \in Q \\ u \neq 1}} (f_{a_0}(u) - c_{a_0})^2.$$

*If  $\widehat{f}(\psi) = \sum_{u \in Q} f_{a_0}(u) \overline{\psi(u)}$ , where  $\psi$  ranges over the characters of  $Q$ , then the flat compensator*

$$f_{a_0}(u) = c_{a_0} \quad (u \neq 1)$$

*is equivalent to the simultaneous lock*

$$\widehat{f}(\psi) = z - c_{a_0} \quad (\psi \neq 1).$$

*Moreover,*

$$\sum_{\psi \neq 1} \left| \widehat{f}(\psi) - (z - c_{a_0}) \right|^2 = h \sum_{\substack{u \in Q \\ u \neq 1}} (f_{a_0}(u) - c_{a_0})^2.$$

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-halfclass-ratio-fourier-lock-router.md.`

The normalization by  $a_0^{-1}$  maps the half-class  $H_s$  bijectively to  $Q$ , because  $a_0$  has the same quadratic character as every element of  $H_s$ . Write

$$f_{a_0}(u) = \begin{cases} z, & u = 1, \\ c_{a_0} + d_u, & u \neq 1, \end{cases} \quad \sum_{u \neq 1} d_u = 0.$$

Substituting this decomposition into  $\sum_{a \in H_s} R_a^2 = (P-1)^2 \sum_{u \in Q} (f_{a_0}(u) - \mu_s)^2$  gives the displayed one-point excess identity; the cross term vanishes because the  $d_u$  sum to zero.

For the Fourier statement, if  $d_u = 0$  for all  $u \neq 1$ , then for every nontrivial  $\psi$

$$\widehat{f}(\psi) = z + c_{a_0} \sum_{u \neq 1} \overline{\psi(u)} = z - c_{a_0}.$$

Conversely, if all nontrivial coefficients equal  $z - c_{a_0}$ , Fourier inversion on the finite abelian group  $Q$  gives  $d_u = 0$  for every  $u \neq 1$ . Applying Parseval to the function  $d_1 = 0$ ,  $d_u = f_{a_0}(u) - c_{a_0}$  for  $u \neq 1$ , gives the final identity.

Thus a flat zero-column compensator is not merely a low-variance event: after normalization by the missing residue it is an all-frequency lock on the quadratic-residue subgroup. The remaining input is the exclusion of persistent locks of this kind, or their registration as ColumnCRT/PDEC or moving-family returns. No unconditional row/column theorem is claimed here.  $\square$

**Theorem 7.28** (Quadratic-twist pair character orbit lock). *Let  $\psi \neq 1$  be a character of the quadratic-residue subgroup  $Q$ , and let  $\chi$  be either extension of  $\psi$  to  $\mathbb{F}_P^*$ . The other extension is  $\chi\chi_P$ . For  $a_0 \in H_s$ , define*

$$T_\chi(P) = \sum_{a \in \mathbb{F}_P^*} \theta(P^2; P, a) \overline{\chi(a)}.$$

*Then the  $Q$ -Fourier coefficient in the preceding theorem satisfies*

$$\widehat{f}(\psi) = \chi(a_0) \frac{T_\chi(P) + sT_{\chi\chi_P}(P)}{2}.$$

*Consequently the flat compensator condition is equivalent to the simultaneous orbit lock*

$$\chi(a_0) \frac{T_\chi(P) + sT_{\chi\chi_P}(P)}{2} = \theta(P^2; P, a_0) - c_{a_0} \quad (\psi \neq 1).$$

*Equivalently,*

$$\sum_{\psi \neq 1} \left| \chi(a_0) \frac{T_\chi(P) + sT_{\chi\chi_P}(P)}{2} - (\theta(P^2; P, a_0) - c_{a_0}) \right|^2 = h \sum_{\substack{u \in Q \\ u \neq 1}} (f_{a_0}(u) - c_{a_0})^2.$$

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-halfclass-twist-pair-character-lock-router.md.`

Since  $\chi$  restricts to  $\psi$  on  $Q$ , for  $a = a_0 u$  with  $u \in Q$  we have  $\overline{\psi(u)} = \chi(a_0) \overline{\chi(a)}$ . Also the indicator of  $H_s$  is  $(1 + s\chi_P(a))/2$ . Therefore

$$\widehat{f}(\psi) = \chi(a_0) \sum_{a \in H_s} \theta(P^2; P, a) \overline{\chi(a)} = \chi(a_0) \frac{T_\chi(P) + sT_{\chi\chi_P}(P)}{2}.$$



Substituting this identity into the ratio-Fourier lock theorem gives the orbit lock and Parseval identity.

This reformulation is important for the global route: the obstruction is not a single exceptional character and not a scalar energy excess. A persistent zero-column compensator must synchronize every nontrivial character of  $Q$ , or equivalently every quadratic-twist pair of Dirichlet characters modulo  $P$ , onto the same  $a_0$ -dependent phase orbit. The remaining input is

**QuadraticTwistPairCharacterOrbitLockExclusionOrColumnCRTPDEC.**

No unconditional row/column theorem is claimed here. □

**Theorem 7.29** (Log-prime degeneration of exact half-class orbit locks). *For  $a \in \mathbb{F}_P^*$ , put*

$$S_a(P) = \{\ell \leq P^2 : \ell \text{ prime}, \ell \neq P, \ell \equiv a \pmod{P}\}$$

*and  $\theta_a = \sum_{\ell \in S_a(P)} \log \ell$ . If  $a \neq b$  and  $\theta_a = \theta_b$ , then  $S_a(P) = S_b(P) = \emptyset$ .*

*Consequently, for  $P \geq 7$ , an exact punctured half-class flatness condition*

$$\theta(P^2; P, a_0 u) = c \quad (u \in Q, u \neq 1)$$

*can hold only in the degenerate case  $c = 0$  and*

$$S_{a_0 u}(P) = \emptyset \quad (u \in Q, u \neq 1).$$

*Thus the exact quadratic-twist pair orbit lock from the preceding theorem is routed to*

**PuncturedHalfClassZeroSupportDegeneracyExclusionOrColumnCRTPDEC.**

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-halfclass-log-independence-degeneracy-router.md.`

If  $\theta_a = \theta_b$ , then

$$\sum_{\ell \in S_a(P)} \log \ell - \sum_{\ell \in S_b(P)} \log \ell = 0.$$

Equivalently,

$$\prod_{\ell \in S_a(P)} \ell = \prod_{\ell \in S_b(P)} \ell.$$

Unique factorization forces the two prime multisets to be identical. Distinct nonzero residues modulo  $P$  are disjoint prime supports, so both supports must be empty.

For  $P \geq 7$ , the punctured set  $\{u \in Q : u \neq 1\}$  contains at least two elements. Exact flatness makes all corresponding  $\theta$ -values equal. Applying the first paragraph to any pair and then to a fixed representative against every other punctured residue gives empty support throughout the punctured half-class and  $c = 0$ .

Therefore a positive exact flat compensator cannot be the persistent obstruction. The only remaining exact-lock exit is the much sharper zero-support degeneracy: after removing  $a_0$ , the same half-class has no prime arrivals up to  $P^2$ . This theorem does not exclude that degeneracy and does not prove the row/column theorem unconditionally. □

**Theorem 7.30** (Half-class mass versus single-residue capacity). *Let*

$$\Theta_s(P) = \sum_{\chi_P(a)=s} \theta(P^2; P, a) = \frac{T_0(P) + sT_\chi(P)}{2}, \quad s \in \{\pm 1\}.$$

For every  $a \in \mathbb{F}_P^*$ ,

$$\theta(P^2; P, a) \leq P \log(P^2) = 2P \log P.$$

Consequently a punctured half-class zero-support degeneracy can occur only if

$$\min_{s=\pm 1} \Theta_s(P) \leq 2P \log P.$$

Equivalently, the sufficient exclusion gate is

$$T_0(P) \left( 1 - \frac{|T_\chi(P)|}{T_0(P)} \right) > 4P \log P.$$

Thus the latest exact-lock exit is routed to

*QuadraticHalfClassMassBeatsSingleResidueCapacityAtP2,*

or else to a named ColumnCRT/PDEC capacity-failure registration.

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-halfclass-single-residue-capacity-router.md.`

In a fixed nonzero residue  $a \bmod P$ , the integers not exceeding  $P^2$  have the form  $a + kP$ ,  $0 \leq k \leq P - 1$ . Hence at most  $P$  prime terms can occur in this residue before the square barrier, and each carries weight at most  $\log(P^2)$ . This gives the displayed single-residue capacity bound.

If a punctured half-class zero-support degeneracy occurs in  $H_s$ , then all prime weight in that half-class is concentrated in the single unpunctured residue  $a_0$ . Therefore

$$\Theta_s(P) = \theta(P^2; P, a_0) \leq 2P \log P.$$

Taking the contrapositive gives the half-class capacity exclusion. Finally, since

$$\min_s \Theta_s(P) = \frac{T_0(P) - |T_\chi(P)|}{2},$$

the condition is exactly

$$T_0(P) \left( 1 - \frac{|T_\chi(P)|}{T_0(P)} \right) > 4P \log P.$$

This is a strict narrowing of the previous support-degeneracy exit, but it is not yet a proof of the required global half-class mass lower bound. No unconditional row/column theorem is claimed here.  $\square$

**Theorem 7.31** (Capacity margin as a logarithmic-over-linear projection gap). *The half-class single-residue capacity gate*

$$\min_{s=\pm 1} \Theta_s(P) > 2P \log P$$

is equivalent to

$$1 - \frac{|T_\chi(P)|}{T_0(P)} > \frac{4P \log P}{T_0(P)}.$$

In particular, if a principal Chebyshev lower bound

$$T_0(P) \geq c_0(P)P^2$$

is available, then it is enough to prove

$$1 - \frac{|T_\chi(P)|}{T_0(P)} > \frac{4 \log P}{c_0(P)P}.$$

Thus any persistent failure of the capacity gate, in the presence of a  $P^2$ -scale principal mass lower bound, forces an ultra-near-one quadratic projection defect

$$\frac{|T_\chi(P)|}{T_0(P)} \geq 1 - O\left(\frac{\log P}{P}\right).$$

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-halfclass-capacity-margin-factorization-router.md.`

The identity

$$\min_s \Theta_s(P) = \frac{T_0(P) - |T_\chi(P)|}{2}$$

was proved in the preceding theorem. Therefore

$$\min_s \Theta_s(P) > 2P \log P$$

is equivalent to

$$\frac{T_0(P) - |T_\chi(P)|}{2} > 2P \log P,$$

and division by  $T_0(P) > 0$  gives the stated projection-gap threshold. Substituting  $T_0(P) \geq c_0(P)P^2$  gives the logarithmic-over-linear sufficient threshold.

This reformulation exposes the scale of the remaining obstruction. The capacity gate does not need a fixed-size half-class margin: a gap of order  $\log P/P$ , after the principal Chebyshev lower bound is in place, is already enough. Conversely, a counterexample surviving this gate must make the quadratic projection almost equal to the principal mass. The remaining input is

`QuadraticProjectionGapBeatsLogOverPThresholdAtSquareScale,`

or a named registration of the ultra-near-one projection defect as a ColumnCRT/PDEC/moving-family exit. No unconditional row/column theorem is claimed here.  $\square$

**Theorem 7.32** (Low CRT insufficiency and the large splitting residual). *Let*

$$\Theta_s(P) = \sum_{\substack{\ell \leq P^2, \ell \neq P \\ \chi_P(\ell)=s}} \log \ell, \quad s \in \{+1, -1\},$$

where  $\chi_P(\ell) = (\ell/P)$ . *Split*

$$L_s(P) = \sum_{\substack{\ell < P \\ \chi_P(\ell)=s}} \log \ell, \quad G_s(P) = \sum_{\substack{P < \ell \leq P^2 \\ \chi_P(\ell)=s}} \log \ell.$$

Then  $\Theta_s(P) = L_s(P) + G_s(P)$ , and the projection-gap capacity gate is equivalent to

$$\min_s \Theta_s(P) > 2P \log P.$$

Moreover the complete low CRT layer satisfies the elementary bound

$$L_+(P) + L_-(P) = \vartheta(P-1) \leq P \log P < 2P \log P.$$

Consequently the low CRT layer alone cannot close the capacity gate. If the capacity gate fails, then for some sign  $s$

$$G_s(P) \leq 2P \log P.$$

Conversely, the large-splitting lower bound

$$\min_s G_s(P) > 2P \log P$$

is sufficient to close the capacity gate.

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-quadratic-projection-large-splitting-router.md.`

The decomposition  $\Theta_s = L_s + G_s$  is only the partition of the prime sum at the modulus  $P$ . The equivalence between the projection-gap gate and  $\min_s \Theta_s(P) > 2P \log P$  is the preceding theorem.

For the low layer, every prime counted in  $L_+ + L_-$  is  $< P$ , hence there are fewer than  $P$  summands and each summand is at most  $\log P$ . Thus  $L_+ + L_- \leq P \log P$ . This is strictly below the minority-mass threshold  $2P \log P$ , so the low CRT period can be entirely absorbed by the failure allowance and cannot by itself force the required contradiction.

If the capacity gate fails, choose  $s$  with  $\Theta_s(P) = \min_t \Theta_t(P) \leq 2P \log P$ . Since  $L_s(P) \geq 0$ , we have  $G_s(P) \leq 2P \log P$ . Conversely, if both large layers satisfy  $G_s(P) > 2P \log P$ , then  $\Theta_s(P) \geq G_s(P) > 2P \log P$  for both signs, which closes the capacity gate. Thus the remaining self-contained attack is

`LargePrimeQuadraticSplittingMinorityMassBeatsTwoPLogPAtSquareScale,`

or the registration of a persistent large-splitting desert as a Siegel/ColumnCRT/PDEC exit. No unconditional row/column theorem is claimed here.  $\square$

**Theorem 7.33** (Large splitting as a beta-gap and zero-packet budget). *Let*

$$G_s(P) = \sum_{\substack{P < \ell \leq P^2 \\ \chi_P(\ell) = s}} \log \ell, \quad H_0(P) = G_+(P) + G_-(P), \quad H_\chi(P) = G_+(P) - G_-(P).$$

*Then the large-splitting gate*

$$\min_s G_s(P) > 2P \log P$$

*is equivalent to*

$$1 - \frac{|H_\chi(P)|}{H_0(P)} > \frac{4P \log P}{H_0(P)}.$$

*Consequently, any explicit-formula estimate of the form*

$$\frac{|H_\chi(P)|}{H_0(P)} \leq R_\beta(P) + E_{\text{zero}}(P)$$

*closes the large-splitting gate whenever*

$$R_\beta(P) + E_{\text{zero}}(P) < 1 - \frac{4P \log P}{H_0(P)}.$$

*For the real-zero shadow model*

$$R_\beta(P) = \frac{P^{2\beta} - P^\beta}{\beta H_0(P)}, \quad \beta = 1 - \delta,$$

*the critical no-residual value of  $\delta$  is asymptotic to  $2/P$  when  $H_0(P) \sim P^2$ .*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-large-splitting-beta-gap-router.md`.

The first equivalence is the identity

$$\min_s G_s(P) = \frac{H_0(P) - |H_\chi(P)|}{2}.$$

Thus  $\min_s G_s(P) > 2P \log P$  is exactly

$$H_0(P) - |H_\chi(P)| > 4P \log P,$$

and division by  $H_0(P) > 0$  gives the stated projection-gap form.

The explicit-formula budget statement is purely algebraic: if the normalized character projection is bounded by  $R_\beta + E_{\text{zero}}$ , then it is enough for that upper bound to be below the allowed ratio  $1 - 4P \log P / H_0(P)$ .

For the scale calculation, put  $x = P^2$ ,  $\beta = 1 - \delta$ , and  $H_0(P) \sim P^2$ . Near  $\delta = 0$ ,

$$\frac{P^{2\beta} - P^\beta}{\beta H_0(P)} = 1 - \delta(2 \log P - 1) + o(\delta \log P) + o(1/P).$$

The required missing ratio is

$$\frac{4P \log P}{H_0(P)} \sim \frac{4 \log P}{P}.$$

Equating these first-order quantities gives

$$\delta \sim \frac{4 \log P / P}{2 \log P - 1} \sim \frac{2}{P}.$$

Hence a persistent failure must come either from a real zero within the  $1/P$  scale, or from a non-real zero packet, endpoint, or prime-power residual consuming the same slack. The remaining self-contained target is

`SquareScaleQuadraticBetaGapAndZeroPacketResidualBudgetAtP2,`

or the named exit

`UltraCloseRealZeroOrCoherentZeroPacketLargeSplittingPDEC.`

No unconditional row/column theorem is claimed here. □

**Theorem 7.34** (Page sparsity for ultra-close beta-gap carriers). *Assume the Landau–Page uniqueness input in the following form: for primitive real characters of conductor  $q \leq Q$ , at most one character has a real zero*

$$\beta > 1 - \frac{c_{\text{Page}}}{\log Q}.$$

*Let  $E(Q)$  be the set of prime moduli  $P \leq Q$  for which the large-splitting beta-gap failure produces a real-zero carrier*

$$\beta_P > 1 - \frac{C(P)}{P}, \quad C(P) \leq C_*.$$

*Then*

$$|E(Q) \cap ((C_*/c_{\text{Page}}) \log Q, Q]| \leq 1.$$

*In particular, after this external Page input and its constant matching are accepted, a positive-density or fixed-period CRT family of such ultra-close real-zero carriers cannot be the persistent obstruction.*

*Proof.* The certificate is recorded in

`docs/monograph/prime-matrix-beta-gap-page-sparsity-router.md.`

If  $P > (C_*/c_{\text{Page}}) \log Q$ , then

$$\frac{C(P)}{P} \leq \frac{C_*}{P} < \frac{c_{\text{Page}}}{\log Q}.$$

Hence every zero counted by  $E(Q)$  in this range satisfies

$$\beta_P > 1 - \frac{C(P)}{P} > 1 - \frac{c_{\text{Page}}}{\log Q},$$

and so lies in the Page uniqueness region for conductors  $q \leq Q$ . The Page input allows at most one such real-zero carrier. Therefore the high part of  $E(Q)$  has cardinality at most one.

This eliminates only the multi-carrier CRT reproduction shape. It does not internalize the Page constant, exclude a moving singleton exceptional carrier, or control non-real zero packets, endpoint terms, or prime-power residuals. The remaining target is

`PageExceptionalSingletonCarrierOrNonrealZeroPacketResidualBudget,`

with fallback

`MovingPageSingletonCarrierPDECOrCoherentZeroPacket.`

No unconditional row/column theorem is claimed here. □

**Theorem 7.35** (Terminal-square downstream frontier synchronization). *The terminal-row square-phase bridge does not leave the row/column program at the older square-phase long-block interface. In the current archive, that interface has already been synchronized with the half-grid and phase-band routers, with the localized P-CRT/AP-zero-packet route, and with the global CRT signed-payload route. Thus*

*SquarePhaseSpecialPhaseLongBlockPDECExclusion*

*is an upstream alias rather than the deepest active terminal-row frontier.*

*Status and reference.* The synchronization certificate is recorded in

`docs/monograph/prime-matrix-terminal-square-phase-downstream-frontier-sync-router.md.`

It reads the existing terminal CRT atom certificate, the terminal square-phase bridge, the half-grid boundary-word router, the no-slot phase-band router, the localized  $P$ -CRT/Linnik-2 barrier, the rank-one/AP-zero-packet routers, the Page-sparsity beta-gap router, and the global CRT signed-payload router.

The result is a bookkeeping theorem: the terminal-square interface is now aligned with the deeper consolidated basis consisting of AP zero-packet residuals, signed-payload construction, same-set PDEC input, and the recorded global incidence/rate/D-structure inputs. No new unconditional contradiction is asserted; the row/column theorem remains open until those downstream inputs are closed or independently accepted.  $\square$

**Theorem 7.36** (Common source declaration packet for payload and ExactUV). *In the strict internal source lane, the signed-payload obstruction and the ActualEmitterExactUV incidence obstruction have the same missing pre-Cauchy source root. More precisely, the current archive routes*

*NoncircularAtomicBasisWordSignedCoefficientOriginIdentityBeforePushforward*

and

*ActualEmitterSourceDomainEntropyLedger*  
 $\wedge$  *ExactUVMapFixedPairPolylogFiberBoundLedger*

to the single packet

*PreCauchyActualNoncanonicalEmitterSourceDeclarationPacket.*

*Status and reference.* The certificate is recorded in

docs/monograph/  
prime-matrix-strict-source-declaration-payload-exactuv-  
unification-router.md.

The packet must be declared before Cauchy,  $\Phi$ , payment, or any zero-row recovery. Its fields are: an actual noncanonical source declaration line, primitive summand rows, basis-word/signed-coefficient origin identities, prepushforward  $\alpha/\delta$  equality, source-domain entropy, fixed exact  $(u, v)$  fiber bounds, and a named return partition for absent declarations, fiber collapse, entropy deficit, sign conflicts, local-factor zeros, over-budget rows, and canonical leakage.

This is a routing and consolidation statement. The packet has not been proved in the present corpus. The AP zero-packet branch and same-set PDEC branch remain open, as do the model/rate ledgers and D-structure/Rankin acceptance. Therefore no unconditional row/column theorem is claimed here.  $\square$

**Theorem 7.37** (Downstream split of the common source packet). *The packet*

*PreCauchyActualNoncanonicalEmitterSourceDeclarationPacket*

*does not remain a single black box in the strict source lane. In the current archive it splits into a signed-payload lane and an ExactUV lane. The signed-payload lane is synchronized to*

*BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows,*

*while the ExactUV lane remains*

*ActualEmitterSourceDomainEntropyLedger*  
 $\wedge$  *ExactUVMapFixedPairPolylogFiberBoundLedger.*

*Status and reference.* The certificate is recorded in

`docs/monograph/prime-matrix-strict-source-declaration-downstream-sync-router.md.`

The ordinary joint-constructor route is rejected as a proof route because its expansion returns through the alpha side, same-row identity, and row-level origin table to the signed-source fixed point. The anti-split route therefore requires an atomic pre-Cauchy rows declaration with built-in word/coefficient pairing; the current archive further reduces that target to the built-in signed coefficient pairing closed form above.

This theorem records a downstream synchronization only. The built-in pairing closed form and the ExactUV entropy/fiber bounds are not proved here, and the global rate and D-structure/Rankin gates remain open.  $\square$

**Theorem 7.38** (Signed-lane cycle closure). *In the strict source lane, the current signed-payload route is a closed dependency cycle:*

$$\begin{aligned}
 & \text{PreCauchyActualNoncanonicalEmitterSourceDeclarationPacket} \\
 & \rightarrow \text{BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows} \\
 & \rightarrow \text{ExactAtomicJointBranchTraceSignedCoefficientFormulaOrReturn} \\
 & \rightarrow \text{AtomicSignedPayloadTraceConstructorBeforeAssignmentOrReturn} \\
 & \rightarrow \text{NoncircularAtomicBasisWordSignedCoefficientOriginIdentityBeforePushforward} \\
 & \rightarrow \text{PreCauchyActualNoncanonicalEmitterSourceDeclarationPacket}.
 \end{aligned}$$

Consequently no node in this lane is currently an independent self-contained proof of the signed payload.

*Status and reference.* The certificate is recorded in

`docs/monograph/prime-matrix-strict-signed-lane-cycle-closure-router.md.`

The first arrow is the downstream split of the common source packet. The second is the built-in pairing frontier. The third and fourth are the atomic branch-trace and atomic-payload routers: the visible branch trace supplies coordinates but not the signed payload. The final arrow is the source declaration unification, where the noncircular origin identity returns to the same common packet.

Thus the signed lane is diagnostic rather than terminal. A proof must add a new primitive signed payload or trace formula, prove a well-founded terminal descent, or use an independently proved same-set PDEC scope certificate. The ExactUV entropy/fiber, rate-preservation, and D-structure/Rankin gates remain open.  $\square$

**Theorem 7.39** (New primitive payload alignment to the source atom). *The exit*

*NewPrimitiveAtomicSignedPayloadOrTraceFormulaArtifact*

*is not an independent terminal name in the present archive. If it is to break the signed-lane cycle, it must supply a pre-Cauchy actual source object in the same formal unit and must include the three source-rank/no-collapse atoms*

$$\begin{aligned}
 & \text{ActualPreCauchySourceDomainAbsoluteEntropyLedger}, \\
 & \text{CompletePrimitiveEmitterKeyPartitionLedger}, \\
 & \text{FixedKeyExactUVLocalMultiplicity01Ledger}.
 \end{aligned}$$

*Otherwise it is only a renaming of an existing trace, payload, origin, or common-packet node, or else it leaves the lane through terminal descent, PDEC, or an external spectral input.*



*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-strict-new-primitive-payload-
source-atom-alignment-router.md.
```

The audit imports the signed-lane cycle closure, the atomic trace/payload frontier, the source declaration unification, and the final exact-source alignment. The visible trace supplies coordinates but no signed coefficient or local factor. The origin identity alone returns to the common source packet. Thus a genuinely new primitive artifact must name the actual pre-Cauchy source and control its exact  $(u, v)$  fibers before any Cauchy,  $\Phi$ , payment, or terminal recovery step.

The deterministic target is therefore the source-rank/no-collapse package. Those three atoms are not proved in the present corpus, and the external spectral and D-structure/Rankin acceptance routes remain open. No unconditional row/column theorem is claimed here.  $\square$

**Theorem 7.40** (Source entropy downstream cycle guard). *The source-domain atom*

*ActualPreCauchySourceDomainAbsoluteEntropyLedger*

*is not a terminal self-contained proof point in the present archive. Its recorded downstream route is*

*ActualPreCauchySourceDomainAbsoluteEntropyLedger*  
 $\rightarrow$  *AcyclicSeedPrimitiveRowSignedCoefficientLawBeforePushforward*  
 $\rightarrow$  *AcyclicSeedPreCauchyBasisWeightSourceFormulaForPrimitiveRows*  
 $\rightarrow$  *AcyclicSeedInternalArithmeticBasisExpansionBeforeCauchy*  
 $\rightarrow$  *AcyclicSeedNoncanonicalPreCauchyBasisAlphabetLedger,*

*and continuing from the basis alphabet enters the signed coordinate-source cycle. That cycle cannot count as a proof of source entropy.*

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-strict-source-entropy-
downstream-cycle-sync-router.md.
```

The imported cycle guard records that the word-coordinate formula, signed slot, coefficient assignment, coefficient value map, basis-word origin, row-level generation table, signed row emitter, primitive coefficient law, basis-weight source, internal basis expansion, basis alphabet, basis-word generation, source-tuple constructor, and basis-word formula form a closed dependency loop. This loop only shows mutual definition inside the present formalism. It does not produce an independent pre-Cauchy primitive basis and coefficient source.

Thus the current source-entropy branch needs a new acyclic primitive basis/coefficient source input, or else it must return through the terminal descent/PDEC/external spectral exits. The complete-key, fixed-key multiplicity, and D-structure/Rankin gates also remain open.  $\square$

**Theorem 7.41** (Unified cycle-cut and terminal-descent frontier). *In the present archive the two immediate exits from the source-entropy cycle guard do not remain independent terminal proof points. The recorded reduction is*

*ActualPreCauchySourceDomainAbsoluteEntropyLedger*  
 $\rightarrow$  *AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput*  
 $\quad$  *OR AcyclicNoncanonicalTerminalReturnWellFoundedDescentCertificate*  
 $\rightarrow$  *AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput*  
 $\rightarrow$  *AcyclicSameSetScopeMatchForDirectPDECcapDualCertificate*  
 $\quad$  *OR NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact*

```

AcyclicNoncanonicalTerminalReturnWellFoundedDescentCertificate
-> AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
OR NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
OR IndependentActualSourceBridgeNotFactoredThrough
ExactUVPairEnergyOrJointConstructorLoop.

```

Moreover, the internal PDEC same-set branch is saturated in the current self-contained corpus, so the first productive internal atom of the remaining joint route is

```

PreCauchyJointWordCoefficientEmitterDeclarationLine
ForActualNoncanonicalSourceTuple.

```

*Status and reference.* The certificate is recorded in

```

docs/monograph/prime-matrix-strict-cyclecut-terminal-descent-
unified-frontier-router.md.

```

The seed-cycle-cut branch has already rejected sequential word-first or coefficient-first splitting and returns to the signed-source fixed point unless a new joint emitter is supplied. The terminal-descent branch forms the recorded macrocycle

TERMINAL → SOURCE → PAIR → JOINT → TERMINAL,

so it is not a well-founded descent proof. The same-set PDEC arm remains a valid conditional or external input, but within the present strict internal corpus it no longer gives an independent non-cyclic exit.

Thus the current internal joint route must begin before Cauchy, payment, and  $\Phi$ -pushforward with a declaration line for the same actual noncanonical source tuple. That line is still unproved; the row formula, word/coefficient identity, no-downstream-return ledger, complete/fixed-key controls, ExactUV, rate-preservation, and D-structure/Rankin gates remain open. Hence this is a frontier compression, not an unconditional row/column proof.  $\square$

**Theorem 7.42** (Downstream anti-split synchronization of the unified frontier). *The ordinary joint declaration line obtained in the unified frontier is not a new non-cyclic proof point. In the present archive it synchronizes to the ordinary explicit joint constructor, and that route returns to the signed-source fixed point unless it is replaced by an anti-split atomic declaration. The current downstream synchronization is*

```

PreCauchyJointWordCoefficientEmitterDeclarationLineForActualNoncanonicalSourceTuple
-> ExplicitJointAlphaDeltaPrimitiveWordCoefficientConstructorRuleForActualNoncanonicalSourceTuple
-> signed-source fixed point unless replaced
AntiSplit route
-> AtomicPreCauchyJointRowsFormulaWithBuiltInWordCoefficientPairing
-> BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows
ExactUV gate
-> ActualEmitterSourceDomainEntropyLedger
AND ExactUVMapFixedPairPolylogFiberBoundLedger.

```

*Status and reference.* The certificate is recorded in

```

docs/monograph/prime-matrix-strict-cyclecut-unified-antisplit-
downstream-sync-router.md.

```

The joint declaration/constructor synchronization shows that a plain declaration line only selects the ordinary explicit joint alpha/delta rule. The direct constructor audit shows that this ordinary route expands through the alpha-side and same-row origin tables back to the signed-source fixed point. Consequently the declaration cannot be used as a proof merely by renaming it.

The anti-split firewall requires an atomic pre-Cauchy row declaration whose rows already contain the word/coefficient pairing and the pre-pushforward payload. The atomic-row audit then identifies the first remaining signed field as the built-in signed coefficient/pairing closed form for atomic joint rows. In parallel, the ExactUV incidence gate has been decomposed into actual source-domain entropy and fixed exact-pair polylog fiber bounds. None of these three inputs is proved in the present corpus, and the complete/fixed-key, model, rate-preservation, and D-structure/Rankin gates remain open.  $\square$

**Theorem 7.43** (Anti-split trace cycle and ExactUV atomization). *The built-in signed pairing frontier does not terminate at an ordinary branch trace. The next recorded synchronization is*

```
BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows
-> ExactAtomicJointBranchTraceSignedCoefficientFormulaOrReturn
-> signed-lane dependency cycle
-> NewPrimitiveAtomicSignedPayloadOrTraceFormulaArtifact
    OR terminal descent OR same-set PDEC.
```

*In parallel, the ExactUV side is atomized as*

```
ActualEmitterSourceDomainEntropyLedger
-> ActualPreCauchySourceDomainEntropyFromSignedRowsAndRowMassLedger
-> AcyclicSeedPrimitiveRowSignedCoefficientLawBeforePushforward
ExactUVMapFixedPairPolyLogFiberBoundLedger
-> RegisteredCompletePrimitiveEmitterKeyPartitionPolylogLedger
    AND FixedKeyExactUVLocalMultiplicity01Ledger.
```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-strict-antisplit-trace-exactuv-
atomized-frontier-router.md.
```

The built-in pairing audit shows that signed coefficients require odd orientation and local-factor data, so unsigned row skeletons and parity shadows do not determine the pairing. The exact atomic branch trace would provide that data only conditionally. The signed-lane cycle audit then shows that the trace, payload constructor, origin identity, and common source declaration packet form a dependency cycle. Hence the trace cannot close the proof unless a genuinely new primitive payload/trace artifact, a well-founded terminal descent, or a same-set PDEC certificate is supplied.

The ExactUV incidence audit is independent of this signed pairing cycle. Its source-domain side is reduced to signed row-mass entropy and then to the primitive signed row coefficient law. Its fixed-pair fiber side is reduced to a registered complete key partition and fixed-key local multiplicity. These atoms, together with the model, rate-preservation, and D-structure/Rankin gates, remain unproved in the present corpus.  $\square$

**Theorem 7.44** (Post-antisplit source-rank convergence). *After the anti-split trace/ExactUV atomization, the recorded new-primitive and terminal-descent exits are not independent terminal proof points. The current synchronization is*

```
NewPrimitiveAtomicSignedPayloadOrTraceFormulaArtifact
-> ActualPreCauchySourceDomainRankAndExactUVNoCollapseLedger,
AcyclicNoncanonicalTerminalReturnWellFoundedDescentCertificate
-> ActualPreCauchySourceDomainRankAndExactUVNoCollapseLedger,
ActualPreCauchySourceDomainRankAndExactUVNoCollapseLedger
-> PointwiseSameFormalUnitPrimitiveAlphaDeltaKernelTableWithNonzeroRankCertificate.
```

*The first internal atom after this convergence is*

```
AlphaRowAnchorPhaseEmissionFormulaLedger,
```

*with parallel obligations*

*IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger  
SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows.*

*Status and reference.* The certificate is recorded in

docs/monograph/prime-matrix-strict-post-antisplit-source-rank-  
convergence-router.md.

The new-primitive payload audit already shows that a purported new signed payload only breaks the signed-lane cycle if it carries a pre-Cauchy actual source object, source entropy, a complete key partition, and fixed-key exact-UV local multiplicity. This is exactly the source-rank/no-collapse package. The terminal descent synchronization reaches the same package through its noncanonical leaf analysis. Thus neither exit supplies a third independent closure mechanism.

The source-rank package has three visible faces: source-domain entropy, registered complete keys, and fixed-key multiplicity. The entropy face returns to the primitive signed row coefficient law; the registered-key face requires an actual emitter source table; and the fixed-key face requires a same-unit rank/multiplicity certificate. These faces meet at the same formal-unit pointwise alpha/delta kernel table. The table is still unproved; in particular the alpha row anchor/phase emission formula, the independent pre-Cauchy arithmetic identity, and the same-unit exact-UV rank/multiplicity certificate remain open, as do same-set PDEC, external spectral input, model, rate-preservation, and D-structure/Rankin gates.  $\square$

**Theorem 7.45** (Post-antisplit alpha return to terminal leaves). *The apparent post-antisplit alpha-row target is not an independent final frontier. The current synchronization is*

*PointwiseSameFormalUnitPrimitiveAlphaDeltaKernelTableWithNonzeroRankCertificate  
-> AlphaRowAnchorPhaseEmissionFormulaLedger  
AND IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger  
AND SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows,  
alpha/weight legs -> PDEC\_CAP\_OR\_INTERNAL\_CleanKLS\_LargeSieve,  
rank/multiplicity leg -> the same primitive table,  
NonrecursivePointwisePrimitiveKernelTableConstructionWithoutTerminalReturn  
-> ExplicitJointAlphaDeltaPrimitiveWordCoefficientConstructorRuleForActualNoncanonicalSourceTuple  
-> signed-source fixed point -> terminal leaf firewall.*

*Thus the active strict self-contained basis is*

*(AcyclicTerminalCanonicalLockToCanonicalSourceBoundary  
OR NoncanonicalFullSComplementLegalClosureMode)  
AND ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem  
AND RatePreservationLedger\_FOR\_moving\_atom\_packet  
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.*

*Status and reference.* The certificate is recorded in

docs/monograph/prime-matrix-strict-post-antisplit-alpha-terminal-  
leaf-sync-router.md.

The alpha formula leg has already been synchronized to the terminal PDEC/CleanKLS gate: its unsigned carry-shell and phase parts give only a skeleton, while the signed lift and overload branches return to terminal capacity/model ledgers. The signed-weight identity leg returns to the same terminal gate through the moving-block taxonomy. The exact-UV rank/multiplicity leg cannot stand alone because it requires the same primitive source table. Hence a three-leg attack is a fixed point unless a nonrecursive pointwise table is supplied at once.

The nonrecursive table field contract then identifies the first productive field as an explicit joint alpha/delta constructor rule. Existing joint constructor expansions proceed through alpha-side, same-row origin, and row-level source tables back to the signed-source fixed point; without a new

formula artifact this route enters the terminal leaf firewall. The currently active terminal leaf has been reduced to canonical-lock or noncanonical legal mode. Neither is proved in the present corpus, and ExactUV incidence, rate-preservation, and D-structure/Rankin remain independent open gates.  $\square$

**Theorem 7.46** (Post-alpha terminal leaf latest noncycle synchronization). *The post-alpha terminal leaf is not the deepest active frontier after importing the existing self-contained and noncycle audits. The synchronized chain is*

```

AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
OR NoncanonicalFullSComplementLegalClosureMode
-> AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
    OR (AcyclicPreCauchyNoncanonicalPrimitiveSourceSeedAndReturn
        AND IndependentExactPairL2EnergyOrMaxAtomBoundForAcyclicSeed),
pair-energy old spine -> source-entropy fixed point
-> ExplicitJointAlphaDeltaPrimitiveWordCoefficientConstructorRuleForActualNoncanonicalSourceTuple
-> signed-source fixed point or terminal descent
-> TERMINAL-SOURCE-PAIR-JOINT macrocycle.

```

Thus the current strict internal noncycle basis is

```

(NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
OR ProveActualFullSNonAPSourceIsCanonicalRIHBuchstab
OR FullSNonAPStrengthenedSourceAntiAtomForActualSource
OR AcyclicSameSetScopeMatchForDirectPDECAPDualCertificate)
AND ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem
AND ExplicitModelGapAndFiniteDPRCLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

*Status and reference.* The certificate is recorded in

```

docs/monograph/prime-matrix-strict-post-alpha-terminal-leaf-
latest-noncycle-sync-router.md.

```

The latest self-contained audit reduces the noncanonical legal mode to the actual-source entropy line and then to an acyclic seed plus independent pair energy input. The pair-energy noncycle audit shows that the present pair-energy/rate-packet/terminal route returns to the source-entropy fixed point and therefore cannot prove that target. The first productive field is the explicit joint alpha/delta constructor, but its direct expansion returns to the signed-source fixed point unless a new actual joint formula artifact is supplied.

The terminal descent alternative has also been audited: through the terminal leaf, source, pair-energy, and joint-constructor routes it returns to itself, forming the TERMINAL-SOURCE-PAIR-JOINT macrocycle. The only strict noncycle ways out currently recorded are a new joint formula, an actual-source bridge before the ExactUV/pair/joint loop, or a fresh same-set PDEC scope certificate. ExactUV bounded multiplicity, model/DPRC margin, rate preservation, and the D-structure/Rankin promotion gate remain independent open inputs. Hence this is a synchronization and cycle-elimination result, not an unconditional proof of the row/column theorem.  $\square$

**Theorem 7.47** (Post-alpha noncycle synchronization to terminal three atoms). *The post-alpha noncycle basis is not allowed to stop at NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact. After importing the branch-trace, signed-payload, source-declaration, signed-lane, and current-global-frontier audits, the present strict frontier is*

```

(AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
OR A1CleanBranchCanonicalSourceAdmission
OR ActualNoncanonicalCleanCoreMovingAtomExclusion)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

*The next direct attack target is*

*IndependentNonterminalMovingAtomExclusionForActualNoncanonicalCleanCore.*

*Status and reference.* The certificate is recorded in

`docs/monograph/prime-matrix-strict-post-alpha-noncycle-to-terminal-three-atoms-sync-router.md.`

The earlier new-joint formula target is absorbed by the global CRT branch-trace audit into an exact atomic branch trace. The signed-payload audit then shows that visible CRT coordinates do not generate the needed orientation, local factor, or signed coefficient payload. The source-declaration and signed-lane audits close that payload line into a dependency cycle through the common pre-Cauchy source declaration packet. Finally, the current-global frontier audit removes the trace/source/terminal/pair-mass self-return routes, leaving only the three displayed terminal atoms. None of these atoms, and not the D-structure/Rankin promotion gate, is proved here; this is a sharper frontier synchronization, not the unconditional row/column proof.  $\square$

**Theorem 7.48** (Terminal atoms to alpha-return bridge synchronization). *The terminal three-atom frontier is still not the deepest active strict frontier. After importing the moving-atom entropy normal form, the post-Mertens nonrecursive synchronization, the same formal-unit kernel identity attack, the pointwise primitive table contract, the terminal leaf firewall, and the alpha-return firewall, the old moving-atom line is absorbed as*

*ActualNoncanonicalCleanCoreMovingAtomExclusion*  
 $\rightarrow$  *ExactCleanCoreFullSNonAPWFDSsourceEntropy*  
 $\rightarrow$  *SameFormalUnitPreCauchyAlphaDeltaKernelIdentityWithSignedPhiAndFiberDispersion*  
 $\rightarrow$  *PointwiseSameFormalUnitPrimitiveAlphaDeltaKernelTableWithNonzeroRankCertificate*  
 $\rightarrow$  *NonrecursivePointwisePrimitiveKernelTableConstructionWithoutTerminalReturn*  
 $\rightarrow$  *terminal leaf firewall.*

*The current strict self-contained frontier is therefore*

*(AcyclicTerminalCanonicalLockToCanonicalSourceBoundary*  
 $\text{OR IndependentActualSourceBridgeNotFactoredThroughAlphaReturn})$   
 $\text{AND (SelfContainedExplicitZetaZeroFreeRegionThetaEnvelopeXGe20000}$   
 $\text{AND SelfContainedMeisselMertensConstantIntervalLedgerAt20000})$   
 $\text{AND SelfContainedDStructureTailLog4FiniteRankinReplacementPackage.}$

*Status and reference.* The certificate is recorded in

`docs/monograph/prime-matrix-strict-terminal-atoms-to-alpha-return-bridge-sync-router.md.`

The A1 branch is only a scoped canonical statement and does not close the unrestricted noncanonical branch. The moving-atom atom is equivalent to an actual clean-core source entropy target; post-Mertens and kernel-identity audits then force the same formal-unit pointwise primitive table. That table cannot be assembled by separate alpha, weight, and rank legs: the field contract requires a single nonrecursive primitive table, while the old joint constructor expansion returns to the signed-source fixed point and then to the terminal leaf firewall.

The terminal leaf firewall has already reduced current materialized PDEC and sparse instances to future admission disciplines, leaving canonical-lock or the noncanonical legal mode. The non-canonical legal mode filters to an actual-source bridge, but the old pointwise/alpha descent route returns through PDEC/CleanKLS to the same terminal family. Hence that route is a backedge, not a well-founded descent. The remaining strict inputs are the displayed canonical-lock or independent actual-source bridge, together with the explicit Mertens/theta tail and D-structure/Rankin replacement gates. This is not an unconditional proof of the row/column theorem.  $\square$

**Theorem 7.49** (Alpha-return bridge to concrete terminal split). *The frontier*

*AcyclicTerminalCanonicalLockToCanonicalSourceBoundary  
OR IndependentActualSourceBridgeNotFactoredThroughAlphaReturn*

*can be sharpened one more level. The independent actual-source arm reduces to an A1 scoped source-admission statement or exact clean-core source entropy; the A1 statement is not a global contradiction, and the entropy arm requires a nonrecursive seed plus an independent pair-energy bound. The canonical-lock arm reduces to a mismatch/DLS/PDEC split. Thus the concrete terminal split is*

*((AcyclicPreCauchyNoncanonicalPrimitiveSourceSeedAndReturn  
AND IndependentExactPairL2EnergyOrMaxAtomBoundForAcyclicSeed)  
OR SelfContainedKuznetsovDLSLargeSieveInequalityForAcyclicCleanBlocks  
OR (PDEC\_CAP\_OR\_INTERNAL\_CleanKLS\_LargeSieve  
AND ExplicitModelGapAndFiniteDPRCLedger))).*

*The next direct attack target is*

*IndependentExactPairL2EnergyOrMaxAtomBoundForAcyclicSeed.*

*Status and reference.* The certificate is recorded in

*docs/monograph/prime-matrix-strict-alpha-return-bridge-to-  
concrete-terminal-split-sync-router.md.*

On the independent actual-source arm, the clean A1 admission route is absorbed as a scoped canonical branch statement and cannot close the unrestricted noncanonical complement. The remaining entropy route is guarded by the nonrecursive pair-energy firewall: one may not prove the pair mass bound using the moving-atom or source-entropy conclusion that it is supposed to prove. It therefore requires both an acyclic pre-Cauchy noncanonical source seed and an independent exact pair L2 or max-atom estimate.

On the canonical-lock arm, direct attack gives only scoped absorption in the equality case. In the mismatch case the signed-source and alpha-weight routes return to the terminal family, while the clean analytic exit is a genuine self-contained Kuznetsov/DLS large-sieve inequality and the combinatorial exit is PDEC/CleanKLS together with the explicit model/DPRC margin. None of these three concrete alternatives is proved in the present corpus, and the rate preservation and D-structure/Rankin gates remain open. Hence this is a frontier synchronization, not an unconditional row/column proof.  $\square$

**Theorem 7.50** (Pair-energy to seed-coordinate cycle synchronization). *The pair-energy target singled out by the alpha-return split does not admit a new internal descent through the present signed-source machinery. Seed-only information and qualitative projection dissipation do not give a log-power rate; the old ExactUV/source-entropy pair-energy spine returns to the same source-entropy target; and the joint-constructor route enters the signed coordinate-source cycle unless a genuinely acyclic primitive basis/coefficient source input is supplied. Since the seed exists/does-not exist dichotomy has already been routed to the acyclic terminal family, the current pair-energy frontier is sharpened to*

*SelfContainedExactPairEnergyLargeSieveInequalityForAcyclicSeed  
OR SelfContainedKuznetsovDLSLargeSieveInequalityForAcyclicCleanBlocks  
OR (PDEC\_CAP\_OR\_INTERNAL\_CleanKLS\_LargeSieve  
AND ExplicitModelGapAndFiniteDPRCLedger).*

*The first term is the direct non-source-cycle pair-energy large-sieve obligation; the other two are the concrete terminal alternatives already exposed by the canonical-lock/direct-terminal audits.*

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-strict-pair-energy-to-seed-
coordinate-cycle-sync-router.md.
```

The router imports the independent pair-energy attack, the pair-energy noncycle reconciliation, the rate-bearing large-pair packet split, the terminal recurrence firewall, the source-entropy downstream cycle guard, the acyclic seed coordinate-source cycle guard, the seed-terminal fusion ledger, and the terminal canonical-lock direct attack. These ledgers together show that every currently documented internal pair-energy descent either lacks a rate, reuses the theorem it is supposed to prove, or returns to the signed coordinate-source cycle and terminal family. Hence the remaining productive work is exactly a direct pair-energy large-sieve inequality, a self-contained Kuznetsov/DLS inequality, or a PDEC/CleanKLS bound with the explicit model margin. None of these three is proved here, and the rate-preservation and D-structure/Rankin gates remain open; this is therefore a sharper frontier, not an unconditional row/column proof.  $\square$

**Theorem 7.51** (Pair-energy diagonal peeling terminal reduction). *The direct exact-pair energy large-sieve atom is not an independent third frontier after diagonal peeling. Any such L2/max-atom estimate must first control the same formal-unit diagonal mass of a single exact pair; otherwise a delta packet supported on one exact pair violates every log-power no-heavy target. Failure of this diagonal ledger is precisely a rate-bearing large pair packet and therefore returns to the already materialized terminal trident. After the diagonal and low-dimensional returns are removed, the off-diagonal exact-pair bilinear form is the clean Kuznetsov/DLS object. With the existing KZ and PDEC/model ledgers imported, the strict frontier is therefore reduced to*

```
AcyclicNCBLKActualBlockNonconcentrationOrStrengthenedSourceAntiAtom
OR (AcyclicSameSetScopeMatchForDirectPDECcapDualCertificate
AND HighSegmentModelGapAlpha043C3AnalyticLedger).
```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-strict-pair-energy-diagonal-
peeling-terminal-reduction-router.md.
```

The router decomposes the direct pair-energy large-sieve statement into a diagonal no-heavy ledger and a clean off-diagonal large-sieve ledger. The single-pair delta model proves that the diagonal ledger is a necessary input, not a consequence of a black-box large-sieve bound. If that ledger fails, the existing pair-energy packetization sends the object to the rate-bearing packet terminal split. If the diagonal ledger is peeled away, the remaining off-diagonal bilinear form is exactly the clean Kloosterman/Kuznetsov-DLS problem, whose self-contained frontier has already been reduced to acyclic NC-BLK/source anti-atom. The PDEC/CleanKLS branch is simultaneously split into same-set PDEC scope matching, while the explicit model ledger has reduced its finite part to a high-segment analytic margin. This removes the direct pair-energy atom as an independent disjunct, but it does not prove either NC-BLK/source anti-atom or the PDEC/model branch, nor the rate-preservation and D-structure/Rankin gates.  $\square$

**Theorem 7.52** (NCBLK/source anti-atom source-root frontier). *After the pair-energy diagonal has been peeled, the NCBLK/source anti-atom label is not the sharpest remaining self-contained target. The generic well-factorable anti-atom route is blocked by the moving-delta model. The actual acyclic route must first produce a forward pre-Cauchy source-root packet: a declaration line before Cauchy/payment, primitive signed rows, built-in word/coefficient pairing, actual source-domain entropy, fixed exact  $(u, v)$  fiber control, and a same formal-unit exact-pair no-heavy or L2 energy ledger. If this packet is not produced forward, the existing taxonomy routes the failure to the moving-block/global PDEC-sparse terminal. Therefore the current strict frontier is*



```
ForwardAcyclicPreCauchySourceRootPacketOrNamedReturn
OR (GlobalPDECOrSparseTerminalExclusion
    AND HighSegmentModelGapAlpha043C3AnalyticLedger)
OR (AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
    AND HighSegmentModelGapAlpha043C3AnalyticLedger).
```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-strict-ncblk-source-
antiatom-frontier-sync-router.md.
```

The router imports the pair-energy diagonal-peeling frontier, the strict acyclic NCBLK deduplication, the actual-source support seed split, the hypothetical zero-row seed no-go, the independent pre-Cauchy identity taxonomy, the moving-block return, and the source-declaration downstream split. These inputs show that NCBLK/source anti-atom remains open, but using the NCBLK name as the direct target now hides the first productive field: the forward source-root packet. The global PDEC/sparse terminal, the direct PDEC same-set scope, the high-segment model margin, rate preservation, and the D-structure/Rankin gate remain unproved. This is a frontier sharpening, not an unconditional row/column theorem.  $\square$

**Theorem 7.53** (Forward source-root terminal-cycle synchronization). *The forward source-root packet singled out by the NCBLK/source anti-atom frontier is not an independent non-circular proof route in the present corpus. When the common packet downstream, signed-lane cycle, new primitive source-rank convergence, cycle-cut/terminal descent, antisplit ExactUV atomization, pointwise primitive alpha/delta kernel table, alpha-row unsigned skeleton, signed-lift return, and anchor-collar overload return are imported, the documented source-root route returns to the terminal capacity gate*

```
PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve.
```

*Hence the sharpened strict frontier is*

```
(AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
OR NonCircularSelfContainedKuznetsovDLSLargeSieveWithoutNCBLKSourceRootReuse)
AND HighSegmentModelGapAlpha043C3AnalyticLedger,
```

*together with the still independent rate-preservation and D-structure/Rankin acceptance gates.*

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-strict-forward-source-root-
terminal-cycle-sync-router.md.
```

The router records the following status flags:

```
forward_source_root_packet_proved=false
forward_source_root_independent_after_router=false
row_column_unconditional_closed=false.
```

Its positive content is only that the currently documented source-root subroutes are exhausted as non-circular descents: the signed lane is a self-cycle, the new primitive and terminal descent meet at the same source-rank/pointwise-kernel front, the unsigned alpha skeleton is already the geometric part, and the signed lift and overload lines return to the terminal gate. Therefore a CleanKLS/Kuznetsov-DLS closure that reuses NCBLK/source-root would be circular. The remaining productive alternatives are a direct same-set PDEC cap dual certificate or a Kuznetsov-DLS large-sieve proof that does not reuse this source-root loop, plus the high-segment model ledger and the rate and D-structure gates. This is a cycle obstruction and frontier synchronization, not a proof of the row/column theorem.  $\square$

**Theorem 7.54** (Post-source-root PDEC scope saturation). *The direct PDEC arm left by the forward source-root synchronization is already saturated in the current internal corpus. Same-set PDEC protocol and the canonical-source capacity boundary may be used as tools, but the strict acyclic non-canonical branch still lacks a proof that its terminal certificate has the same formal unit, the same bad-window set, the same  $U\_CRT/L\_PDEC$  pushforward, and the same mass convention as the canonical same-set certificate. Following the branch through canonical-lock or ordinary Kuznetsov/DLS returns to scoped canonical absorption, mismatch/DLS, or the terminal-family cycle. Thus, unless a new PDEC scope certificate or an accepted external DIBFI no-projection input is supplied, the internal non-circular frontier becomes*

```
(NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
OR NonCircularSelfContainedKuznetsovDLSLargeSieveWithoutNCBLKSourceRootReuse)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-strict-post-source-root-pdec-scope-
saturation-sync-router.md.
```

It records

```
post_source_root_pdec_scope_active=true
pdec_scope_branch_saturated_in_current_internal_corpus=true
pdec_scope_proved=false
noncircular_kuznetsov_dls_without_source_root_reuse_proved=false
new_explicit_joint_constructor_formula_artifact_present=false
row_column_unconditional_closed=false.
```

The positive information is a routing exclusion: direct PDEC is not a proved non-circular exit inside the present material. The conditional PDEC line is kept as a possible new scope certificate, and the external DIBFI line is kept as an external input; neither is supplied here. The non-circular Kuznetsov/DLS arm is likewise open because the documented KZ route would otherwise reuse the NCBLK/source-root loop. Hence this step narrows the active internal frontier to a new explicit joint alpha/delta constructor or a genuinely non-circular KZ/DLS proof, with model, rate, and D-structure gates still open. It is not an unconditional row/column proof.  $\square$

**Theorem 7.55** (Post-PDEC new-joint noncycle synchronization). *The new explicit joint alpha/delta constructor line left by the post-PDEC scope audit is saturated in the current internal corpus. The documented joint/antisplit route only reaches exact atomic branch trace; branch trace only supplies visible coordinates; the missing signed coefficient and local factor require an atomic signed payload constructor; and the signed-payload line is already part of the signed-lane/source-rank/alpha terminal cycle. Thus the old internal new-joint route is not a non-circular exit. Unless a new joint/payload/PDEC/external input is supplied, the remaining internal non-circular frontier is*

```
NonCircularSelfContainedKuznetsovDLSLargeSieveWithoutNCBLKSourceRootReuse
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-strict-post-pdec-new-joint-
noncycle-sync-router.md.
```

It records

```

new_joint_current_internal_route_saturated=true
new_explicit_joint_constructor_formula_artifact_present=false
exact_atomic_branch_trace_formula_proved=false
atomic_signed_payload_constructor_proved=false
new_primitive_artifact_independent_present=false
noncircular_kuznetsov_dls_without_source_root_reuse_proved=false
row_column_unconditional_closed=false.

```

The positive information is a non-circularity audit. Finite CRT phase replication does not create orientation, local factor, or signed coefficient data. A new primitive would have to carry a pre-Cauchy actual source-rank and no-collapse package from outside the signed-lane cycle; no such package is supplied here. Hence this step deletes the old new-joint self-proof route and moves the direct attack to a Kuznetsov/DLS large-sieve proof that does not reuse NCBLK/source-root, with the high-segment model, rate preservation, and D-structure gates still open. It is not an unconditional row/column proof.  $\square$

**Theorem 7.56** (Post-new-joint KZ no-cycle gate). *The non-circular KZ/DLS gate left by the post-new-joint synchronization is not closed by the currently documented KZ route. The windowed DLS formal layer and the KZ-A through KZ-D spectral spine may be imported, but the documented KZ-E log-saving route factors through NCBLK/source-antiatom projection and then returns to the source-root or terminal cycle. Since the current gate explicitly requires no NCBLK/source-root reuse, that route cannot count as a non-circular proof. The remaining internal frontier is therefore*

```

AcyclicKZEWellFactorableDispersionLogSavingWithoutNCBLKProjection
AND HighSegmentModelGapAlpha043C3AnalyticLedge
AND RatePreservationLedge_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgeIndependentAcceptance.

```

*Status and reference.* The certificate is recorded in

```

docs/monograph/prime-matrix-strict-post-new-joint-kz-
nocycle-gate-sync-router.md.

```

It records

```

latest_kz_nocycle_gate_active=true
windowed_dls_formal_layer_imported=true
kz_abcd_spine_imported=true
existing_kz_e_route_factors_through_ncblk=true
ncblk_projection_forbidden_for_nocycle_gate=true
noncircular_kuznetsov_dls_without_source_root_reuse_proved=false
kz_e_direct_log_saving_without_ncblk_projection_proved=false
row_column_unconditional_closed=false.

```

The positive information is a gate refinement. KZ-A–KZ-D no longer hide the main obstruction; the obstruction is the direct KZ-E well-factorable dispersion saving. If the saving is obtained only by projecting to NCBLK/source anti-atoms, it reuses the cycle that the gate was designed to exclude. Thus the next direct attack is a KZ-E log-saving theorem that does not pass through NCBLK projection, or an explicitly external no-projection DI/BFI/Kuznetsov certificate. This is not an unconditional row/column proof.  $\square$

**Theorem 7.57** (Post-KZ-E direct source bridge). *The direct KZ-E no-projection gate is not a free generic WFD theorem. In the current corpus the generic self-contained WFD route is refuted by the moving-delta model, while the canonical-restricted route is closed only after the source is locked, before Cauchy and dispersion, to the canonical Rosser–Iwaniec/Buchstab decision-tree coefficient. Since noncanonical moving-atom or source-antiatom returns are terminal returns rather than a KZ no-cycle proof, the internal KZ-E direct gate is reduced to*

```

A1CleanBranchCanonicalSourceAdmission
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

*Status and reference.* The certificate is recorded in

```

docs/monograph/prime-matrix-strict-post-kze-direct-
source-bridge-sync-router.md.

```

It records

```

kze_direct_no_projection_gate_active=true
self_contained_taxonomy_imported=true
canonical_provenance_closed_only_for_canonical_branch=true
source_lock_scoped_not_global=true
actual_source_bridge_obstruction_imported=true
noncanonical_moving_atom_not_kz_nocycle_proof=true
kze_direct_current_internal_route_reduced_to_source_admission=true
a1_clean_branch_canonical_source_admission_proved=false
row_column_unconditional_closed=false.

```

The positive information is a source-bridge refinement. The canonical branch may use the internal support chain only after the actual clean A1 branch is admitted to the canonical source branch. The generic complement remains external or terminal; it cannot be silently upgraded, and a return through moving atoms would reuse the terminal route rather than prove KZ-E directly. Thus the next direct attack is source admission, with an external no-projection DI/BFI/Kuznetsov certificate retained only as a conditional input. This is not an unconditional row/column proof.  $\square$

**Theorem 7.58** (Post-source-admission macrocycle synchronization). *The A1 source-admission target left by the KZ-E direct source bridge does not supply a new non-circular descent in the present corpus. The canonical equality case is absorbed only as a scoped canonical-source case. The non-canonical mismatch case is routed through T1 mismatch forcing, signed-lift failure registration, alpha signed-weight downstream synchronization, the PDEC/CleanKLS terminal gate, new-joint saturation, and the KZ no-cycle gate; that route returns to the KZ-E source bridge and hence to A1 source admission. Thus the currently documented internal route forms the macrocycle*

```

A1CleanBranchCanonicalSourceAdmission
-> T1_classifier -> mismatch/signed_lift
-> PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve
-> new-joint_saturation -> non-circular_KZ/DLS
-> KZ-E_direct_source_bridge
-> A1CleanBranchCanonicalSourceAdmission.

```

*Consequently the remaining non-circular break basis is*

```

(AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
OR NewPrimitiveJointPayloadArtifactOutsideSignedLaneCycle
OR ExactExternalDIBFIKuznetsovNoProjectionCertificate_FOR_NONCIRCULAR_KZ_ONLY)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

*Status and reference.* The certificate is recorded in

```

docs/monograph/prime-matrix-strict-post-source-admission-
macrocycle-sync-router.md.

```

It records

```

post_kze_source_admission_active=true
a1_source_admission_scoped_not_global=true
t1_mismatch_no_silent_exit_imported=true
signed_lift_failure_return_only_registers=true
alpha_weight_downstream_returns_to_terminal=true
internal_pdec_clean_kls_cycle_imported=true
nonrecursive_breaker_hits_seed_cycle=true
new_joint_and_kz_return_to_a1_gate=true
a1_pdec_kz_macrocycle_detected=true
row_column_unconditional_closed=false.

```

The positive information is a non-circularity audit. It proves that the current A1/PDEC/KZ internal route returns to its starting gate and therefore cannot be counted as a proof of the row/column contradiction. It does not exclude the outside-cycle inputs listed in the theorem, nor does it close the high-segment model, rate-preservation, or D-structure gates.  $\square$

**Theorem 7.59** (Row-gap supply and phase cycle-cut audit). *Fix an odd prime  $P$  and  $1 \leq k < P$ , and consider the row interval*

$$I_k = \{kP + a : 1 \leq a < P\}.$$

*If this interval contains no prime, then each element of  $I_k$  has a prime divisor  $q < P$ . The correct uniform root split, however, is*

```

low-root primes:  q <= floor(sqrt(kP)),
near-root primes: floor(sqrt(kP)) < q <= floor(sqrt(kP+P-1)).

```

*Thus it is too strong to assert that the primes below  $\sqrt{kP}$  alone must cover the row. The zero-row obligation is instead the exact finite covering condition that the low-root residue classes together with the near-root slots cover every column  $a = 1, \dots, P-1$ . Raw covering capacity with multiplicity is not a contradiction; the active non-circular inputs are therefore a uniform low-root sifted-residue deficit after near-root slots, or an adjacent-prime  $Q_1, Q_2$  CRT transport defect forcing a stable short same-label return. The synchronized basis is*

```

(UniformLowRootSiftedResidueDeficitAfterNearRootSlots
OR AdjacentPrimeQ1Q2CRTTransportDefectOrStableShortReturn
OR AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
OR ExactExternalPrimeGapSqrtBarrierCertificate_FOR_ROW_GAP_ONLY)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-row-gap-supply-phase-cycle-cut-router.md.
```

It records

```

row_gap_model_pinned=true
correct_divisor_supply_law=true
low_root_only_claim_rejected=true
capacity_only_contradiction_rejected=true
crt_period_mirror_not_contradiction=true
low_root_deficit_after_near_root_slots_proved=false
q1q2_transport_defect_or_stable_short_return_proved=false
row_column_unconditional_closed=false.

```

The positive information is the correction of the supply law and the separation of two possible non-circular inputs. If  $Q_1 < kP$  and  $Q_2 > (k+1)P$  are adjacent primes around the putative gap, CRT mirror symmetry by itself only gives a mirror covered block. It becomes a contradiction only after one proves a stable same-formal-unit same-label short return, or proves that all instabilities enter registered PDEC/SAE/ColumnCRT defects. Those assertions are not proved here.  $\square$

**Theorem 7.60** (Low-root sifted deficit frontier). *The row-gap low-root deficit target is equivalent to a sharp short-interval rough-residue lower bound. The near-root contribution is confined to*

$$\sqrt{kP} < q \leq \sqrt{kP + P - 1},$$

*and each such prime contributes only one residue class in the row. Hence the near-root side has an elementary capacity envelope. The open point is not that envelope, but the following non-circular lower bound:*

*NonCircularShortIntervalRoughResidueLowerBoundLengthPLevelSqrtkP.*

*Mertens density gives only a heuristic for this lower bound; it cannot be substituted for a pointwise row estimate. The current synchronized basis is*

```
(NonCircularShortIntervalRoughResidueLowerBoundLengthPLevelSqrtkP
OR AdjacentPrimeQ1Q2CRTTransportDefectOrStableShortReturn
OR AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

*Status and reference.* The certificate is recorded in

`docs/monograph/prime-matrix-lowroot-sifted-deficit-frontier-router.md.`

It records

```
exact_deficit_identity_closed=true
near_root_slot_upper_bound_elementary=true
mertens_heuristic_not_proof=true
jacobsthal_barrier_identified=true
noncircular_short_interval_rough_residue_lower_bound_proved=false
row_column_unconditional_closed=false.
```

The exact identity is that, inside  $I_k$ , a slot not covered by any prime up to  $\sqrt{kP + P - 1}$  must itself be prime. Thus a positive uncovered slot is precisely the desired row prime. This identity explains why the remaining rough-residue lower bound is not a harmless auxiliary estimate: it is the hard row-prime content in sieve form. This step therefore narrows the frontier but does not close it.  $\square$

**Theorem 7.61** (Short-interval rough-residue barrier). *The direct target*

*NonCircularShortIntervalRoughResidueLowerBoundLengthPLevelSqrtkP*

*is not an independent internal lower-sieve input in the self-contained row/column proof. For  $I_k = \{kP + a : 1 \leq a < P\}$ ,  $1 \leq k < P$ , a slot uncovered by all primes  $q \leq \sqrt{kP + P - 1}$  is necessarily prime, since  $kP + a < P^2$ . Conversely, a prime slot is uncovered by every such  $q$ . After subtracting the near-root-only slots*

$$\sqrt{kP} < q \leq \sqrt{kP + P - 1},$$

*the same equivalence remains: the positive deficit is exactly a row prime. Thus a proof of this direct rough-residue lower bound is already a sqrt-length prime-gap proof for the row. It may be registered as an external strong input, but it cannot be used as a non-circular internal black box. The current internal non-circular basis is therefore*

```
(AdjacentPrimeQ1Q2CRTTransportDefectOrStableShortReturn
OR AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

*Status and reference.* The certificate is recorded in

docs/monograph/prime-matrix-short-interval-rough-residue-barrier-router.md.

It records

```
full_root_rough_equals_prime_in_row=true
near_root_subtraction_keeps_equivalence=true
mertens_average_insufficient=true
periodwide_jacobsthal_shortcut_rejected=true
noncircular_internal_use_rejected=true
noncircular_short_interval_rough_residue_lower_bound_proved=false
row_column_unconditional_closed=false.
```

Mertens density, finite samples, and period-wide Jacobsthal intuition do not give the required point-wise row estimate. The next internal attack is the adjacent-prime  $Q_1, Q_2$  CRT transport defect, with seed cycle-cut and same-set PDEC scope matching kept as parallel non-circular routes. This is a barrier classification, not a final unconditional closure.  $\square$

**Theorem 7.62** ( $Q_1/Q_2$  transport latest non-circular synchronization). *The adjacent-prime  $Q_1, Q_2$  transport branch is not an independent pure-CRT terminal for the row/column proof. The existing  $Q_2$ -stage ladder shows that endpoint-stable replay is impossible: in the full  $Q_2$ -wheel, the two endpoint copies are respectively divisible by  $Q_1$  and  $Q_2$ . A persistent closed carrier block without prime endpoints is a ColumnCRT/PDEC formal unit; a controlled fresh endpoint tail is SAE; and an uncontrolled aperture explosion or support motion must be submitted as an explicit moving-family/PDEC schema. Hence the non-circular residue of the  $Q_1/Q_2$  branch is exact-source dispersion, external spectral input, or the already registered seed/PDEC/new-joint strict frontier:*

```
((NonterminalExactUVFiberAperiodicityEstimateForActualPreCauchySource
OR ExternalDIBFIKuznetsouDispersionTheoremMatch)
OR AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
OR AcyclicSameSetScopeMatchForDirectPDECcapDualCertificate)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

*If the existing seed/PDEC saturation synchronizers are imported, the strict internal branch is further reduced to*

```
NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

*Status and reference.* The certificate is recorded in

docs/monograph/prime-matrix-q1q2-transport-latest-noncycle-sync-router.md.

It records

```
q2_endpoint_stable_replay_impossible=true
persistent_closed_carrier_routes_to_columncrt_pdec=true
controlled_fresh_endpoint_tail_routes_to_sae=true
unnamed_aperture_explosion_forbidden=true
pure_crt_global_phase_contradiction_blocked=true
q1q2_branch_reduced_to_exact_source_or_external=true
q1q2_transport_defect_or_stable_short_return_proved_as_global_contradiction=false
row_column_unconditional_closed=false.
```

The stable-replay CRT contradiction is therefore closed only for that subcase. The proof still lacks a theorem forcing the global counterexample chain into an exact-source nonconcentration contradiction, a new joint constructor, or an accepted external dispersion theorem. Thus the row/column theorem remains open.  $\square$

**Theorem 7.63** (Exact-UV fiber latest non-circular synchronization). *The target*

*NonterminalExactUVFiberAperiodicityEstimateForActualPreCauchySource*

*is not a pure CRT position or phase-rigidity problem. It is an actual-source rank/no-collapse problem inside one formal unit. The formal implication now recorded is*

*ActualPreCauchySourceDomainAbsoluteEntropyLedger  
AND RegisteredCompletePrimitiveEmitterKeyPartitionPolylogLedger  
AND FixedKeyExactUVLocalMultiplicity01Ledger  
implies NonterminalExactUVFiberAperiodicityEstimateForActualPreCauchySource.*

*After atomization, the self-contained internal basis is*

*((AcyclicSeedPrimitiveRowSignedCoefficientLawBeforePushforward  
AND SameFormalUnitPrimitiveRowMassNormalizationAndNoHeavyRowLedger  
AND PrimitiveRowSupportLowerBoundBeforeExactUVProjectionLedger)  
AND (ActualNoncanonicalPrimitiveEmitterSourceTableLedger  
AND CompleteEmitterTraceKeyBudgetLedger  
AND SignLocalFactorRefinementNoCancellationLedger  
AND OverBudgetOrUnregisteredReturnLedger)  
AND FixedKeyExactUVLocalMultiplicity01Ledger)  
AND HighSegmentModelGapAlpha043C3AnalyticLedger  
AND RatePreservationLedger\_FOR\_moving\_atom\_packet  
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.*

*The next direct attack is*

*AcyclicSeedPrimitiveRowSignedCoefficientLawBeforePushforward,*

*with the actual source table, fixed-key local multiplicity, new joint formula, and external dispersion route kept in parallel.*

*Status and reference.* The certificate is recorded in

`docs/monograph/prime-matrix-exactuv-fiber-latest-noncycle-sync-router.md.`

It records

```
preterminal_fiber_atomization_imported=true
deterministic_source_atom_implication_closed=true
source_entropy_reduced_to_signed_rows=true
complete_key_reduced_to_actual_source_table=true
fixed_pair_fiber_formal_inequality_closed=true
map_rank_equivalent_to_bounded_incidence=true
nonterminal_exactuv_fiber_aperiodicity_proved=false
row_column_unconditional_closed=false.
```

Thus this step closes only the deterministic source-atom implication and the misleading pure-CRT exit. It does not prove the source entropy, the complete key partition, fixed-key local multiplicity, the model tail, rate preservation, or D-structure/Rankin promotion. The row/column theorem remains open.  $\square$

**Theorem 7.64** (Inverse-alignment  $P+1$  row reduction check). *In the row multiplier notation used in this manuscript, the row  $x = P$  is the one-indexed  $P+1$  row and is the square-anchored interval*

$$P^2 + r, \quad 1 \leq r < P.$$

*The older inverse-alignment files already prove the equivalence*

*NoZeroRowAtXEqualsP.PlusOneRowAfterSquare  
if and only if PrimeInFirstHalfAfterPrimeSquareForEveryPrimeP.*



They also prove that  $X_0(P) > P$  is equivalent to the absence of zero rows for all  $1 \leq x \leq P$ . However, the old files do not prove that the single assertion  $x = P$  is nonzero implies that every  $1 \leq x < P$  is nonzero. Such an implication would require a separate non-circular transfer:

```
EarlyZero( $x < P$ )
implies ZeroAtXEqualsP OR NamedReturn(PDEC/SAE/ColumnCRT/source-rank).
```

Thus the new branch target is

```
AcyclicEarlyZeroToSquareAnchorPhaseTransferOrNamedReturn,
```

while the exact-UV source-rank frontier still keeps

```
AcyclicSeedPrimitiveRowSignedCoefficientLawBeforePushforward
```

as a parallel direct attack.

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-inverse-alignment-pplus1-row-reduction-check-router.md.
```

It records

```
old_minrep_equivalence_closed=true
pplus1_row_is_x_equals_p=true
x_equals_p_no_cover_equivalent_to_first_half_prime_square=true
pplus1_nonzero_suffices_for_all_early_rows_proved=false
row_column_unconditional_closed=false.
```

The obstruction is phase-theoretic: different early multipliers  $x$  give different covering phases  $\rho_q(x) = -xP \bmod q$ . The square anchor  $x = P$  is one special phase line and does not automatically dominate the whole family  $1 \leq x < P$ . Therefore this check sharpens the old route but does not close the row/column theorem.  $\square$

**Theorem 7.65** (Early-to-square phase-transfer split). *The abstract transfer target from an early zero row to the square anchor is not kept as an unnamed obligation. Suppose that an early zero row  $x_0 < P$  exists. Moving from  $x_0$  to  $P$ , either every row remains zero up to  $x = P$ , in which case the square-anchor row is itself zero, or there is a first break row  $y \leq P$ . At that row  $y - 1$  is still zero and  $y$  has a nonempty released-column set. The local phase motion is*

$$\rho_q(t) = -tP \pmod{q}, \quad \rho_q(t+1) = \rho_q(t) - P \pmod{q}.$$

*Thus the first break is a unit phase-slip boundary defect. Under the previous early-zero phase-defect schema it is a named PDEC/SAE/ColumnCRT/LocalSurvivor return, not a fourth exit. Under the additional condition that  $x = P$  is not a zero row, the transfer branch is reduced to*

```
FirstBreakPhaseSlipNamedReturnExclusion.
```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-early-to-square-phase-transfer-split-router.md.
```

It records

```
contiguous_zero_block_dichotomy_closed=true
persist_to_square_implies_square_zero=true
first_break_release_set_nonempty=true
unit_phase_slip_formula_closed=true
phase_slip_schema_admission_imported=true
transfer_proved_as_contradiction=false
row_column_unconditional_closed=false.
```

This closes the unnamed-transfer ambiguity, but it does not exclude the first-break named terminal. The row/column theorem remains open until that terminal is excluded, or the parallel signed-source/source-rank frontier is closed.  $\square$

**Theorem 7.66** (First-break phase-slip LCM width barrier). *Let the first break from an early zero block occur at row  $y \leq P$ , so that row  $y - 1$  is still zero and row  $y$  has a nonempty released-column set. Fix a released formal unit and let  $\Lambda$  be its active carrier prime labels. If the same released phase pattern reappears at row  $y + d$  with the same carrier labels, then for every  $q \in \Lambda$*

$$\rho_q(y + d) = \rho_q(y), \quad \rho_q(t) = -tP \pmod{q},$$

and hence  $dP \equiv 0 \pmod{q}$ . Since  $q < P$  is prime and  $q \neq P$ , this is equivalent to  $q \mid d$ . Therefore

$$\text{lcm}(\Lambda) \mid d.$$

The remaining row support before the square anchor has width  $H = P - y$ . Thus if  $\text{lcm}(\Lambda) > H$ , no nonzero fixed-carrier replay can occur before the square anchor. If  $\text{lcm}(\Lambda) \leq H$ , the replay is a small-LCM fixed-residue event and must be routed to ColumnCRT/PDEC. If the carrier labels move, the branch is a moving-carrier phase-slip PDEC/SAE return. Consequently the previous first-break terminal is reduced to

```
SmallLCMColumnCRTPDECExclusion
AND NonreplaySparseFirstBreakSAESummability
AND MovingCarrierPhaseSlipPDECExclusion.
```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-phase-slip-lcm-barrier-router.md.
```

It records

```
fixed_carrier_lcm_replay_period_closed=true
square_support_width_sharpened_to_p_minus_y=true
lcm_exceeds_width_no_fixed_replay=true
firstbreak_phase_slip_named_return_exclusion_proved=false
row_column_unconditional_closed=false.
```

This is a non-circular sharpening of the first-break named terminal: it uses only the forward phase motion from row  $y$  to the square anchor. It does not exclude the small-LCM ColumnCRT/PDEC branch, the nonreplay sparse SAE branch, or the moving-carrier PDEC branch. The inverse-alignment return therefore becomes

```
NoZeroRowAtXEqualsP_PlusOneRowAfterSquare
AND SmallLCMColumnCRTPDECExclusion
AND NonreplaySparseFirstBreakSAESummability
AND MovingCarrierPhaseSlipPDECExclusion.
```

The row/column theorem remains open.  $\square$

**Theorem 7.67** (First-break small-LCM rank-pressure compression). *In the fixed small-LCM branch of the first-break phase-slip return, the replay width is positive,  $H = P - y \geq 1$ . If  $y = P$ , there is no nonzero replay row before the square anchor and the branch returns to the square-anchor/SAE boundary. Let  $\Lambda$  be the active carrier-prime label set. Since these labels are distinct primes,*

$$\text{lcm}(\Lambda) = \prod_{q \in \Lambda} q \leq H.$$

Consequently for every threshold  $B > 1$ ,

$$|\{q \in \Lambda : q > B\}| \leq \left\lfloor \frac{\log H}{\log B} \right\rfloor.$$

In particular, two carriers  $q > \sqrt{H}$  cannot occur in the same fixed small-LCM formal unit. Therefore a small-LCM first-break branch cannot be a free superposition of many medium/high carrier phases. Any remaining pressure must be routed to one of three lower-level exits:

*LowCarrierFixedResidueColumnCRTPDECEExclusion*  
*AND LowCarrierNonpersistentSparseSAESummability*  
*AND HighCarrierRankDeficitCapacityBoundOrSingletonSAE.*

*Status and reference.* The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-small-lcm-rank-pressure-router.md.`

It records

`distinct_prime_product_law_closed=true`  
`small_lcm_rank_pressure_closed=true`  
`two_large_carrier_sqrt_barrier_closed=true`  
`small_lcm_branch_excluded=false`  
`row_column_unconditional_closed=false.`

This closes only the deterministic rank-pressure structure of the small-LCM branch. It does not exclude persistent low-carrier fixed residues, nonpersistent sparse SAE mass, or the high-carrier rank-deficit capacity obligation. The inverse-alignment return is updated to

`NoZeroRowAtXEqualsP_PlusOneRowAfterSquare`  
*AND LowCarrierFixedResidueColumnCRTPDECEExclusion*  
*AND LowCarrierNonpersistentSparseSAESummability*  
*AND HighCarrierRankDeficitCapacityBoundOrSingletonSAE*  
*AND NonreplaySparseFirstBreakSAESummability*  
*AND MovingCarrierPhaseSlipPDECEExclusion.*

The row/column theorem remains open. □

**Theorem 7.68** (First-break low-carrier fixed-residue AP skeleton). *In the low-carrier fixed-residue branch, fix a carrier prime  $q < P$  and a column residue  $a \bmod q$ . On a row  $t$ , the covering condition is*

$$tP + c \equiv 0 \pmod{q}, \quad c \equiv a \pmod{q}.$$

*Since  $P$  is invertible modulo  $q$ , this is equivalent to the single row class*

$$t \equiv -aP^{-1} \pmod{q}.$$

*Therefore in the post-break row interval  $I_y = \{y, \dots, P-1\}$ , whose length is  $H = P - y$ , one fixed residue cell  $(q, a)$  can occupy at most*

$$\lceil H/q \rceil$$

*row positions. With a fixed low-carrier threshold  $B$ , the number of available residue cells is at most  $\sum_{q \leq B} q$ . Thus the low-carrier fixed-residue branch is no longer an unstructured ColumnCRT label; it is a finite AP table. The branch is reduced to*

*LowCarrierResidueAPEnvelopeCapacityComparison*  
*AND DenseLowCarrierResidueTablePDECEExclusion*  
*AND SparseLowCarrierResidueCellSAESummability.*

*Status and reference.* The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-low-carrier-residue-ap-router.md.

It records

```
residue_to_row_ap_formula_closed=true
single_residue_ap_envelope_closed=true
low_carrier_residue_table_finite_closed=true
ap_envelope_capacity_comparison_proved=false
low_carrier_fixed_residue_excluded=false
row_column_unconditional_closed=false.
```

This step supplies the exact AP skeleton and the terminal routing. It does not prove that the required counterexample pressure exceeds the AP envelope, nor does it exclude the dense low-carrier residue-table PDEC or the sparse cell SAE branch. The row/column theorem remains open.  $\square$

**Theorem 7.69** (First-break low-carrier AP exact envelope). *Let  $T \subset \{(q, a) : q \leq B, a \bmod q\}$  be a low-carrier residue table and let  $I_y = \{y, \dots, P-1\}$  have length  $H = P - y$ . Define*

$$U_T(I_y) = \sum_{(q,a) \in T} \#\{t \in I_y : t \equiv -aP^{-1} \pmod{q}\}.$$

*Then every selected cell satisfies*

$$\#\{t \in I_y : t \equiv -aP^{-1} \pmod{q}\} \leq \lceil H/q \rceil.$$

*Moreover, for each fixed  $q$ ,*

$$\sum_{a \bmod q} \#\{t \in I_y : t \equiv -aP^{-1} \pmod{q}\} = H.$$

*Consequently*

$$U_T(I_y) \leq \sum_{(q,a) \in T} \lceil H/q \rceil, \quad U_T(I_y) \leq H \cdot \#\{q : \exists a (q, a) \in T\}.$$

*Thus the formal supply side of `LowCarrierResidueAPEnvelopeCapacityComparison` is exact, and the remaining branch is reduced to*

*ActualLowCarrierRowIncidenceDemandLowerBound  
AND LowCarrierAPEnvelopeStrictGapOrDenseTablePDEC.*

*Status and reference.* The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-low-carrier-ap-envelope-router.md.

It records

```
cell_count_formula_closed=true
single_cell_sharp_bound_closed=true
all_residues_exact_mass_closed=true
selected_table_envelope_closed=true
ap_capacity_comparison_proved=false
row_column_unconditional_closed=false.
```

This closes only the exact AP envelope. It does not prove the actual row-incidence demand lower bound, and it does not exclude the dense-table PDEC alternative. The row/column theorem remains open.  $\square$

**Theorem 7.70** (First-break actual-demand source cut). *In the first-break branch, the imported release-set statement gives only a unit source-side demand:*

$$R_y \neq \emptyset \implies D_y \geq 1.$$

*This is a genuine actual lower bound, but it is not enough to beat the AP envelope, since a single low-carrier residue cell can already support one incidence whenever it meets  $I_y$ . Therefore the gate `ActualLowCarrierRowIncidenceDemandLowerBound` cannot be closed by the nonemptiness of  $R_y$  alone. It is reduced to*

*FirstBreakReleaseMassAmplificationOrSingletonSAE  
AND LowCarrierActualPaymentInjectionWithoutEnvelopeReuse.*

*Status and reference.* The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-actual-demand-source-cut-router.md.`

It records

```
firstbreak_release_set_imported=true
unit_release_demand_lower_bound_closed=true
unit_demand_not_gap_sufficient=true
no_envelope_recycling_guard=true
release_mass_amplification_proved=false
low_carrier_payment_injection_proved=false
actual_low_carrier_row_incidence_demand_proved=false
row_column_unconditional_closed=false.
```

The demand lower bound must be proved from the zero-row/first-break source side, not by recycling AP-envelope saturation. If source mass does not amplify, the event is a singleton or sparse SAE; if amplified mass cannot be injected into the low-carrier AP table, it returns to the high-rank, moving-carrier, PDEC, or SAE exits. The row/column theorem remains open.  $\square$

**Theorem 7.71** (First-break release mass zero-block ladder). *Assume the square-anchor branch does not take over and let  $y$  be the first nonzero row after an early zero row  $x_0 < P$ . Then  $[x_0, y - 1]$  is a zero-row block of length  $L = y - x_0$ . For every  $t \in [x_0, y - 1]$  and every  $1 \leq c < P$ , there is a carrier  $q < P$  with*

$$q \mid tP + c.$$

*Hence the source-side cover-obligation mass in the block is exactly*

$$L(P - 1).$$

*The boundary release at row  $y$  only gives a unit event; any amplification must therefore come from this zero-block cover ledger. For any threshold  $A \geq 1$ , the branch is reduced to the two obligations*

*ShortZeroBlockSingletonSAESummability  
AND LongZeroBlockCoverMassTransferToAPDemandOrPDEC.*

*Status and reference.* The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-release-mass-zero-block-ladder-router.md.`

It records

```
contiguous_zero_block_imported=true
boundary_only_amplification_blocked=true
zero_block_cover_obligation_mass_closed=true
block_length_dichotomy_closed=true
short_zero_block_singleton_sae_proved=false
long_zero_block_mass_transfer_proved=false
release_mass_amplification_proved=false
row_column_unconditional_closed=false.
```

This step localizes the only possible amplification source. It does not prove short-block SAE summability, and it does not prove that long-block cover obligations transfer noncircularly into post-break AP demand; failed transfer is routed to persistent cover-history PDEC/ColumnCRT/SAE. The row/column theorem remains open.  $\square$

**Theorem 7.72** (Long zero-block cover-mass transfer routing). *Assume the long zero-block branch of the first-break ladder. Let  $B = [x_0, y - 1]$  and let*

$$O_B = \{(t, c) : x_0 \leq t < y, 1 \leq c < P\}.$$

*Then  $|O_B| = L(P - 1)$ , where  $L = y - x_0$ . Each cover obligation  $(t, c) \in O_B$  admits a carrier  $q < P$  with  $q \mid tP + c$ , hence a history key*

$$\kappa = (q, a), \quad a \equiv c \pmod{q}, \quad t \equiv -aP^{-1} \pmod{q}.$$

*Importing the no-loss return accounting ledger, these obligations cannot disappear into an unnamed sink. They are routed either to a stable low-carrier/residue payment table, or to a named history-switch PDEC/ColumnCRT/MovingCarrier/SAE return. Therefore*

```
LongZeroBlockCoverMassTransferToAPDemandOrPDEC
→ ZeroBlockHistoryProjectionNoLossLedger
AND StableLowCarrierPaymentTableOrHistorySwitchPDEC
AND PostBreakAPDemandInjectionFromStableHistory.
```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-long-zero-block-mass-transfer-router.md.
```

It records

```
zero_block_cover_obligation_mass_imported=true
history_projection_key_defined=true
no_loss_return_accounting_imported=true
stable_table_or_named_switch_route_registered=true
history_switch_pdec_excluded=false
postbreak_ap_demand_injection_proved=false
long_zero_block_mass_transfer_proved=false
row_column_unconditional_closed=false.
```

This step closes only the unnamed-escape route for long zero-block mass. It does not prove that stable history injects into post-break actual demand, and it does not exclude the named history-switch terminals. The row/column theorem remains open.  $\square$

**Theorem 7.73** (Stable history AP-arrival dichotomy). *Assume the stable low-carrier/residue history branch. For a history key  $\kappa = (q, a)$ , the associated rows are exactly*

$$t \equiv -aP^{-1} \pmod{q}.$$

*Let  $t_*$  be the last row in the zero block  $B = [x_0, y - 1]$  carrying this same key. Then the next row with the same key is  $t_* + q$ . By maximality of  $t_*$ , one has  $t_* + q \geq y$ . Hence, in the post-break support  $I_y = [y, P - 1]$ , the exact dichotomy is*

$$t_* + q \in I_y \iff q \leq P - 1 - t_*, \quad t_* + q \geq P \iff q \geq P - t_*.$$

*Consequently*

*PostBreakAPDemandInjectionFromStableHistory*  
 $\rightarrow$  *StableHistoryAPSuccessorDichotomyLedger*  
*AND ArrivalCandidateActualDemandInjectionWithoutEnvelopeReuse*  
*AND TerminalNonarrivalLargeStepEscapePDECOrSAE.*

*Status and reference.* The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-stable-history-ap-arrival-router.md.

It records

```
stable_history_route_imported=true
ap_row_class_formula_closed=true
last_prebreak_hit_successor_closed=true
arrival_nonarrival_dichotomy_closed=true
terminal_nonarrival_large_step_registered=true
arrival_candidate_registered=true
arrival_candidate_actual_demand_proved=false
terminal_nonarrival_excluded=false
postbreak_ap_demand_injection_proved=false
row_column_unconditional_closed=false.
```

This step proves only the AP successor and arrival/nonarrival dichotomy. An arrival is still only a candidate for actual demand, and a terminal nonarrival remains a named large-step PDEC/SAE exit. The row/column theorem remains open.  $\square$

**Theorem 7.74** (Arrival candidate unit incidence and quotient discipline). *Assume an arrival candidate with source history tag  $(q, a)$  and*

$$y \leq t_{\text{next}} \leq P - 1, \quad t_{\text{next}} \equiv -aP^{-1} \pmod{q}.$$

*Then it gives one source-tagged low-carrier AP row-incidence*

$$(source\ tag, t_{\text{next}}, q, a) \mapsto incidence(t_{\text{next}}, q, a),$$

*without using any AP-envelope saturation. However, raw source-tagged arrivals are not an aggregate demand lower bound. They must be quotiented by*

$$\pi : source\text{-tagged arrivals} \longrightarrow (t_{\text{next}}, q, a),$$

*and the usable actual demand is the number of distinct incidences in  $\text{im } \pi$ . Multiple sources mapping to the same incidence are not deleted; they are quotient, weighted, duplicate/collision, PDEC, ColumnCRT, or SAE returns. Consequently*

*ArrivalCandidateActualDemandInjectionWithoutEnvelopeReuse*  
 $\rightarrow$  *SourceTaggedArrivalUnitIncidenceLedger*  
*AND DistinctArrivalQuotientDemandLowerBoundOrCollisionPDEC*  
*AND ArrivalCollisionOrDuplicatePaymentReturnLedger.*

*Status and reference.* The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-arrival-candidate-actual-demand-router.md.

It records

```
arrival_candidate_imported=true
no_envelope_recycling_guard_imported=true
source_tagged_unit_incidence_closed=true
arrival_quotient_map_defined=true
no_loss_collision_return_imported=true
arrival_collision_return_registered=true
distinct_arrival_quotient_lower_bound_proved=false
arrival_collision_return_excluded=false
arrival_candidate_actual_demand_injection_proved=false
row_column_unconditional_closed=false.
```

This step closes the unit source-tagged incidence injection and the quotient discipline only. It does not prove a large distinct-arrival demand lower bound, and it does not exclude duplicate/collision returns. The row/column theorem remains open.  $\square$

**Theorem 7.75** (Arrival quotient fiber envelope). *Let  $A$  be the set of source-tagged arrivals and*

$$\pi : A \longrightarrow (t_{\text{next}}, q, a)$$

*be the arrival quotient map. Fix an image point  $(t, q, a)$ . Any preimage source row  $s$  must lie in the zero block  $B = [x_0, y - 1]$  and satisfy*

$$s \equiv t \pmod{q},$$

*so the row factor is at most  $\lceil L/q \rceil$ . Any source column  $c$  must satisfy  $1 \leq c < P$  and  $c \equiv a \pmod{q}$ , so the column factor is at most  $\lceil (P-1)/q \rceil$ . Therefore*

$$|\pi^{-1}(t, q, a)| \leq \left\lceil \frac{L}{q} \right\rceil \left\lceil \frac{P-1}{q} \right\rceil.$$

*Consequently, with*

$$F(t, q, a) = \left\lceil \frac{L}{q} \right\rceil \left\lceil \frac{P-1}{q} \right\rceil,$$

*one has the weighted image lower bound*

$$|\text{im } \pi| \geq \sum_{u \in A} \frac{1}{F(\pi(u))}.$$

*Thus*

```
DistinctArrivalQuotientDemandLowerBoundOrCollisionPDEC
→ ArrivalQuotientFiberMultiplicityEnvelopeLedger
AND ArrivalRawSourceMassAfterNonarrivalRemoval
AND WeightedArrivalImageLowerBoundFromFiberEnvelope
AND HighFiberArrivalCollisionPDECOrDenseLowCarrierReturn.
```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-arrival-quotient-fiber-router.md.
```

It records

```
source_tagged_unit_incidence_imported=true
arrival_quotient_map_imported=true
fiber_row_factor_closed=true
fiber_column_factor_closed=true
arrival_fiber_envelope_closed=true
weighted_image_lower_bound_formula_closed=true
raw_arrival_mass_after_nonarrival_removal_proved=false
high_fiber_concentration_registered=true
high_fiber_collision_pdec_excluded=false
distinct_arrival_quotient_lower_bound_proved=false
row_column_unconditional_closed=false.
```

This step closes the fiber envelope and the weighted image lower-bound formula only. It does not prove enough raw arrival mass after terminal nonarrival is removed, and it does not exclude high-fiber collision or dense low-carrier returns. The row/column theorem remains open.  $\square$



**Theorem 7.76** (Arrival raw-mass layer balance). *Let  $S$  be the stable source-tagged history layer after no-loss projection and named history-switch removal. For  $u \in S$ , let  $q(u)$  be its carrier and let  $t_*(u) \leq y - 1$  be the last row in the zero block carrying the same history key. The AP successor dichotomy gives*

$$t_{\text{next}}(u) = t_*(u) + q(u),$$

and partitions

$$S = A \sqcup E,$$

where  $A$  is the source-tagged arrival layer and  $E$  is the terminal nonarrival layer. Put  $H = P - y$ . If  $q(u) \leq H$ , then

$$t_*(u) + q(u) \leq y - 1 + H = P - 1,$$

so  $u \in A$ . If  $u \in E$ , then

$$q(u) \geq P - t_*(u) \geq P - (y - 1) = H + 1.$$

Consequently terminal nonarrival removes only the large-step tail  $q > H$ , and

$$|A| \geq |S_{\leq H}|, \quad S_{\leq H} = \{u \in S : q(u) \leq H\}.$$

Thus

```
ArrivalRawSourceMassAfterNonarrivalRemoval
→ ArrivalNonarrivalSourceLayerBalanceLedger
AND LowStepStableHistoryAlwaysArrivesLedger
AND RawArrivalMassLowerBoundFromLowStepHistory
AND LowStepStableHistoryMassLowerBoundOrLargeStepTailEscapePDEC.
```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-arrival-raw-mass-layer-router.md.
```

It records

```
raw_arrival_mass_imported=true
ap_successor_dichotomy_imported=true
source_layer_universe_defined=true
arrival_nonarrival_source_layer_balance_closed=true
low_step_always_arrives_closed=true
terminal_escape_tail_cutoff_closed=true
raw_arrival_lower_bound_formula_closed=true
low_step_stable_history_mass_lower_bound_proved=false
large_step_tail_escape_registered=true
large_step_tail_escape_excluded=false
raw_arrival_mass_after_nonarrival_removal_proved=false
row_column_unconditional_closed=false.
```

This step closes the exact low-step cutoff and layer-balance formula only. It does not prove that  $S_{\leq H}$  has enough mass, and it does not exclude large-step tail escape. The row/column theorem remains open.  $\square$

**Theorem 7.77** (Low-step/tail capacity envelope). *Let  $B = [x_0, y - 1]$ ,  $L = y - x_0$ , and  $H = P - y$ . In the stable history layer, write*

$$S = S_{\leq H} \sqcup S_{> H}, \quad S_{\leq H} = \{u \in S : q(u) \leq H\}.$$

*The preceding layer shows that  $S_{\leq H}$  is contained in the arrival layer. For  $u \in S_{> H}$  to be terminal nonarrival, its last zero-block hit  $t_*(u)$  must satisfy*

$$t_*(u) + q(u) \geq P, \quad t_*(u) \in B.$$

Hence, for a fixed  $q > H$ ,

$$t_*(u) \in T_q := B \cap [P - q, y - 1], \quad |T_q| \leq \min(L, q - H).$$

For each fixed  $q$  and  $t \in T_q$ , the source column must satisfy

$$c \equiv -tP \pmod{q}, \quad 1 \leq c < P,$$

so there are at most  $\lceil (P-1)/q \rceil$  such columns. Therefore the non-duplicated large-step tail satisfies the explicit envelope

$$|S_{>H}| \leq C_{\text{tail}} := \sum_{H < q < P} |B \cap [P - q, y - 1]| \left\lceil \frac{P-1}{q} \right\rceil,$$

and consequently, outside duplicate or tail-saturation returns,

$$|S_{\leq H}| \geq |S| - C_{\text{tail}}.$$

Thus

```
LowStepStableHistoryMassLowerBoundOrLargeStepTailEscapePDEC
→ StableHistoryLowTailMassPartitionLedger
AND LargeStepTailTerminalWindowEnvelopeLedger
AND LowStepStableMassLowerBoundFromTotalMinusTailEnvelope
AND StableTotalMinusTailEnvelopeGapOrLargeStepTailSaturationPDEC.
```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-lowstep-tail-capacity-router.md.
```

It records

```
lowstep_mass_imported=true
raw_layer_balance_imported=true
stable_low_tail_partition_closed=true
terminal_tail_row_window_closed=true
tail_column_multiplicity_closed=true
large_step_tail_envelope_closed=true
lowstep_mass_from_total_minus_tail_closed=true
large_step_tail_no_small_lcm_replay_imported=true
stable_total_minus_tail_envelope_gap_proved=false
large_step_tail_saturation_pdec_excluded=false
lowstep_stable_history_mass_lower_bound_proved=false
row_column_unconditional_closed=false.
```

This step closes the terminal tail row-window and column-multiplicity envelope only. It does not prove that  $|S| - C_{\text{tail}}$  is large enough, and it does not exclude tail saturation PDEC/SAE. The row/column theorem remains open.  $\square$

**Theorem 7.78** (Stable tail gap functional). *Let  $O_B = \{(t, c) : x_0 \leq t < y, 1 \leq c < P\}$  be the zero-block obligation domain. Then*

$$|O_B| = L(P - 1).$$

*By no-loss accounting, after history-switch and duplicate returns are named,*

$$O_B = S \sqcup R_{\text{named}},$$

*where  $S$  is the stable source layer and  $R_{\text{named}}$  is the named return layer. Hence*

$$|S| \geq L(P - 1) - R_{\text{named}}.$$

Combining this with the tail envelope  $C_{\text{tail}}$  from the preceding theorem gives the explicit low-step lower bound

$$|S_{\leq H}| \geq L(P - 1) - R_{\text{named}} - C_{\text{tail}}.$$

Thus

```
StableTotalMinusTailEnvelopeGapOrLargeStepTailSaturationPDEC
→ ZeroBlockCoverObligationMassLedger
AND StableSourceTotalAfterNamedReturnsLedger
AND ExplicitStableTailGapFunctionalLedger
AND PositiveStableTailGapOrNamedReturnMassPDEC.
```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-stable-tail-gap-router.md.
```

It records

```
tail_gap_imported=true
zero_block_obligation_mass_imported=true
no_loss_accounting_imported=true
stable_total_after_named_returns_closed=true
tail_envelope_imported=true
explicit_tail_gap_functional_closed=true
positive_stable_tail_gap_proved=false
named_return_mass_pdec_excluded=false
stable_total_minus_tail_envelope_gap_proved=false
row_column_unconditional_closed=false.
```

This step closes only the explicit gap functional. It does not prove that the gap is positive or sufficient, and it does not exclude named return mass or tail saturation PDEC/SAE. The row/column theorem remains open.  $\square$

**Theorem 7.79** (Tail gap normalization). *Let  $H = P - y$  and write every large-step tail carrier as  $q = H + r$ . Then*

$$P - q = y - r.$$

*For the zero block  $B = [x_0, y - 1]$  of length  $L$ , the terminal row window from the preceding theorem becomes*

$$B \cap [P - q, y - 1] = B \cap [y - r, y - 1], \quad |B \cap [P - q, y - 1]| = \min(L, r).$$

*Consequently the tail envelope has the one-dimensional form*

$$C_{\text{tail}} = \sum_{\substack{q=H+r < P \\ q \text{ prime}}} \min(L, r) \left\lceil \frac{P-1}{H+r} \right\rceil,$$

*and is dominated by the integer envelope*

$$C_{\text{tail}} \leq \sum_{1 \leq r < P-H} \min(L, r) \left\lceil \frac{P-1}{H+r} \right\rceil.$$

*Thus the remaining gap separates as*

$$G' = L(P - 1) - C_{\text{tail}}, \quad G = G' - R_{\text{named}}.$$

Hence

```

PositiveStableTailGapOrNamedReturnMassPDEC
→ TailIndexChangeOfVariablesLedger
AND ExactPrimeTailEnvelopeOneDimensionalLedger
AND StableTailGapNamedReturnSeparationLedger
AND NormalizedTailGapPositiveOrDenseTailReturnPDEC.

```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-normalization-router.md.
```

It records

```

positive_gap_imported=true
explicit_gap_functional_imported=true
tail_index_change_of_variables_closed=true
exact_prime_tail_envelope_closed=true
all_integer_dominating_envelope_closed=true
named_return_separation_closed=true
normalized_positive_gap_proved=false
dense_tail_return_pdec_excluded=false
positive_stable_tail_gap_proved=false
row_column_unconditional_closed=false.

```

This step closes only the  $q = H + r$  normalization and the one-dimensional tail functional. It does not prove the normalized gap positive, and it does not exclude dense tail, tail saturation, or named-return mass PDEC/SAE. The row/column theorem remains open.  $\square$

**Theorem 7.80** (Integer margin split for the tail gap). *In the notation of the preceding theorem, define the all-integer tail envelope*

$$C_{\text{all}} := \sum_{1 \leq r < y} \min(L, r) \left\lceil \frac{P-1}{P-y+r} \right\rceil$$

and the integer margin

$$G_{\text{int}} = L(P-1) - C_{\text{all}}.$$

Since the prime tail is a sub-sum of the integer tail,

$$C_{\text{tail}} \leq C_{\text{all}}, \quad L(P-1) - C_{\text{tail}} \geq G_{\text{int}}.$$

Therefore  $G_{\text{int}} > R_{\text{named}}$  implies

$$L(P-1) - C_{\text{tail}} - R_{\text{named}} > 0.$$

Moreover, if  $P$  is odd and  $y \leq \lfloor P/2 \rfloor$ , then

$$P - y + r > (P-1)/2, \quad \left\lceil \frac{P-1}{P-y+r} \right\rceil \leq 2$$

for every  $1 \leq r < y$ . Since  $1 \leq L \leq y$ ,

$$C_{\text{all}} \leq 2 \sum_{1 \leq r < y} \min(L, r) = L(2y - L - 1),$$

and hence

$$G_{\text{int}} \geq L(P - 2y + L) > 0.$$

Thus a pure tail envelope cannot saturate the zero-block obligation in the early-half support branch. The remaining obstruction must be a late-support dense tail or named-return mass. Hence

*NormalizedTailGapPositiveOrDenseTailReturnPDEC*  
 $\rightarrow$  *NormalizedIntegerTailMarginFunctionalLedger*  
*AND PrimeTailDominatedByIntegerTailEnvelopeLedger*  
*AND EarlyHalfSupportTailCannotSaturateLemma*  
*AND IntegerMarginPositiveBranchCriterionLedger*  
*AND IntegerTailMarginPositiveOrLateSupportDenseTailNamedReturnPDEC.*

*Status and reference.* The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-integer-margin-router.md.

It records

```

normalized_tail_gap_imported=true
one_dimensional_tail_imported=true
integer_margin_functional_closed=true
prime_tail_dominated_by_integer_tail_closed=true
positive_branch_criterion_closed=true
early_half_support_tail_cannot_saturate_closed=true
integer_margin_positive_globally_proved=false
late_support_dense_tail_named_return_pdec_excluded=false
normalized_positive_gap_proved=false
row_column_unconditional_closed=false.

```

This step closes only the integer-margin domination and the early-half pure-tail non-saturation branch. It does not prove the global integer margin positive after named returns, and it does not exclude late-support dense-tail or named-return mass PDEC/SAE. The row/column theorem remains open.  $\square$

**Theorem 7.81** (Late-support collar formula for the tail gap). *Assume the late-support branch  $y > (P - 1)/2$ , and write*

$$P = 2m + 1, \quad H = P - y, \quad D = 2y - P, \quad E = y - m - 1 = (D - 1)/2.$$

*For  $r > E$  and  $r < y - 1$ , one has  $H + r > m$  and  $H + r < P - 1$ , hence*

$$\left\lceil \frac{P - 1}{H + r} \right\rceil = 2.$$

*At the endpoint  $r = y - 1$ , one has  $H + r = P - 1$ , hence the ceiling is 1. Thus every excess above two-unit tail multiplicity is concentrated in the short core  $1 \leq r \leq E$ . Put*

$$X_{\text{core}} := \sum_{1 \leq r \leq E} \min(L, r) \left( \left\lceil \frac{P - 1}{P - y + r} \right\rceil - 2 \right).$$

*Then the all-integer tail envelope decomposes exactly as*

$$C_{\text{all}} = L(2y - L - 1) + X_{\text{core}} - \min(L, y - 1),$$

*and therefore*

$$G_{\text{int}} = L(P - 1) - C_{\text{all}} = L(L - D) + \min(L, y - 1) - X_{\text{core}}.$$

*Consequently*

$$L(L - D) + \min(L, y - 1) > X_{\text{core}} + R_{\text{named}}$$

*implies the positive gap  $G > 0$ . Hence*

```

IntegerTailMarginPositiveOrLateSupportDenseTailNamedReturnPDEC
→ LateSupportExcessCoordinateLedger
AND RegularTailTwoUnitEndpointDefectLedger
AND LateCoreExcessFunctionalLedger
AND LateCollarMarginExactFormulaLedger
AND DeepLateShortCollarOrCoreExcessNamedReturnPDEC.

```

*Status and reference.* The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-late-collar-router.md.

It records

```

late_dense_imported=true
late_coordinate_closed=true
regular_tail_endpoint_defect_closed=true
late_core_excess_functional_closed=true
late_margin_exact_formula_closed=true
positive_late_margin_criterion_closed=true
deep_late_collar_excluded=false
core_excess_named_return_pdec_excluded=false
late_dense_tail_named_return_proved=false
row_column_unconditional_closed=false.

```

This step closes only the exact late-support collar formula. It does not exclude the deep-late short collar, and it does not control  $X_{\text{core}}$  or  $R_{\text{named}}$ . The row/column theorem remains open.  $\square$

**Theorem 7.82** (Deep-late collar and core quotient layers). *Let  $x_0 = y - L$ ,  $H = P - y$ , and  $D = 2y - P$ . Then the deep-late short collar condition is equivalent to a self-mirror collar containment:*

$$L \leq D \iff y - L \geq y - D \iff x_0 \geq H \iff [x_0, y - 1] \subseteq [H, y - 1].$$

If  $L > D$ , set

$$M = L(L - D) + \min(L, y - 1).$$

The late-collar formula gives

$$G_{\text{int}} = M - X_{\text{core}}.$$

Therefore  $X_{\text{core}} + R_{\text{named}} < M$  implies the positive gap  $G > 0$ . If a counterexample remains in the cross-collar branch, then necessarily

$$X_{\text{core}} + R_{\text{named}} \geq M.$$

Moreover, with  $h = H + r$  and

$$k(h) = \left\lceil \frac{P-1}{h} \right\rceil - 2,$$

the core excess decomposes by quotient layers

$$I_k = \{h : \lceil (P-1)/h \rceil = k+2\},$$

namely

$$X_{\text{core}} = \sum_{k \geq 1} k \sum_{h \in I_k} \min(L, h - H).$$

Hence

```

DeepLateShortCollarOrCoreExcessNamedReturnPDEC
→ DeepLateCollarMirrorContainmentLedger
AND CrossCollarPositiveMarginCriterionLedger
AND CoreExcessQuotientLayerDecompositionLedger
AND CoreExcessLayerConcentrationOrNamedReturnPDEC
AND SelfMirrorDeepLateCollarOrCoreLayerExcessNamedReturnPDEC.

```

*Status and reference.* The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-deep-collar-layer-router.md.`

It records

```
deep_collar_imported=true
mirror_collar_equivalence_closed=true
cross_collar_positive_margin_closed=true
core_layer_decomposition_closed=true
core_or_named_consumption_closed=true
self_mirror_collar_excluded=false
core_layer_concentration_excluded=false
deep_late_closed=false
row_column_unconditional_closed=false.
```

This step closes only the collar-containment equivalence, the cross-collar margin criterion, and the quotient-layer decomposition of  $X_{\text{core}}$ . It does not exclude the self-mirror collar branch or quotient core-layer concentration. The row/column theorem remains open.  $\square$

**Theorem 7.83** (Sieved defect form of the self-mirror tail gap). *Let*

$$w(q) = \min(L, q - H) \left\lceil \frac{P-1}{q} \right\rceil, \quad H < q < P.$$

*Write*

$$C_{\text{all}} = \sum_{H < q < P} w(q), \quad C_{\text{tail}} = \sum_{\substack{H < q < P \\ q \text{ prime}}} w(q),$$

*and*

$$D_{\text{np}} = \sum_{\substack{H < q < P \\ q \text{ not prime}}} w(q).$$

*Then*

$$C_{\text{tail}} = C_{\text{all}} - D_{\text{np}}.$$

*Let*  $z = \lfloor \sqrt{P-1} \rfloor$  *and*

$$W_z = \prod_{\substack{\ell \leq z, \ell \text{ prime}}} \ell.$$

*For*  $z < q < P$ ,

$$q \text{ prime} \iff (q, W_z) = 1.$$

*Therefore*

$$C_{\text{tail}} = \sum_{\substack{H < q \leq z \\ q \text{ prime}}} w(q) + \sum_{\substack{z < q < P \\ (q, W_z) = 1}} w(q).$$

*Combining this with the late-collar formula gives the true sieved gap*

$$G = L(L - D) + \min(L, y - 1) - X_{\text{core}} + D_{\text{np}} - R_{\text{named}}.$$

*In the self-mirror branch*  $L \leq D$ , *this becomes*

$$G = L(L - D) + L - X_{\text{core}} + D_{\text{np}} - R_{\text{named}}.$$

*Consequently, if*

$$L(L - D) + \min(L, y - 1) - X_{\text{core}} + D_{\text{np}} > R_{\text{named}},$$

*then the positive gap is closed. If not, the remaining obstruction is a weighted  $\sqrt{P}$ -rough CRT over-density or named-return mass:*

```

SelfMirrorDeepLateCollarOrCoreLayerExcessNamedReturnPDEC
→ PrimeTailNonprimeDefectExactLedger
AND SqrtRoughPrimeTailIdentityLedger
AND EndpointParityNonprimeDefectLowerBoundLedger
AND SievedTailGapMarginFunctionalLedger
AND SievedPositiveGapCriterionLedger
AND WeightedRoughTailCRTDefectOrNamedReturnPDEC
AND SelfMirrorSievedTailGapOrWeightedRoughCRTDefectPDEC.

```

*Status and reference.* The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-sieved-defect-router.md.

It records

```

self_mirror_imported=true
nonprime_defect_exact_closed=true
sqrt_rough_prime_tail_identity_closed=true
endpoint_parity_defect_lower_bound_closed=true
sieved_margin_functional_closed=true
sieved_positive_gap_criterion_closed=true
weighted_rough_crt_defect_excluded=false
self_mirror_sieved_gap_proved=false
row_column_unconditional_closed=false.

```

This step removes the purely integer-tail saturation interpretation. It does not yet prove that  $D_{\text{np}}$  always dominates  $X_{\text{core}}$  and  $R_{\text{named}}$ . The remaining branch is the weighted rough CRT defect, and the row/column theorem remains open.  $\square$

**Theorem 7.84** (Least-prime-factor deletion form of the weighted rough tail). *Keep*

$$w(q) = \min(L, q - H) \left\lceil \frac{P-1}{q} \right\rceil, \quad z = \lfloor \sqrt{P-1} \rfloor.$$

*Separate the finite small tail*

$$D_{\text{small}} = \sum_{\substack{H < q \leq z \\ q \text{ not prime}}} w(q).$$

*For  $z < q < P$ , every non-prime  $q$  has a unique least prime factor  $\ell \leq z$ . Hence the large non-prime defect has the disjoint decomposition*

$$D_{\text{large}} = \sum_{\ell \leq z} B_{\ell}, \quad B_{\ell} = \sum_{\substack{z < q < P \\ \ell = \text{lpf}(q)}} w(q),$$

*and*

$$D_{\text{np}} = D_{\text{small}} + D_{\text{large}}.$$

*If*

$$W_{<\ell} = \prod_{p < \ell, p \text{ prime}} p,$$

*then each cell has the CRT form*

$$B_{\ell} = \sum_{\substack{z/\ell < n < P/\ell \\ (n, W_{<\ell})=1}} w(\ell n).$$



Equivalently, with  $W_u = \prod_{p \leq u} p$  and

$$S_u = \sum_{\substack{z < q < P \\ (q, W_u) = 1}} w(q),$$

the prefix sieve telescopes by

$$S_{\ell-} - S_\ell = B_\ell, \quad S_0 - S_z = \sum_{\ell \leq z} B_\ell.$$

Let

$$\Phi = L(L - D) + \min(L, y - 1) - X_{\text{core}}.$$

The true gap is therefore

$$G = \Phi + D_{\text{small}} + \sum_{\ell \leq z} B_\ell - R_{\text{named}}.$$

Thus  $G > 0$  follows from

$$\Phi + D_{\text{small}} + \sum_{\ell \leq z} B_\ell > R_{\text{named}}.$$

If a counterexample remains, then

$$\sum_{\ell \leq z} B_\ell \leq R_{\text{named}} - \Phi - D_{\text{small}},$$

which is a disjoint least-prime-factor deletion debt. Hence

```
SelfMirrorSievedTailGapOrWeightedRoughCRTDefectPDEC
→ SmallTailFiniteNonprimeDefectLedger
AND LargeTailLeastPrimeFactorPartitionLedger
AND RoughPrefixDeletionTelescopingLedger
AND LeastPrimeFactorCRTDeletionCellLedger
AND SievedGapLPFDeletionFunctionalLedger
AND DyadicLPFDeletionDebtOrNamedReturnPDEC
AND LPFDeletionDebtOrRoughPrefixOverdensityPDEC.
```

*Status and reference.* The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-lpf-deletion-router.md.
```

It records

```
weighted_rough_target_imported=true
small_tail_finite_nonprime_defect_closed=true
large_tail_lpf_partition_closed=true
rough_prefix_deletion_telescoping_closed=true
lpf_crt_deletion_cell_closed=true
lpf_sieved_gap_functional_closed=true
dyadic_lpf_deletion_debt_excluded=false
lpf_deletion_debt_proved_impossible=false
row_column_unconditional_closed=false.
```

This step removes overlap ambiguity from the weighted rough branch by assigning each large non-prime tail point to its unique least prime factor. It does not prove that the resulting dyadic deletion layers always pay the gap. The row/column theorem remains open.  $\square$

**Theorem 7.85** (LPF dyadic cofactor interval reduction). *The remaining least-prime-factor deletion debt can be localized to explicit dyadic/quotient/cofactor CRT cells. More precisely, the open hard point*

*LPFDeletionDebtOrRoughPrefixOverdensityPDEC*

is reduced to

*DyadicLPFDeletionLayerPartitionLedger*  
*AND DyadicDebtLocalizationForAnyBudgetVectorLedger*  
*AND RampSaturatedTailWeightSplitLedger*  
*AND QuotientLayerCofactorIntervalLedger*  
*AND CofactorIntervalEndpointFormulaLedger*  
*AND RoughCofactorIntervalCRTSupportLedger*  
*AND DyadicRoughCofactorIntervalDebtOrNamedReturnPDEC*  
*AND DyadicRoughCofactorIntervalDebtOrColumnCRTPDEC.*

*Reduction ledger.* For each dyadic least-prime-factor layer let

$$B_Y = \sum_{\substack{Y < \ell \leq 2Y \\ \ell \text{ prime}}} B_\ell, \quad B_{\text{tot}} = \sum_Y B_Y.$$

Thus, for any proposed layer budget vector  $A_Y$ , if

$$B_{\text{tot}} \leq T < \sum_Y A_Y,$$

then some layer satisfies  $B_Y < A_Y$ . This is only finite localization; it is not an analytic lower bound.

Within one layer write

$$w(q) = u(q)v(q), \quad u(q) = \min(L, q - H), \quad v(q) = \left\lceil \frac{P-1}{q} \right\rceil.$$

The  $u$ -factor splits the support into the ramp region  $H < q < H + L$  and the saturated region  $q \geq H + L$ . Next decompose by

$$Q_j = \{q : \lceil (P-1)/q \rceil = j\}.$$

Put  $A = P - 1$ . Then

$$q_{j,\min} = \lfloor A/j \rfloor + 1, \quad q_{j,\max} = \begin{cases} A, & j = 1, \\ \lfloor A/(j-1) \rfloor, & j \geq 2. \end{cases}$$

After intersecting with  $z < q < P$ , the ramp cell has

$$q_{\min} = \max(z + 1, H + 1, q_{j,\min}), \quad q_{\max} = \min(P - 1, H + L - 1, q_{j,\max}),$$

and the saturated cell has

$$q_{\min} = \max(z + 1, H + L, q_{j,\min}), \quad q_{\max} = \min(P - 1, q_{j,\max}).$$

For fixed  $\ell$  and  $q = \ell n$ , this gives the integer cofactor interval

$$n_{\min} = \lceil q_{\min}/\ell \rceil, \quad n_{\max} = \lfloor q_{\max}/\ell \rfloor.$$

The LPF condition becomes  $(n, W_{<\ell}) = 1$ . Hence each cell has the explicit CRT form

$$B_{Y,\sigma,j} = \sum_{\substack{Y < \ell \leq 2Y \\ \ell \text{ prime}}} \sum_{\substack{n_{\min} \leq n \leq n_{\max} \\ (n, W_{<\ell})=1}} u(\ell n)j,$$

and the dyadic layer reassembles exactly as

$$B_Y = \sum_{\sigma \in \{\text{ramp}, \text{sat}\}} \sum_j B_{Y, \sigma, j}.$$

The remaining open task is to exclude insufficient rough-cofactor support in these cells, or to route that insufficiency to a named ColumnCRT/PDEC/SAE exit. The row/column theorem is therefore still not unconditionally closed. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-lpf-dyadic-cofactor-router.md.`

□

**Theorem 7.86** (Cofactor-LPF cover reduction). *The remaining dyadic rough-cofactor support debt is equivalent to an explicit cofactor least-prime-factor cover debt. The open hard point*

*DyadicRoughCofactorIntervalDebtOrColumnCRTPDEC*

*is reduced to*

*CofactorCellWeightedEnvelopeLedger  
AND RoughSupportQuotaCriterionLedger  
AND CofactorLeastPrimeFactorPartitionLedger  
AND CofactorLPFCRTCellLedger  
AND CofactorCoverPrimeProductWidthLedger  
AND LocalCofactorLPFCoverDebtOrFiniteAtomPDEC  
AND CofactorLPFCoverDebtOrProductWidthColumnCRTPDEC.*

*Reduction ledger.* Fix a dyadic/quotient/cofactor cell  $C = (Y, \sigma, j)$ . Put

$$\alpha_{\ell, n} = j \cdot u(\ell n).$$

Let

$$F_C = \sum_{\ell \in Y} \sum_{n \in I_{\ell, j, \sigma}} \alpha_{\ell, n},$$

$$B_C = \sum_{\ell \in Y} \sum_{\substack{n \in I_{\ell, j, \sigma} \\ (n, W_{<\ell})=1}} \alpha_{\ell, n}, \quad E_C = F_C - B_C.$$

Here  $F_C$  is the full integer envelope,  $B_C$  is the rough cofactor support, and  $E_C$  is the mass deleted by smaller cofactor factors. Therefore, for a local budget  $A_C$ ,

$$B_C < A_C \iff E_C > F_C - A_C.$$

Thus rough support debt is exactly cofactor small-factor over-cover.

For every deleted cofactor, let  $r = \text{lpf}(n) < \ell$ . The least-prime-factor partition is disjoint:

$$E_C = \sum_{\ell \in Y} \sum_{r < \ell} E_{r \leftarrow \ell}, \quad E_{r \leftarrow \ell} = \sum_{\substack{n \in I_{\ell, j, \sigma} \\ r = \text{lpf}(n)}} \alpha_{\ell, n}.$$

Each component has the CRT form

$$n = rm, \quad (m, W_{<r}) = 1, \quad \lceil n_{\min}/r \rceil \leq m \leq \lfloor n_{\max}/r \rfloor.$$

This step uses only the unique least prime factor of an integer; it does not use any short-interval prime existence input.

If a local cover debt uses the distinct cofactor primes  $R_C$ , its complete phase word has period

$$M_R = \prod_{r \in R_C} r.$$

When  $M_R$  exceeds the cell support width, exact replay cannot be a local short-period drift; it must be recorded as a finite atom, a ColumnCRT/PDEC exit, or a deeper moving-support exit. This does not yet exclude the local cofactor-LPF cover debt. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-cofactor-lpf-cover-router.md.`

□

**Theorem 7.87** (Cofactor-LPF dyadic pressure reduction). *The cofactor-LPF cover debt can be reduced to a product-width dichotomy and a dyadic lower-prime pressure problem. The open hard point*

*CofactorLPFCoverDebtOrProductWidthColumnCRTPDEC*

*is reduced to*

*CofactorCoverExcessThresholdLedger  
AND ActiveCofactorPrimeProductDichotomyLedger  
AND DyadicCofactorPrimePressurePartitionLedger  
AND OverfullDyadicRLayerLocalizationLedger  
AND FixedRToMIntervalEndpointLedger  
AND RLayerRoughMCRTSupportLedger  
AND SmallProductActiveCoverConcentrationPDEC  
AND DyadicCofactorLPFPPressureOrSmallProductColumnCRTPDEC.*

*Reduction ledger.* For a fixed cell  $C$ , put

$$H_C = F_C - A_C.$$

The previous ledger gives

$$B_C < A_C \iff E_C > H_C.$$

Thus a local counterexample requires explicit over-cover:

$$E_C > H_C.$$

Let

$$R_C = \{r \text{ prime} : E_r(C) > 0\}, \quad M_C = \prod_{r \in R_C} r.$$

If  $M_C$  exceeds the support width of  $C$ , the full cover word has a period larger than the cell itself; exact replay must be treated as a product-width ColumnCRT/PDEC or finite-atom exit. If not, the cover is confined to a small-product active set.

Now decompose the active  $r$ -primes dyadically:

$$E_C = \sum_Z E_Z(C), \quad M_C = \prod_Z M_Z.$$

For any proposed dyadic allowance  $U_Z$ , if

$$\sum_Z U_Z \leq H_C$$

while  $E_C > H_C$ , then some dyadic layer satisfies

$$E_Z(C) > U_Z.$$

This is a finite pigeonhole localization of over-cover pressure.

Finally, for fixed  $r$ , the cofactor interval becomes

$$n = rm, \quad m_{\min} = \lceil n_{\min}/r \rceil, \quad m_{\max} = \lfloor n_{\max}/r \rfloor,$$

with the residual CRT condition

$$(m, W_{<r}) = 1.$$

Thus the remaining pressure is a lower-prime rough- $m$  CRT support problem, not an invocation of the original row-gap statement. This reduction does not exclude dyadic  $r$ -layer overpressure or small-product active-cover concentration. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-cofactor-lpf-dyadic-pressure-router.md.`

□

**Theorem 7.88** (Cofactor-LPF single- $r$  pressure reduction). *The small-product dyadic  $r$ -pressure branch can be localized to a single cofactor prime and then to a fixed  $(r, \ell)$  rough- $m$  CRT cell. The open hard point*

*DyadicCofactorLPFPressureOrSmallProductColumnCRTPDEC*

*is reduced to*

*ActivePrimeCardinalityFromProductLedger  
AND SmallZFiniteAtomBoundaryLedger  
AND SingleCofactorPrimePressureLocalizationLedger  
AND FixedRSourceEllPartitionLedger  
AND FixedREllRoughMCRTEllCellLedger  
AND FixedPairProductWidthColumnCRTEllCellLedger  
AND SingleRSmallProductPressurePDEC  
AND SingleCofactorPrimePressureOrFixedPairMCRTEllCellCRTPDEC.*

*Reduction ledger.* In a dyadic  $r$ -layer  $Z < r \leq 2Z$ , let

$$R_Z(C) = \{r : Z < r \leq 2Z, E_r(C) > 0\}, \quad M_Z = \prod_{r \in R_Z(C)} r.$$

The boundary  $Z < 2$  is a finite small-prime atom. For  $Z \geq 2$ , if the small-product branch gives  $M_Z \leq W_C$ , where  $W_C$  is the cell support width, then each active  $r$  is  $> Z$ , hence

$$|R_Z(C)| \leq K_Z := \lfloor \log W_C / \log Z \rfloor.$$

If the dyadic layer is over budget,

$$E_Z(C) > U_Z,$$

then some active cofactor prime satisfies

$$E_r(C) > U_Z/K_Z.$$

Thus the pressure is not merely a layer total; it localizes to a single  $r$ .

For fixed  $r$ ,

$$E_r(C) = \sum_{\substack{\ell \in Y \\ \ell > r}} E_{r \leftarrow \ell}(C).$$

Applying the same finite localization to the  $\ell$ -sources sends any remaining overpressure to one fixed pair  $(r, \ell)$ . On this pair,

$$q = \ell r m, \quad m_{\min} = \lceil n_{\min}/r \rceil, \quad m_{\max} = \lfloor n_{\max}/r \rfloor,$$

and

$$(m, W_{<r}) = 1, \quad \alpha_{\ell, rm} = j u(\ell r m).$$

This is a fixed-pair rough- $m$  CRT cell. The reduction does not exclude single- $r$  or fixed-pair pressure; it only removes the remaining small-product dyadic ambiguity. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-cofactor-lpf-single-r-pressure-router.md`.

□

**Theorem 7.89** (Fixed-pair second-LPF descent reduction). *The fixed-pair rough- $m$  pressure branch admits a strict scale descent. The open hard point*

*SingleCofactorPrimePressureOrFixedPairMCRTColumnCRTPDEC*

*is reduced to*

*FixedPairPressureImportedLedger  
AND FixedPairMSupportStrictDescentLedger  
AND MEqualsOneFiniteAtomLedger  
AND SecondCofactorLeastPrimeFactorPartitionLedger  
AND SecondLPFRoughTCRTCellLedger  
AND SecondLPFProductWidthColumnCRTExitLedger  
AND ResidualSupportWidthStrictDecreaseNoCycleLedger  
AND SecondLPFDescentPressureOrTripleMCRTColumnCRTPDEC.*

*Reduction ledger.* On a fixed pair  $(r, \ell)$ , the cell is

$$q = \ell r m, \quad m_{\min} \leq m \leq m_{\max}, \quad (m, W_{<r}) = 1,$$

with weight

$$\beta_m = j u(\ell r m).$$

The case  $m = 1$  is the single point  $q = \ell r$ , hence a finite atom.

For  $m > 1$ , let  $s = \text{lpf}(m)$ . Since  $(m, W_{<r}) = 1$ , one has

$$s \geq r.$$

Thus

$$m = st, \quad (t, W_{<s}) = 1,$$

with

$$t_{\min} = \lceil m_{\min}/s \rceil, \quad t_{\max} = \lfloor m_{\max}/s \rfloor.$$

This gives the disjoint second-LPF partition

$$E_{r \leftarrow \ell} = E_{m=1} + \sum_{s \geq r} E_{s \leftarrow r, \ell}.$$

Let  $w_m = m_{\max} - m_{\min} + 1$  and

$$w_t = t_{\max} - t_{\min} + 1.$$

For fixed  $s$ ,

$$w_t \leq \lceil w_m/s \rceil \leq \lceil w_m/r \rceil.$$

In the non-finite branch  $r \geq 2$ , so  $w_t < w_m$  whenever  $w_m > 1$ ; if  $w_m \leq 1$ , the cell is already finite. Therefore repeated descent cannot cycle at the same support scale.

If the active second-level primes  $S$  have product

$$M_S = \prod_{s \in S} s$$

larger than  $w_m$ , the second-level phase word has period exceeding the original  $m$ -support and must be registered as a ColumnCRT/PDEC or finite atom exit. This step does not exclude the resulting second-LPF/triple rough- $t$  pressure; it records a strict no-cycle reduction. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-fixed-pair-second-lpf-descent-router.md.`

□

**Theorem 7.90** (Iterated LPF rank-budget reduction). *The second-LPF/triple rough residual branch is not a free recurrent failure mode. The open hard point*

*SecondLPFDescentPressureOrTripleMCRTCColumnCRTPDEC*

*is reduced to*

*SecondLPFTriplePressureImportedLedger  
AND IteratedLPFOrderedRoughResidualChainLedger  
AND LPFSupportProductReciprocityInvariantLedger  
AND LPFDepthRankBudgetLedger  
AND IteratedLPFCRTWordCellLedger  
AND IteratedLPFProductWidthColumnCRTExitLedger  
AND TerminalResidualFiniteAtomLedger  
AND IteratedLPFWellFoundedNoCycleLedger  
AND RankBudgetedIteratedLPFMovingFamilyOrColumnCRTPDEC.*

*Rank-budget ledger.* After the fixed-pair second-LPF step, fix  $(r, \ell, s)$ . The cell has

$$q = \ell rm, \quad m = st, \quad s = \text{lpf}(m) \geq r, \quad (t, W_{<s}) = 1.$$

Starting from the residual  $n_0 = t$ , recursively define, whenever  $n_i > 1$ ,

$$a_i = \text{lpf}(n_i), \quad n_i = a_i n_{i+1}, \quad (n_{i+1}, W_{<a_i}) = 1.$$

The roughness condition gives an ordered LPF word

$$r \leq s \leq a_1 \leq a_2 \leq \cdots,$$

with repetitions allowed.

Let

$$A_h = s \prod_{i \leq h} a_i.$$

If the original  $m$ -support has width  $w_m$ , then after fixing this LPF word the residual support has width at most

$$\left\lceil \frac{w_m}{A_h} \right\rceil.$$

Thus each added least-prime-factor enlarges the composite CRT period while shrinking the residual support by the same product scale. If the active word modulus product exceeds  $w_m$ , the same phase word can only contribute an isolated atom in the original support and is registered as a ColumnCRT/PDEC or finite-atom exit.

On the non-finite branch  $r \geq 2$ . If no product-width exit has occurred, then  $A_h \leq w_m$ . Since every factor in the word is at least  $r$ , the total factor depth  $d$ , including the initial second-level factor  $s$ , satisfies

$$d \leq \lfloor \log_r w_m \rfloor.$$

Consequently the triple branch cannot hide an infinite same-scale LPF tower. The remaining obstruction is a rank-budgeted iterated-LPF moving family, not an unstructured triple pressure. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-iterated-lpf-rank-budget-router.md.

□

**Theorem 7.91** (LPF word-entropy and first-moving-coordinate reduction). *The rank-budgeted iterated-LPF moving-family branch admits a finite word partition and a first-motion reduction. The open hard point*

*RankBudgetedIteratedLPFMovingFamilyOrColumnCRTPDEC*

*is reduced to*

*RankBudgetedMovingFamilyImportedLedger  
AND IteratedLPFWordSignaturePartitionLedger  
AND LPFWordEntropyFiniteCapLedger  
AND AggregatePressureToSingleLPFWordLedger  
AND FixedLPFWordColumnCRTExitLedger  
AND FirstMovingLPFCoordinateLedger  
AND MovingCoordinateSupportReciprocityLedger  
AND NoAnonymousRankBudgetedMovingFamilyLedger  
AND FirstMovingLPFCoordinatePressureOrWordMotionColumnCRTPDEC.*

*Word-motion ledger.* By uniqueness of least prime factors, each residual lies in a unique ordered word

$$\omega = (s, a_1, \dots, a_d), \quad r \leq s \leq a_1 \leq \dots \leq a_d,$$

with product

$$A(\omega) = s \prod_{i \leq d} a_i.$$

If  $A(\omega) > W$ , where  $W$  is the original  $m$ -support width, the word has already crossed the product-width boundary and is registered as a ColumnCRT/PDEC or finite atom. Otherwise

$$A(\omega) \leq W.$$

Since all factors in the non-finite branch are at least  $r \geq 2$ , the total factor depth obeys

$$d + 1 \leq \lfloor \log_r W \rfloor.$$



Thus the active word set  $\Omega(P, C)$  is finite and has an explicit entropy budget  $H(W, r)$ .

The pressure decomposes over disjoint word cells:

$$E(\Omega) = \sum_{\omega \in \Omega} E_{\omega}.$$

If  $E(\Omega) > U$ , then one word satisfies

$$E_{\omega} > \frac{U}{|\Omega|}.$$

Hence any remaining excess must be carried by a single LPF word, not by an anonymous moving-family reservoir.

If that pressure-carrying word has stable prime coordinates and stable phase residues along the family, it is a fixed MCRT word and is routed to the fixed word ColumnCRT/PDEC or finite-atom exit. If not, there is a first moving coordinate  $\mu$ . Let  $A_{\text{prefix}}$  be the product of the stable word prefix before  $\mu$ . After  $\mu$  is imposed, the residual support width is bounded by

$$\left\lceil \frac{W}{A_{\text{prefix}}\mu} \right\rceil.$$

When  $A_{\text{prefix}}\mu > W$ , this is again a product-width ColumnCRT/PDEC or finite-atom exit. Otherwise the remaining obstruction is the named first-moving LPF coordinate pressure. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-lpf-word-entropy-motion-router.md.`

□

**Theorem 7.92** (First-moving LPF coordinate reduction). *The first-moving LPF coordinate branch has a dyadic product-width reduction. The open hard point*

*FirstMovingLPFCoordinatePressureOrWordMotionColumnCRTPDEC*

*is reduced to*

*FirstMovingLPFCoordinatePressureImportedLedger  
AND StablePrefixProductSupportLedger  
AND FirstMovingCoordinateEffectiveWidthLedger  
AND LowMovingCoordinateFiniteAtomLedger  
AND FirstMovingCoordinateDyadicPartitionLedger  
AND MovingCoordinateActiveProductWidthExitLedger  
AND MovingCoordinateActiveCountBoundLedger  
AND SingleMovingCoordinatePressureLocalizationLedger  
AND FixedMovingCoordinateDegeneratesToColumnCRTLedger  
AND NoAnonymousFirstMovingCoordinatePoolLedger  
AND SingleMovingLPFCoordinateDriftOrPrefixColumnCRTPDEC.*

*Coordinate ledger.* Let  $\mu$  be the first LPF coordinate that moves in the pressure-carrying word. The word prefix before  $\mu$  is stable; write its product as

$$A_{\text{prefix}}.$$

If  $W$  is the original  $m$ -support width, define the effective coordinate width

$$W_{\text{prefix}} = \left\lceil \frac{W}{A_{\text{prefix}}} \right\rceil.$$

When  $\mu > W_{\text{prefix}}$ , the composite period

$$A_{\text{prefix}}\mu$$

already exceeds  $W$ , so the branch is a product-width ColumnCRT/PDEC or a finite atom. The low coordinate case  $\mu \leq 2$  is also finite.

For the remaining coordinates, partition into dyadic layers

$$B < \mu \leq 2B, \quad B = 2^b, \quad B \geq 2, \quad B \leq W_{\text{prefix}}.$$

Fix such a layer and let  $M_B$  be its active coordinate set. If

$$\prod_{\mu \in M_B} \mu > W_{\text{prefix}},$$

then the active coordinate word has period exceeding the effective support and is again a Column-CRT/PDEC or finite-atom exit. Otherwise, since every  $\mu > B$ ,

$$|M_B| \leq \left\lfloor \frac{\log W_{\text{prefix}}}{\log B} \right\rfloor.$$

Thus the active coordinate pool has a finite logarithmic capacity.

The layer pressure decomposes over disjoint coordinate cells:

$$E_B = \sum_{\mu \in M_B} E_\mu.$$

If  $E_B > U_B$ , then some coordinate satisfies

$$E_\mu > \frac{U_B}{|M_B|}.$$

If that coordinate is stable along the family, the supposed first motion degenerates to a fixed word coordinate and is routed to the prefix ColumnCRT/PDEC or finite-atom exit. If it is not stable, the remaining obstruction is a single moving LPF coordinate drift. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-first-moving-lpf-coordinate-router.md.

□

**Theorem 7.93** (Single-moving LPF coordinate scale-escape reduction). *The single moving LPF coordinate drift has a scale-escape reduction. The open hard point*

*SingleMovingLPFCoordinateDriftOrPrefixColumnCRTPDEC*

*is reduced to*

*SingleMovingLPFCoordinateDriftImportedLedger  
AND SingleMovingCoordinateStablePrefixLedger  
AND SingleMovingCoordinateDyadicScaleLedger  
AND BoundedScaleDriftDegeneratesToFixedCoordinateLedger  
AND UnboundedCoordinateScaleEscapeLedger  
AND PostMovingCoordinateSupportDescentLedger  
AND SameScaleCoordinateCycleExcludedLedger  
AND SparseCoordinateDriftSAERegistrationLedger  
AND NoAnonymousSingleCoordinateDriftLedger  
AND ScaleEscapingSingleCoordinateDescentOrSparseDriftSAEColumnCRTPDEC.*

*Scale-escape ledger.* Import the remaining branch in which a single LPF coordinate  $\mu$  moves with the family. The prefix before  $\mu$  is stable; with original support width  $W$ , set

$$W_{\text{prefix}} = \left\lfloor \frac{W}{A_{\text{prefix}}} \right\rfloor.$$

Assign to  $\mu$  its dyadic scale

$$B(\mu) = 2^{\lfloor \log_2 \mu \rfloor}, \quad B(\mu) \leq \mu \leq 2B(\mu).$$

If  $B(\mu)$  is bounded on an infinite subfamily, then  $\mu$  ranges over only finitely many primes. By the infinite pigeonhole principle, an infinite subfamily has fixed  $\mu$ . That branch is no longer a moving coordinate drift; it is the already named fixed-coordinate ColumnCRT/PDEC or finite-atom exit. Therefore any genuine single-coordinate drift must satisfy

$$B(\mu) \rightarrow \infty$$

on a cofinal subfamily.

After imposing the coordinate  $\mu$ , the residual support width satisfies

$$W_{\text{after}} \leq \left\lceil \frac{W_{\text{prefix}}}{\mu} \right\rceil \leq \left\lceil \frac{W_{\text{prefix}}}{B(\mu)} \right\rceil.$$

On every non-finite active layer  $B(\mu) \geq 2$ , so this gives a strict support descent before any same-scale cycle can recur. Thus the anonymous single-coordinate drift is removed: bounded scales degenerate to fixed coordinates, true moving scales descend, and nonpersistent drift is registered as a sparse drift/SAE exit.

This does not yet exclude scale-escaping coordinate descent, and it does not prove global summability of the sparse drift/SAE mass. The row/column theorem therefore remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-single-moving-lpf-coordinate-scale-escape-router.md.

□

**Theorem 7.94** (Scale-escape support-clock reduction). *The scale-escaping single-coordinate descent has an integer support clock. The open hard point*

*ScaleEscapingSingleCoordinateDescentOrSparseDriftSAEColumnCRTPDEC*

*is reduced to*

*ScaleEscapingSingleCoordinateDescentImportedLedger  
AND ScaleEscapeIntegerSupportClockLedger  
AND ScaleEscapeHalvingClockDescentLedger  
AND FiniteDepthScaleEscapePerFiberLedger  
AND TerminalWidthOneFiniteAtomLedger  
AND ScaleLadderProductWidthColumnCRTExitLedger  
AND PersistentScaleLadderSignatureRegistrationLedger  
AND SparseScaleLadderSAERegistrationLedger  
AND NoCyclicScaleEscapeDescentLedger  
AND NoAnonymousScaleEscapeDescentLedger  
AND PersistentScaleLadderSignatureOrSparseScaleLadderSAEColumnCRTPDEC.*

*Support-clock ledger.* Let  $W_i$  be the effective residual support width after  $i$  scale-escaping coordinates have been imposed. Define the integer clock

$$K(W) = \lceil \log_2 \max(W, 1) \rceil.$$

Thus  $K(W) = 0$  exactly at terminal width  $W \leq 1$ , which is a finite atom or an already named boundary.

For a genuine scale-escaping coordinate at step  $i$ , its dyadic scale  $B_i$  satisfies  $B_i \geq 2$ . The previous scale-escape ledger gives

$$W_{i+1} \leq \left\lceil \frac{W_i}{B_i} \right\rceil \leq \left\lceil \frac{W_i}{2} \right\rceil.$$

If  $W_i \geq 2$ , then

$$K(W_{i+1}) \leq K(W_i) - 1.$$

Hence a fixed counterexample fiber has at most  $K(W_0)$  scale-escape steps. No scale-escape descent can cycle back to the same support-clock layer.

If the descent reaches  $W \leq 1$ , the branch is terminal. If the dyadic scale ladder satisfies

$$\prod_i B_i > W_0,$$

then the composite period exceeds the original support and the branch is a product-width Column-CRT/PDEC or finite-atom exit. Therefore an infinite obstruction that is not absorbed by terminal width, product width, or nonpersistent sparse mass must carry a named ordered scale-ladder signature with its phase data.

This removes the anonymous scale-escape descent and the possibility of an internal scale-escape cycle. It does not yet exclude persistent scale-ladder signatures, and it does not prove global summability of the sparse scale-ladder SAE mass. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-scale-escape-support-clock-router.md.`

□

**Theorem 7.95** (Scale-ladder word-entropy reduction). *The persistent scale-ladder signature branch has a finite dyadic word entropy budget. The open hard point*

*PersistentScaleLadderSignatureOrSparseScaleLadderSAEColumnCRTPDEC*

*is reduced to*

*PersistentScaleLadderSignatureImportedLedger  
AND ScaleLadderDyadicWordPartitionLedger  
AND ScaleLadderProductBudgetLedger  
AND ScaleLadderWordEntropyFiniteCapLedger  
AND AggregatePersistentPressureToSingleScaleWordLedger  
AND FixedScaleLadderMCRTCColumnCRTExitLedger  
AND FirstMovingScaleLadderPhaseCoordinateLedger  
AND SparseScaleLadderSAECarriedForwardLedger  
AND NoAnonymousPersistentScaleLadderSignatureLedger  
AND FirstMovingScaleLadderPhaseDriftOrSparseScaleLadderSAEColumnCRTPDEC.*

*Scale-word ledger.* Write the persistent dyadic scale ladder as

$$B_i = 2^{b_i}, \quad b_i \geq 1, \quad \sigma = (b_1, \dots, b_d).$$

Its scale product is

$$A_\sigma = \prod_i B_i = 2^{\sum_i b_i}.$$

If  $A_\sigma > W_0$ , the composite period already exceeds the initial support and the branch is a product-width ColumnCRT/PDEC or finite atom. Otherwise

$$\sum_i b_i \leq K_0 = \lceil \log_2 \max(W_0, 1) \rceil.$$

The number of ordered positive-integer scale words is therefore finite:

$$N_{\text{ladder}}(K_0) = 1 + \sum_{n=1}^{K_0} 2^{n-1} \leq 2^{K_0}.$$

The persistent pressure decomposes over these scale-word cells,

$$E_{\text{total}} = \sum_{\sigma} E_{\sigma}.$$

If  $E_{\text{total}} > U$ , then at least one scale word satisfies

$$E_{\sigma} > \frac{U}{N_{\text{ladder}}(K_0)}.$$

Thus a persistent scale-ladder obstruction cannot remain an anonymous signature reservoir.

For the pressure-carrying word  $\sigma$ , if all actual prime coordinates and phase residues stabilize on an infinite subfamily, the branch is a fixed MCRT word and is routed to ColumnCRT/PDEC or a finite atom. If not, there is a first moving prime coordinate or phase residue inside the fixed scale word. That is the remaining named hard point. The sparse scale-ladder SAE outlet is only carried forward here; its global summability is not proved. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-scale-ladder-word-entropy-router.md.`

□

**Theorem 7.96** (Scale-ladder finite-slot lock). *Inside a fixed dyadic scale word, persistent first-moving phase drift is impossible. The open hard point*

*FirstMovingScaleLadderPhaseDriftOrSparseScaleLadderSAEColumnCRTPDEC*

*is reduced to*

*FirstMovingScaleLadderPhaseDriftImportedLedger  
AND FixedScaleWordSlotLedger  
AND FinitePrimeChoicesPerScaleSlotLedger  
AND FiniteResidueChoicesPerPrimeSlotLedger  
AND FiniteActualScaleLadderAtomSetLedger  
AND InfinitePigeonholeStableActualScaleLadderLedger  
AND NoPersistentFirstMovingScaleLadderPhaseDriftLedger  
AND StableActualScaleLadderMCRTColumnCRTExitLedger  
AND SparseScaleLadderSAECarriedForwardAfterFiniteSlotLockLedger  
AND NoAnonymousFirstMovingScaleLadderPhaseDriftLedger  
AND SparseScaleLadderSAESummabilityOrStableActualLadderColumnCRTPDEC.*

*Finite-slot ledger.* Fix the pressure-carrying dyadic scale word

$$\sigma = (b_1, \dots, b_d),$$

so each  $b_i$  is fixed. The actual prime in slot  $i$  must lie in the finite dyadic set

$$Q_i = \{q \text{ prime} : 2^{b_i} \leq q < 2^{b_i+1}\}.$$

For a fixed  $q \in Q_i$ , the phase residue has only  $q$  possible values. Hence the number of actual prime-residue ladder atoms under  $\sigma$  is finite, with the crude bound

$$N_{\text{actual}}(\sigma) \leq \prod_i \sum_{q \in Q_i} q < \infty.$$

If a fixed scale word carries persistent pressure along an infinite subfamily, the infinite pigeonhole principle gives an infinite subfamily on which the entire actual prime-residue ladder atom is fixed. Such a stable actual ladder is a fixed MCRT cell and is routed to the ColumnCRT/PDEC or finite-atom exit.

Therefore a first moving prime coordinate or phase residue cannot be a persistent obstruction inside a fixed scale word. It is either fixed on an infinite subfamily, or it is nonpersistent and is registered as sparse scale-ladder SAE. This does not prove global summability of that sparse mass, and it does not exclude the stable actual ladder ColumnCRT/PDEC exit. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-scale-ladder-finite-slot-lock-router.md.`

□

**Theorem 7.97** (Stable-ladder Fourier/PDEC bridge). *The stable actual ladder ColumnCRT exit can be written as an explicit finite Fourier/PDEC input. The open hard point*

*SparseScaleLadderSAESummabilityOrStableActualLadderColumnCRTPDEC*

*is reduced to*

*StableActualLadderOrSparseSAEImportedLedger*  
*AND StableActualLadderFiniteGroupLedger*  
*AND StableActualLadderZeroMeanCellFunctionLedger*  
*AND StableActualLadderExactEccessIdentityLedger*  
*AND StableActualLadderFourierPDECBridgeLedger*  
*AND StableActualLadderNontrivialCharacterLowerBoundLedger*  
*AND NoAnonymousStableActualLadderColumnCRTExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterFourierBridgeLedger*  
*AND SparseScaleLadderSAESummabilityOrStableActualLadderFourierPDECCap.*

*Finite-group Fourier ledger.* A stable actual ladder fixes a finite list of prime-residue pairs

$$((q_i, a_i))_{i=1}^d.$$

Work on the finite product group

$$G = \prod_{i=1}^d \mathbb{Z}/q_i\mathbb{Z}, \quad N = |G| = \prod_{i=1}^d q_i,$$

with

$$\tau(n) = (n \bmod q_i)_i, \quad a = (a_i)_i.$$

This product-group form does not require the displayed primes to be distinct. Define the zero-mean cell function

$$F_a(g) = 1_{\{a\}}(g) - \frac{1}{N}.$$

Then for every support set  $S$  the stable-ladder excess is exactly

$$E_a(S) = \sum_{n \in S} F_a(\tau(n)) = |\{n \in S : \tau(n) = a\}| - \frac{|S|}{N}.$$

Expanding  $F_a$  in the character group  $\widehat{G}$ , the trivial character coefficient vanishes. Hence

$$E_a(S) = \sum_{\chi \in \widehat{G}, \chi \neq 1} \widehat{F}_a(\chi) S_\chi, \quad S_\chi = \sum_{n \in S} \chi(\tau(n)).$$

For the point-mass-minus-average function one has  $|\widehat{F}_a(\chi)| = 1/N$  for nontrivial  $\chi$ . Therefore, if  $E_a(S) > 0$ , then some nontrivial character satisfies

$$|S_\chi| \geq \frac{N E_a(S)}{N-1}.$$

Thus a stable actual ladder cannot remain an anonymous ColumnCRT outlet: any positive ladder excess forces a named nontrivial Fourier correlation. What remains is the explicit stable-ladder Fourier/PDEC cap, or the separate global summability of the sparse scale-ladder SAE outlet. This theorem does not prove either remaining input, and the row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-fourier-pdec-router.md.`

□

**Theorem 7.98** (Stable-ladder pivot-fiber PDEC localization). *The stable-ladder Fourier/PDEC cap has a primitive one-coordinate fiber form. The open hard point*

*SparseScaleLadderSAESummabilityOrStableActualLadderFourierPDECCap*

*is reduced to*

*StableActualLadderFourierCapImportedLedger  
AND StableLadderCharacterFactorizationLedger  
AND StableLadderNontrivialPivotCoordinateLedger  
AND StableLadderComplementFiberPartitionLedger  
AND StableLadderGlobalCharacterToPivotFiberLocalizationLedger  
AND StableLadderPrimitivePivotCharacterCorrelationLedger  
AND NoAnonymousStableLadderFourierCapLedger  
AND SparseScaleLadderSAECarriedForwardAfterPivotFiberLedger  
AND SparseScaleLadderSAESummabilityOrStableActualLadderPrimitiveFiberPDECCap.*

*Pivot-fiber ledger.* Let

$$G = \prod_{i=1}^d \mathbb{Z}/q_i \mathbb{Z}, \quad N = |G| = \prod_{i=1}^d q_i.$$

Every character of  $G$  factors as

$$\chi(g) = \prod_{i=1}^d \chi_i(g_i).$$

If  $\chi$  is nontrivial, choose a pivot coordinate  $j$  with  $\chi_j \neq 1$ . Write

$$G_{-j} = \prod_{i \neq j} \mathbb{Z}/q_i \mathbb{Z}, \quad |G_{-j}| = \frac{N}{q_j}.$$

For  $h \in G_{-j}$ , split the support set by the complementary coordinates,

$$S_h = \{n \in S : \tau_{-j}(n) = h\},$$

and set

$$A_h = \sum_{n \in S_h} \chi_j(n \bmod q_j).$$

Then the global character sum decomposes exactly as

$$S_\chi = \sum_{h \in G_{-j}} \chi_{-j}(h) A_h.$$

Since  $|\chi_{-j}(h)| = 1$ , the triangle inequality gives

$$|S_\chi| \leq \sum_{h \in G_{-j}} |A_h| \leq |G_{-j}| \max_{h \in G_{-j}} |A_h|.$$

Thus if  $|S_\chi| \geq \Lambda$ , some complementary-coordinate fiber satisfies

$$|A_h| \geq \frac{\Lambda}{|G_{-j}|}.$$

Taking the lower bound from the preceding Fourier bridge,

$$\Lambda = \frac{N E_a(S)}{N-1},$$

one obtains the primitive fiber threshold

$$|A_h| \geq \frac{q_j E_a(S)}{N-1}.$$

Equivalently, the global stable-ladder Fourier cap is witnessed by a single nontrivial coordinate character on one fixed complementary fiber. On the cyclic factor  $\mathbb{Z}/q_j \mathbb{Z}$ , such a character has the form

$$r \mapsto \exp(2\pi i c_j r / q_j), \quad c_j \not\equiv 0 \pmod{q_j}.$$

So the remaining cap is a primitive pivot-fiber PDEC statement, not an anonymous global Fourier outlet.

This theorem does not prove that primitive cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem therefore remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-pivot-fiber-pdec-router.md.`

□

**Theorem 7.99** (Stable-ladder residue-count PDEC localization). *The primitive pivot-fiber character cap has an equivalent residue-count imbalance form. The open hard point*



*SparseScaleLadderSAESummabilityOrStableActualLadderPrimitiveFiberPDECCap*

is reduced to

*StableActualLadderPrimitiveFiberPDECImportedLedger*  
*AND StableLadderPivotFiberResidueCountVectorLedger*  
*AND StableLadderPivotFiberZeroMeanDeviationLedger*  
*AND StableLadderPivotFiberCharacterToResidueDeviationLedger*  
*AND StableLadderPivotFiberResidueImbalanceLocalizationLedger*  
*AND StableLadderPivotFiberResidueImbalanceThresholdLedger*  
*AND NoAnonymousPrimitiveFiberCharacterPDECExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterResidueCountLedger*  
*AND SparseScaleLadderSAESummabilityOrStableActualLadderResidueCountPDECCap.*

*Residue-count ledger.* Keep the pivot coordinate  $j$ , the modulus  $q_j$ , and the complementary fiber  $S_h$  from the preceding reduction. Let

$$L = |S_h|, \quad M_r = |\{n \in S_h : n \equiv r \pmod{q_j}\}|, \quad r \in \mathbb{Z}/q_j\mathbb{Z},$$

and define the centered residue-count deviations

$$D_r = M_r - \frac{L}{q_j}.$$

Then  $\sum_r D_r = 0$ . For the nontrivial pivot character  $\chi_j$ , the primitive fiber sum is

$$A_h = \sum_{n \in S_h} \chi_j(n \bmod q_j) = \sum_r M_r \chi_j(r).$$

Since  $\sum_r \chi_j(r) = 0$ , the average part vanishes and

$$A_h = \sum_r D_r \chi_j(r).$$

Thus

$$|A_h| \leq \sum_r |D_r| \leq q_j \max_r |D_r|.$$

The preceding pivot-fiber theorem gives the lower bound

$$|A_h| \geq \frac{q_j E_a(S)}{N-1}.$$

Consequently at least one residue class  $r$  satisfies

$$\left| M_r - \frac{L}{q_j} \right| \geq \frac{E_a(S)}{N-1}.$$

Therefore the primitive fiber character outlet is not anonymous: it is a specific residue-count imbalance inside a fixed complementary-coordinate fiber.

This theorem does not prove the residue-count PDEC cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-residue-count-pdec-router.md.`

□

**Theorem 7.100** (Stable-ladder positive-surplus PDEC localization). *The signed residue-count imbalance cap has a positive-surplus form. The open hard point*

*SparseScaleLadderSAESummabilityOrStableActualLadderResidueCountPDECCap*

is reduced to

*StableActualLadderResidueCountPDECImportedLedger  
AND StableLadderResidueDeviationSignDichotomyLedger  
AND StableLadderResidueDeviationZeroSumTransferLedger  
AND StableLadderPositiveResidueSurplusLocalizationLedger  
AND StableLadderPositiveResidueSurplusThresholdLedger  
AND NoAnonymousSignedResidueImbalanceExitLedger  
AND SparseScaleLadderSAECarriedForwardAfterPositiveSurplusLedger  
AND SparseScaleLadderSAESummabilityOrStableActualLadderPositiveResidueSurplusPDECCap.*

*Positive-surplus ledger.* Keep the fixed complementary fiber and write

$$D_r = M_r - \frac{L}{q_j}, \quad \sum_r D_r = 0.$$

The preceding residue-count localization gives a residue  $r$  with

$$|D_r| \geq \delta, \quad \delta = \frac{E_a(S)}{N-1}.$$

If  $D_r \geq \delta$ , the desired positive surplus is already present. If  $D_r \leq -\delta$ , then zero sum gives

$$\sum_{s \neq r} D_s = -D_r \geq \delta.$$

There are  $q_j - 1$  residue classes with  $s \neq r$ , so one of them satisfies

$$D_s \geq \frac{\delta}{q_j - 1}.$$

Thus in all cases there is a residue class  $s$  with

$$M_s - \frac{L}{q_j} \geq \frac{E_a(S)}{(N-1)(q_j-1)}.$$

Hence an absolute signed residue imbalance can be routed to a positive overload residue class, with only the explicit  $q_j - 1$  dilution in the threshold. This theorem does not prove the positive residue-surplus PDEC cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-positive-surplus-pdec-router.md.

□

**Theorem 7.101** (Stable-ladder occupancy dichotomy). *The positive residue-surplus cap has an integer occupancy form. The open hard point*

*SparseScaleLadderSAESummabilityOrStableActualLadderPositiveResidueSurplusPDECCap*

is reduced to

```

StableActualLadderPositiveResidueSurplusImportedLedger
AND StableLadderOverloadedCellIntegerOccupancyLedger
AND StableLadderPositiveSurplusSingletonOrPairDichotomyLedger
AND StableLadderSingletonSurplusAtomRegistrationLedger
AND StableLadderSameCellPairCongruenceLedger
AND StableLadderSameCellPairPeriodMultipleLedger
AND NoAnonymousPositiveResidueSurplusExitLedger
AND SparseScaleLadderSAECarriedForwardAfterOccupancyDichotomyLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderSingletonSurplusOrCellPairPeriodPDECCap.

```

*Occupancy-dichotomy ledger.* Import the positive overload residue class from the preceding localization:

$$M_s - \frac{L}{q_j} \geq \eta, \quad \eta = \frac{E_a(S)}{(N-1)(q_j-1)} > 0.$$

Here  $M_s$  is a nonnegative integer occupancy number inside the fixed complementary fiber. Hence  $M_s \geq 1$ . There are exactly two cases:

$$M_s = 1 \quad \text{or} \quad M_s \geq 2.$$

The first case is recorded as a singleton surplus atom. This theorem does not prove summability of all such singleton atoms.

In the second case, two distinct points  $n_1 < n_2$  lie in the same complete stable-ladder cell. Let

$$W = \text{lcm}_i(q_i)$$

be the period of the ladder coordinates. Equality of all cell coordinates gives

$$n_2 - n_1 \equiv 0 \pmod{q_i} \quad \text{for every } i,$$

and therefore

$$W \mid (n_2 - n_1), \quad n_2 - n_1 > 0.$$

Consequently any upstream width bound  $H < W$  rules out the pair branch at once; without such a bound the remaining obstruction is the explicit same-cell period-pair PDEC cap.

Thus the positive residue-surplus outlet is no longer anonymous: it is either a singleton surplus atom or a same-cell period-pair outlet. The theorem does not prove the singleton summability cap, it does not prove the same-cell period-pair cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-occupancy-dichotomy-router.md.
```

□

**Theorem 7.102** (Stable-ladder singleton/pair width reduction). *The singleton-or-period-pair outlet has a low-fiber/long-width form. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableActualLadderSingletonSurplusOrCellPairPeriodPDECCap
```

*is reduced to*

```

StableActualLadderSingletonOrPairPeriodImportedLedger
AND StableLadderSingletonSurplusLowFiberQuotaLedger
AND StableLadderSameCellPairSupportWidthDichotomyLedger
AND StableLadderShortSupportPairExclusionLedger
AND StableLadderLongWidthSameCellPeriodPairRegistrationLedger
AND NoAnonymousSingletonOrPairPeriodExitLedger
AND SparseScaleLadderSAECarriedForwardAfterSingletonPairWidthLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderLowFiberSingletonSurplusOrLongWidthCellPairPDECCap.

```

*Singleton/pair width ledger.* For the singleton branch, combine  $M_s = 1$  with the positive overload

$$M_s - \frac{L}{q_j} \geq \eta > 0.$$

This gives

$$1 - \frac{L}{q_j} > 0, \quad L < q_j.$$

Thus a singleton surplus atom cannot live in a high pivot fiber  $L \geq q_j$ . It is a low-fiber singleton event and remains to be summed or excluded there.

For the pair branch, the preceding occupancy theorem gives two points  $n_1 < n_2$  in the same stable-ladder cell. With

$$W = \text{lcm}_i(q_i),$$

there is an integer  $t \geq 1$  such that

$$n_2 - n_1 = tW \geq W.$$

If  $H$  is the diameter of any support carrying both points, then

$$H \geq n_2 - n_1 \geq W.$$

Consequently the short-support case  $H < W$  excludes the pair branch. The only remaining pair obstruction is a named long-width same-cell period-pair PDEC cap.

Therefore the singleton-or-period-pair outlet is not anonymous: it is either a low-fiber singleton outlet or a long-width same-cell period-pair outlet. This theorem does not prove the low-fiber singleton summability cap, it does not prove the long-width period-pair cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-singleton-pair-width-router.md.`

□

**Theorem 7.103** (Stable-ladder low-fiber phase reduction). *The low-fiber singleton outlet has an isolated-or-anti-pivot form. The open hard point*

*`SparseScaleLadderSAESummabilityOrStableActualLadderLowFiberSingletonSurplusOrLongWidthCellPairPDECCap`*

*is reduced to*

*`StableActualLadderLowFiberSingletonOrLongPairImportedLedger`  
`AND StableLadderLowFiberOccupancyOneOrMultiDichotomyLedger`  
`AND StableLadderFiberIsolatedSingletonAtomLedger`  
`AND StableLadderRepeatedPivotModulusForcesIsolationLedger`  
`AND StableLadderComplementFiberAntiPivotPairLedger`  
`AND StableLadderComplementFiberPeriodAndAntiPivotPhaseLedger`  
`AND StableLadderComplementFiberPairWidthDichotomyLedger`  
`AND StableLadderLongFullCellPairCarriedForwardLedger`  
`AND NoAnonymousLowFiberSingletonExitLedger`  
`AND SparseScaleLadderSAECarriedForwardAfterLowFiberPhaseLedger`  
`AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrAntiPivotComplementPairOrLongFullCellPairPDECCap`*

*Low-fiber phase ledger.* The preceding theorem leaves the low-fiber singleton branch with

$$M_s = 1, \quad 1 \leq L < q_j.$$

Hence

$$L = 1 \quad \text{or} \quad 2 \leq L < q_j.$$

The case  $L = 1$  is a singleton atom isolated inside its fixed complementary fiber.

Assume next that  $2 \leq L < q_j$ . Take the point in the singleton residue class  $s$ , and take another point in the same complementary fiber. These two points have identical residues in every coordinate  $i \neq j$ , but their pivot residues modulo  $q_j$  are distinct because  $M_s = 1$ . Let

$$W_{-j} = \text{lcm}_{i \neq j}(q_i),$$

with  $W_{-j} = 1$  when the complementary list is empty. Then the two points  $n_1 < n_2$  satisfy

$$W_{-j} \mid (n_2 - n_1), \quad q_j \nmid (n_2 - n_1).$$

If  $q_j \mid W_{-j}$ , the fixed complementary fiber already fixes the pivot residue modulo  $q_j$ , so the alternative  $2 \leq L < q_j$  is impossible and the branch collapses to  $L = 1$ .

If  $H$  is the support diameter carrying this complementary-fiber pair, then

$$H \geq n_2 - n_1 \geq W_{-j}.$$

Thus  $H < W_{-j}$  excludes the anti-pivot complement-pair branch; otherwise it is a named complement-pair PDEC cap. The previous long full-cell pair branch, with  $H \geq W = \text{lcm}_i(q_i)$ , is carried forward unchanged.

Therefore the low-fiber singleton outlet is no longer anonymous: it is either a fiber-isolated singleton atom, or a complement-period aligned but pivot anti-aligned pair. This theorem does not prove summability of isolated singletons, it does not prove the anti-pivot complement-pair cap, it does not prove the long full-cell pair cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-low-fiber-phase-router.md.

□

**Theorem 7.104** (Stable-ladder pair quotient-phase reduction). *The anti-pivot complement-pair outlet and the long full-cell pair outlet have a single quotient-phase form. The open hard point*

*SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrAntiPivotComplementPairOrLongFullCellPairPDECCap*

*is reduced to*

*StableActualLadderIsolatedSingletonOrPairImportedLedger*  
*AND StableLadderIsolatedSingletonCarriedForwardLedger*  
*AND StableLadderComplementBasePeriodLedger*  
*AND StableLadderPairDifferenceQuotientNormalizationLedger*  
*AND StableLadderPivotPhaseQuotientPeriodLedger*  
*AND StableLadderAntiPivotPairNonzeroQuotientPhaseLedger*  
*AND StableLadderFullCellPairZeroQuotientPhaseLedger*  
*AND StableLadderPairQuotientWidthEnvelopeLedger*  
*AND StableLadderQuotientNoWrapOrFullPeriodDichotomyLedger*  
*AND NoAnonymousAntiPivotOrFullPairExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterPairQuotientPhaseLedger*  
*AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrPairQuotientPhaseCyclePDECCap.*

*Pair quotient-phase ledger.* The isolated singleton atom is carried forward as its own summability/cap outlet. For the pair outlets, use the complementary-coordinate period

$$B = W_{-j} = \text{lcm}_{i \neq j}(q_i),$$

with  $B = 1$  for an empty complementary list. Every complement-fiber pair has a unique quotient

$$d = n_2 - n_1 = tB, \quad t \geq 1.$$

Set

$$g = \gcd(q_j, B), \quad R = \frac{q_j}{g}.$$

Then the pivot phase  $d \bmod q_j$  is determined exactly by  $t \bmod R$ . Indeed  $q_j \mid tB$  if and only if  $R \mid t$ .

Therefore the anti-pivot complement-pair branch is precisely the nonzero quotient phase branch

$$t \not\equiv 0 \pmod{R},$$

whereas the full-cell pair branch is the zero quotient phase branch

$$t \equiv 0 \pmod{R}, \quad t = Ru.$$

If  $H$  is the support diameter, then

$$1 \leq t \leq \left\lfloor \frac{H}{B} \right\rfloor.$$

Thus the quotient interval either has no complete pivot phase period,  $\lfloor H/B \rfloor < R$ , or contains a complete quotient phase cycle,  $\lfloor H/B \rfloor \geq R$ . This is the remaining explicit quotient-phase PDEC/cap input.

The two pair outlets are therefore no longer separate anonymous exits; they are the nonzero and zero residues of the same quotient variable modulo  $R$ . This theorem does not prove summability of isolated singletons, it does not prove the quotient phase cycle cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-pair-quotient-phase-router.md.

□

**Theorem 7.105** (Stable-ladder quotient short-arc/mean reduction). *The quotient-phase cycle outlet has a non-cyclic short-arc/mean decomposition. The open hard point*

*SparseScaleLaddersSAESummabilityOrStableActualLadderIsolatedSingletonOrPairQuotientPhaseCyclePDECCap*

*is reduced to*

*StableActualLadderQuotientPhaseCycleImportedLedger*  
*AND StableLadderIsolatedSingletonCarriedForwardAfterArcMeanLedger*  
*AND StableLadderQuotientSpanParameterLedger*  
*AND StableLadderQuotientNoWrapZeroPhaseExclusionLedger*  
*AND StableLadderQuotientShortArcClusterLedger*  
*AND StableLadderFullCycleEuclideanDecompositionLedger*  
*AND StableLadderFullCycleFormalPhaseMeanLedger*  
*AND StableLadderFullCycleResidualTailArcLedger*  
*AND NoAnonymousQuotientPhaseCycleExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterArcMeanLedger*  
*AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrQuotientShortArcClusterOrPhaseCycleMeanPDECCap.*

*Short-arc/mean ledger.* Keep the notation of the pair quotient-phase reduction:

$$B = W_{-j}, \quad g = \gcd(q_j, B), \quad R = q_j/g, \quad d = n_2 - n_1 = tB.$$

If  $H$  is the support diameter, set

$$T = \lfloor H/B \rfloor.$$

Then every quotient-pair witness satisfies  $1 \leq t \leq T$ .

If  $T < R$ , the set  $\{1, \dots, T\}$  is a proper non-wrapping arc in  $\mathbf{Z}/R\mathbf{Z}$ , and it contains no zero phase. Hence the full-cell branch is impossible inside the no-wrap case; any actual pair in that case is registered as a quotient short-arc cluster.

If  $T \geq R$ , write

$$T = aR + s, \quad a = \lfloor T/R \rfloor \geq 1, \quad 0 \leq s < R.$$

The first  $aR$  formal quotient parameters decompose into  $a$  complete cycles, so each pivot phase occurs exactly  $a$  times in the formal quotient parameter interval. The residual  $s$  parameters again form a proper no-wrap tail arc. Thus a persistent actual excess cannot hide in an anonymous quotient cycle: after subtracting the complete-cycle formal mean it must appear either as a phase-cycle actual-mean cap problem, or as a tail short-arc cluster.

This theorem does not prove the short-arc cluster cap, it does not prove the phase-cycle actual-mean cap, it does not prove summability of isolated singletons, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-quotient-phase-arc-mean-router.md.`

□

**Theorem 7.106** (Stable-ladder quotient arc Fourier reduction). *The quotient short-arc cluster and the phase-cycle actual-mean outlet have a single finite Fourier form. The open hard point*

*SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrQuotientShortArcClusterOrPhaseCycleMeanPDECCap*

*is reduced to*

*StableLadderQuotientShortArcOrMeanImportedLedger*  
*AND StableLadderIsolatedSingletonCarriedForwardAfterQuotientFourierLedger*  
*AND StableLadderQuotientCircleGroupLedger*  
*AND StableLadderQuotientActualPhaseLoadMeasureLedger*  
*AND StableLadderQuotientArcDiscrepancyFunctionalLedger*  
*AND StableLadderQuotientDirichletKernelIdentityLedger*  
*AND StableLadderQuotientNontrivialFrequencyLowerBoundLedger*  
*AND StableLadderPhaseMeanCapAsPointArcLedger*  
*AND NoAnonymousShortArcOrPhaseMeanExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterQuotientFourierLedger*  
*AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrQuotientArcFourierPDECCap.*

*Quotient arc Fourier ledger.* Let

$$C_R = \mathbf{Z}/R\mathbf{Z}, \quad R = q_j / \gcd(q_j, B), \quad d = n_2 - n_1 = tB.$$

All pivot phases of the quotient-pair witnesses are determined by  $t \bmod R$ . Let  $\mu(r)$  be the number of actual quotient witnesses with  $t \equiv r \pmod{R}$ . Subtract the relevant actual-load model or complete cycle mean and write

$$\nu(r) = \mu(r) - \text{model}(r), \quad \sum_{r \in C_R} \nu(r) = 0.$$

This centering is the formal-to-actual guard: the model is not used as actual load unless the actual-counting ledger has supplied it.

For any arc  $A \subset C_R$ , set

$$\Delta(A) = \sum_{r \in A} \nu(r).$$

A quotient short-arc cluster is exactly a proper arc with  $\Delta(A) > 0$ . The phase-cycle actual-mean cap is the point-arc case  $A = \{r_0\}$ .

Use the Fourier transform

$$\widehat{f}(h) = \sum_{r \in C_R} f(r) e^{-2\pi i h r / R}.$$

Since  $\widehat{\nu}(0) = 0$ , Fourier inversion gives the exact identity

$$\Delta(A) = \frac{1}{R} \sum_{h=1}^{R-1} \widehat{\nu}(h) \widehat{\mathbf{1}_A}(-h).$$

Let

$$\Lambda_R(A) = \sum_{h=1}^{R-1} \left| \widehat{\mathbf{1}_A}(h) \right|.$$

If  $\Delta(A) > 0$ , the triangle inequality forces a non-trivial frequency

$$|\widehat{\nu}(h)| \geq \frac{R\Delta(A)}{\Lambda_R(A)} \quad (1 \leq h \leq R-1).$$

For  $A = \{r_0\}$ , one has  $\Lambda_R(A) = R-1$ . Thus both the short-arc cluster and the phase-cycle actual-mean outlet are no longer anonymous: they are quotient arc Fourier/PDEC inputs.

This theorem does not prove the quotient arc Fourier/PDEC cap, it does not prove summability of isolated singletons, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-quotient-arc-fourier-router.md.`

□

**Theorem 7.107** (Stable-ladder quotient Fourier lift reduction). *The quotient arc Fourier outlet lifts to an endpoint bilinear Fourier outlet on the original pair variables. The open hard point*

`SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrQuotientArcFourierPDECcap`

*is reduced to*

`StableLadderQuotientArcFourierImportedLedger`  
`AND StableLadderIsolatedSingletonCarriedForwardAfterFourierLiftLedger`  
`AND StableLadderNontrivialQuotientFrequencyForcesRGreaterOneLedger`  
`AND StableLadderQuotientCoprimeFactorizationLedger`  
`AND StableLadderQuotientInverseLiftLedger`  
`AND StableLadderQuotientCharacterToPivotDifferenceLedger`  
`AND StableLadderLiftedCharacterNontrivialityLedger`  
`AND StableLadderEndpointBilinearPhaseFactorizationLedger`  
`AND StableLadderCenteredModelKernelSeparatedLedger`  
`AND NoAnonymousQuotientArcFourierExitLedger`  
`AND SparseScaleLadderSAECarriedForwardAfterFourierLiftLedger`  
`AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrQuotientEndpointBilinearFourierPDECcap.`

*Fourier lift ledger.* Let  $d = n_2 - n_1 = tB$ , and put

$$g = \gcd(q_j, B), \quad B = gB_0, \quad q_j = gR.$$

Then  $\gcd(B_0, R) = 1$ . If the quotient Fourier branch contains a non-trivial frequency  $h$ , then  $1 \leq h \leq R-1$ , so  $R > 1$ . Choose

$$uB_0 \equiv 1 \pmod{R}.$$

Since  $d/g = tB_0$ , one has

$$t \equiv u(d/g) \pmod{R}.$$



Set  $\beta \equiv hu \pmod{R}$ . For every actual pair witness,

$$e_R(ht) = e_R(hu d/g) = e_{q_j}(\beta d).$$

Here  $\beta \not\equiv 0 \pmod{R}$ , because  $h \not\equiv 0 \pmod{R}$  and  $u$  is invertible modulo  $R$ . Thus the lifted character is non-trivial on the quotient phase.

Finally, using  $d = n_2 - n_1$ ,

$$e_{q_j}(\beta(n_2 - n_1)) = e_{q_j}(\beta n_2) \overline{e_{q_j}(\beta n_1)}.$$

The centered coefficient from the quotient arc Fourier certificate is therefore an actual endpoint bilinear phase sum minus its explicit model or Dirichlet-kernel term. The formal model is not being counted as actual load; it is separated as a known kernel.

This theorem does not prove the endpoint bilinear Fourier/PDEC cap, it does not prove summability of isolated singletons, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-quotient-fourier-lift-router.md.`

□

**Theorem 7.108** (Stable-ladder endpoint bilinear balance reduction). *The endpoint bilinear Fourier outlet has a finite matrix balance form. The open hard point*

*SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrQuotientEndpointBilinearFourierPDECcap*

*is reduced to*

*StableLadderEndpointBilinearFourierImportedLedger*  
*AND StableLadderIsolatedSingletonCarriedForwardAfterBilinearBalanceLedger*  
*AND StableLadderEndpointPairMatrixRegisteredLedger*  
*AND StableLadderEndpointCenteredKernelTotalZeroLedger*  
*AND StableLadderEndpointMarginalBalancedDecompositionLedger*  
*AND StableLadderEndpointBilinearPhasePairingLedger*  
*AND StableLadderEndpointBilinearTriangleTrichotomyLedger*  
*AND StableLadderEndpointRowMarginalFourierExitLedger*  
*AND StableLadderEndpointColumnMarginalFourierExitLedger*  
*AND StableLadderEndpointBalancedCoreEnergyLowerBoundLedger*  
*AND NoAnonymousEndpointBilinearFourierExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterBilinearBalanceLedger*  
*AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrEndpointMarginalFourierOrBalancedBilinearEnergy*

*Endpoint bilinear balance ledger.* Let  $G = \mathbf{Z}/q_j\mathbf{Z}$ . For the actual pair witnesses supplied by the previous quotient-Fourier lift, record the endpoint residues

$$x \equiv n_1 \pmod{q_j}, \quad y \equiv n_2 \pmod{q_j}.$$

Let  $M(x, y)$  be the actual witness-counting matrix. Subtract the explicit model or Dirichlet-kernel term already separated by the previous ledger and write

$$K(x, y) = M(x, y) - \text{Model}(x, y), \quad \sum_{x, y} K(x, y) = 0.$$

This is only a centered kernel identity; the model is not counted as actual load.

On the active endpoint domains  $X, Y$ , define

$$\rho(x) = \sum_{y \in Y} K(x, y), \quad \sigma(y) = \sum_{x \in X} K(x, y).$$

The centered ANOVA decomposition is

$$K(x, y) = \frac{\rho(x)}{|Y|} + \frac{\sigma(y)}{|X|} + K_0(x, y),$$

where

$$\sum_{y \in Y} K_0(x, y) = 0, \quad \sum_{x \in X} K_0(x, y) = 0.$$

If  $X = Y = G$ , non-trivial characters annihilate the pure row and column pieces. If a proper endpoint aperture remains, these pieces are exactly row or column marginal Fourier outlets.

For a non-trivial endpoint character

$$\phi_\beta(z) = e^{2\pi i \beta z / q_j},$$

the previous hard point has the form

$$S_\beta = \sum_{x \in X, y \in Y} K(x, y) \overline{\phi_\beta(x)} \phi_\beta(y).$$

Substituting the decomposition gives

$$S_\beta = S_{\text{row}}(\beta) + S_{\text{col}}(\beta) + S_{\text{bal}}(\beta).$$

Thus if  $|S_\beta| \geq \eta$ , then at least one of the three alternatives

$$|S_{\text{row}}(\beta)| \geq \eta/3, \quad |S_{\text{col}}(\beta)| \geq \eta/3, \quad |S_{\text{bal}}(\beta)| \geq \eta/3$$

holds. The first two alternatives are endpoint marginal Fourier/PDEC inputs. For the balanced alternative, Cauchy–Schwarz gives

$$|S_{\text{bal}}(\beta)| \leq \sqrt{|X||Y|} \|K_0\|_{\text{HS}},$$

and therefore

$$\|K_0\|_{\text{HS}}^2 \geq \frac{\eta^2}{9|X||Y|}.$$

The anonymous endpoint bilinear cap is therefore replaced by endpoint marginal Fourier caps or a balanced-core energy cap.

This theorem does not prove the endpoint marginal Fourier/PDEC cap, it does not prove the balanced bilinear energy/PDEC cap, it does not prove summability of isolated singletons, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-bilinear-balance-router.md.`

□

**Theorem 7.109** (Stable-ladder endpoint energy packet reduction). *The endpoint marginal Fourier outlet and the balanced bilinear energy outlet have a common dyadic packet form. The open hard point*

*SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrEndpointMarginalFourierOrBalancedBilinearEnergyPDEC*  
is reduced to

*StableLadderEndpointMarginalOrBalancedEnergyImportedLedger*  
*AND StableLadderIsolatedSingletonCarriedForwardAfterEnergyPacketLedger*  
*AND StableLadderEndpointMarginalFourierParsevalVarianceLedger*  
*AND StableLadderEndpointMarginalVarianceDyadicPacketLedger*  
*AND StableLadderBalancedCoreEnergyImportedLedger*  
*AND StableLadderBalancedCoreEnergyDyadicCellPacketLedger*  
*AND StableLadderEndpointEnergyPacketSignedHalfLedger*  
*AND StableLadderEndpointDyadicEnergyPacketRegisteredLedger*  
*AND NoAnonymousEndpointMarginalOrBalancedEnergyExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterEndpointEnergyPacketLedger*  
*AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrEndpointDyadicEnergyPacketPDECcap.*

*Endpoint energy packet ledger.* Consider first a row endpoint marginal  $\rho$  on  $\mathbf{Z}/q_j\mathbf{Z}$ . If the endpoint marginal Fourier outlet gives a non-trivial frequency with

$$|\widehat{\rho}(\beta)| \geq \eta,$$

then Parseval gives

$$\sum_x |\rho(x)|^2 \geq \frac{\eta^2}{q_j}.$$

The same argument applies to a column marginal  $\sigma$ . Hence every marginal Fourier outlet is already a one-dimensional endpoint variance outlet.

The balanced branch from the previous ledger is already an energy outlet:

$$\|K_0\|_{\text{HS}}^2 \geq E,$$

where  $K_0$  has zero row and column marginals. Thus both possible outlets are finite  $L^2$ -energy statements: either on endpoint residues or on endpoint cells.

For any finite real-valued deviation function  $F$  on a finite support  $\Omega$ , let

$$E_F = \sum_{z \in \Omega} |F(z)|^2.$$

Partition the non-zero values of  $|F|$  into dyadic layers. If  $L$  is the number of non-empty layers, then one layer with scale  $\lambda$  satisfies

$$\lambda^2 \#\{z : \lambda < |F(z)| \leq 2\lambda\} \geq \frac{E_F}{L}.$$

Applied to  $F = \rho$  or  $F = \sigma$ , this gives a one-dimensional endpoint marginal packet. Applied to  $F = K_0$ , this gives a two-dimensional balanced endpoint-cell packet.

Finally split the selected layer by sign. Since the deviations are real, at least one sign contributes half the selected square energy. Therefore the remaining obstruction can be registered as a signed dyadic endpoint energy packet, carrying its dimension, scale, support, sign, and energy lower bound.

This theorem does not prove the endpoint dyadic energy packet/PDEC cap, it does not prove summability of isolated singletons, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-energy-packet-router.md.

□

**Theorem 7.110** (Stable-ladder endpoint packet autocorrelation reduction). *The endpoint dyadic energy packet outlet reduces to either a singleton packet atom or a non-zero endpoint displacement autocorrelation. The open hard point*

*SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrEndpointDyadicEnergyPacketPDECCap*

is reduced to

*StableLadderEndpointDyadicEnergyPacketImportedLedger*  
*AND StableLadderIsolatedSingletonCarriedForwardAfterPacketAutocorrelationLedger*  
*AND StableLadderEndpointPacketFiniteGroupLedger*  
*AND StableLadderEndpointPacketSignedSupportLedger*  
*AND StableLadderEndpointPacketSingletonAtomLedger*  
*AND StableLadderEndpointPacketNonzeroPairCountLedger*  
*AND StableLadderEndpointPacketDisplacementPigeonholeLedger*  
*AND StableLadderEndpointPacketWeightedAutocorrelationLedger*  
*AND StableLadderEndpointPacketDimensionPreservedLedger*  
*AND NoAnonymousEndpointDyadicEnergyPacketExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterPacketAutocorrelationLedger*  
*AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrEndpointPacketSingletonAtomSAEOrEndpointDisplacement*

*Endpoint packet autocorrelation ledger.* The previous ledger produces a signed dyadic packet on a finite endpoint group

$$H = \mathbf{Z}/q_j\mathbf{Z} \quad \text{or} \quad H = (\mathbf{Z}/q_j\mathbf{Z})^2.$$

It has a scale  $\lambda$ , a sign  $\epsilon \in \{\pm 1\}$ , and support

$$S = \{z \in H : \lambda < \epsilon F(z) \leq 2\lambda\}.$$

Here  $F$  is either an endpoint marginal deviation or the balanced endpoint cell deviation.

If  $|S| = 1$ , the packet is a singleton endpoint atom. This branch is not closed here; it is merely no longer anonymous.

Assume now that  $|S| = m \geq 2$ . Define

$$C_S(\delta) = \#\{z \in S : z + \delta \in S\}.$$

Every ordered pair of distinct elements of  $S$  contributes one non-zero displacement, hence

$$\sum_{\delta \neq 0} C_S(\delta) = m(m-1).$$

Since  $H$  is finite, there is a non-zero displacement  $\delta$  such that

$$C_S(\delta) \geq \frac{m(m-1)}{|H|-1}.$$

On the signed dyadic support, all products have the same sign and satisfy

$$F(z)F(z+\delta) \geq \lambda^2.$$

Therefore the weighted autocorrelation obeys

$$A_F(\delta) = \sum_{z, z+\delta \in S} F(z)F(z+\delta) \geq \lambda^2 C_S(\delta).$$

For dimension one this is an endpoint-residue displacement; for dimension two it is an endpoint-cell displacement. Thus the packet has become an explicit same-sign displacement autocorrelation.

This theorem does not prove the endpoint packet singleton atom/SAE branch, it does not prove the endpoint displacement autocorrelation/PDEC cap, it does not prove summability of isolated singletons, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-packet-autocorrelation-router.md.

□

**Theorem 7.111** (Stable-ladder endpoint displacement orbit reduction). *The endpoint displacement autocorrelation outlet reduces to a cyclic adjacency packet on one translation orbit. The open hard point*

*SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrEndpointPacketSingletonAtomSAEOrEndpointDisplacement*  
is reduced to

```
StableLadderEndpointSingletonOrDisplacementAutocorrelationImportedLedger
AND StableLadderEndpointSingletonAtomSAEUnifiedLedger
AND StableLadderEndpointDisplacementAutocorrelationImportedLedger
AND StableLadderEndpointNonzeroDisplacementOrderLedger
AND StableLadderEndpointDisplacementOrbitPartitionLedger
AND StableLadderEndpointAutocorrelationOrbitDecompositionLedger
AND StableLadderEndpointWeightedOrbitPigeonholeLedger
AND StableLadderEndpointOrbitCyclicAdjacencyPacketLedger
AND StableLadderEndpointOrbitDimensionPreservedLedger
AND NoAnonymousEndpointDisplacementAutocorrelationExitLedger
AND SparseScaleLadderSAECarriedForwardAfterDisplacementOrbitLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointTranslationOrbitAdjacencyPDECCap.
```

*Endpoint displacement orbit ledger.* The singleton branches from the previous ledger are first unified as endpoint singleton atoms. This unification does not prove their summability; it only prevents two names for the same single-point obstruction.

Now consider the displacement autocorrelation branch. Let  $H$  be the finite endpoint group and let  $\delta \neq 0$  be the displacement. Put

$$r = \text{ord}_H(\delta) > 1.$$

The translation  $T_\delta z = z + \delta$  partitions  $H$  into cycles

$$O = a + \langle \delta \rangle = \{a + t\delta : t \in \mathbf{Z}/r\mathbf{Z}\}.$$

The set of all cycles is  $\Omega = H/\langle \delta \rangle$ , with  $|\Omega| = |H|/r$ .

The weighted autocorrelation from the previous ledger decomposes exactly along these cycles:

$$A_F(\delta) = \sum_{O \in \Omega} A_O(\delta),$$

where

$$A_O(\delta) = \sum_{t \in \mathbf{Z}/r\mathbf{Z}} F(a + t\delta) F(a + (t+1)\delta) \mathbf{1}_{a+t\delta \in S} \mathbf{1}_{a+(t+1)\delta \in S}.$$

Hence, if  $A_F(\delta)$  is the active lower-bound obstruction, then some orbit satisfies

$$A_O(\delta) \geq \frac{A_F(\delta)}{|\Omega|}.$$

On that orbit the obstruction is a signed dyadic adjacency packet on the cycle  $\mathbf{Z}/r\mathbf{Z}$ . In dimension one this is an endpoint-residue cycle; in dimension two it is an endpoint-cell cycle. Thus the full-group autocorrelation has been reduced to one explicit CRT translation orbit.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the endpoint translation orbit adjacency/PDEC cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-displacement-orbit-router.md.

□

**Theorem 7.112** (Stable-ladder endpoint orbit run-boundary reduction). *The endpoint translation-orbit adjacency outlet reduces to a long same-sign arc outlet or to a boundary-flux outlet. The open hard point*

*SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointTranslationOrbitAdjacencyPDECCap* is reduced to

```
StableLadderEndpointTranslationOrbitAdjacencyImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterOrbitRunLedger
AND StableLadderEndpointOrbitSignedCycleModelLedger
AND StableLadderEndpointOrbitDyadicEdgeCountLedger
AND StableLadderEndpointOrbitRunPartitionLedger
AND StableLadderEndpointOrbitAdjacencyRunBoundaryIdentityLedger
AND StableLadderEndpointOrbitRunBoundaryBudgetDichotomyLedger
AND StableLadderEndpointOrbitLongSameSignArcRegistrationLedger
AND StableLadderEndpointOrbitBoundaryFluxRegistrationLedger
AND NoAnonymousEndpointTranslationOrbitAdjacencyExitLedger
AND SparseScaleLadderSAECarriedForwardAfterOrbitRunBoundaryLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitLongSameSignArcSAEOrEndpointOrbit
```

*Endpoint orbit run-boundary ledger.* The endpoint singleton atom outlet and the sparse scale-ladder SAE outlet are carried forward. Consider the translation-orbit adjacency branch from the previous ledger. On the selected orbit write

$$C_r = \mathbf{Z}/r\mathbf{Z}, \quad z_t = a + t\delta.$$

Let  $u_t$  be the active indicator of  $z_t \in S$ , and let  $\sigma_t = \text{sgn } F(z_t)$ . A same-sign active edge is

$$e_t = 1 \iff u_t = u_{t+1} = 1, \quad \sigma_t = \sigma_{t+1}.$$

Let  $E = \sum_t e_t$ .

Because the packet is dyadic, there is a scale  $\lambda$  such that each active endpoint satisfies  $\lambda \leq |F(z_t)| < 2\lambda$ . Therefore the weighted adjacency mass  $W$  and the unweighted same-sign edge count obey

$$\lambda^2 E \leq W \leq 4\lambda^2 E.$$

Thus the capacity part of the translation-orbit adjacency obstruction may be tracked by  $E$ , up to an absolute dyadic constant.

Partition the active same-sign consecutive vertices into maximal cyclic runs. Let  $n$  be the number of active vertices and let  $b$  be the number of cuts, equivalently the number of runs unless the whole cycle is one same-sign run. If the whole cycle is one same-sign run, then the long-arc outlet is already present with length  $r$ . Otherwise the exact run identity is

$$E = n - b.$$

Consequently, for every boundary budget  $B \geq 1$ , either  $b > B$ , which is registered as the boundary-flux outlet, or  $b \leq B$ , in which case a pigeonhole bound gives a same-sign run of length

$$L_{\max} \geq \frac{n}{b} \geq \frac{E}{B}.$$

The anonymous translation-orbit adjacency outlet has therefore been replaced by two explicit controlled exits: a long same-sign arc on  $C_r$ , or many cycle cuts carrying boundary flux.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the long same-sign arc SAE outlet, it does not prove the boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-run-boundary-router.md.

□

**Theorem 7.113** (Stable-ladder endpoint orbit signed Fourier reduction). *The endpoint long-arc/boundary-flux outlet reduces to a full-cycle mean atom or to a signed Fourier cap on the endpoint orbit. The open hard point*

*SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitLongSameSignArcSAEOrEndpointOrbitBoun*

*is reduced to*

*StableLadderEndpointOrbitRunBoundaryImportedLedger*  
*AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterSignedFourierLedger*  
*AND StableLadderEndpointOrbitSignedIndicatorSequenceLedger*  
*AND StableLadderEndpointOrbitFullCycleMeanAtomRegistrationLedger*  
*AND StableLadderEndpointOrbitLongArcCenteredDiscrepancyLedger*  
*AND StableLadderEndpointOrbitArcDirichletKernelLedger*  
*AND StableLadderEndpointOrbitBoundaryDerivativeSupportLedger*  
*AND StableLadderEndpointOrbitBoundaryDerivativeFourierLedger*  
*AND StableLadderEndpointOrbitSignedFourierCapRegistrationLedger*  
*AND NoAnonymousEndpointOrbitLongArcBoundaryExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterOrbitSignedFourierLedger*  
*AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbit*

*Endpoint orbit signed Fourier ledger.* The endpoint singleton atom outlet and sparse scale-ladder SAE outlet are carried forward. On the endpoint orbit  $C_r = \mathbf{Z}/r\mathbf{Z}$ , define

$$g_t = \sigma_t u_t \in \{-1, 0, 1\}, \quad \bar{g} = \frac{1}{r} \sum_t g_t, \quad \nu_t = g_t - \bar{g}.$$

First suppose the long same-sign arc outlet is active. Take  $I$  to be the maximal cyclic run and let  $\epsilon \in \{\pm 1\}$  be its sign, so  $g_t = \epsilon$  for every  $t \in I$ . If  $I = C_r$ , this is registered as a full-cycle mean atom. Otherwise  $\epsilon \sum_t g_t \leq r - 1$ , because the maximal run has at least one endpoint outside  $I$  not equal to  $\epsilon$ . Hence

$$\Delta_\epsilon(I) = \epsilon \sum_{t \in I} \nu_t = |I| - \frac{|I|}{r} \epsilon \sum_t g_t \geq \frac{|I|}{r} > 0.$$

Since  $\nu$  has zero mean, Fourier inversion gives

$$\Delta_\epsilon(I) = \frac{\epsilon}{r} \sum_{h \neq 0} \widehat{g}(h) \widehat{\mathbf{1}_I}(-h).$$

Thus, with  $\Lambda_r(I) = \sum_{h \neq 0} |\widehat{\mathbf{1}_I}(h)|$ , some nonzero frequency satisfies

$$|\widehat{g}(h)| \geq \frac{r \Delta_\epsilon(I)}{\Lambda_r(I)}.$$

Now suppose the boundary-flux outlet is active with  $b$  cuts. Put

$$Dg_t = g_{t+1} - g_t.$$

Each cut gives a nonzero cyclic difference, so

$$\sum_t |Dg_t|^2 \geq b.$$

Parseval and the identity  $\widehat{Dg}(h) = (e^{2\pi i h/r} - 1)\widehat{g}(h)$  give

$$\sum_{h \neq 0} |(e^{2\pi i h/r} - 1)\widehat{g}(h)|^2 = r \sum_t |Dg_t|^2 \geq rb.$$

Therefore some  $h \neq 0$  satisfies

$$|(e^{2\pi i h/r} - 1)\widehat{g}(h)| \geq \sqrt{\frac{rb}{r-1}}.$$

Both active branches have become signed Fourier data on the endpoint orbit, except for the explicit full-cycle mean atom.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the signed Fourier PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-signed-fourier-router.md.

□

**Theorem 7.114** (Stable-ladder endpoint orbit conductor-character reduction). *The endpoint-orbit signed Fourier outlet reduces to a primitive conductor character packet. The open hard point*

*SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSi*

*is reduced to*

*StableLadderEndpointOrbitSignedFourierImportedLedger*  
*AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterConductorLedger*  
*AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardLedger*  
*AND StableLadderEndpointOrbitNonzeroFrequencyConductorLedger*  
*AND StableLadderEndpointOrbitFrequencyKernelQuotientLedger*  
*AND StableLadderEndpointOrbitKernelFiberCollapseLedger*  
*AND StableLadderEndpointOrbitPrimitiveConductorCharacterPacketLedger*  
*AND StableLadderEndpointOrbitDerivativeMultiplierAbsorbedLedger*  
*AND NoAnonymousEndpointOrbitSignedFourierExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterOrbitConductorLedger*  
*AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSi*

*Endpoint orbit conductor-character ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. Consider the signed Fourier branch on  $C_r = \mathbf{Z}/r\mathbf{Z}$ . Let  $h$  be the nonzero frequency. Put

$$d = \gcd(h, r), \quad m = r/d, \quad h_0 = h/d.$$

Then  $m > 1$  and  $\gcd(h_0, m) = 1$ . The character factors through the conductor quotient:

$$e^{-2\pi i h t/r} = e^{-2\pi i h_0 (t \bmod m)/m}.$$

Its kernel in  $C_r$  is

$$\{0, m, 2m, \dots, (d-1)m\},$$



of size  $d$ , and the quotient is  $C_m$ .

Collapse the signed load along these kernel fibers:

$$G_s = \sum_{u=0}^{d-1} g_{s+um}, \quad s \in \mathbf{Z}/m\mathbf{Z}.$$

Then the Fourier coefficient has the exact primitive-conductor form

$$\widehat{g}(h) = \sum_{s \bmod m} G_s e^{-2\pi i h_0 s/m}.$$

Because  $h_0$  is coprime to  $m$ , this is a primitive additive character packet on  $C_m$ .

If the previous lower bound came from the derivative Fourier branch, then

$$|(e^{2\pi i h/r} - 1)\widehat{g}(h)| \geq M$$

implies

$$|\widehat{g}(h)| \geq M/2,$$

since  $|e^{2\pi i h/r} - 1| \leq 2$ . Thus the long-arc Fourier and boundary derivative-Fourier branches both enter the same primitive conductor-character outlet, up to an absolute constant already recorded in the ledger.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the primitive conductor-character PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-conductor-character-router.md`.

□

**Theorem 7.115** (Stable-ladder endpoint orbit standard-character reduction). *The primitive conductor-character outlet reduces to a standard first-harmonic packet on the same conductor. The open hard point*

*SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitPr*

*is reduced to*

*StableLadderEndpointOrbitPrimitiveConductorCharacterImportedLedger*  
*AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterStandardCharacterLedger*  
*AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterStandardCharacterLedger*  
*AND StableLadderEndpointOrbitConductorUnitAutomorphismLedger*  
*AND StableLadderEndpointOrbitStandardPhaseCoordinateLedger*  
*AND StableLadderEndpointOrbitAutomorphicLoadPermutationLedger*  
*AND StableLadderEndpointOrbitFirstHarmonicIdentityLedger*  
*AND StableLadderEndpointOrbitLoadNormsAndSupportPreservedLedger*  
*AND NoAnonymousPrimitiveConductorFrequencyExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterStandardCharacterLedger*  
*AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb*

*Endpoint orbit standard-character ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. Consider the remaining primitive conductor packet on  $C_m = \mathbf{Z}/m\mathbf{Z}$ :

$$\sum_{s \bmod m} G_s e^{-2\pi i h_0 s/m}, \quad (h_0, m) = 1.$$

Let  $u$  be the inverse of  $h_0$  modulo  $m$ . Put

$$a = h_0 s \bmod m, \quad S_a = G_{ua \bmod m}.$$

Since multiplication by  $h_0$  is a permutation of  $C_m$ , this is an exact change of variables. Hence

$$\sum_{s \bmod m} G_s e^{-2\pi i h_0 s/m} = \sum_{a \bmod m} S_a e^{-2\pi i a/m}.$$

The permutation preserves support size,  $L^1$  and  $L^2$  mass, the total mean, and every lower-bound constant in the Fourier outlet. Thus the primitive frequency  $h_0$  is not a separate terminal obstruction: it is only a coordinate choice on the conductor group. The remaining outlet is the standard first-harmonic conductor packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the standard first-harmonic PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-standard-character-router.md.`

□

**Theorem 7.116** (Stable-ladder endpoint orbit axis-lobe reduction). *The standard first-harmonic conductor outlet reduces to an axis-lobe weighted surplus packet. The open hard point*

*SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSt*

*is reduced to*

*StableLadderEndpointOrbitStandardFirstHarmonicImportedLedger*  
*AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterAxisLobeLedger*  
*AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterAxisLobeLedger*  
*AND StableLadderEndpointOrbitComplexFirstHarmonicAmplitudeLedger*  
*AND StableLadderEndpointOrbitAxisProjectionDichotomyLedger*  
*AND StableLadderEndpointOrbitAxisSignChoiceLedger*  
*AND StableLadderEndpointOrbitTrigonometricLobeWeightLedger*  
*AND StableLadderEndpointOrbitAxisLobeWeightedSurplusPacketLedger*  
*AND NoAnonymousStandardFirstHarmonicExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterAxisLobeLedger*  
*AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSt*

*Endpoint orbit axis-lobe ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. The remaining standard first-harmonic packet has the form

$$F = \sum_{a \bmod m} S_a e^{-2\pi i a/m}, \quad |F| \geq M.$$

Its coordinate projections are

$$\Re F = \sum_a S_a \cos(2\pi a/m), \quad \Im F = - \sum_a S_a \sin(2\pi a/m).$$

Since  $|F|^2 = (\Re F)^2 + (\Im F)^2$ , one of these two real projections has absolute value at least  $M/\sqrt{2}$ . Define

$$\phi_0(a) = \cos(2\pi a/m), \quad \phi_1(a) = -\sin(2\pi a/m).$$

Thus for some  $j \in \{0, 1\}$  and some sign  $\epsilon \in \{\pm 1\}$ ,

$$\sum_a S_a \epsilon \phi_j(a) \geq M/\sqrt{2}.$$

Split the chosen trigonometric weight into positive and negative lobes:

$$w_a = \max(\epsilon \phi_j(a), 0), \quad v_a = \max(-\epsilon \phi_j(a), 0).$$

Then  $w$  and  $v$  are supported on opposite axis half-circles and

$$\sum_a S_a (w_a - v_a) \geq M/\sqrt{2}.$$

Hence the complex first-harmonic cap has been converted to a real axis-lobe weighted surplus cap with explicit axis, sign, and half-circle weights.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the axis-lobe weighted-surplus PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-axis-lobe-router.md.

□

**Theorem 7.117** (Stable-ladder endpoint orbit single-lobe reduction). *The axis-lobe weighted-surplus outlet reduces to a single-lobe signed surplus packet. The open hard point*

*SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitAxis*

*is reduced to*

*StableLadderEndpointOrbitAxisLobeWeightedSurplusImportedLedger  
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterSingleLobeLedger  
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterSingleLobeLedger  
AND StableLadderEndpointOrbitTwoLobeSurplusDecompositionLedger  
AND StableLadderEndpointOrbitHalfThresholdLossLedger  
AND StableLadderEndpointOrbitSingleLobeSignChoiceLedger  
AND StableLadderEndpointOrbitSingleLobeHalfCircleSupportLedger  
AND StableLadderEndpointOrbitSingleLobeSignedSurplusPacketLedger  
AND NoAnonymousAxisLobeDifferenceExitLedger  
AND SparseScaleLadderSAECarriedForwardAfterSingleLobeLedger  
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitAxis*

*Endpoint orbit single-lobe ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. The axis-lobe outlet has the form

$$\sum_a S_a (w_a - v_a) \geq L, \quad w_a \geq 0, \quad v_a \geq 0.$$

Put

$$A = \sum_a S_a w_a, \quad B = \sum_a S_a v_a.$$

Then  $A - B \geq L$ . If  $A \geq L/2$ , choose  $W = w$  and  $\eta = +1$ . If  $A < L/2$ , then  $-B > L/2$ , so choose  $W = v$  and  $\eta = -1$ . In either case,

$$\sum_a \eta S_a W_a \geq L/2.$$

The chosen  $W$  is one of the two trigonometric lobe weights, hence

$$0 \leq W_a \leq 1$$

and its support is a single axis half-circle. Thus the two-lobe difference has been converted, with only an absolute factor 2, into a single-lobe signed surplus packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the single-lobe signed-surplus PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-single-lobe-router.md.`

□

**Theorem 7.118** (Stable-ladder endpoint orbit single-arc reduction). *The single-lobe signed-surplus outlet reduces to a single-arc signed surplus packet. The open hard point*

*`SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSi`*

*is reduced to*

*`StableLadderEndpointOrbitSingleLobeSignedSurplusImportedLedger`  
`AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterSingleArcLedger`  
`AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterSingleArcLedger`  
`AND StableLadderEndpointOrbitSingleLobeLayerCakeIdentityLedger`  
`AND StableLadderEndpointOrbitLobeSuperlevelArcSupportLedger`  
`AND StableLadderEndpointOrbitLayerCakeArcPigeonholeLedger`  
`AND StableLadderEndpointOrbitSingleArcSignedSurplusPacketLedger`  
`AND NoAnonymousSingleLobeWeightExitLedger`  
`AND SparseScaleLadderSAECarriedForwardAfterSingleArcLedger`  
`AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb`*

*Endpoint orbit single-arc ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. The single-lobe outlet has the form

$$\sum_a \eta S_a W_a \geq L, \quad 0 \leq W_a \leq 1.$$

The layer-cake identity gives, pointwise,

$$W_a = \int_0^1 \mathbf{1}_{\{W_a \geq t\}} dt.$$

Therefore

$$\sum_a \eta S_a W_a = \int_0^1 \sum_{a: W_a \geq t} \eta S_a dt.$$

If every level set had signed mass below  $L$ , the integral would be below  $L$ . Hence for some  $t \in [0, 1]$ ,

$$\sum_{a: W_a \geq t} \eta S_a \geq L.$$

Because  $W$  is a single trigonometric lobe supported on an axis half-circle, each nonempty superlevel set

$$A_t = \{a \bmod m : W_a \geq t\}$$

is a cyclic arc inside that half-circle. Thus the continuous lobe weight has been replaced by one explicit single-arc signed surplus packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the single-arc signed-surplus PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-single-arc-router.md.`

□

**Theorem 7.119** (Stable-ladder endpoint orbit arc endpoint-potential reduction). *The single-arc signed-surplus outlet reduces to either the full-cycle mean atom outlet or an endpoint-potential gap packet. The open hard point*

*SparseScaleLaddersSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSi*

*is reduced to*

*StableLadderEndpointOrbitSingleArcSignedSurplusImportedLedger*  
*AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterArcEndpointPotentialLedger*  
*AND StableLadderEndpointOrbitArcSignedLoadSequenceLedger*  
*AND StableLadderEndpointOrbitArcMeanContributionDichotomyLedger*  
*AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterArcEndpointPotentialLedger*  
*AND StableLadderEndpointOrbitCenteredArcSurplusLedger*  
*AND StableLadderEndpointOrbitCenteredPrefixPotentialLedger*  
*AND StableLadderEndpointOrbitArcEndpointPotentialGapLedger*  
*AND StableLadderEndpointOrbitArcEndpointPotentialPacketLedger*  
*AND NoAnonymousSingleArcInteriorSurplusExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterArcEndpointPotentialLedger*  
*AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSi*

*Endpoint orbit arc endpoint-potential ledger.* The endpoint singleton atom outlet and the sparse scale-ladder SAE outlet are carried forward. The single-arc outlet has the form

$$\sum_{a \in A} X_a \geq L, \quad X_a = \eta S_a,$$

where  $A = [u, v)$  is a cyclic arc in  $C_m$ . Put

$$\mu = \frac{1}{m} \sum_{a \in C_m} X_a, \quad Y_a = X_a - \mu.$$

Then

$$\sum_{a \in A} X_a = |A|\mu + \sum_{a \in A} Y_a.$$

If the mean contribution satisfies  $|A|\mu \geq L/2$ , then, since  $|A| \leq m$ ,

$$\sum_{a \in C_m} X_a = m\mu \geq L/2,$$

and the packet is registered as a full-cycle mean atom outlet. Otherwise the centered part satisfies

$$\sum_{a \in A} Y_a \geq L/2.$$

Define the centered prefix potential on a lift of the orbit by

$$F(0) = 0, \quad F(j+1) = F(j) + Y_j.$$

Because  $\sum_{C_m} Y_a = 0$ , this potential is cyclic,  $F(m) = 0$ . For the arc  $A = [u, v)$ ,

$$\sum_{a \in A} Y_a = F(v) - F(u).$$

Thus the non-mean branch forces a pair of arc endpoints whose centered potential gap is at least  $L/2$ . The anonymous single-arc interior surplus has therefore been replaced by the named endpoint-potential PDEC/cap outlet, or by the full-cycle mean atom outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the endpoint-potential PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-arc-endpoint-potential-router.md`.

□

**Theorem 7.120** (Stable-ladder endpoint orbit potential-variation reduction). *The endpoint-potential outlet reduces to a signed-variation packet on actual orbit increments. The open hard point*

*is reduced to*

```
StableLadderEndpointOrbitArcEndpointPotentialImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterPotentialVariationLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterPotentialVariationLedger
AND StableLadderEndpointOrbitCenteredPotentialIncrementLedger
AND StableLadderEndpointOrbitEndpointGapDirectedArcLedger
AND StableLadderEndpointOrbitPositiveVariationLowerBoundLedger
AND StableLadderEndpointOrbitNegativeVariationLowerBoundLedger
AND StableLadderEndpointOrbitSignedVariationPacketLedger
AND NoAnonymousEndpointPotentialGapExitLedger
AND SparseScaleLadderSAECarriedForwardAfterPotentialVariationLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbit
```

*Endpoint orbit potential-variation ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. The endpoint-potential outlet has the form

$$F(v) - F(u) \geq G, \quad G = L/2.$$

The centered potential increments are

$$Y_j = F(j+1) - F(j), \quad \sum_{j \in C_m} Y_j = 0.$$

Let  $I = [u, v)$  and  $J = [v, u)$  be the two directed cyclic arcs. Then

$$\sum_{j \in I} Y_j = F(v) - F(u) \geq G$$

and, by the zero total increment condition,

$$\sum_{j \in J} Y_j = F(u) - F(v) \leq -G.$$

Hence the positive variation on  $I$  and the negative variation on  $J$  obey

$$\sum_{j \in I} (Y_j)_+ \geq G, \quad \sum_{j \in J} (Y_j)_- \geq G.$$

Thus the anonymous two-endpoint potential gap has been converted into a signed-variation obligation carried by actual adjacent orbit increments.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the signed-variation PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-potential-variation-router.md.

□

**Theorem 7.121** (Stable-ladder endpoint orbit variation run-boundary reduction). *The signed-variation outlet reduces to an increment-run surplus outlet or to a variation-boundary flux outlet. The open hard point*

*SparseScaleLaddersSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSi*

*is reduced to*

*StableLadderEndpointOrbitSignedVariationImportedLedger*  
*AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterVariationRunBoundaryLedger*  
*AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterVariationRunBoundaryLedger*  
*AND StableLadderEndpointOrbitVariationSignedSideChoiceLedger*  
*AND StableLadderEndpointOrbitPositiveIncrementEdgeSetLedger*  
*AND StableLadderEndpointOrbitVariationRunPartitionLedger*  
*AND StableLadderEndpointOrbitVariationRunBoundaryBudgetDichotomyLedger*  
*AND StableLadderEndpointOrbitIncrementRunSurplusPacketLedger*  
*AND StableLadderEndpointOrbitVariationBoundaryFluxPacketLedger*  
*AND NoAnonymousSignedVariationExitLedger*  
*AND SparseScaleLadderSAECarriedForwardAfterVariationRunBoundaryLedger*  
*AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSi*

*Endpoint orbit variation run-boundary ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. The signed-variation outlet gives a directed support  $K$ , a sign  $\sigma \in \{\pm 1\}$ , and

$$\sum_{j \in K} (\sigma Y_j)_+ \geq G.$$

Let

$$E_+ = \{j \in K : \sigma Y_j > 0\}.$$

Partition  $E_+$  into maximal consecutive runs  $R_1, \dots, R_b$  along the directed orbit. Given any boundary budget  $B \geq 1$ , there are two cases. If  $b > B$ , the packet has high sign-switch boundary flux and is registered as the variation-boundary flux PDEC/cap outlet. If  $b \leq B$ , then by the pigeonhole principle one run satisfies

$$\sum_{j \in R_h} \sigma Y_j \geq G/B.$$

This is the increment-run surplus outlet. Thus the anonymous signed variation has been converted into a named long same-sign increment run or a named boundary-flux packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the increment-run surplus SAE outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-variation-run-boundary-router.md.

□

**Theorem 7.122** (Stable-ladder endpoint orbit increment-run amplitude drift ledger). *The endpoint-orbit increment-run surplus outlet admits the following non-cyclic refinement:*

```
SparseScaleLaddersSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitIn
⇒ StableLadderEndpointOrbitIncrementRunSurplusImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterIncrementRunAmplitudeDriftLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterIncrementRunAmplitudeDriftLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterIncrementRunAmplitudeDriftLedger
AND StableLadderEndpointOrbitIncrementRunSameSignCoordinateLedger
AND StableLadderEndpointOrbitIncrementRunMassLowerBoundLedger
AND StableLadderEndpointOrbitIncrementRunAmplitudeThresholdDichotomyLedger
AND StableLadderEndpointOrbitIncrementEdgeSpikePacketLedger
AND StableLadderEndpointOrbitLongBoundedIncrementDriftPacketLedger
AND NoAnonymousIncrementRunSurplusExitLedger
AND SparseScaleLadderSAECarriedForwardAfterIncrementRunAmplitudeDriftLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbit
```

*Endpoint orbit increment-run amplitude-drift ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The low-boundary branch from the previous ledger gives a consecutive run  $R$  and a sign  $\sigma$  such that, with

$$Z_j = \sigma Y_j > 0 \quad (j \in R),$$

one has

$$\sum_{j \in R} Z_j \geq H, \quad H = G/B.$$

Fix any amplitude threshold  $\Lambda > 0$ . If

$$\max_{j \in R} Z_j \geq \Lambda,$$

then the packet contains a named increment edge-spike atom. Otherwise

$$0 < Z_j < \Lambda \quad (j \in R),$$

and the mass lower bound forces

$$|R| \geq H/\Lambda.$$

For thresholds with large  $H/\Lambda$ , this is a long bounded-increment monotone drift packet. Thus the anonymous increment-run surplus outlet has been converted into either an increment edge-spike atom or a long bounded-increment drift packet, while the high-boundary variation-flux outlet remains a named carried-forward outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the increment edge-spike SAE outlet, it does not prove the long bounded-increment drift PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in



docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-increment-run-amplitude-drift-router.md.

□

**Theorem 7.123** (Stable-ladder endpoint orbit edge-spike mean-singleton ledger). *The endpoint-orbit increment edge-spike outlet is absorbed by the named singleton and full-cycle mean outlets:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitIn
⇒ StableLadderEndpointOrbitIncrementEdgeSpikeImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterEdgeSpikeMeanSingletonLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterEdgeSpikeMeanSingletonLedger
AND StableLadderEndpointOrbitLongBoundedIncrementDriftCarriedForwardAfterEdgeSpikeMeanSingletonLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterEdgeSpikeMeanSingletonLedger
AND StableLadderEndpointOrbitEdgeSpikeCenteredLoadExpansionLedger
AND StableLadderEndpointOrbitEdgeSpikeHalfThresholdDichotomyLedger
AND StableLadderEndpointOrbitEdgeSpikeSingletonAtomAbsorptionLedger
AND StableLadderEndpointOrbitEdgeSpikeFullCycleMeanAtomAbsorptionLedger
AND NoAnonymousIncrementEdgeSpikeExitLedger
AND SparseScaleLadderSAECarriedForwardAfterEdgeSpikeMeanSingletonLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

*Endpoint orbit edge-spike mean-singleton ledger.* The long bounded-increment drift outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The edge-spike branch gives an index  $j$ , a sign  $\sigma$ , and a threshold  $\Lambda > 0$  such that

$$\sigma Y_j \geq \Lambda.$$

Here  $Y_j$  is the centered point load used in the previous ledgers, so

$$Y_j = X_j - \mu,$$

where  $X_j$  is the uncentered signed point load and  $\mu$  is the full-cycle mean. Hence

$$\sigma X_j - \sigma \mu \geq \Lambda.$$

The elementary half-threshold dichotomy gives

$$\sigma X_j \geq \Lambda/2 \quad \text{or} \quad -\sigma \mu \geq \Lambda/2.$$

In the first case the spike is already an endpoint singleton atom. In the second case it is already a full-cycle mean atom. Thus the edge-spike outlet does not remain as an independent anonymous terminal; it is absorbed by the two already named outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the long bounded-increment drift PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-edge-spike-mean-singleton-router.md.

□

**Theorem 7.124** (Stable-ladder endpoint orbit long-drift dyadic plateau ledger). *The endpoint-orbit long bounded-increment drift outlet admits the following non-cyclic dyadic refinement:*

```

SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitLo
⇒ StableLadderEndpointOrbitLongBoundedIncrementDriftImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterLongDriftDyadicPlateauLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterLongDriftDyadicPlateauLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterLongDriftDyadicPlateauLedger
AND StableLadderEndpointOrbitLongDriftPositiveBoundedRunLedger
AND StableLadderEndpointOrbitLongDriftDyadicAmplitudePartitionLedger
AND StableLadderEndpointOrbitLongDriftAmplitudeDepthBudgetDichotomyLedger
AND StableLadderEndpointOrbitLongDriftHeavyDyadicBandLedger
AND StableLadderEndpointOrbitDyadicBandPlateauRunPartitionLedger
AND StableLadderEndpointOrbitDyadicPlateauRunBoundaryDichotomyLedger
AND StableLadderEndpointOrbitComparableAmplitudePlateauDriftPacketLedger
AND StableLadderEndpointOrbitAmplitudeDepthPacketLedger
AND NoAnonymousLongBoundedIncrementDriftExitLedger
AND SparseScaleLadderSAECarriedForwardAfterLongDriftDyadicPlateauLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitLo

```

*Endpoint orbit long-drift dyadic plateau ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The long bounded-increment drift branch gives a consecutive run  $R$  such that

$$0 < Z_j < \Lambda \quad (j \in R), \quad \sum_{j \in R} Z_j \geq H.$$

For  $l \geq 0$ , set

$$A_l = \{j \in R : 2^{-(l+1)}\Lambda \leq Z_j < 2^{-l}\Lambda\}.$$

Since  $R$  is finite, only finitely many  $A_l$  are nonempty. Given a depth budget  $D$ , if the number of nonempty dyadic bands is greater than  $D$ , the packet is registered as an amplitude-depth PDEC/cap outlet. Otherwise, by pigeonhole, some band satisfies

$$\sum_{j \in A_l} Z_j \geq H/D.$$

Partition this heavy dyadic band into maximal consecutive plateau runs inside  $R$ . Given a plateau boundary budget  $B$ , if the number of plateau runs is greater than  $B$ , this is a boundary-flux outlet. Otherwise some plateau run  $P_s$  satisfies

$$\sum_{j \in P_s} Z_j \geq H/(DB).$$

All increments in  $P_s$  lie in the same dyadic band and are comparable within a factor less than two. Thus the anonymous long bounded-increment drift outlet has been converted into a comparable-amplitude plateau drift, an amplitude-depth packet, or a boundary-flux packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the comparable-amplitude plateau drift PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-long-drift-dyadic-plateau-router.md.

□

**Theorem 7.125** (Stable-ladder endpoint orbit plateau-ramp ledger). *The comparable-amplitude plateau-drift outlet admits the following non-cyclic ramp refinement:*

```

SparseScaleLaddersSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitCo
⇒ StableLadderEndpointOrbitComparableAmplitudePlateauDriftImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterPlateauRampLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterPlateauRampLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterPlateauRampLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterPlateauRampLedger
AND StableLadderEndpointOrbitComparablePlateauSameSignLoadLedger
AND StableLadderEndpointOrbitPlateauLengthBudgetDichotomyLedger
AND StableLadderEndpointOrbitShortPlateauEdgeSpikeAbsorptionLedger
AND StableLadderEndpointOrbitLongPlateauMonotoneRampPacketLedger
AND NoAnonymousComparableAmplitudePlateauDriftExitLedger
AND SparseScaleLadderSAECarriedForwardAfterPlateauRampLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb

```

*Endpoint orbit plateau-ramp ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The comparable-amplitude plateau branch gives a consecutive arc  $P_s$ , a sign  $\sigma$ , and a scale  $a > 0$  such that

$$Z_j = \sigma Y_j > 0, \quad a \leq Z_j < 2a \quad (j \in P_s), \quad \sum_{j \in P_s} Z_j \geq H_0.$$

Fix a length budget  $L_0 \geq 1$ . If  $|P_s| \leq L_0$ , then

$$\max_{j \in P_s} Z_j \geq H_0/L_0.$$

This is an edge-spike at the induced threshold. By the already recorded centered-load expansion  $Y_j = X_j - \mu$  and the half-threshold dichotomy, this short-plateau branch is absorbed by the endpoint singleton atom outlet or by the full-cycle mean atom outlet.

It remains to consider  $|P_s| > L_0$ . On this branch the prefix potential

$$S(t) = \sum_{i \leq t} \sigma Y_i$$

increases monotonically along the plateau arc and its total increase across  $P_s$  is at least  $H_0$ . Thus the anonymous comparable-amplitude plateau drift has been converted into a long monotone plateau-ramp potential packet. The next direct task is to exclude this ramp potential packet, or to show that it necessarily enters the already named amplitude-depth or boundary-flux controlled exits.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the plateau-ramp potential PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-plateau-ramp-router.md.

□

**Theorem 7.126** (Stable-ladder endpoint orbit plateau-return mirror ledger). *The plateau-ramp potential outlet admits the following non-cyclic return-mirror refinement:*

```

SparseScaleLaddersSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitPL
⇒ StableLadderEndpointOrbitPlateauRampPotentialImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterPlateauReturnMirrorLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterPlateauReturnMirrorLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterPlateauReturnMirrorLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterPlateauReturnMirrorLedger

```

```

AND StableLadderEndpointOrbitPositivePlateauRampImportedModelLedge
AND StableLadderEndpointOrbitFullCycleZeroSumReturnObligationLedge
AND StableLadderEndpointOrbitNegativeReturnMassLowerBoundLedge
AND StableLadderEndpointOrbitReturnMassDyadicAmplitudePartitionLedge
AND StableLadderEndpointOrbitReturnAmplitudeDepthDichotomyLedge
AND StableLadderEndpointOrbitHeavyReturnDyadicBandLedge
AND StableLadderEndpointOrbitReturnPlateauRunPartitionLedge
AND StableLadderEndpointOrbitReturnRunBoundaryDichotomyLedge
AND StableLadderEndpointOrbitOppositeSignPlateauRampPairPacketLedge
AND NoAnonymousPlateauRampPotentialExitLedge
AND SparseScaleLadderSAECarriedForwardAfterPlateauReturnMirrorLedge
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb

```

*Endpoint orbit plateau-return mirror ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The plateau-ramp branch gives a consecutive arc  $P$  and a sign  $\sigma$  such that

$$Z_j = \sigma Y_j > 0 \quad (j \in P), \quad \sum_{j \in P} Z_j \geq H_0.$$

Because  $Y$  is the centered orbit load, the full-cycle identity gives

$$\sum_{\text{cycle}} \sigma Y_j = 0.$$

Thus the complementary arc must carry return mass. With

$$W_j = (-\sigma Y_j)_+$$

on the complement of  $P$ , we have

$$\sum W_j \geq H_0.$$

Partition this return mass into dyadic amplitude bands. Given a depth budget  $D$ , if the number of nonempty return bands is greater than  $D$ , the packet enters the amplitude-depth PDEC/cap outlet. Otherwise one return band carries mass at least  $H_0/D$ . Partition that heavy return band into maximal consecutive return plateau runs. Given a boundary budget  $B$ , if the number of return runs is greater than  $B$ , the packet enters the variation-boundary flux outlet. Otherwise some negative return plateau  $Q$  satisfies

$$\sum_{j \in Q} W_j \geq H_0/(DB).$$

Consequently a plateau-ramp potential cannot remain as a one-sided anonymous growth packet: it is either an amplitude-depth packet, a boundary-flux packet, or a paired positive-ramp/negative-return mirror packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the opposite-sign plateau-ramp pair PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-plateau-return-mirror-router.md.

□

**Theorem 7.127** (Stable-ladder endpoint orbit plateau-pair phase ledger). *The opposite-sign plateau-ramp pair outlet admits the following non-cyclic phase refinement:*

```

SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitOp
⇒ StableLadderEndpointOrbitOppositeSignPlateauRampPairImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterPlateauPairPhaseLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterPlateauPairPhaseLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterPlateauPairPhaseLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterPlateauPairPhaseLedger
AND StableLadderEndpointOrbitOppositeSignPlateauPairModelLedger
AND StableLadderEndpointOrbitPlateauPairCyclicGapDecompositionLedger
AND StableLadderEndpointOrbitPlateauPairGapBudgetDichotomyLedger
AND StableLadderEndpointOrbitNearContactOppositeSignBoundaryPacketLedger
AND StableLadderEndpointOrbitPhaseSeparatedBipolarPlateauPairPacketLedger
AND NoAnonymousOppositeSignPlateauRampPairExitLedger
AND SparseScaleLadderSAECarriedForwardAfterPlateauPairPhaseLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb

```

*Endpoint orbit plateau-pair phase ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The opposite-sign plateau pair gives two disjoint consecutive arcs  $P$  and  $Q$ , with a sign  $\sigma$ , such that

$$\sigma Y_j > 0 \quad (j \in P), \quad -\sigma Y_j > 0 \quad (j \in Q).$$

The two arcs determine two cyclic gaps  $g_1, g_2$  between them. Fix a gap budget  $E$ . If

$$\min(g_1, g_2) \leq E,$$

then the positive and negative plateau packets meet through a short bridge; this is recorded as a near-contact opposite-sign boundary packet and is carried by the variation-boundary flux outlet. Otherwise

$$g_1 > E, \quad g_2 > E,$$

so the two packets form a genuinely phase-separated bipolar plateau pair.

Thus the opposite-sign pair is no longer anonymous. It is either a near-contact boundary packet, or a phase-separated bipolar packet with explicit cyclic separation.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the phase-separated bipolar plateau pair PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-plateau-pair-phase-router.md.

□

**Theorem 7.128** (Stable-ladder endpoint orbit bipolar-shelf ledger). *The phase-separated bipolar plateau-pair outlet admits the following non-cyclic bridge refinement:*

```

SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitPh
⇒ StableLadderEndpointOrbitPhaseSeparatedBipolarPlateauPairImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterBipolarShelfLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterBipolarShelfLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterBipolarShelfLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterBipolarShelfLedger
AND StableLadderEndpointOrbitPhaseSeparatedBipolarPairModelLedger
AND StableLadderEndpointOrbitBipolarPrefixPotentialCoordinateLedger
AND StableLadderEndpointOrbitBipolarBridgeDecompositionLedger
AND StableLadderEndpointOrbitBridgeCancellationOrShelfDichotomyLedger
AND StableLadderEndpointOrbitBridgeCancellationPacketLedger
AND StableLadderEndpointOrbitLongPotentialShelfPacketLedger
AND NoAnonymousPhaseSeparatedBipolarPlateauPairExitLedger
AND SparseScaleLadderSAECarriedForwardAfterBipolarShelfLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb

```

*Endpoint orbit bipolar-shelf ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The phase-separated bipolar pair gives a positive plateau  $P$ , a negative plateau  $Q$ , and a sign  $\sigma$ , with both cyclic gaps between them greater than  $E$ . Define the directed prefix potential

$$S(t) = \sum_{i \leq t} \sigma Y_i.$$

The arc  $P$  raises  $S$  by at least the packet height  $H$ , while  $Q$  returns  $S$  by at least the same scale. Consider the bridge from the end of  $P$  to the start of  $Q$ . If this bridge changes  $S$  downward by at least  $H/2$  before the packet reaches  $Q$ , then the bridge itself is a named bridge-cancellation packet. Otherwise  $S$  remains displaced by at least  $H/2$  throughout a bridge of length greater than  $E$ , giving a long potential shelf. The bridge from  $Q$  back to  $P$  is treated in the same way with the sign reversed.

Thus a phase-separated bipolar pair is not an anonymous terminal. It must produce either explicit bridge cancellation or a long potential shelf, unless it has already entered amplitude-depth or variation-boundary flux.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the long potential shelf PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-bipolar-shelf-router.md.

□

**Theorem 7.129** (Stable-ladder endpoint orbit long-potential-shelf static-bias ledger). *The long-potential-shelf outlet admits the following non-cyclic internal refinement:*

```

SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitLongPotentialShelfImportedLedger
⇒ StableLadderEndpointOrbitLongPotentialShelfImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterShelfStaticBiasLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterShelfStaticBiasLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterShelfStaticBiasLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterShelfStaticBiasLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterShelfStaticBiasLedger
AND StableLadderEndpointOrbitLongPotentialShelfModelLedger
AND StableLadderEndpointOrbitShelfPotentialFloorLedger
AND StableLadderEndpointOrbitShelfIncrementBalanceLedger
AND StableLadderEndpointOrbitShelfOscillationBoundaryFluxDichotomyLedger
AND StableLadderEndpointOrbitShelfPositiveDriftAmplitudeDepthDichotomyLedger
AND StableLadderEndpointOrbitShelfNegativeCancellationDichotomyLedger
AND StableLadderEndpointOrbitStaticPotentialShelfBiasPacketLedger
AND NoAnonymousLongPotentialShelfExitLedger
AND SparseScaleLadderSAECarriedForwardAfterShelfStaticBiasLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitLongPotentialShelfImportedLedger

```

*Endpoint orbit long-potential-shelf static-bias ledger.* The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A long potential shelf gives an interval  $I$  of length  $L > E$  and a sign  $\sigma$  such that

$$\sigma S(t) \geq H/2 \quad (t \in I).$$

Writing  $d_t = S(t+1) - S(t)$ , the internal motion on  $I$  has only three named ways to avoid becoming static. If the boundary switches or total variation exceed the active budget, it is a variation-boundary flux packet. If same-sign drift keeps increasing the potential through a dyadic depth threshold, it is an amplitude-depth packet. If opposite-sign increments cancel a fixed part of the height before the opposite plateau is reached, it is a bridge-cancellation packet.

If none of these three alternatives occurs, the shelf has a one-sided potential floor, low oscillation, no deep same-sign drift, and no early opposite-sign cancellation. This remaining case is precisely the named static potential shelf bias packet. Thus a long potential shelf is no longer an anonymous terminal.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the static potential shelf bias PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-long-potential-shelf-static-bias-router.md`.

□

**Theorem 7.130** (Internal line-by-line audit). *The author-side line-by-line audit has found no new item classified as a mathematical block. The remaining blocks are referee blocks: Prime Matrix final promotion requires independent acceptance of the D-structure/Tail-log4/finite-verification interface, and RH final promotion requires independent acceptance of all controlled exits in the RH contradiction-field manuscript.*

*Proof.* The internal audit matrix assigns one of four labels to each link. No link is a math-block.

The terminal entries PM-16 and RH-16 remain referee-blocks. An author-side check cannot replace independent referee verification. Therefore the internal audit is complete, but final unconditional theorem promotion is not. □

**Remark 7.131** (No overclaim principle). The monograph may state the shared contradiction-field method and the completed audit closures. It must not state unconditional proofs of RH or of the prime-matrix row/column theorem until the remaining structured inputs have independent acceptance.





## Chapter 8

# Current Status and Honest Boundary

The purpose of this monograph draft is to make the shared structure visible and auditable. It is not yet a final unconditional theorem manuscript. The precise statuses are maintained in `docs/monograph/claim-status-table.md`. Any future promotion to final theorem status requires independent verification of every reduced structured input and every controlled exit.